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An Introduction to Univalent Functions

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Rosihan M. Ali · See Keong Lee · V. Ravichandran

An Introduction to Univalent Functions

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With the mercy of God and His blessings.

—Rosihan M. Ali

To my parents, siblings, teachers and friends.

—See Keong Lee

To my teachers at National College, Trichy—especially Profs. K. Raghavan, R. Ganapathy, M. A. Gopalan, and A. Krishnamoorthy—who nurtured my love of mathematics.

—V. Ravichandran

Preface

This book is an exposition of analytic complex-valued functions that are univalent (one-to-one) in a planar domain. When the domain is simply connected and not the entire complex plane, the Riemann mapping theorem asserts that these functions may be assumed to take the form $f(z) = z + \sum_{n=2}^{\infty} a_n z^n$ in the unit disk $\mathbb{D}$. The works of P. Koebe in 1907 and L. Bieberbach in the early-twentieth century are among the pioneering works which inspired the development of univalent function theory.

These functions f are often grouped based on geometric considerations. Notably, these include the domain $f(\mathbb{D})$ which is convex, starlike (with respect to the origin), or close-to-convex. Indeed, these are the three major subclasses of univalent functions treated in this book. The geometric characterizations of these functions have useful equivalent analytic criteria, which are formulated in terms of normalized analytic functions with positive real part in $\mathbb{D}$. Satisfying the analytic criterion to be convex or starlike also leads to the univalence of the function. Bounded analytic functions form another important subclass which has a close relation to the subclasses of geometric univalent functions. A principal tool often used in function theory is the method of differential subordination. This tool is applied in the book to classes of analytic functions with positive real part, bounded functions, and Caratheódory functions. In particular, differential subordination is used to deduce sufficient conditions for functions to be convex or starlike.

When a subclass contains functions that map $\mathbb{D}$ onto non-convex (or non-starlike) domains, we seek to find the largest subdisk in $\mathbb{D}$ so that every function in the class maps the subdisk onto a convex domain. Such a radius problem, which includes the radius of starlikeness and the radius of close-to-convexity, is also given emphasis in this book.

An analytic univalent function $f(z) = z + \sum_{n=2}^{\infty} a_n z^n$ in the unit disk $\mathbb{D}$ strongly influenced its coefficients a_n. Bieberbach in 1916 proved that the second coefficient of every such function satisfies $|a_2| \leq 2$. It is remarkable that such a simple result has profound implications on the growth and distortion estimates, as well as exact covering radius, for every analytic univalent function. Since equality for the second coefficient bound is attained by the Koebe function $f(z) = z/(1-z)^2 = z + \sum_{n=2}^{\infty} n z^n$, Bieberbach set forth his famous conjecture that $|a_n| \leq n$ with equality for the Koebe

function (or its rotations). His conjecture stood for more than six decades before L. de Branges finally proved its validity in 1985. Interestingly, his proof used tools in topics outside classical geometric function theory which helped expand the interactions and the body of knowledge in the field.

In this book, bounds of coefficients for functions in the major classes of univalent functions are shown to be sharp with explicit description of the extremal functions. In the opposite direction, monotonic conditions on the coefficients are obtained which ensure that a function is close-to-convex. Coefficient problems related to the Fekete-Szegö inequality, the Hankel determinant and the Bohr inequality for bounded functions are also investigated.

This book is intended for early career researchers and students. It places emphasis on clarity to the fundamentals underlying the rich history of geometric function theory. Recent pertinent advances have also been included. In addition, we have endeavored to provide a proof to practically every result in the book. Thus the book is self-contained with a minimal prerequisite required. A good grounding in an advanced undergraduate complex analysis course would be adequate.

In writing the book, we have drawn from our wide experience as co-authors of scores of research articles on several different topics in the field. This book is another outcome from our continued collaboration, which arose from facilitating courses in geometric function theory to research scholars at Universiti Sains Malaysia, the University of Delhi, and the National Institute of Technology Tiruchirappalli over several years. Certain parts of this book have also been previously published as survey articles in the Newsletter of the Ramanujan Mathematical Society. We are hopeful that this book will be useful for an easy passage into the subject of univalent function theory.

We wish to express our deepest gratitude to several individuals who have provided us with their kind assistance and continued encouragement. We are indebted to Dr. Sumit Nagpal and his research team for their meticulous reading of the manuscript and invaluable feedback; in fact, they have perused the draft for their series of seminars. We thank Mr. K. Durai, Ms. Neenu Jose, Ms. A. Lakshmipriya and Ms. S. Rajalakshmi for proof-reading the manuscript. Our appreciation also goes to Drs. Asha Sebastian, Adiba Naz, Vibha Madaan, Priya G. Krishnan, S. Madhumitha, R. Kanaga, Somya, and Shalu Yadav for assistance in initial typesetting. The authors thank the anonymous referee for the brief useful comments on the manuscript. Finally, our heartfelt gratitude to Mr. Shamim Ahmad, Publishing Editor and Mr. Kamesh Senthilkumar, Production Editor at Springer, for their dedicated patience and support.

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Nomenclature

$\mathcal{H}[a,\ n]$	Class of all analytic functions of the form $f(z) = a + \sum_{k=n}^{\infty} a_k z^k$
$[z_1,\ z_2]$	The line segment joining z_1 and z_2
$\mathcal{CC}$	Class of close-to-convex functions
$\mathcal{CC}$	Class of close-to-star functions
$\mathcal{CC}_\alpha$	Class of close-to-convex functions with parameter α
$\mathcal{CCS}$	Class of functions close-to-convex wrt symmetric points
$\mathcal{CV}$	Class of convex functions
$\mathcal{CV}(\alpha)$	Class of convex functions of order α
$\mathcal{CVS}$	Class of functions convex wrt symmetric points
γk	Logarithmic coefficients of f
κ	The Koebe function $\kappa(z) = z/(1-z)^2$
$\mathbb{D}$	The unit disk
$\mathbb{D}(a;\ r)$	Disk of radius r centered at $z = a$
$\mathcal{A}$	The class of analytic functions f on $\mathbb{D}$ normalized by $f(0) = 0 = f'(0) - 1$
$\mathcal{B}$	Bounded analytic functions
$\mathcal{B}_0$	Bounded functions or Schur functions
$\mathcal{H}^2$	The Hardy-Hilbert space
$\mathcal{K}(\alpha,\ \beta)$	Kaplan classes
$\mathcal{P}$	The class of functions with positive real part
$\mathcal{P}(\alpha)$	The class of functions with real part $> \alpha$
$\mathcal{Q}$	Functions with nice boundary
$\mathcal{R}$	The class of functions whose derivative has positive real part
$\mathcal{S}$	Subclass of $\mathcal{A}$ consisting of univalent functions
$\mathcal{S}^{(n)}$	Class of n-fold symmetric univalent functions
$\prec$	Is subordinate to
$\Psi_n(\Omega)$	Class of admissible functions
$\Psi_n(\Omega,\ q)$	Class of admissible functions
$\mathcal{ST}$	Class of starlike functions
$\mathcal{ST}(\alpha)$	Class of starlike functions of order α
$\mathcal{STS}$	Class of functions starlike wrt symmetric points

φ_a	The Möbius transformation $(a - z)/(1 - \overline{a}z)$
$f\,,\ z$	Schwarzian derivative of f
K_w	Curvature of image of a curve
$M_p(r, f)$	Integral mean $M_p(r, f)$ of f

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Chapter 1
Analytic Functions

Abstract A function differentiable at each point in an open subset G of the complex plane is said to be analytic in G. Sections 1.1 and 1.2 review basic concepts and characteristics of analytic functions. Sections 1.3 and 1.4 discuss the space of continuous functions and analytic functions. The chapter concludes with a proof of the Riemann mapping theorem.

1.1 Analytic Functions

Denote by $\mathbb{C}$ the set consisting of all complex numbers and the set $\mathbb{C}_\infty := \mathbb{C} \cup \{\infty\}$ be the extended complex plane. For $a \in \mathbb{C}$ and $r > 0$, the sets

$$\mathbb{D}(a; r) := \{z \in \mathbb{C} : |z - a| < r\} \quad \text{and} \quad \overline{\mathbb{D}}(a; r) := \{z \in \mathbb{C} : |z - a| \leqslant r\}$$

are, respectively, called the *open disk* and the *closed disk* of radius r centered at $z = a$. Its boundary $\partial\mathbb{D}(a; r)$ is the circle $\partial\mathbb{D}(a; r) := \{z \in \mathbb{C} : |z - a| = r\}$. Let $\mathbb{D}_r := \mathbb{D}(0; r)$. Then the set $\mathbb{D} := \mathbb{D}_1$ is the *open unit disk* in $\mathbb{C}$. A subset G of $\mathbb{C}$ is called open if, associated to each $a \in G$, there is a $r = r(a) > 0$ such that $\mathbb{D}(a; r) \subset G$. A *neighborhood* of a point $a \in \mathbb{C}$ is an open set that contains the point a. The intersection of all open sets containing A is its *interior*, denoted by int A. A set A is *open* if and only if $A = \text{int } A$. The arbitrary union and the finite intersection of open sets are open.

A subset F of $\mathbb{C}$ is *closed* if its complement $\mathbb{C} \setminus F$ is open. By de Morgan's laws, it follows that an arbitrary intersection and finite union of closed sets are closed. A complex number $a \in \mathbb{C}$ is a *limit point* of a set $A \subset \mathbb{C}$ if every open disk $\mathbb{D}(a; r)$, with $r > 0$, contains a point of A distinct from a. The set of all limit points of A is denoted by A', and the *closure* $\overline{A}$ of A is defined by $\overline{A} = A \cup A'$. It is a known fact that a subset F of $\mathbb{C}$ is closed if and only if it contains all its limit points: $F' \subset F$ or, equivalently, $F = \overline{F}$.

A set $A \subset \mathbb{C}$ is *bounded* if there is a number M such that $|z| \leqslant M$ for all $z \in A$. A closed and bounded subset of $\mathbb{C}$ is known to be a *compact* subset of $\mathbb{C}$. If a subset of $\mathbb{C}$ cannot be written as a union of two non-empty disjoint open sets, then it is a

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connected subset of $\mathbb{C}$. It can be shown that an open subset G of $\mathbb{C}$ is connected if and only if for each pair of points $a, b \in G$, there is a polygonal path from a to b that lies entirely inside G. Indeed, the line segments that form the polygonal path can be chosen to be parallel to either the real or the imaginary axes. An open connected set is called a region.

A sequence $\langle z_n \rangle$ of complex numbers *converges* to a number $z \in \mathbb{C}$, written $\lim_{n\to\infty} z_n = z$ (or $z_n \to z$ as $n \to \infty$), if, for each $\epsilon > 0$, there is a natural number $N = N(\epsilon)$ such that

$$|z_n - z| < \epsilon \quad \text{for all } n \geqslant N.$$

It is a *Cauchy sequence* if, for each $\epsilon > 0$, there is a natural number $N = N(\epsilon)$ such that

$$|z_n - z_m| < \epsilon \quad \text{for all } n, m \geqslant N.$$

A sequence in $\mathbb{C}$ is convergent if and only if it is a Cauchy sequence. An infinite series (or simply a series) $\sum_{k=0}^{\infty} z_k$ *converges* if the sequence of partial sums $\langle \sum_{k=0}^{n} z_k \rangle$ converges; otherwise, it is said to diverge. The series $\sum_{k=0}^{\infty} z_k$ is *absolutely convergent* if $\sum_{k=0}^{\infty} |z_k|$ is convergent. Evidently, every absolutely convergent series is convergent.

A sequence $\langle f_n \rangle$ of functions $f_n : G \to \mathbb{C}$ is said to *converge pointwise* to a function $f : G \to \mathbb{C}$ if, for each $z \in G$ and $\epsilon > 0$, there is a natural number $N = N(\epsilon, z)$ such that

$$|f_n(z) - f(z)| < \epsilon \quad \text{for all } n \geqslant N.$$

For most purposes, we need a stronger notion wherein the choice of N depends only on ϵ and is independent of the point z. The sequence $\langle f_n \rangle$ of functions $f_n : G \to \mathbb{C}$ is said to *converge uniformly* to a function $f : G \to \mathbb{C}$ if, for each $\epsilon > 0$, there is a natural number $N = N(\epsilon)$ such that

$$|f_n(z) - f(z)| < \epsilon \quad \text{for all } n \geqslant N \text{ and for all } z \in G.$$

For the series of functions $\sum_{k=0}^{\infty} f_k(z)$, uniform convergence is defined in terms of the uniform convergence of the sequence of partial sums $\langle \sum_{k=0}^{n} f_k(z) \rangle$.

Theorem 1.1.1 (Weierstrass M-test) *For each $n \in \mathbb{N}_0 := \mathbb{N} \cup \{0\}$, let $f_n : G \to \mathbb{C}$ be a bounded function satisfying $|f_n(z)| \leqslant M_n$ for all $z \in G$ where M_n is a constant. If the series $\sum_{k=0}^{\infty} M_k$ converges, then the series $\sum_{k=0}^{\infty} f_k(z)$ converges uniformly on G.*

A power series centered at $z = a$ is the series $\sum_{k=0}^{\infty} a_k (z-a)^k$. Every power series is convergent in some disk centered at $z = a$ and divergent outside it; the radius of such disk is called the *radius of convergence*. The radius of convergence R of the power series $\sum_{k=0}^{\infty} a_k (z-a)^k$ is given by

$$\frac{1}{R} = \limsup_{k\to\infty} |a_k|^{1/k}.$$

The power series converges uniformly in the disk $\overline{\mathbb{D}}(a; r)$ for each r with $0 < r < R$. Here the *limit superior* of a sequence $\langle a_n \rangle$ of real numbers is given by

$$\limsup_{k\to\infty} a_k = \lim_{k\to\infty} (\sup\{a_k, a_{k+1}, a_{k+2}, \dots\}).$$

The limit superior always exists. However, that is not true for $\lim_{k\to\infty} |a_k|^{-1/k}$. If it exists, it equals the radius of convergence. Similarly, the limit $\lim_{k\to\infty}(|a_k|/|a_{k+1}|)$ whenever it exists yields the radius of convergence of the power series $\sum_{k=0}^{\infty} a_k(z - a)^k$.

Let G be an open subset of $\mathbb{C}$ and $z_0 \in G$. The *limit* of a function $f : G \to \mathbb{C}$ as $z \to z_0$ is the number l, written as

$$\lim_{z\to z_0} f(z) = l \quad \text{or} \quad f(z) \to l \text{ as } z \to z_0,$$

if, for a given $\epsilon > 0$, there is a $\delta = \delta(\epsilon, z_0) > 0$ such that

$$|f(z) - l| < \epsilon \text{ for every } z \in G \text{ with } 0 < |z - z_0| < \delta.$$

The number l is unique whenever it exists. The limit of a function can exist at z_0 even if the function is undefined at $z = z_0$. The function f is *continuous at the point* $z = z_0$ if $\lim_{z\to z_0} f(z) = f(z_0)$, or equivalently, for a given $\epsilon > 0$, there is a $\delta = \delta(\epsilon, z_0) > 0$ such that

$$|f(z) - f(z_0)| < \epsilon \quad \text{whenever } z \in G,\ |z - z_0| < \delta.$$

The function f is *continuous in* G if it is continuous at every point of G. The function f is *uniformly continuous in* G if, for a given $\epsilon > 0$, there is a $\delta = \delta(\epsilon) > 0$ such that

$$|f(z) - f(w)| < \epsilon \quad \text{whenever } z, w \in G \text{ and } |z - w| < \delta.$$

A continuous function $f : G \to \mathbb{C}$ is called a *branch of logarithm on the open set* G (or a branch of $\log z$ on G) if $\mathrm{e}^{f(z)} = z$ for all $z \in G$. Since the exponential function is non-vanishing, it is clear that $\mathrm{e}^{f(z)} = z$ cannot hold for $z = 0$. Thus, a branch of logarithm cannot be defined on any open set containing the origin. If there exists a branch f of $\log z$ on G, then the totality of all branches of $\log z$ are the functions $f + 2\pi k\mathrm{i}$, $k \in \mathbb{Z}$. For $a \in \mathbb{C}$, the power function z^a is defined by $z^a = \mathrm{e}^{a \log z}$, where $\log z$ is a branch of logarithm on G. For example, when a is not an integer, z^a is defined in $\mathbb{D} \setminus \{z : -1 < z \leqslant 0\}$ but not in $\mathbb{D}$.

The *derivative* of the function $f : G \to \mathbb{C}$ at a point $z_0 \in G$, denoted by $f'(z_0)$, is given by

$$f'(z_0) = \lim_{z\to z_0} \frac{f(z) - f(z_0)}{z - z_0} = \lim_{\Delta z\to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

whenever the limit exists. The function $f : G \to \mathbb{C}$ is *differentiable at* z_0 if it has a derivative at z_0. The function is analytic at a point z_0 if it is differentiable at all points in some neighborhood of the point z_0; it is analytic in a domain G if it is differentiable at every point of G. A function analytic in $\mathbb{C}$ is called an *entire function*.

The Looman-Menchoff theorem (see [7]) states that a function $f : G \to \mathbb{C}$, given by $f(z) = u(x, y) + \mathrm{i}v(x, y)$, where $z = x + \mathrm{i}y$ is differentiable in an open set $G \subset \mathbb{C}$ if it is continuous in G and the Cauchy-Riemann equations $u_x = v_y$ and $u_y = -v_x$ hold throughout G. A function $u : G \to \mathbb{R}$ is *harmonic* in G if all the first-order and second-order partial derivatives of u exist and u satisfies the Laplace equation $u_{xx} + u_{yy} = 0$. It follows from Cauchy-Riemann equations that the real and imaginary parts of an analytic function are harmonic.

A *path* in a region $G \subseteq \mathbb{C}$ is a continuous function $\gamma : [a, b] \to G$ for some interval $[a, b] \subseteq \mathbb{R}$. If $\gamma'(t)$ exists for each $t \in [a, b]$, and $\gamma' : [a, b] \to \mathbb{C}$ is a continuous function, then γ is a *smooth path*. A path γ is *piecewise smooth* if there is a partition of $[a, b]$, $a = t_0 < t_1 < \ldots < t_n = b$ such that γ is smooth on each subinterval $[t_{j-1}, t_j]$, $1 \leqslant j \leqslant n$. The function $\gamma : [a, b] \to G$ is differentiable at $z = a$ if the right derivative exists at $z = a$ and it is differentiable at b if the left derivative exists at $z = b$. Let $\gamma : [a, b] \to G$ be a smooth path and that for some $t_0 \in (a, b)$, $\gamma'(t_0) \neq 0$. Then γ has a tangent line at $z_0 = \gamma(t_0)$. This line goes through z_0 in the direction of $\gamma'(t_0)$. The slope of this tangent is $\tan(\arg \gamma'(t_0))$. If γ_1 and γ_2 are two smooth paths with $\gamma_1(t_1) = \gamma_2(t_2) = z_0$, $\gamma_1'(t_1) \neq 0$, $\gamma_2'(t_2) \neq 0$, then the angle θ between the paths γ_1, γ_2 is

$$\theta = \arg \gamma_2'(t_2) - \arg \gamma_1'(t_1).$$

Let G be an open set and $z_0 \in G$. If a function $f : G \to \mathbb{C}$ is analytic and $f'(z_0) \neq 0$, then it preserves the angle between two paths intersecting at z_0. Such mappings are *conformal*.

A *Möbius transformation* $M : \mathbb{C}_\infty \to \mathbb{C}_\infty$ defined by

$$M(z) = \frac{az + b}{cz + d}, \quad ad - bc \neq 0$$

is one-to-one and onto. This could be shown by proving that the inverse is the Möbius transformation given by

$$M^{-1}(z) = \frac{dz - b}{-cz + a}.$$

Here are some important examples of Möbius transformations:

- *Translation*: $S(z) = z + a$,
- *Dilation*: $S(z) = az$,
- *Rotation by angle* α: $S(z) = \mathrm{e}^{i\alpha} z$,
- *Inversion*: $S(z) = 1/z$.

The general equation of "circles and straight lines" on $\mathbb{C}$ has the form

$$\alpha|z|^2 + \overline{\beta}z + \beta\overline{z} + \gamma = 0.$$

If $w = 1/z$, then the equation above becomes

$$\alpha|1/w|^2 + \overline{\beta}(1/w) + \beta(1/\overline{w}) + \gamma = 0$$

or

$$\gamma|w|^2 + \beta w + \overline{\beta}\overline{w} + \alpha = 0.$$

This equation also represents "circles and straight lines". Indeed, every translation, dilation, rotation and inversion maps "circles and straight lines" onto "circles and straight lines". This is true for any Möbius transformation too since a Möbius transformation is a composition of translation, dilation, rotation and inversion. A Möbius transformation can have at most two fixed points unless it reduces to the identity transformation. As a consequence, it follows that a Möbius transformation is uniquely determined by its action on any three points in $\mathbb{C}_\infty$. Let $z_2, z_3, z_4 \in \mathbb{C}_\infty$. Let $S : \mathbb{C}_\infty \to \mathbb{C}_\infty$ be defined by

$$S(z) := \left(\frac{z - z_3}{z - z_4}\right) / \left(\frac{z_2 - z_3}{z_2 - z_4}\right), \quad z_2, z_3, z_4 \in \mathbb{C}.$$

If any of the numbers z_2, z_3 or z_4 equals ∞, S is defined by taking limiting value. If $z_2 = \infty$, $z_3 = \infty$ or $z_4 = \infty$, then S is given, respectively, by

$$S(z) := \frac{z - z_3}{z - z_4}, \quad S(z) := \frac{z_2 - z_4}{z - z_4}, \quad \text{or} \quad S(z) := \frac{z - z_3}{z_2 - z_3}.$$

Clearly, the mapping S given above is a Möbius transformation and

$$S(z_2) = 1, \quad S(z_3) = 0, \quad S(z_4) = \infty.$$

The mapping S is the only Möbius transformation with this property.

We use this mapping to define the *cross-ratio* of $z_1, z_2, z_3, z_4 \in \mathbb{C}_\infty$. The cross-ratio of four numbers $z_1, z_2, z_3, z_4 \in \mathbb{C}_\infty$, denoted by (z_1, z_2, z_3, z_4), is the image of z_1 under the unique Möbius transformation which takes z_2 to 1, z_3 to 0 and z_4 to ∞. If z_2, z_3, z_4 are distinct points and T is any Möbius transformation, then

$$(z_1, z_2, z_3, z_4) = (Tz_1, Tz_2, Tz_3, Tz_4).$$

for any point z_1. If z_2, z_3, z_4 are distinct points of $\mathbb{C}_\infty$, and w_2, w_3, w_4 are also distinct points in $\mathbb{C}_\infty$, then there is only one Möbius transformation S such that $S(z_2) = w_2$, $S(z_3) = w_3$, $S(z_4) = w_4$. The transformation $w = S(z)$ is given implicitly by

$$(w, w_2, w_3, w_4) = (z, z_2, z_3, z_4).$$

Let Γ be a circle through the points z_2, z_3, z_4. The points $z, z^* \in \mathbb{C}_\infty$ are symmetric with respect to Γ if

$$(z^*, z_2, z_3, z_4) = \overline{(z, z_2, z_3, z_4)}.$$

If Γ is a straight line, then z, z^* are symmetric with respect to Γ if they are on the different sides of the line Γ and $|z^* - z_3| = |z - z_3|$ where z_3 is any point on Γ. If Γ is the circle $\{z \in \mathbb{C} : |z - a| = R\}$, then a point z^* on the half line from a through z is symmetric to z if $|z - a|\,|z^* - a| = R^2$. If a Möbius transformation T takes a circle Γ onto the circle Λ, then any pair of points symmetric with respect to Γ are mapped by T onto a pair of points symmetric with respect to Λ. An orientation Γ is a triplet of points (z_1, z_2, z_3) where $z_1, z_2, z_3 \in \Gamma$. The left side of Γ with respect to the orientation (z_1, z_2, z_3) is the set $\{z \in \mathbb{C} : \mathrm{Im}(z, z_1, z_2, z_3) > 0\}$ and the right side of Γ with respect to the orientation (z_1, z_2, z_3) is the set $\{z \in \mathbb{C} : \mathrm{Im}(z, z_1, z_2, z_3) < 0\}$. If a Möbius transformation T takes a circle Γ onto the circle Λ, then T maps the left and right sides of the circle Γ with respect to an orientation (z_1, z_2, z_3), respectively, onto the left and right sides of Λ with respect to the orientation $(T(z_1), T(z_2), T(z_3))$. This can be used to get Möbius transformation mapping half-planes or disks onto half-planes or disks.

We shall often use the Möbius transformation $\varphi_a : \overline{\mathbb{D}} \to \overline{\mathbb{D}}$ given by

$$\varphi_a(z) = \frac{a - z}{1 - \overline{a}z}, \quad |a| < 1. \tag{1.1.1}$$

Note that the mapping is defined because $|\overline{a}z| = |a|\,|z| < 1$ for $z \in \overline{\mathbb{D}}$ and $|a| < 1$ and hence $1 - \overline{a}z \neq 0$.

Theorem 1.1.2 *For $|a| < 1$, the mapping $\varphi_a : \overline{\mathbb{D}} \to \overline{\mathbb{D}}$ defined by (1.1.1) is a bijection, and the inverse of φ_a is φ_a. Also, the mapping φ_a maps $\mathbb{D}$ onto $\mathbb{D}$, $\partial\mathbb{D}$ onto $\partial\mathbb{D}$, $\varphi_a(a) = 0$, $\varphi_a'(0) = |a|^2 - 1$, and $\varphi_a'(a) = (|a|^2 - 1)^{-1}$.*

Proof It maps $\mathbb{D}$ into $\mathbb{D}$ can be seen from the equation

$$1 - |\varphi_a(z)|^2 = \frac{(1 - |a|^2)(1 - |z|^2)}{|1 - \overline{a}z|^2}.$$

It also follows from this equation that the mapping φ_a maps $\partial\mathbb{D}$ onto $\partial\mathbb{D}$. If $\varphi_a(z_1) = \varphi_a(z_2)$, then $(z_2 - z_1)(1 - |a|^2) = 0$ and hence $z_1 = z_2$ proving the injectivity of φ_a. If $w = \varphi_a(z)$, then solving $w = (a - z)/(1 - \overline{a}z)$ yields $z = (a - w)/(1 - \overline{a}w) = \varphi_a(w) \in \overline{\mathbb{D}}$ whenever $w \in \overline{\mathbb{D}}$, proving the surjectivity of φ_a. Thus, the mapping is a bijection, and the inverse of φ_a is φ_a. Clearly, $\varphi_a(a) = 0$. Evidently,

$$\varphi_a'(z) = \frac{|a|^2 - 1}{(1 - \overline{a}z)^2}.$$

It follows then that $\varphi_a'(0) = |a|^2 - 1$, and $\varphi_a'(a) = (|a|^2 - 1)^{-1}$. □

1.2 Complex Integration

A function $\gamma : [a, b] \to \mathbb{C}$ is of *bounded variation* if there exists a constant $M > 0$ such that, for each partition $P = \{a = t_0 < t_1 < \ldots < t_n = b\}$ of $[a, b]$,

$$v(\gamma, P) = \sum_{k=1}^{n} |\gamma(t_k) - \gamma(t_{k-1})| \leqslant M.$$

For a function γ of bounded variation, its *total variation*, denoted by $V(\gamma)$, is

$$V(\gamma) = \sup \{v(\gamma, P) : P \text{ is a partition of } [a, b]\}.$$

Clearly, $V(\gamma) \leqslant M < \infty$. If $\gamma : [a, b] \to \mathbb{C}$ is piecewise smooth, then γ is of bounded variation and

$$V(\gamma) = \int_a^b |\gamma'(t)| \mathrm{d}t.$$

For a smooth path γ, we define the integral of f along γ by

$$\int_\gamma f(z) \mathrm{d}z := \int_a^b f(\gamma(t))\gamma'(t) \mathrm{d}t.$$

A path $\gamma : [a, b] \to \mathbb{C}$ is *rectifiable* if it is a function of bounded variation and its trace $\{\gamma\}$ is the image of $[a, b]$; that is, $\{\gamma\} := \{\gamma(t) : a \leqslant t \leqslant b\}$. Let γ be a rectifiable path in $\mathbb{C}$ and suppose that F_n and F are continuous on $\{\gamma\}$. If F_n converges to F uniformly on $\{\gamma\}$ as $n \to \infty$, then

$$\int_\gamma F(z) \mathrm{d}z = \lim_{n \to \infty} \int_\gamma F_n(z) \mathrm{d}z.$$

Theorem 1.2.1 (Cauchy's Integral Formula) *If $f : G \to \mathbb{C}$ is analytic and $\overline{\mathbb{D}}(a, r) \subset G$, then*

$$f^{(n)}(a) = \frac{n!}{2\pi \mathrm{i}} \int_\gamma \frac{f(w)}{(w - a)^{n+1}} \mathrm{d}w, \quad n \geqslant 0,$$

where $\gamma(t) = a + r\mathrm{e}^{\mathrm{i}t}$, $0 \leqslant t \leqslant 2\pi$.

If $f : G \to \mathbb{C}$ is analytic in G and $a \in G$, then $f(z) = \sum_{k=0}^{\infty} a_k (z - a)^k$ for $|z - a| < R$ where $R = d(a, G) := \inf\{|a - z| : z \in G\}$. It is not difficult to see that the coefficient a_k is given by $a_k = f^{(k)}(a)/k!$. Two functions f and g analytic in a region G are identical if and only if the set $\{z \in G : f(z) = g(z)\}$ has a limit point in G.

Liouville's theorem [6, Theorem 3.4, p. 77] states that every bounded entire function is constant. We say that a function f assumes the value w if $f(z) = w$ for some z and it omits the value w otherwise. A bounded function omits all the values outside a disk and the little Picard theorem gives a complete generalization of Liouville's theorem. The Little Picard theorem states that an entire function that omits two values is a constant.

If γ is a closed rectifiable curve in $\mathbb{C}$, then for $a \notin \{\gamma\}$,

$$n(\gamma; a) = \frac{1}{2\pi \mathrm{i}} \int_{\gamma} \frac{\mathrm{d}z}{z - a},$$

is called the *index* of γ with respect to a or the *winding number* of γ around a. If γ is a closed rectifiable curve in $\mathbb{C}$, then the function $a \to n(\gamma; a)$ is continuous on $G = \mathbb{C} \setminus \{\gamma\}$. Therefore, the winding number $n(\gamma; a)$ is constant on each component of G and it vanishes on the unbounded component of G.

A curve γ is *homologous to zero* in an open set G, written $\gamma \approx 0$, if $n(\gamma; a) = 0$ for all points $a \in \mathbb{C} \setminus G$. Let $\gamma_0,\ \gamma_1 : [0, 1] \to G$ be two closed rectifiable curves in a region G. Then γ_0 is *homotopic* to γ_1 in G, written $\gamma_0 \sim \gamma_1$, if there is a continuous function $\Gamma : [0, 1] \times [0, 1] \to G$ such that (i) $\Gamma(s, 0) = \gamma_0(s)$ and $\Gamma(s, 1) = \gamma_1(s)$ for all $0 \leqslant s \leqslant 1$, and (ii) $\Gamma(0, t) = \Gamma(1, t)$ for all $0 \leqslant t \leqslant 1$. If γ is a closed rectifiable curve in G, then γ is *homotopic to zero*, written $\gamma \sim 0$, if γ is homotopic to a constant curve. An open set $G \subset \mathbb{C}$ is *simply connected* if it is connected and every closed curve in G is homotopic to zero. Equivalently, an open connected set $G \subset \mathbb{C}$ is simply connected if its complement $\mathbb{C}_\infty \setminus G$ is a connected subset of $\mathbb{C}_\infty$. It is well-known [6, Theorem 2.2, p. 202] that a connected subset G of $\mathbb{C}$ is simply connected if, for every non-vanishing analytic function f, there is an analytic function g on G such that $f(z) = \mathrm{e}^{g(z)}$, or equivalently, if there is a function h analytic in G such that $f(z) = (h(z))^2$; the functions g and h are, respectively, called the analytic logarithm and the analytic square root of the function f.

Theorem 1.2.2 (Cauchy's Theorem) *If $\gamma_0,\ \gamma_1$ are two closed rectifiable curves in G and $\gamma_0 \sim \gamma_1$, then*

$$\int_{\gamma_0} f(z)\mathrm{d}z = \int_{\gamma_1} f(z)\mathrm{d}z$$

for every analytic function f on G.

In particular, if $f : G \to \mathbb{C}$ is an analytic function and γ is a closed rectifiable curve in the open set G such that $\gamma \sim 0$, then $\int_\gamma f(z)\mathrm{d}z = 0$. In the case of a simply connected open subset G of $\mathbb{C}$, it follows that $\int_\gamma f(z)\mathrm{d}z = 0$ for every analytic function f on G and for every closed rectifiable curve γ in G.

Theorem 1.2.3 (Morera's Theorem) *If $f : G \to \mathbb{C}$ is continuous, and $\int_T f(z)\mathrm{d}z = 0$ for every triangular path T inside G, then f is analytic in G.*

A point a is an *isolated singularity* of a function f if f is analytic in $\mathbb{D}(a; r) \setminus \{a\}$ for some $r > 0$ but not in $\mathbb{D}(a; r)$. An isolated singularity at a of a function f is a *removable singularity* if there is an analytic function $g : \mathbb{D}(a; r) \to \mathbb{C}$ such that $f(z) = g(z)$ for all $z \in \mathbb{D}(a; r) \setminus \{a\}$. An isolated singularity a of a function f is a *pole* of f if $|f(z)| \to \infty$ as $z \to a$. A function analytic except for poles is called a *meromorphic function*. If we define $f(a) = \infty$ for each pole a, then f is a mapping from an open subset G to the extended complex plane $\mathbb{C}_\infty$. Using stereographic projection, $\mathbb{C}_\infty$ can be identified with the unit sphere in $\mathbb{R}^3$. This gives a metric defined by

$$d(z, w) = \frac{2|z - w|}{\sqrt{(1 + |z|^2)(1 + |w|^2)}} \quad \text{and} \quad d(z, \infty) = \frac{2}{\sqrt{1 + |z|^2}}.$$

In this metric, every meromorphic function defined on G is continuous on G.

Theorem 1.2.4 (Argument Principle) *Let G be a region. If f is a meromorphic function on G with zeros $a_1, a_2, \ldots, a_m$ and poles $p_1, p_2, \ldots, p_l$ (repeated according to multiplicities), γ is a closed rectifiable curve in G which does not pass through neither $a_1, a_2, \ldots, a_m$ nor $p_1, p_2, \ldots, p_l$ and if $\gamma \approx 0$, then*

$$\frac{1}{2\pi \mathrm{i}} \int_\gamma \frac{f'(z)}{f(z)} \mathrm{d}z = \sum_{k=1}^{m} n(\gamma; a_k) - \sum_{k=1}^{l} n(\gamma; p_k).$$

Let Z_f and P_f be, respectively, the number of zeros and poles of a function f inside a circle γ counted according to their multiplicities. Then

$$\frac{1}{2\pi \mathrm{i}} \int_\gamma \frac{f'(z)}{f(z)} \mathrm{d}z = m - l = Z_f - P_f.$$

In addition to the hypothesis in Theorem 1.2.4, assume that $g : G \to \mathbb{C}$ is analytic, then it can be shown that

$$\frac{1}{2\pi \mathrm{i}} \int_\gamma \frac{g(z) f'(z)}{f(z)} \mathrm{d}z = \sum_{k=1}^{m} n(\gamma; a_k) g(a_k) - \sum_{k=1}^{l} n(\gamma; p_k) g(p_k).$$

Theorem 1.2.5 (Rouché's Theorem) *Let f and g be meromorphic in some open set containing $\overline{\mathbb{D}}(a; R)$ with no zeros or poles on the circle $|z - a| = R$. If*

$$|f(z) + g(z)| < |f(z)| + |g(z)| \quad \text{on} \quad |z - a| = R,$$

then $Z_f - P_f = Z_g - P_g$.

If f and g are analytic in some open set containing $\overline{\mathbb{D}}(a; R)$ with no zeros on the circle $|z - a| = R$ and

$$|f(z)+g(z)|<|f(z)|+|g(z)| \quad \text{on} \quad |z-a|=R,$$

then f and g have the same number of zeros inside $|z-a|=R$. This gives the classical Rouché's theorem.

Corollary 1.2.6 (Classical Rouché's Theorem) *Let f and g be analytic in some open set containing $\overline{\mathbb{D}}(a;R)$ with no zeros on the circle $|z-a|=R$. If*

$$|f(z)+g(z)|<|f(z)| \quad \textit{on} \quad |z-a|=R,$$

then f and g have the same number of zeros inside $|z-a|=R$.

Theorem 1.2.7 *If $f:G\to\mathbb{C}$ is analytic and one-to-one, then $f'(z)\neq 0$ for all $z\in G$.*

Proof Assume on the contrary that $f'(z_0)=0$ for some $z_0\in G$. Since zeros of analytic functions are isolated, and $f'(z_0)=0$, there is a real number $\delta>0$ such that $f'(z)$ has no zeros except z_0 in $\mathbb{D}(z_0;2\delta)\subset G$.

Since $f'(z_0)=0$, it follows that z_0 is a zero of $h(z)=f(z)-f(z_0)$ of order $n\geqslant 2$. Since f is one-to-one, h does not vanish in $G\setminus\{z_0\}$. Let

$$m:=\min\{|h(z)|:|z-z_0|=\delta\}.$$

A continuous function attains its extremum on any compact set, whence $m>0$. If $0<|a|<m$ and $g(z)=a+f(z_0)-f(z)$, then

$$|h(z)+g(z)|=|a|<m\leqslant|h(z)|$$

on $|z-z_0|=\delta$. The only zero of h in the disk $\mathbb{D}(z_0;\delta)$ is z_0 and it is of order n. Rouché's theorem shows that g has n zeros $z_1,z_2,\ldots,z_n$ in the disk $\mathbb{D}(z_0;\delta)$. Since $g'(z)=-f'(z)\neq 0$ in $\mathbb{D}(z_0;\delta)\setminus\{z_0\}$, these zeros are simple zeros. Therefore, $g(z_k)=0$ for all $k=1,2,\ldots,n$ and hence $f(z_k)=f(z_0)+a$ for all k. Thus, $f(z_1)=\cdots=f(z_n)$, which contradicts the one-to-oneness of f. □

The Open Mapping Theorem states that if f is a non-constant analytic function in an open set G, then $f(G)$ is open.

Theorem 1.2.8 (Maximum Modulus Theorem) *Let G be a region. If a function $f:G\to\mathbb{C}$ is analytic and there is an $a\in G$ such that $|f(a)|\geqslant|f(z)|$ for all $z\in G$, then the function f is a constant function. In other words, no non-constant analytic function defined in a region G attains its maximum modulus inside G.*

Proof Let $\Omega=f(G)$ and put $\alpha=f(a)$. Clearly $\alpha=f(a)\in f(G)=\Omega$. We first claim that $\alpha\in\partial\Omega$. For each $\omega\in\Omega$, there is a $z\in G$ with $\omega=f(z)$ and therefore, by the given hypothesis, we have $|\alpha|=|f(a)|\geqslant|f(z)|=|\omega|$ for all $\omega\in\Omega$. Suppose $\alpha\in\operatorname{int}\Omega$. Then, there exists $\rho>0$ such that $D(\alpha,\rho)\subset\Omega$. Clearly there exists

$\omega \in D(\alpha, \rho)$ such that $|\omega| > |\alpha|$. This contradicts $|\alpha| \geqslant |\omega|$ for all $\omega \in \Omega$. Thus, $\alpha \notin \operatorname{int} \Omega$ and hence $\alpha \in \Omega \setminus \operatorname{int} \Omega = \partial\Omega$.

Since $\alpha \in \Omega$ and $\alpha \in \partial\Omega$, it follows that $\Omega \cap \partial\Omega \neq \emptyset$. If Ω is open, then $\Omega \cap \partial\Omega = (\operatorname{int} \Omega) \cap \partial\Omega = \emptyset$, contradicting $\Omega \cap \partial\Omega \neq \emptyset$. Thus, Ω cannot be open. Since G is a region and $f(G) = \Omega$ is not open, the open mapping theorem implies that the function f is constant. □

Corollary 1.2.9 (Minimum Modulus Theorem) *Let G be a region. If a function $f : G \to \mathbb{C}$ is analytic and there is an $a \in G$ with $|f(a)| \leqslant |f(z)|$ for all $z \in G$, then either $f(a) = 0$ or the function f is a constant.*

Proof If $f(a) \neq 0$, then $g = 1/f$ is analytic in G and

$$|g(a)| = 1/|f(a)| \geqslant 1/|f(z)| = |g(z)|$$

for all $z \in G$. By applying Theorem 1.2.8, we see that g is constant and hence $f = 1/g$ is also a constant. □

Theorem 1.2.10 (Maximum Modulus Theorem for Bounded Domain)
Let $G \subset \mathbb{C}$ be a bounded open set. If a function $f : \overline{G} \to \mathbb{C}$ is continuous on $\overline{G}$ and is analytic in G, then the maximum modulus of the function f on $\overline{G}$ is attained on the boundary of G; that is,

$$\max\{|f(z)| : z \in \overline{G}\} = \max\{|f(z)| : z \in \partial G\}.$$

Proof We may assume that the function f is non-constant. Since G is bounded, $\operatorname{diam} G = \sup\{|z - w| : z, w \in G\} < \infty$. Since $\operatorname{diam} G = \operatorname{diam} \overline{G}$, $\overline{G}$ is also bounded. Thus $\overline{G}$ is compact. Since a continuous function on a compact set attains its maximum or minimum inside it, there exists $a \in \overline{G}$ such that $|f(a)| = \max\{|f(z)| : z \in \overline{G}\}$ or $|f(a)| \geqslant |f(z)|$ for all $z \in \overline{G}$. Since f is non-constant, Theorem 1.2.8 ensures that $a \notin G$. Thus, $a \in \overline{G} \setminus G = \partial G$. Hence $\max\{|f(z)| : z \in \overline{G}\} = |f(a)| = \max\{|f(z)| : z \in \partial G\}$. □

Corollary 1.2.11 (Minimum Modulus Theorem for Bounded Domain) *Let $G \subset \mathbb{C}$ be a bounded open set. If a function $f : \overline{G} \to \mathbb{C}$ is continuous on $\overline{G}$ and analytic in G, then either f has a zero in $\overline{G}$ or the minimum modulus of f on $\overline{G}$ is attained on the boundary of G.*

As a simple application, the following theorem is readily deduced which will be often used.

Theorem 1.2.12 *If $p : \mathbb{D} \to \mathbb{C}$ is analytic, non-constant, and satisfies* $\operatorname{Re} p(z) \geqslant 0$ *for all $z \in \mathbb{D}$, then* $\operatorname{Re} p(z) > 0$ *for all $z \in \mathbb{D}$.*

Proof Since $\left|e^{p(z)}\right| = e^{\operatorname{Re} p(z)} \geqslant e^0 = 1$, it follows that

$$\min\left\{\left|e^{p(z)}\right| : z \in \mathbb{D}\right\} \geqslant 1.$$

Assume that $\operatorname{Re} p(a) = 0$ for some $a \in \mathbb{D}$. Then

$$\left|e^{p(a)}\right| = e^{\operatorname{Re} p(a)} = 1$$

and hence

$$\min\left\{\left|e^{p(z)}\right| : z \in \mathbb{D}\right\} = 1 = \left|e^{p(a)}\right|.$$

Thus the minimum modulus theorem implies that the function e^p is a constant and hence the function p is constant. By hypothesis, the function p is non-constant, and so $\operatorname{Re} p(z) > 0$ for all $z \in \mathbb{D}$. □

1.3 Space of Continuous Functions

Let (Ω, d) be a metric space. A sequence $\langle a_n \rangle$ of points in Ω *converges* to an element $a \in \Omega$, written $a_n \to a$, if, for each $\epsilon > 0$, there is a natural number $N = N(\epsilon)$ such that $d(a_n, a) < \epsilon$ for all $n \geqslant N$. It is a *Cauchy sequence* if, for each $\epsilon > 0$, there is a natural number $N = N(\epsilon)$ such that $d(a_n, a_m) < \epsilon$ for all $n, m \geqslant N$. Every convergent sequence is a Cauchy sequence but the converse might not hold. A metric space (Ω, d) is *complete* if every Cauchy sequence in Ω converges to some element in Ω.

Let G be an open subset of $\mathbb{C}$ and (Ω, d) be a complete metric space. Denote by $C(G, \Omega)$ the class of all continuous functions from G to Ω. Every analytic function defined on G belongs to $C(G, \mathbb{C})$ and every meromorphic function defined on G belongs to $C(G, \mathbb{C}_\infty)$. Since constant functions are continuous, the class $C(G, \Omega)$ is non-empty. However, when G is connected and Ω is a subset of integers, it contains only constant functions. Let $G = \cup_{k=1}^{\infty} K_k$, where each K_k is compact and contained in the interior of K_{k+1}: $K_k \subseteq \operatorname{int} K_{k+1}$. Define $\rho : C(G, \Omega) \times C(G, \Omega) \to \mathbb{R}$ by

$$\rho(f, g) = \sum_{k=1}^{\infty} \left(\tfrac{1}{2}\right)^k \frac{\rho_k(f, g)}{1 + \rho_k(f, g)}, \tag{1.3.1}$$

where

$$\rho_k(f, g) = \sup\{d(f(z), g(z)) : z \in K_k\}.$$

Then, ρ is a metric, and it makes the space $C(G, \Omega)$ a complete metric space.

Lemma 1.3.1 *Let $G \subset \mathbb{C}$ be an open set, and (Ω, d) be a complete metric space. Let the metric ρ on $C(G, \Omega)$ be defined by (1.3.1). (a) If $\epsilon > 0$ is given, then there is $\delta > 0$ and a compact set $K \subset G$ such that, for $f, g \in C(G, \Omega)$,*

$$\sup\{d(f(z), g(z)) : z \in K\} < \delta \Longrightarrow \rho(f, g) < \epsilon.$$

(b) Conversely, if $\delta > 0$ and a compact set $K \subseteq G$ are given, then there is an $\epsilon > 0$ such that, for $f, g \in C(G, \Omega)$,

$$\rho(f, g) < \epsilon \Longrightarrow \sup\{d(f(z), g(z)) : z \in K\} < \delta.$$

Proof (a) Let K_k be the compact set used in defining the metric ρ, and $G = \cup_{k=1}^{\infty} K_k$. For given $\epsilon > 0$, choose a positive integer p such that $\sum_{k=p+1}^{\infty} (1/2)^k < \epsilon/2$ and put $K = K_p$. Choose $\delta > 0$ such that $t/(1+t) < \frac{\epsilon}{2}$ whenever $0 \leqslant t < \delta$. Suppose $f, g \in C(G, \Omega)$ satisfy $\sup\{d(f(z), g(z)) : z \in K\} < \delta$. Since $K_1 \subset K_2 \subset K_3 \subset \cdots \subset K_p = K$, then

$$\rho_k(f, g) = \sup\{d(f(z), g(z)) : z \in K_k\} \leqslant \sup\{d(f(z), g(z)) : z \in K\} < \delta$$

for $k = 1, 2, \ldots, p$. The choice of δ shows that, for $k = 1, 2, \ldots, p$,

$$\frac{\rho_k(f, g)}{1 + \rho_k(f, g)} < \frac{\epsilon}{2}.$$

Therefore,

$$\begin{aligned}\rho(f, g) &= \sum_{k=1}^{\infty} \left(\tfrac{1}{2}\right)^k \frac{\rho_k(f, g)}{1 + \rho_k(f, g)} < \sum_{k=1}^{p} \left(\tfrac{\epsilon}{2}\right)\left(\tfrac{1}{2}\right)^k + \sum_{k=p+1}^{\infty} \left(\tfrac{1}{2}\right)^k \\ &= \tfrac{1}{2}\epsilon \cdot \left(1 - \left(\tfrac{1}{2}\right)^p\right) + \sum_{k=p+1}^{\infty} \left(\tfrac{1}{2}\right)^k < \tfrac{1}{2}\epsilon + \tfrac{1}{2}\epsilon = \epsilon.\end{aligned}$$

(b) Let K, δ be given. Since $G = \cup_{k=1}^{\infty} K_k = \cup_{k=1}^{\infty} \operatorname{int} K_k$, and K is compact, then $K \subseteq K_p$ for some $p \geqslant 1$. For this p,

$$\rho_p(f, g) = \sup\{d(f(z), g(z)) : z \in K_p\} \geqslant \sup\{d(f(z), g(z)) : z \in K\}. \quad (1.3.2)$$

Let $\epsilon > 0$ be chosen such that $s/(1-s) < \delta$ whenever $0 \leqslant s < 2^p\epsilon$. Then, replacing s by $t/(1+t)$, we see that $t < \delta$ whenever $0 < t/(1+t) < 2^p\epsilon$. If $\rho(f, g) < \epsilon$, then

$$\left(\tfrac{1}{2}\right)^p \frac{\rho_p(f, g)}{1 + \rho_p(f, g)} \leqslant \rho(f, g) < \epsilon$$

and so, by the choice of ϵ, $\rho_p(f, g) < \delta$. From (1.3.2), it follows that

$$\sup\{d(f(z), g(z)) : z \in K\} < \delta.$$

□

Using this lemma, we can show that a set $\mathcal{O} \subseteq C(G, \Omega)$ is open in $(C(G, \Omega), \rho)$ if and only if, for each $f \in \mathcal{O}$, there is a compact set K and $\delta > 0$ such that $\{g : \sup\{d(f(z), g(z)) : z \in K\} < \delta\} \subset \mathcal{O}$. Thus, the collection of open sets in $(C(G, \Omega), \rho)$ is independent of the choice of the sets $\{K_n\}$ used in defining the metric ρ.

Theorem 1.3.2 *Let $G \subset \mathbb{C}$ be an open set. Let f_n and f be functions in $C(G, \mathbb{C})$. Then, $f_n \to f$ in $C(G, \mathbb{C})$ if and only if $f_n \to f$ uniformly on each compact subset of G.*

Proof Let $\langle f_n \rangle$ be a sequence in $(C(G, \mathbb{C}), \rho)$ convergent to f. Let K be a compact subset of G and $\delta > 0$. Using Lemma 1.3.1, choose $\epsilon > 0$ such that, for $f, g \in C(G, \mathbb{C})$,

$$\rho(f, g) < \epsilon \Rightarrow \sup\{d(f(z), g(z)) : z \in K\} < \delta. \tag{1.3.3}$$

For this $\epsilon > 0$, there is an $N > 0$ such that $\rho(f_n, f) < \epsilon$ for all $n \geqslant N$. Therefore, using the implication in (1.3.3), we see that for $n \geqslant N$, $\sup\{d(f_n(z), f(z)) : z \in K\} < \delta$. Thus, $f_n \to f$ uniformly on K.

Conversely, let $f_n \to f$ uniformly on compact subsets of G. Let $\epsilon > 0$ be given. Using Lemma 1.3.1, there exists a compact set K and $\delta > 0$ such that $\sup\{d(f(z), g(z)) : z \in K\} < \delta \Rightarrow \rho(f, g) < \epsilon$. Since $f_n \to f$ uniformly on compact sets, there is $N > 0$ such that $\sup\{d(f_n(z), f(z)) : z \in K\} < \delta$ for all $n \geqslant N$. Therefore $\rho(f_n, f) < \epsilon$ for all $n \geqslant N$. Thus, $f_n \to f$ in $C(G, \mathbb{C})$. □

Since closed disks are compact, one-way implication in the following corollary is trivial.

Corollary 1.3.3 *Let $G \subset \mathbb{C}$ be an open set. Let f_n and f be functions in $C(G, \mathbb{C})$. Then, $f_n \to f$ in $C(G, \mathbb{C})$ if and only if $f_n \to f$ uniformly on each closed disk inside G.*

Proof For the converse, assume that $f_n \to f$ uniformly on each closed disk inside G. Let K be a compact subset of G. Since K is disjoint from the closed set ∂G, it follows that $R := d(K, \partial G) > 0$. Then, for each $a \in K$, the disk $\mathbb{D}(a; R) \subset G$. Thus, the collection $\{\mathbb{D}(a; R) : a \in K\}$ is an open cover of K and, therefore, $K \subset \cup_{k=1}^{n} \mathbb{D}(a_k; R)$ for some $a_1, a_2, \ldots, a_n \in K$.

Let $\epsilon > 0$ be given. For each k, there is $N_k > 0$ such that $|f_n(z) - f(z)| < \epsilon$ for each $z \in \mathbb{D}(a_k, R)$ and for all $n \geqslant N_k$. Let $N = \max\{N_1, \ldots, N_n\}$. For each $z \in K$, there is an m such that $z \in \mathbb{D}(a_m; R)$ and so $|f_n(z) - f(z)| < \epsilon$ for all $n \geqslant N$. Therefore, $f_n(z) \to f(z)$ uniformly on K. By previous theorem, it follows that $f_n \to f$ in $C(G, \mathbb{C})$. □

A set $\mathcal{F} \subset C(G, \Omega)$ is *normal* if each sequence in $\mathcal{F}$ has a subsequence which converges to a function f in $C(G, \Omega)$. The limit of the subsequence is not required to be in $\mathcal{F}$. If the limit belongs to $\mathcal{F}$, then $\mathcal{F}$ is said to be (sequentially) compact. A compact family is normal and so it is sometimes called a compact normal family. Since $C(G, \Omega)$ is a metric space with metric ρ given by (1.3.1), sequential compactness of $\mathcal{F}$ is equivalent to compactness of $\mathcal{F}$.

Theorem 1.3.4 *Let $G \subset \mathbb{C}$ be an open set, and let (Ω, d) be a complete metric space. A set $\mathcal{F} \subset C(G, \Omega)$ is normal if and only if its closure $\overline{\mathcal{F}}$ is compact.*

Proof Let $\overline{\mathcal{F}}$ be compact. Let $\langle f_n \rangle \subseteq \mathcal{F}$ be any sequence. Then $\langle f_n \rangle \subseteq \overline{\mathcal{F}}$ and $\overline{\mathcal{F}}$ is sequentially compact. Therefore, there is a function $f \in \overline{\mathcal{F}} \subseteq C(G, \Omega)$ and a subsequence $\langle f_{n_k} \rangle$ of $\langle f_n \rangle$ such that $f_{n_k} \to f$. Hence, $\mathcal{F}$ is normal.

Let $\mathcal{F}$ be normal. We show that $\overline{\mathcal{F}}$ is compact. Let $\langle f_n \rangle$ be a sequence in $\overline{\mathcal{F}}$. Since $f_n \in \overline{\mathcal{F}}$, there is $g_n \in \mathcal{F}$ such that $\rho(g_n, f_n) < 1/n$ for each n. Since $\langle g_n \rangle$ is a sequence in $\mathcal{F}$ it has a subsequence $\langle g_{n_k} \rangle$ converging to g in $C(G, \Omega)$. Therefore, there exists a positive integer N such that $\rho(g_{n_k}, g) < \epsilon/2$ for $n_k \geqslant N$. It follows that

$$\rho(f_{n_k}, g) \leqslant \rho(f_{n_k}, g_{n_k}) + \rho(g_{n_k}, g) \leqslant \frac{\epsilon}{2} + \frac{\epsilon}{2} \text{ for } n_k \geqslant \max\{N, 2/\epsilon\}.$$

Thus, f_{n_k} converges to the function g in $C(G, \Omega)$. Since $\overline{\mathcal{F}}$ is closed, the limit function $g \in \overline{\mathcal{F}}$. This proves that $\overline{\mathcal{F}}$ is sequentially compact and hence compact. □

A subset Y of a metric space X is *totally bounded* if for every $\epsilon > 0$, there are points $y_1, y_2, \ldots, y_n \in Y$ such that

$$Y \subset \cup_{k=1}^{n} B(y_k, \epsilon).$$

Here $B(a, \epsilon)$ denotes the ball of radius ϵ centered at a:

$$B(a, \epsilon) = \{x \in X : d(x, a) < \epsilon\}.$$

A set $\mathcal{F} \subseteq C(G, \Omega)$ is *equicontinuous* at a point $z_0 \in G$ if and only if for every $\epsilon > 0$, there is a $\delta > 0$ such that $d(f(z), f(z_0)) < \epsilon$ for all $f \in \mathcal{F}$, and for all $z \in G$ with $|z - z_0| < \delta$. The set $\mathcal{F}$ is *equicontinuous* over a set $E \subseteq G$ if and only if for every $\epsilon > 0$, there is a $\delta > 0$ such that $d(f(z), f(z')) < \epsilon$ for all $f \in \mathcal{F}$, and for all $z, z' \in E$ with $|z - z'| < \delta$.

Theorem 1.3.5 *Let G be an open subset of $\mathbb{C}$ and let (Ω, d) be a complete metric space. A set $\mathcal{F} \subseteq C(G, \Omega)$ is normal if and only if, for every compact set $K \subseteq G$ and $\delta > 0$, there are functions $f_1, f_2, \ldots, f_n \in \mathcal{F}$ such that, for $f \in \mathcal{F}$, there is at least one k, $1 \leqslant k \leqslant n$, with*

$$\sup\{d(f(z), f_k(z)) : z \in K\} < \delta.$$

Proof Suppose $\mathcal{F}$ is normal. Let $K \subseteq G$ be compact, and $\delta > 0$ be given. By Lemma 1.3.1, there is $\epsilon > 0$ such that

$$\rho(f, g) < \epsilon \Rightarrow \sup\{d(f(z), g(z)) : z \in K\} < \delta$$

whenever $f, g \in C(G, \Omega)$. Since $\mathcal{F}$ is normal, $\overline{\mathcal{F}}$ is compact and therefore, $\overline{\mathcal{F}}$ is totally bounded. By Theorem 1.3.4, $\mathcal{F}$ is also totally bounded. Thus, there are functions $f_1, f_2, \ldots, f_n \in \mathcal{F}$ such that

$$\begin{aligned}\mathcal{F} &\subseteq \cup_{k=1}^{n} B(f_k, \epsilon) = \cup_{k=1}^{n} \{f : \rho(f, f_k) < \epsilon\} \\ &\subseteq \cup_{k=1}^{n} \{f : \sup\{d(f(z), f_k(z)) : z \in K\} < \delta\}.\end{aligned}$$

This shows that, if $f \in \mathcal{F}$, there is k, $1 \leqslant k \leqslant n$, such that

$$\sup\{d(f(z), f_k(z)) : z \in K\} < \delta.$$

Conversely, assume that for every compact set $K \subseteq G$ and $\delta > 0$, there are f_1, $f_2, \ldots, f_n \in \mathcal{F}$ such that

$$\sup\{d(f(z), f_k(z)) : z \in K\} < \delta. \tag{1.3.4}$$

We need to show that $\mathcal{F}$ is normal. By Theorem 1.3.4, it will follow if $\overline{\mathcal{F}}$ is compact. Recall that a metric space is compact if and only if it is complete and totally bounded. Since $\overline{\mathcal{F}}$ is a closed subset of the complete metric space $C(G, \Omega)$, $\overline{\mathcal{F}}$ is complete. Thus, we need to just prove that $\mathcal{F}$ is totally bounded. Let $\epsilon > 0$. Then, by Lemma 1.3.1, there exists a compact set $K \subseteq G$, and $\delta > 0$ such that

$$\sup\{d(f(z), g(z)) : z \in K\} < \delta \Rightarrow \rho(f, g) < \epsilon.$$

Applying the hypothesis to this K, and $\delta > 0$, we get $f_1, f_2, \ldots, f_n \in \mathcal{F}$ such that, for $f \in \mathcal{F}$, there is k, $1 \leqslant k \leqslant n$, satisfying

$$\sup\{d(f(z), f_k(z)) : z \in K\} < \delta.$$

Therefore, $\rho(f, f_k) < \epsilon$ and hence $f \in B_\rho(f_k, \epsilon)$. Thus, $\mathcal{F} \subseteq \cup_{k=1}^{n} B_\rho(f_k, \epsilon)$. Since $\mathcal{F}$ is totally bounded, it follows that $\overline{\mathcal{F}}$ is totally bounded. □

Theorem 1.3.6 (Arzela–Ascoli Theorem) *Let $G \subset \mathbb{C}$ be open and (Ω, d) be a complete metric space. A set $\mathcal{F} \subseteq C(G, \Omega)$ is normal if and only if $\mathcal{F}$ is equicontinuous at each point of G and the set $\{f(z) : f \in \mathcal{F}\}$ has compact closure in Ω for each $z \in G$.*

Proof Assume that $\mathcal{F}$ is normal. We first show that $\mathcal{F}_z = \{f(z) : f \in \mathcal{F}\}$ for each fixed $z \in G$ has compact closure. First note that the function $F : C(G, \Omega) \to \Omega$ defined by $F(f) = f(z)$ for each fixed $z \in G$ is continuous. For, if $\delta > 0$ is given and $K = \{z\}$, by Lemma 1.3.1, there is an $\epsilon > 0$ such that $\rho(g, f) < \epsilon \Rightarrow d(g(z), f(z)) < \delta$ or $d(F(g), F(f)) < \delta$. Since continuous functions map compact sets to compact sets, $F(\overline{\mathcal{F}})$ is compact. But, $\mathcal{F}_z = F(\mathcal{F}) \subseteq F(\overline{\mathcal{F}})$ and, therefore, $\overline{\mathcal{F}}_z \subseteq F(\overline{\mathcal{F}})$. Being closed subset of a compact set $F(\overline{\mathcal{F}})$, the set $\overline{\mathcal{F}}_z$ is compact.

We now show that $\mathcal{F}$ is equicontinuous at each point of G. Let $z_0 \in G$ be fixed and $\epsilon > 0$ be given. Choose $R > 0$ so that $K = \overline{\mathbb{D}(z_0; R)} \subset G$. Then K is compact. By Theorem 1.3.5, there are functions $f_1, f_2, \ldots, f_n \in \mathcal{F}$ such that, for each $f \in \mathcal{F}$, there is at least one f_k with

$$\sup\{d(f(z), f_k(z)) : z \in K\} < \epsilon/3. \tag{1.3.5}$$

Since each f_k is continuous, there is δ_k with $0 < \delta_k < R$ such that

$$d(f_k(z), f_k(z_0)) < \epsilon/3$$

for all $z \in K$ with $|z - z_0| < \delta_k$. Let $\delta = \min\{\delta_k : 1 \leqslant k \leqslant n\}$. Then,

$$d(f_k(z), f_k(z_0)) < \epsilon/3$$

for all $1 \leqslant k \leqslant n$ and for all $z \in K$ with $|z - z_0| < \delta$. Let $z \in K$ satisfy $|z - z_0| < \delta$, and let $f \in \mathcal{F}$. Choose k satisfying the inequality (1.3.5). Then

$$\begin{aligned} d(f(z), f(z_0)) &\leqslant d(f(z), f_k(z)) + d(f_k(z), f_k(z_0)) + d(f_k(z_0), f(z_0)) \\ &< \tfrac{\epsilon}{3} + \tfrac{\epsilon}{3} + \tfrac{\epsilon}{3} = \epsilon. \end{aligned}$$

Therefore $\mathcal{F}$ is equicontinuous at z_0.

The proof of the converse is omitted; see Conway [6, pp. 148–150] for a proof. □

A family $\mathcal{F} \subset C(G, \mathbb{C})$ is *uniformly bounded* on $A \subset G$ if there exists a non-negative real number M such that $|f(z)| \leqslant M$ for all $z \in A$ and for all $f \in \mathcal{F}$. The family $\mathcal{F}$ is *locally bounded* on G if, for each $a \in G$, there is a $r > 0$ such that $\mathcal{F}$ is uniformly bounded on $\mathbb{D}(a; r)$.

Clearly, any uniformly bounded family $\mathcal{F} \subset C(G, \mathbb{C})$ is locally bounded and a locally bounded family is not necessarily uniformly bounded. However, a locally bounded family is uniformly bounded on every compact subset of G as shown in the following lemma:

Lemma 1.3.7 *A family $\mathcal{F} \subset C(G, \mathbb{C})$ is locally bounded on G if and only if it is uniformly bounded on each compact subset $K \subset G$.*

Proof If $\mathcal{F}$ is uniformly bounded on each compact subset $K \subset G$, and $a \in G$, choose $r > 0$ such that $\overline{\mathbb{D}}(a; r) \subset G$. Then $\mathcal{F}$ is uniformly bounded in $\overline{\mathbb{D}}(a, r)$ and, in particular, $\mathcal{F}$ is uniformly bounded in $\mathbb{D}(a; r)$. Thus, $\mathcal{F}$ is locally bounded on G.

For the converse, assume that the family $\mathcal{F}$ is locally bounded on G. Let $K \subset G$ be a compact set. Since the compact set K is disjoint from the closed set ∂G, it follows that $\alpha := d(K, \partial G) > 0$. Then, for each $a \in K$, the disk $\mathbb{D}(a; \alpha) \subset G$. Thus, the collection $\{\mathbb{D}(a; \alpha) : a \in K\}$ is an open cover of K and, therefore, there are $a_1, a_2, \ldots, a_n \in K$ such that $K \subset \cup_{k=1}^{n} \mathbb{D}(a_k; \alpha)$. Since $\mathcal{F}$ is locally bounded, for each $1 \leqslant k \leqslant n$, there are $r_k < \alpha$ and $M_k \geqslant 0$ such that $|f(z)| \leqslant M_k$ for all $f \in \mathcal{F}$ and for all $z \in \mathbb{D}(a_k; r_k)$. Let $M := \max\{M_1, \ldots, M_n\}$. If $z \in K$, then $z \in \mathbb{D}(a_k; r_k)$ for some k. If $f \in \mathcal{F}$, then $|f(z)| \leqslant M_k \leqslant M$. Thus, $\mathcal{F}$ is uniformly bounded on K. □

1.4 Space of Analytic Functions

For an open subset G of $\mathbb{C}$, let $\mathcal{H}(G)$ denote the subset of $C(G, \mathbb{C})$ consisting of all functions analytic in G. The following theorem shows that $\mathcal{H}(G)$ is a closed subspace of the complete space $C(G, \mathbb{C})$ and hence the space $\mathcal{H}(G)$ is complete.

Theorem 1.4.1 (Weierstrass' Theorem) *If $f_n \in \mathcal{H}(G)$ and $f_n \to f$ in $C(G, \mathbb{C})$, then $f \in \mathcal{H}(G)$. Also, for each $k \geqslant 1$, the sequence of kth derivatives converges to the kth derivative of the limit function: $f_n^{(k)} \to f^{(k)}$ in $\mathcal{H}(G)$.*

Proof We may assume that G is a disk as G is a union of disks. Consider any triangle T inside G. Since T is a compact subset of $\mathbb{C}$, and $f_n \to f$ in $C(G, \mathbb{C})$, it follows that $f_n(z) \to f(z)$ uniformly on T and hence

$$\int_T f_n(z)\mathrm{d}z \to \int_T f(z)\mathrm{d}z.$$

Since each f_n is analytic, $\int_T f_n(z)\mathrm{d}z = 0$ and hence $\int_T f(z) = 0$. Since f is continuous,, it follows from Morera's theorem that f is analytic in G.

It suffices to show that $f_n^{(k)}(z) \to f^{(k)}(z)$ uniformly on each closed disk $\overline{\mathbb{D}}(a; r)$ inside G. Since $\alpha := d(\overline{\mathbb{D}}(a; r), \partial G) > 0$, then $\overline{\mathbb{D}}(a; R) \subset G$ where $R := r + \alpha/2$. Let γ be the circle $|w - a| = R$. Since $f_n \to f$, it follows that $f_n \to f$ uniformly on γ. Thus, for $\epsilon > 0$, there is a positive integer N such that

$$|f_n(w) - f(w)| < \frac{(R - r)^{k+1}\epsilon}{k!R}$$

for all w on γ and for all $n \geqslant N$. By Cauchy's integral formula for the derivatives,

$$f_n^{(k)}(z) - f^{(k)}(z) = \frac{k!}{2\pi \mathrm{i}} \int_\gamma \frac{f_n(w) - f(w)}{(w - z)^{k+1}}\mathrm{d}w$$

for each $z \in \overline{\mathbb{D}}(a; r)$. Therefore, for $n \geqslant N$ and $z \in \overline{\mathbb{D}}(a; r)$,

$$|f_n^{(k)}(z) - f^{(k)}(z)| \leqslant \frac{k!}{2\pi} \frac{(R - r)^{k+1}\epsilon}{k!R(R - r)^{k+1}} 2\pi R = \epsilon.$$

□

If $f_n \in \mathcal{H}(G)$ and the series $\sum_{n=1}^{\infty} f_n$ converges to f in $C(G, \mathbb{C})$, then, by applying the Weierstrass Theorem to the sequence of partial sums, we conclude that $f \in \mathcal{H}(G)$, and moreover, $\sum_{n=1}^{\infty} f_n^{(k)} \to f^{(k)}$ in $\mathcal{H}(G)$ for each $k \in \mathbb{N}$.

Theorem 1.4.2 (Hurwitz's Theorem) *Let G be a region, and let $\overline{\mathbb{D}}(a; r) \subseteq G$. Let $f_n \in \mathcal{H}(G)$, $f_n \to f$ in $\mathcal{H}(G)$ and $f \not\equiv 0$. If $f(z) \neq 0$ on $\partial\overline{\mathbb{D}}(a; r)$, then there is a*

positive integer N such that f_n and f have the same number of zeros in $\mathbb{D}(a, r)$ for each $n \geqslant N$.

Proof Since $f(z) \neq 0$ on $\partial\overline{\mathbb{D}}(a; r)$, and $|f(z)|$ is continuous on the compact set $\partial\overline{\mathbb{D}}(a; r)$,

$$\epsilon := \min\{|f(z)| : z \in \partial\overline{\mathbb{D}}(a; r)\} > 0.$$

Since the convergence in $\mathcal{H}(G)$ is the same as the uniform convergence on compact subsets of G, $f_n \to f$ implies that there is a positive integer N such that, for $n \geqslant N$, we have

$$|f_n(z) - f(z)| < \frac{\epsilon}{2} \quad (z \in \partial\overline{\mathbb{D}}(a; r)).$$

For $n \geqslant N$ and $z \in \partial\overline{\mathbb{D}}(a; r)$, we have

$$|f_n(z) - f(z)| < \frac{\epsilon}{2} < \epsilon \leqslant |f(z)| \leqslant |f_n(z)| + |f(z)|.$$

From the last equation, it is easy to see that $f_n \neq 0$ on $\partial\overline{\mathbb{D}}(a; r)$. Hence, by Rouche's theorem, it follows that f_n and f have the same number of zeros in $\mathbb{D}(a; r)$. □

Corollary 1.4.3 *Let $f_n \in \mathcal{H}(G)$ and $f_n \to f$ in $\mathcal{H}(G)$. If each f_n is non-vanishing in G, then either $f \equiv 0$ or f is non-vanishing in G.*

Proof Assume that $f \not\equiv 0$ and $f(a) = 0$ for some $a \in G$. Since $f \in \mathcal{H}(G)$, its zeros are isolated, and therefore, there is a closed neighborhood $\overline{\mathbb{D}}(a; r) \subset G$ such that f has no zeros in $\overline{\mathbb{D}}(a; r) \setminus \{a\}$. By Hurwitz's theorem, there is an integer N such that f_n and f have the same number of zeros in $\mathbb{D}(a, r)$. Since f has a zero in $\mathbb{D}(a; r)$, it follows that f_n has a zero in $\mathbb{D}(a; r)$, contradicting the hypothesis. □

If a function $f \in \mathcal{H}(G)$ is one-to-one in G, then it is called *univalent* in G. A function f is *locally one-to-one* at z_0 if there is a neighborhood of z_0 in which f is one-to-one. If f is analytic and locally one-to-one at z_0, then it is *locally univalent* at z_0. For univalent functions, we have the following corollary.

Corollary 1.4.4 *Let $f_n \in \mathcal{H}(G)$ and $f_n \to f$ in $\mathcal{H}(G)$. If each f_n is univalent in G, then either f is a constant function or f is univalent in G.*

Proof Let $a, b \in G$ and $a \neq b$. Choose $r > 0$ such that $\overline{\mathbb{D}}(a; r) \subset G$ and $b \notin \overline{\mathbb{D}}(a; r)$. Since $f_n \to f$ in $\mathcal{H}(G)$, it follows that $f_n(z) \to f(z)$ uniformly on $\overline{\mathbb{D}}(a; r)$ and hence $f_n(z) - f_n(b) \to f(z) - f(b)$ uniformly on $\overline{\mathbb{D}}(a; r)$. Clearly $f_n(z) - f_n(b) \neq 0$ for $z \in \overline{\mathbb{D}}(a; r)$ and hence, by Hurwitz's theorem, either $f(z) - f(b) \equiv 0$ on G or $f(z) - f(b) \neq 0$ on $\overline{\mathbb{D}}(a; r)$. In the first case, f is a constant function. In the latter case, $f(a) - f(b) \neq 0$, or f is one-to-one in G. □

Let $\mathcal{S}$ be the class of all univalent functions $f \in \mathcal{H}(\mathbb{D})$ normalized by the conditions $f(0) = 0$ and $f'(0) = 1$.

Corollary 1.4.5 *If $f_n \in \mathcal{S}$ and $f_n \to f$ in $\mathcal{H}(\mathbb{D})$, then $f \in \mathcal{S}$.*

Proof Since $f_n \to f$ in $\mathcal{H}(\mathbb{D})$, then $f_n' \to f'$ in $\mathcal{H}(\mathbb{D})$, and hence $1 = f_n'(0) \to f'(0)$, that is, $f'(0) = 1$. Thus, f is non-constant and from Corollary 1.4.4, f is one-to-one. Clearly $f(0) = 0$ and so $f \in \mathcal{S}$. □

Theorem 1.4.6 (Montel's Theorem) *A family $\mathcal{F} \subset \mathcal{H}(G)$ is normal if and only if it is locally bounded on G.*

Proof Let $\mathcal{F}$ be normal. If $\mathcal{F}$ is not locally bounded on G, then, by Lemma 1.3.7, it is not uniformly bounded on some compact subset $K \subset G$. Then, for each $n \in \mathbb{N}$, there is a function $f_n \in \mathcal{F}$ such that $\sup\{|f_n(z)| : z \in K\} \geqslant n$. If there exists no such f_n, then $|f(z)| < n$ for all $f \in \mathcal{F}$ and for all $z \in K$. Since $\mathcal{F}$ is normal and $f_n \in \mathcal{F}$, there is a subsequence $f_{n_k} \to f$ in $C(G, \mathbb{C})$ and, in particular, $f_{n_k}(z) \to f(z)$ uniformly on K. Thus, there is a natural number N such that, for $n_k \geqslant N$, $|f_{n_k}(z) - f(z)| < 1$ for all $z \in K$. Also, since f is continuous on K, there is a real number $M \geqslant 0$ such that $|f(z)| \leqslant M$ for all $z \in K$. Therefore, for all $n_k \geqslant N$,

$$\begin{aligned} n_k &\leqslant \sup\{|f_{n_k}(z)| : z \in K\} \\ &\leqslant \sup\{|f_{n_k}(z) - f(z)| : z \in K\} + \sup\{|f(z)| : z \in K\} \leqslant 1 + M. \end{aligned}$$

This contradiction shows that $\mathcal{F}$ is locally bounded on G.

We use the Arzela–Ascoli theorem to prove the converse. Since $\mathcal{F}$ is locally bounded on G, for each point $a \in G$, there is $r > 0$ such that $\sup\{|f(z)| : z \in \mathbb{D}(a; r)\} \leqslant M$. In particular, the set $\{f(a) : f \in \mathcal{F}\}$ is bounded and hence its closure is also bounded. Thus, being a closed set, the closure of $\{f(a) : f \in \mathcal{F}\}$ is compact for each $a \in G$.

The proof is complete if we show that the family $\mathcal{F}$ is equicontinuous at each point of G. Let $a \in G$ and $\epsilon > 0$ be given. Since $\mathcal{F}$ is locally bounded on G, choose $r > 0$, $M \geqslant 0$ such that $|f(z)| \leqslant M$ for all $f \in \mathcal{F}$ and for all $z \in \overline{\mathbb{D}}(a; r)$. Let γ be the circle $|w - a| = r$. Then, for $f \in \mathcal{F}$ and for all $z \in \overline{\mathbb{D}}(a; r/2)$, Cauchy's integral formula yields

$$\begin{aligned} f(z) - f(a) &= \frac{1}{2\pi \mathrm{i}} \int_\gamma \frac{f(w)}{w - z} \mathrm{d}w - \frac{1}{2\pi \mathrm{i}} \int_\gamma \frac{f(w)}{w - a} \mathrm{d}w \\ &= \frac{1}{2\pi \mathrm{i}} \int_\gamma \frac{f(w)(z - a)}{(w - z)(w - a)} \mathrm{d}w. \end{aligned}$$

For w on γ and $z \in \overline{\mathbb{D}}(a; r/2)$, we have $|w - a| = r$ and

$$|w - z| \geqslant |w - a| - |a - z| \geqslant r - r/2 = r/2.$$

Therefore,

$$|f(z) - f(a)| \leqslant \frac{1}{2\pi} \frac{M|z-a|2\pi r}{(r/2)r} = \frac{2M}{r}|z-a|.$$

If we choose $\alpha = \min\{r/2, r\epsilon/(2M)\}$, then for each $f \in \mathcal{F}$, and $|z-a| < \alpha$, it follows that $|f(z) - f(a)| < \epsilon$. □

Corollary 1.4.7 *A family $\mathcal{F} \subset \mathcal{H}(G)$ is compact if and only if it is closed and locally bounded on G.*

Proof If $\mathcal{F} \subset \mathcal{H}(G)$ is compact, then it is closed and normal. By the Montel's theorem, it is locally bounded. Conversely, if $\mathcal{F} \subset \mathcal{H}(G)$ is closed and locally bounded, then $\mathcal{F}$ is normal and so $\overline{\mathcal{F}} = \mathcal{F}$ is compact. □

1.5 Riemann Mapping Theorem

Let G_1, G_2 be open subsets of $\mathbb{C}$. Then G_1 is *conformally equivalent* to G_2 if there is a univalent function $f : G_1 \to G_2$ such that $f(G_1) = G_2$. Conformal equivalence is an equivalence relation. By Liouville's theorem, there is no analytic function that maps $\mathbb{C}$ onto a bounded subset of $\mathbb{C}$ and so no bounded domain is conformally equivalent to $\mathbb{C}$.

Consider next the slit domain $S := \mathbb{C} \setminus \{z : z \leqslant 0\}$; this is a simply connected domain in $\mathbb{C}$. The function $f : S \to \mathbb{C}$ given by $f(z) = \sqrt{z}$ is univalent and it maps S onto the right half-plane $H = \{z \in \mathbb{C} : \operatorname{Re} z > 0\}$. The mapping $g : H \to \mathbb{C}$ given by $g(z) = (z-1)/(z+1)$ is univalent and maps H onto $\mathbb{D}$. Thus, the composite mapping $g \circ f$, given by $(g \circ f)(z) = (\sqrt{z}-1)/(\sqrt{z}+1)$ is univalent and maps S onto $\mathbb{D}$. Thus, the slit domain S is conformally equivalent to the disk $\mathbb{D}$ (see Fig. 1.1).

In general, there are univalent functions that map a proper simply connected domain of $\mathbb{C}$ onto $\mathbb{D}$. To prove this, we need the following lemma.

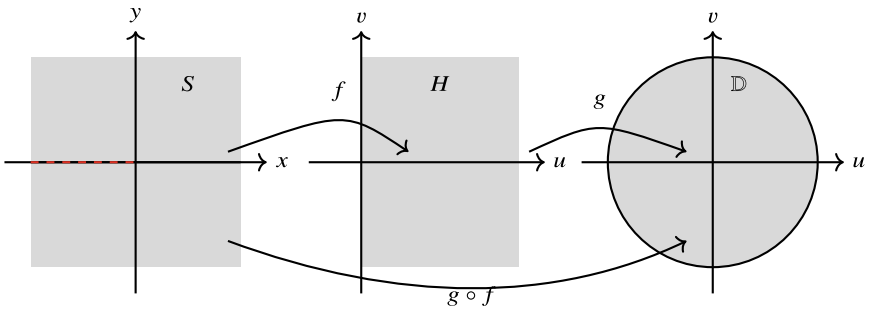


Fig. 1.1 The mapping $(g \circ f)(z) = (\sqrt{z}-1)/(\sqrt{z}+1)$

Lemma 1.5.1 *Let G be a simply connected domain in $\mathbb{C}$ which is not the whole complex plane $\mathbb{C}$ and $a \in G$. Then there is a univalent function $f : G \to \mathbb{D}$ with $f(a) = 0$, $f'(a) > 0$.*

Proof Since G is not the whole complex plane $\mathbb{C}$, choose $b \in \mathbb{C} \setminus G$. Since $z - b \neq 0$ for $z \in G$, and G is simply connected, there is an analytic function $g : \mathbb{G} \to \mathbb{C}$ satisfying

$$g^2(z) = z - b.$$

The function g is non-constant and univalent in G; indeed, if $g(z_1) = \pm g(z_2)$, then $z_1 = z_2$. By the open mapping theorem, $g(G)$ is an open subset of $\mathbb{C}$ and therefore there is an $\alpha > 0$ such that $\overline{\mathbb{D}}(g(a); \alpha) \subset g(G)$.

The function f is constructed as follows. We show that $\overline{\mathbb{D}}(-g(a); \alpha) \cap g(G) = \emptyset$ and find a Möbius transformation T that maps $\mathbb{C} \setminus \overline{\mathbb{D}}(-g(a); \alpha)$ onto the disk $\mathbb{D}$. The composition of these two functions maps G into $\mathbb{D}$. We use an automorphism S of the disk to get a mapping f with desired properties (see Fig. 1.2).

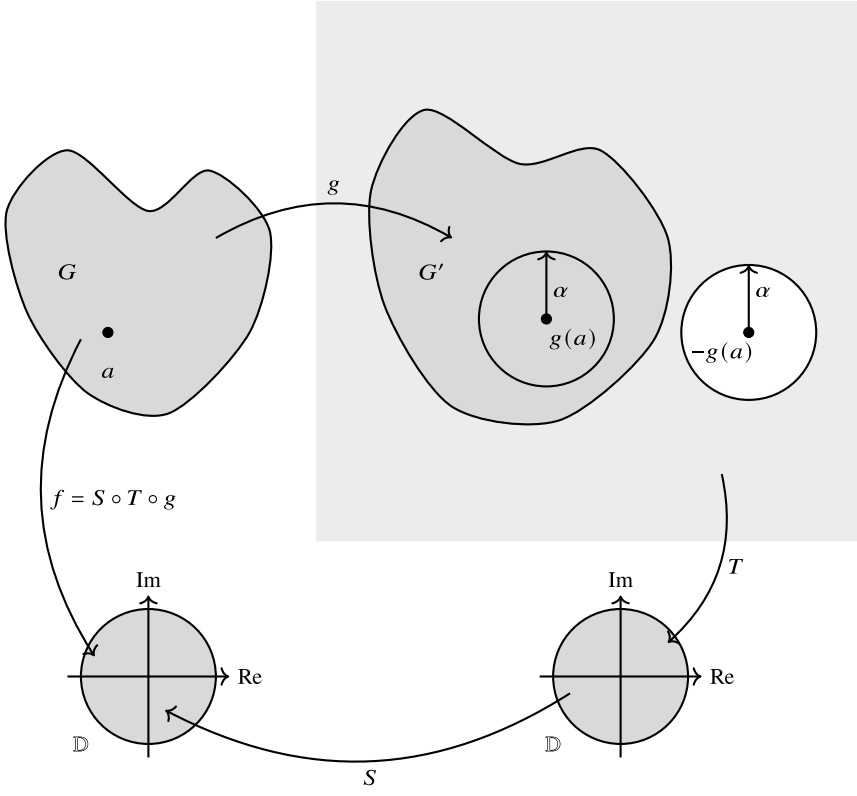


Fig. 1.2 The mapping f as a composition $S \circ T \circ g$

We now show that $\overline{\mathbb{D}}(-g(a);\alpha) \cap g(G) = \emptyset$. If $\overline{\mathbb{D}}(-g(a);\alpha) \cap g(G) \neq \emptyset$, there is $w \in G$ such that $g(w) \in \overline{\mathbb{D}}(-g(a);\alpha)$ or $|g(w) + g(a)| \leqslant \alpha$. This shows that $-g(w) \in \overline{\mathbb{D}}(g(a);\alpha) \subset g(G)$ or $-g(w) = g(z)$ for some $z \in G$. Then, $w = z$ and so $g(w) = 0$ or $w = b$. Since $w \in G$ while $b \in \mathbb{C} \setminus G$, this is clearly a contradiction.

The bilinear transformation

$$T(z) = \frac{\alpha}{z + g(a)}$$

is univalent and maps $\mathbb{C} \setminus \overline{\mathbb{D}}(-g(a);\alpha)$ onto $\mathbb{D}$. Hence, the mapping $h(z) = T(g(z))$ is univalent and maps G into $\mathbb{D}$. Since h is univalent, it follows from Theorem 1.2.7 that $h'(a) \neq 0$ and hence the mapping

$$f(z) = \frac{|h'(a)|}{h'(a)} \frac{h(z) - h(a)}{1 - \overline{h(a)}h(z)} \tag{1.5.1}$$

is well-defined and analytic in G. It has the desired properties. Since h maps G into $\mathbb{D}$ and the mapping S defined by

$$S(z) = \frac{|h'(a)|}{h'(a)} \frac{w - h(a)}{1 - \overline{h(a)}w}$$

maps $\mathbb{D}$ onto $\mathbb{D}$, the function $f = S \circ h$ maps G into $\mathbb{D}$. Clearly, $f(a) = 0$ and

$$f'(a) = \frac{|h'(a)|}{1 - |h(a)|^2} > 0. \tag{1.5.2}$$

Since f is the composition of a univalent function h with the bilinear transformation S, the function f is univalent. □

The function f constructed in Lemma 1.5.1 need not be onto. We now show that we can find a bijection of G onto $\mathbb{D}$ (see Fig. 1.3).

Theorem 1.5.2 (Riemann Mapping Theorem) *Any simply connected domain G in $\mathbb{C}$ which is not the whole complex plane $\mathbb{C}$ is conformally equivalent to the unit disk $\mathbb{D}$. In other words, there is a univalent function $f : G \to \mathbb{D}$ such that $f(G) = \mathbb{D}$. If, for some $a \in G$, $f(a) = 0$ and $f'(a) > 0$, then the mapping f is unique.*

Proof Fix $a \in G$. Consider the class $\mathcal{F}$ of all univalent function $g : G \to \mathbb{D}$ such that $g(G) \subset \mathbb{D}$, $g(a) = 0$ and $g'(a) > 0$. By Lemma 1.5.1, $\mathcal{F} \neq \emptyset$. We show that $\overline{\mathcal{F}} = \mathcal{F} \cup \{0\}$. Since there is at least one $g \in \mathcal{F}$, and $g(z)/n$ converges uniformly to 0 on each compact subset of G, $0 \in \overline{\mathcal{F}}$ and so $\mathcal{F} \cup \{0\} \subset \overline{\mathcal{F}}$. To prove the reverse inclusion, let $g \in \overline{\mathcal{F}}$ and let $g_n \in \mathcal{F}$ and $g_n \to g$ in $\mathcal{H}(G)$. By Corollary 1.4.4, it follows that either g is a constant function or g is univalent. If g is constant, then $g \equiv 0$. For, $g_n \to g$ and $g_n(a) = 0$ implies $g(a) = 0$. If g is univalent, then $g'(a) \neq 0$.

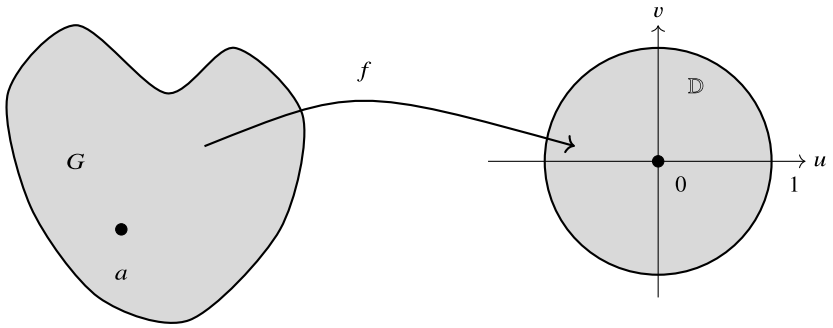


Fig. 1.3 Riemann mapping theorem

But $g_n'(a) > 0$ implies that $g'(a) \geqslant 0$ and so $g'(a) > 0$. Thus, $g \in \mathcal{F}$. In either case, $g \in \mathcal{F} \cup \{0\}$ which proves $\overline{\mathcal{F}} \subset \mathcal{F} \cup \{0\}$.

We shall now complete the proof by showing that there is a function $f \in \mathcal{F}$ such that $f(G) = \mathbb{D}$. If $g_n \to g$ in $\mathcal{H}(G)$, then Weierstrass theorem (Theorem 1.4.1) yields $g_n' \to g'$ and in particular $g_n'(a) \to g'(a)$. This shows that the function $F : \mathcal{H}(G) \to \mathbb{R}$ defined by $F(g) = g'(a)$ is continuous. Since $\mathcal{F}$ is uniformly bounded on G, it is locally bounded on G. By Montel's theorem (Theorem 1.4.6), it follows that $\mathcal{F}$ is normal and so $\overline{\mathcal{F}}$ is compact. Since F is a continuous real valued function on a compact set $\overline{\mathcal{F}}$, it has its maximum in $\overline{\mathcal{F}}$. Let $f \in \overline{\mathcal{F}}$ be the function such that $f'(a) = F(f) \geqslant F(g) = g'(a)$ for all $g \in \overline{\mathcal{F}}$; in other words, f is the function in $\mathcal{F}$ having maximum possible derivative at $z = a$. Since $\mathcal{F} \neq \emptyset$, there is a function $g_1 \in \mathcal{F}$ with $g_1'(a) > 0$. Thus, $f'(a) > 0$. Since $\overline{\mathcal{F}} = \mathcal{F} \cup \{0\}$, it follows that $f \in \mathcal{F}$.

Next it is shown that $f(G) = \mathbb{D}$. Suppose on the contrary that $f(G) \subsetneq \mathbb{D}$. Let $w \in \mathbb{D} \setminus f(G)$ and consider

$$(T_w \circ f)(z) = T_w(f(z)) = \frac{f(z) - w}{1 - \overline{w} f(z)} \tag{1.5.3}$$

where the bilinear transformation $T_w(z) = (z - w)/(1 - \overline{w}z)$ maps $\mathbb{D}$ onto $\mathbb{D}$. The mapping $T_w \circ f$ is analytic, univalent and non-vanishing in G and maps G into $\mathbb{D}$. Since G is simply connected, there is an analytic function $h : G \to \mathbb{C}$ such that $h(z)^2 = (T_w \circ f)(z)$. Since the mapping $T_w \circ f$ is univalent in G and maps G into $\mathbb{D}$, it follows that the mapping h is also univalent in G and maps G into $\mathbb{D}$. Now consider the mapping $g : G \to \mathbb{C}$ defined by the equation (see equation (1.5.1))

$$g(z) = \frac{|h'(a)|}{h'(a)} \frac{h(z) - h(a)}{1 - \overline{h(a)} h(z)}.$$

This function g is also univalent (being a composition of a bilinear transformation with a univalent function) and also maps G into $\mathbb{D}$. It follows from (1.5.3) that

$$2h(a)h'(a) = g'(a) = f'(a)(1 - |w|^2).$$

Since $h(a)^2 = g(a) = -w$, we see from (1.5.2) (with f replaced by g) that

$$g'(a) = \frac{|h'(a)|}{1 - |h(a)|^2} = \frac{f'(a)(1 - |w|^2)}{2\sqrt{|w|}} \frac{1}{1 - |w|} = f'(a) \frac{(1 + |w|)}{2\sqrt{|w|}} > f'(a)$$

as $|w| < 1$. This contradicts the choice of f and hence $f(G) = \mathbb{D}$.

To prove the uniqueness, let f and g be two univalent functions from G onto $\mathbb{D}$ with positive derivatives at $a \in G$ and both map a to the origin. Then, $f \circ g^{-1}$ maps $\mathbb{D}$ onto $\mathbb{D}$, $(f \circ g^{-1})(0) = f(a) = 0$. Thus, $(f \circ g^{-1})(w) = cw$ for some constant c with $|c| = 1$. Here we have used the fact (see Theorem 3.1.4) that the only univalent function that maps $\mathbb{D}$ onto $\mathbb{D}$ with $f(0) = \alpha$ are of the form $c(z - \alpha)/(1 - \overline{\alpha}z)$ where $|c| = 1$. Since g is onto, writing $w = g(z)$, we get $f(z) = cg(z)$. Since $|c| = 1$ and $c = f'(a)/g'(a) > 0$, then $c = 1$ and hence $f = g$. □

The conformal equivalence on the set of all simply connected regions separates them into two equivalence classes; one consisting of the whole complex plane $\mathbb{C}$ alone and the other containing all the proper simply connected regions (and the simplest of these is the unit disk $\mathbb{D}$).

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[1] This chapter is a collection of important results from the seminal book of Conway [6]. There are several other books on complex analysis. The text by Ali and Ravichandran [1] and Churchill and Brown [5] are meant for beginners. For an advanced reading, one may refer to [2, 4, 6, 7, 9, 10]. The books [8, 10] has nice introductions to univalent functions. A proof of the Looman-Menchoff theorem can be found in [7]. The book by Burckel [3] is encyclopedic and derived from his earlier work [4].

Chapter 2
Univalent Functions

Abstract An analytic one-to-one function $f : \mathbb{D} \to \mathbb{C}$ is called a *univalent* function. The nth Taylor coefficient $a_n = f^{(n)}(0)/n!$ of a univalent function f satisfies the sharp inequality $|a_n| \leqslant n$. Bieberbach conjectured this inequality and proved it for $n = 2$. In this chapter, the coefficient result $|a_2| \leqslant 2$ is proved which leads to several significant consequences. Several other basic properties of univalent functions are also explored.

2.1 Univalence and Local Univalence

Recall that a function $f \in \mathcal{H}(G)$ is called *univalent* in G if it is one-to-one in G. The function $f \in \mathcal{H}(G)$ is *locally univalent* at $z_0 \in G$ if there is a neighborhood $\mathbb{D}(z_0; r)$, $r > 0$, in which f is one-to-one. A function univalent in a region G is locally univalent at each point in G, but the converse is not generally true (see Example 2.1.2). The concept of univalence can be extended to meromorphic functions and harmonic functions. If a meromorphic function $f : G \to \mathbb{C}_\infty$ is one-to-one in G, it is called a *meromorphic univalent* function. A meromorphic univalent function can have only one simple pole in G. A harmonic function $f : G \to \mathbb{C}$ which is one-to-one in G is called a *harmonic univalent* function.

Example 2.1.1 Let $G := \mathbb{C} \setminus \{1\}$ and consider the function

$$f(z) = \frac{z}{1 - z}, \quad z \in G.$$

If $f(z_1) = f(z_2)$ for points $z_1, z_2 \in G$, then

$$z_2(1 - z_1) = z_1(1 - z_2)$$

and this immediately reduces to $z_1 = z_2$. Thus, the function f is one-to-one in G. The function f is analytic in G and is therefore univalent in G. Since

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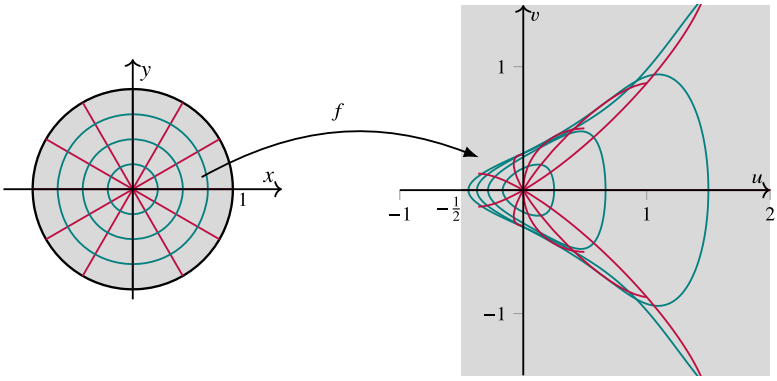


Fig. 2.1 $f(\mathbb{D})$ when $f(z) = z/(1-z)$

$$f'(z) = \frac{1}{(1-z)^2},$$

it is clear that $f'(z) \neq 0$ for all $z \in G$. Figure 2.1 shows the image of the unit disk $\mathbb{D}$ under this function f. The picture on the right shows the image of the concentric circles $|z| = r$ and the radial lines $\arg z = \theta$ in the unit disk.[1]

Example 2.1.2 Consider the function $f : G \to \mathbb{C}$ given by $f(z) = z^2$. Since

$$f(-z) = (-z)^2 = z^2 = f(z),$$

this function is not univalent in any region G containing both z and $-z$ for some point $z \in G$. Any neighborhood of the origin contains both z and $-z$ and therefore the function $f(z) = z^2$ is not even locally univalent at the origin. Notice that $f'(0) = 0$.

Suppose that the region G does not contain z and $-z$ for any z. For example, G could be the upper half-plane $\operatorname{Im} z > 0$, or the right half-plane $\operatorname{Re} z > 0$ or the disk $|z - a| < |a|$. Let $z_1, z_2 \in G$. If $f(z_1) = f(z_2)$, then $z_1^2 = z_2^2$ or

$$(z_1 - z_2)(z_1 + z_2) = 0.$$

Since $z_1, z_2 \in G$, it follows that $z_1 \neq -z_2$ or $z_1 + z_2 \neq 0$. Hence $z_1 = z_2$ and $f(z) = z^2$ is univalent in G. In particular, the function f is locally univalent at all points $z \neq 0$. Notice that $f'(z) = 2z \neq 0$ for $z \neq 0$.

We now show that Theorem 1.2.7 holds for locally univalent functions.

Theorem 2.1.3 *A function $f \in \mathcal{H}(G)$ is locally univalent at $z_0 \in G$ if and only if $f'(z_0) \neq 0$.*

[1] Henceforth, the unit disk will not be shown and the figure for $f(\mathbb{D})$ will contain either the images of the concentric circles $|z| = r$ or the radial lines $\arg z = \theta$ in the unit disk or both.

Proof We first prove the local univalency by assuming $f'(z_0) \neq 0$. Since $f \in \mathcal{H}(G)$, it has a Taylor series given by

$$f(z) = \sum_{k=0}^{\infty} a_k (z - z_0)^k$$

convergent in a disk $\mathbb{D}(z_0; R)$ where $R = d(z_0, \partial G) > 0$. If $z_1, z_2 \in \mathbb{D}(z_0; R)$ and $f(z_1) = f(z_2)$, then

$$\sum_{k=1}^{\infty} a_k \big((z_1 - z_0)^k - (z_2 - z_0)^k\big) = 0.$$

This simplifies to

$$(z_1 - z_2)\left(a_1 + \sum_{k=2}^{\infty} a_k \Big(\sum_{n=1}^{k} (z_1 - z_0)^{k-n} (z_2 - z_0)^{n-1}\Big)\right) = 0. \tag{2.1.1}$$

If $|z_1 - z_0| < \rho$ and $|z_2 - z_0| < \rho$, then

$$\left|a_1 + \sum_{k=2}^{\infty} a_k \Big(\sum_{n=1}^{k} (z_1 - z_0)^{k-n} (z_2 - z_0)^{n-1}\Big)\right| \geqslant |a_1| - \sum_{k=2}^{\infty} k\,|a_k|\,\rho^{k-1}.$$

Since $a_1 = f'(z_0) \neq 0$, choose ρ, $0 < \rho < R$, such that

$$\sum_{k=2}^{\infty} k\,|a_k|\,\rho^{k-1} < |a_1|\,.$$

Then, (2.1.1) implies that $z_1 = z_2$. The function f is therefore univalent in $\mathbb{D}(z_0; \rho)$ and thus f is locally univalent at z_0.

On the other hand, it follows from Theorem 1.2.7 that $f'(z_0) \neq 0$ since the function f is univalent in some disk $\mathbb{D}(z_0; r)$. □

Theorem 2.1.3 gives a necessary and sufficient condition for an analytic function to be locally univalent.

Example 2.1.4 The function $f(z) = \mathrm{e}^z$ has a non-vanishing derivative in $\mathbb{C}$ and therefore it is locally univalent in $\mathbb{C}$. However, the function f is not univalent in $\mathbb{C}$, as

$$f(z + 2n\pi\mathrm{i}) = \mathrm{e}^{z+2n\pi\mathrm{i}} = \mathrm{e}^z = f(z)$$

for any integer n.

We shall now prove a sufficient condition for the univalency of an analytic function defined in a convex region. The set of points

$$[z_1, z_2] := \{(1-t)z_1 + tz_2 : 0 \leqslant t \leqslant 1\}$$

is the line segment joining z_1 and z_2. A region G is *convex* if the line segment $[z_1, z_2]$ joining any two points $z_1, z_2 \in G$ lies entirely in G. A related notion that will be needed later is starlikeness. A domain G is *starlike* with respect to a point $z_0 \in G$ if the line segment $[z, z_0]$ lies inside G for every $z \in G$. A convex domain is starlike with respect to each point in it. We first need the following lemma.

Lemma 2.1.5 *If $f : [a, b] \to \mathbb{R}$ is continuous, and $f(x) > 0$ for all $x \in [a, b]$, then* $\int_a^b f(x)dx > 0$.

If a function $f : \mathbb{R} \to \mathbb{R}$ satisfy $f'(x) > 0$, then it is strictly increasing and therefore one-to-one. The complex analogue is given in the following theorem (see Fig. 2.2) .

Theorem 2.1.6 (Noshiro-Warschawski Theorem) *If a function f is analytic in a convex region G and there is some $\alpha \in \mathbb{R}$ such that* $\mathrm{Re}(\mathrm{e}^{\mathrm{i}\alpha} f'(z)) > 0$ *for all $z \in G$, then f is univalent in G.*

Proof One needs to show that $f(z_1) \neq f(z_2)$ for all $z_1, z_2 \in G$ with $z_1 \neq z_2$. By writing

$$z(t) = (1-t)z_1 + tz_2,$$

it follows that

$$\begin{aligned} f(z_2) - f(z_1) &= \int_{z_1}^{z_2} f'(z)\mathrm{d}z \\ &= (z_2 - z_1)\mathrm{e}^{-\mathrm{i}\alpha} \int_0^1 \mathrm{e}^{\mathrm{i}\alpha} f'((1-t)z_1 + tz_2)\mathrm{d}t \\ &= (z_2 - z_1)\mathrm{e}^{-\mathrm{i}\alpha} \int_0^1 \mathrm{e}^{\mathrm{i}\alpha} f'(z(t))\mathrm{d}t. \end{aligned}$$

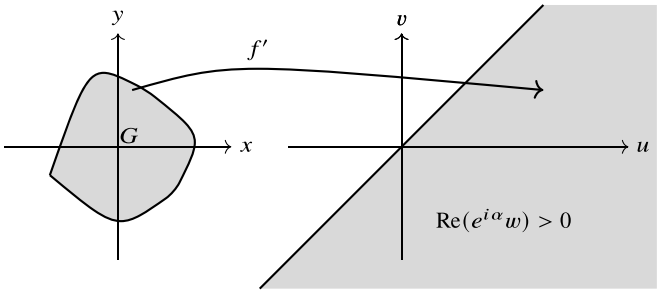


Fig. 2.2 Mapping f' satisfying the hypothesis of the Noshiro–Warschawski theorem

Since $\mathrm{Re}(\mathrm{e}^{\mathrm{i}\alpha} f'(z(t))) > 0$ for all $t \in [0, 1]$, it follows from Lemma 2.1.5 that the real part of the integral $\int_0^1 \mathrm{e}^{\mathrm{i}\alpha} f'(z(t))\mathrm{d}t$ is positive and therefore $f(z_2) \neq f(z_1)$. This proves that f is univalent in G. □

When $f'(z_0) \neq 0$, the Noshiro–Warschawski theorem (Theorem 2.1.6) yields an alternate proof for local univalence of f. By considering $\mathrm{e}^{-\mathrm{i}\arg(f'(z_0))} f'(z)$, we may assume that $\epsilon := f'(z_0) > 0$. Since $f \in \mathcal{H}(G)$, f' is continuous and, therefore, there is a $\delta > 0$ such that

$$|f'(z) - f'(z_0)| < \epsilon = f'(z_0)$$

whenever $|z - z_0| < \delta$. Evidently,

$$\begin{aligned} \mathrm{Re}\, f'(z) &= \mathrm{Re}\, f'(z_0) + \mathrm{Re}(f'(z) - f'(z_0)) \\ &\geqslant f'(z_0) - |f'(z) - f'(z_0)| > 0. \end{aligned}$$

Since $\mathrm{Re}\, f'(z) > 0$ for all $z \in \mathbb{D}(z_0; \delta)$, the function f is univalent in $\mathbb{D}(z_0; \delta)$ by the Noshiro-Warschawski theorem (Theorem 2.1.6).

A closed curve is said to be simple if it does not intersect itself, except that its initial and terminal points coincide. The region bounded by a simple closed curve (also known as a Jordan curve) is called the Jordan interior of the curve. The next theorem gives a sufficient condition for the univalence of a function in a region from its univalence on the boundary.

Theorem 2.1.7 (Darboux's Theorem) *Let G be a region and its boundary $C = \partial G$ be a simple closed rectifiable curve. If f is analytic in a region containing $\overline{G}$ and one-to-one on C, then f is univalent in G.*

Proof The function $w = f(z)$ is a continuous one-to-one correspondence between the points of C and the points of a closed contour $\Gamma := f(C)$. Let Δ be the Jordan interior of Γ. If w_0 is any point of the w-plane which does not lie on Γ, the number of zeros of $f(z) - w_0$ in G is

$$m = \frac{1}{2\pi\mathrm{i}} \int_C \frac{f'(z)}{f(z) - w_0} \mathrm{d}z = \frac{1}{2\pi\mathrm{i}} \int_\Gamma \frac{\mathrm{d}w}{w - w_0}$$

with the direction of integration around Γ being that which corresponds to the positive sense of the description of C. If w_0 lies outside Δ, then $m = 0$, and thus no point within G is mapped to a point outside Δ. On the other hand, if w_0 lies within Δ, then $m = \pm 1$. Since m cannot be negative, the equation $f(z) = w_0$ has precisely one root within G.

We now show that f cannot attain a value w_1 on Γ at any point $z_1 \in G$. Suppose, to the contrary, that $f(z_1) = w_1$ for some $z_1 \in G$. Then a neighborhood of z_1 would be mapped conformally onto a neighborhood of w_1. However, this is not possible, since the neighborhood of w_1 contains points outside Δ. □

A function satisfying the hypothesis of the Riemann mapping theorem is called a *Riemann mapping function*. We are interested in the inverse of Riemann mapping functions, which map the unit disk $\mathbb{D}$ onto simply connected domains. In view of the Riemann mapping theorem, properties of univalent functions on a simply connected region other than the whole complex plane corresponds to properties of univalent functions defined on the unit disk $\mathbb{D}$. For this reason, our focus will be on functions defined on $\mathbb{D}$.

Lemma 2.1.8 *If a function g is univalent in $\mathbb{D}$, then $f : \mathbb{D} \to \mathbb{C}$ given by*

$$f(z) := \frac{g(z) - g(0)}{g'(0)}$$

is also univalent in $\mathbb{D}$.

Proof If g is univalent in a region $\mathbb{D}$, then Theorem 1.2.7, gives $g'(z) \neq 0$ for all $z \in \mathbb{D}$. Since $0 \in \mathbb{D}$, it follows that $g'(0) \neq 0$ and hence the function f is well-defined and analytic in $\mathbb{D}$. If $f(z_1) = f(z_2)$ for $z_1, z_2 \in \mathbb{D}$, then

$$\frac{g(z_1) - g(0)}{g'(0)} = \frac{g(z_2) - g(0)}{g'(0)}.$$

Hence $g(z_1) = g(z_2)$ which yields $z_1 = z_2$ from the univalency of g. Thus f is univalent in $\mathbb{D}$. □

The function f defined in Lemma 2.1.8 satisfies the conditions

$$f(0) = 0, \quad f'(0) = 1,$$

and its Taylor's series has the form

$$f(z) = z + \sum_{k=2}^{\infty} a_k z^k, \quad z \in \mathbb{D}.$$

Definition 2.1.9 For each $n \in \mathbb{N}$, the class $\mathcal{A}_n$ consists of all analytic functions $f : \mathbb{D} \to \mathbb{C}$ of the form $f(z) = z + \sum_{k=n+1}^{\infty} a_k z^k$. The class $\mathcal{A} := \mathcal{A}_1$ consists of all analytic functions $f : \mathbb{D} \to \mathbb{C}$ normalized by $f(0) = 0 = f'(0) - 1$. The class $\mathcal{S}$ is a subclass of $\mathcal{A}$ consisting of univalent functions. The class $\mathcal{A}$ is called the class of *normalized analytic functions*, and the class $\mathcal{S}$ is called the class of *normalized univalent functions*.

Example 2.1.10 The function $f(z) = z + z^2 \in \mathcal{A}$ is not univalent as $f'(z) = 1 + 2z = 0$ at $z = -1/2 \in \mathbb{D}$. The boundary curve is not simple (see Fig. 2.3). However, even if the boundary curve is not simple, the function could still be univalent. Consider $G = \{z \in \mathbb{C} : |z| < 1, \operatorname{Im} z > 0\}$ and the function f on G defined by $f(z) = z^2$. The image of the boundary $\partial G = \{z \in \mathbb{C} : |z| = 1, \operatorname{Im} z > 0\} \cup [-1, 1]$ is a non-simple

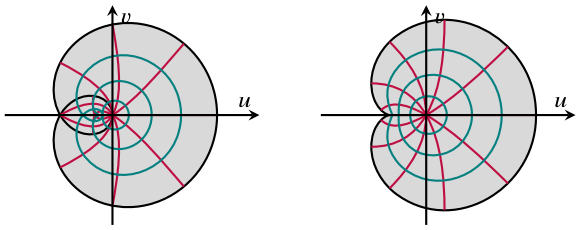


Fig. 2.3 $f(\mathbb{D})$ when $f(z) = z + z^2$ and $f(z) = z + z^2/2$

closed curve that traces the unit circle once from 1 anticlockwise back to 1, then follows the real interval from 1 to 0, and finally returns from 0 to 1. The image of ∂G is not simple, yet f is univalent in G.

Example 2.1.11 The function $g(z) = (1+z)^2$ is univalent in $\mathbb{D}$ but does not belong to $\mathcal{S}$.

To prove univalence, let us assume that $g(z_1) = g(z_2)$ for $z_1, z_2 \in \mathbb{D}$. Then

$$(1+z_1)^2 = (1+z_2)^2$$

or

$$(z_1 + z_2 + 2)(z_1 - z_2) = 0.$$

Clearly

$$|z_1 + z_2| \leqslant |z_1| + |z_2| < 2$$

for $z_1, z_2 \in \mathbb{D}$, so that $z_1 + z_2 + 2 \neq 0$ and hence $z_1 = z_2$. Thus the function g is univalent in $\mathbb{D}$.

Therefore, by Lemma 2.1.8, the function

$$f(z) := \frac{g(z) - g(0)}{g'(0)} = \frac{(1+z)^2 - 1}{2} = z + \frac{1}{2}z^2$$

is in $\mathcal{S}$.

That f is univalent can also be shown by using Theorem 2.1.6. Since

$$\operatorname{Re} f'(z) = \operatorname{Re}(1+z) = 1 - |z| > 0$$

for all $z \in \mathbb{D}$, by Theorem 2.1.6, the function f is univalent in $\mathbb{D}$. The image of the unit disk $\mathbb{D}$ under this function f is shown in Fig. 2.3.

Example 2.1.12 Consider the function $f \in \mathcal{A}$ given by

$$f(z) := \frac{1}{2}\log\left(\frac{1+z}{1-z}\right), \quad z \in \mathbb{D}.$$

For $z \in \mathbb{D}$,

$$\operatorname{Re} \frac{1}{1+z} = \frac{1}{2} + \frac{1}{2} \operatorname{Re} \frac{1-z}{1+z} = \frac{1}{2} + \frac{1}{2} \frac{1-|z|^2}{|1+z|^2} > \frac{1}{2}$$

and similarly

$$\operatorname{Re} \frac{1}{1-z} > \frac{1}{2}.$$

Since

$$\operatorname{Re} f'(z) = \operatorname{Re} \frac{1}{1-z^2} = \frac{1}{2} \left(\operatorname{Re} \frac{1}{1+z} + \operatorname{Re} \frac{1}{1-z} \right) > \frac{1}{2} > 0, \quad z \in \mathbb{D},$$

the function f is univalent in $\mathbb{D}$ by Theorem 2.1.6.

This function f maps $\mathbb{D}$ onto the strip $\{w \in \mathbb{C} : |\operatorname{Im} w| < \pi/4\}$ (see Fig. 2.4).

Theorem 2.1.13 *If* $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{A}$ *satisfies*

$$\sum_{k=2}^{\infty} k|a_k| \leqslant 1, \tag{2.1.2}$$

then $f \in \mathcal{S}$.

Proof We show that the derivative of f satisfying (2.1.2) has positive real part in $\mathbb{D}$. First note that

$$|f'(z) - 1| = \left| \sum_{k=2}^{\infty} k a_k z^{k-1} \right| \leqslant \sum_{k=2}^{\infty} k|a_k|\,|z|^{k-1} < \sum_{k=2}^{\infty} k|a_k| \leqslant 1, \quad z \in \mathbb{D}.$$

Then

$$-\operatorname{Re}(f'(z) - 1) \leqslant |f'(z) - 1| < 1, \quad z \in \mathbb{D},$$

which readily gives $\operatorname{Re} f'(z) > 0$ for all $z \in \mathbb{D}$. By Theorem 2.1.6, it follows that f belongs to $\mathcal{S}$. □

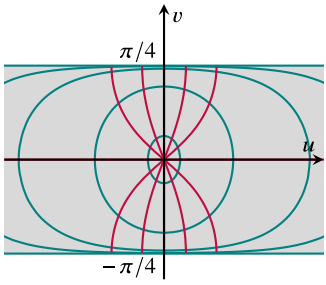


Fig. 2.4 $f(\mathbb{D})$ for $f(z) = (1/2)\log((1+z)/(1-z))$

Example 2.1.14 The function

$$f(z) = z + \frac{1}{n}z^n \in \mathcal{A}$$

satisfies the hypothesis of Theorem 2.1.13, and therefore it belongs to $\mathcal{S}$.

Example 2.1.15 (Koebe function) The Koebe function

$$\kappa(z) = \frac{z}{(1-z)^2} = z + \sum_{k=2}^{\infty} kz^k$$

is univalent in $\mathbb{D}$ and so $\kappa \in \mathcal{S}$. Also, $\kappa'(z) \neq 0$ for all $z \in \mathbb{D}$. Further, $\kappa(\mathbb{D}) = \mathbb{C} \setminus (-\infty, -1/4]$ or in other words, $\kappa(\mathbb{D})$ is the entire complex plane except for a slit along the negative real axis from $-\infty$ to $-1/4$ (see Fig. 2.5).

To see that κ is univalent, let $z_1, z_2 \in \mathbb{D}$ satisfy $\kappa(z_1) = \kappa(z_2)$. Then,

$$\frac{z_1}{(1-z_1)^2} = \frac{z_2}{(1-z_2)^2},$$

or

$$(z_1 - z_2)(1 - z_1 z_2) = 0. \tag{2.1.3}$$

Since $z_1, z_2 \in \mathbb{D}$, it follows that $|z_1 z_2| = |z_1||z_2| < 1$, and hence $1 - z_1 z_2 \neq 0$. Consequently, (2.1.3) yields $z_1 = z_2$. Thus, the function κ is one-to-one in $\mathbb{D}$. Since the function κ is normalized univalent in $\mathbb{D}$, it follows that $\kappa \in \mathcal{S}$.

A calculation shows that

$$\kappa'(z) = \frac{1+z}{(1-z)^3}.$$

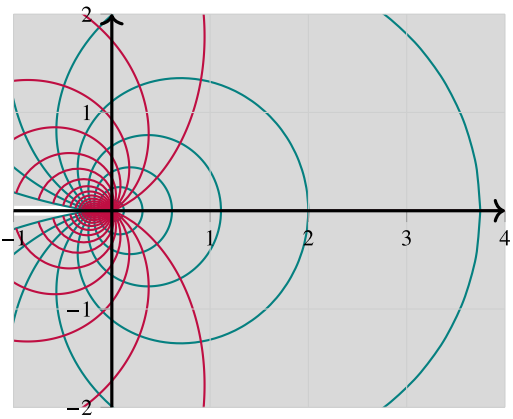


Fig. 2.5 $\kappa(\mathbb{D})$ when $\kappa(z) = z/(1-z)^2$

This function is nonzero in $\mathbb{D}$. Since

$$\kappa(e^{i\theta}) = -\frac{1}{4}\csc^2\frac{\theta}{2} \quad (\theta \in [0, 2\pi]),$$

the image of $\mathbb{D}$ is the region $D := \mathbb{C} \setminus (-\infty, -1/4]$ consisting the entire complex plane except for a slit along the negative real axis from $w = -\infty$ to $w = -\frac{1}{4}$.

The image of $\mathbb{D}$ can also be obtained by writing

$$\kappa(z) = \frac{1}{4}\left(\left(\frac{1+z}{1-z}\right)^2 - 1\right) = (w_3 \circ w_2 \circ w_1)(z)$$

where $w_1(z) = (1+z)/(1-z)$, $w_2(z) = z^2$ and $w_3(z) = (z-1)/4$ (see Fig. 2.6). For $z \in \mathbb{D}$,

$$\operatorname{Re} w_1(z) = \operatorname{Re}\frac{1+z}{1-z} = \operatorname{Re}\frac{(1+z)(1-\overline{z})}{|1-z|^2} = \frac{1-|z|^2}{|1-z|^2} > 0,$$

and so the mapping w_1 maps $\mathbb{D}$ into the right half-plane $\{z \in \mathbb{C} : \operatorname{Re} z > 0\}$. For any $w \in \{z \in \mathbb{C} : \operatorname{Re} z > 0\}$, $w_1(z) = w$ whenever $z = -(1-w)/(1+w)$. Also,

$$|z|^2 = \frac{1+|w|^2-2\operatorname{Re} w}{1+|w|^2+2\operatorname{Re} w} < 1$$

when $\operatorname{Re} w > 0$. Thus, the point z belongs to $\mathbb{D}$ and $w_1(z) = w$ proving that the mapping w_1 maps $\mathbb{D}$ onto the half-plane $\{z \in \mathbb{C} : \operatorname{Re} z > 0\}$. Further, the mapping w_2 clearly maps the half-plane $\{z \in \mathbb{C} : \operatorname{Re} z > 0\}$ onto the slit domain $\mathbb{C} \setminus (-\infty, 0]$ and the mapping w_3 maps this slit domain onto $\mathbb{C} \setminus (-\infty, -1/4]$. Thus, the function κ maps $\mathbb{D}$ onto $D = \mathbb{C} \setminus (-\infty, -1/4]$.

The following is an example of a Möbius transformation that belongs to $\mathcal{S}$.

Example 2.1.16 The function

$$\ell(z) = \frac{z}{1-z}, \quad z \in \mathbb{D},$$

is univalent in $\mathbb{D}$, $\ell(0) = 0$ and $\ell'(0) = 1$. Hence, $\ell \in \mathcal{S}$. Since

$$\operatorname{Re}\ell(z) = -\frac{1}{2} + \frac{1}{2}\operatorname{Re}\frac{1+z}{1-z} = -\frac{1}{2} + \frac{1}{2}\frac{1-|z|^2}{|1-z|^2},$$

it follows that the function ℓ maps $\mathbb{D}$ into the half-plane $\{w : \operatorname{Re} w > -1/2\}$. For any w satisfying $\operatorname{Re} w > -1/2$, the point $z = w/(1+w)$ belongs to unit disk $\mathbb{D}$, since $|z|^2 = |w|^2/|1+w|^2 < 1$ whenever $\operatorname{Re} w > -1/2$. Thus, for each such w, there exists $z \in \mathbb{D}$ such that $\ell(z) = w$. It follows that ℓ maps $\mathbb{D}$ onto the half-plane $\{w : \operatorname{Re} w > -1/2\}$.

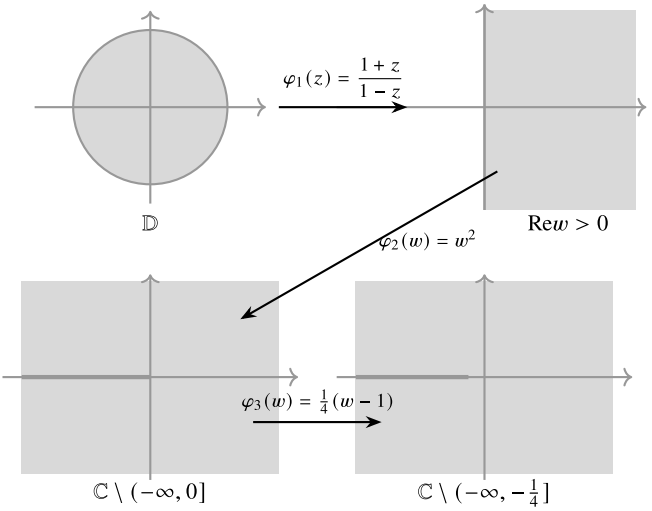


Fig. 2.6 Koebe function as a composition of simple mappings

2.2 Univalence of Transformations

If the functions f and g belong to $\mathcal{A}$, then their sum $f + g$ and product fg do not necessarily belong to $\mathcal{A}$. However, the functions $(f + g)/2$ and fg/z are in $\mathcal{A}$. The following examples show that there exist functions $f, g \in \mathcal{S}$ such that $(f + g)/2 \notin \mathcal{S}$ and $fg/z \notin \mathcal{S}$.

Example 2.2.1 Consider the functions $f, g : \mathbb{D} \to \mathbb{C}$ given by

$$f(z) = \frac{z}{1-z} \text{ and } g(z) = \mathrm{i} f(-\mathrm{i}z) = \frac{z}{1+\mathrm{i}z}.$$

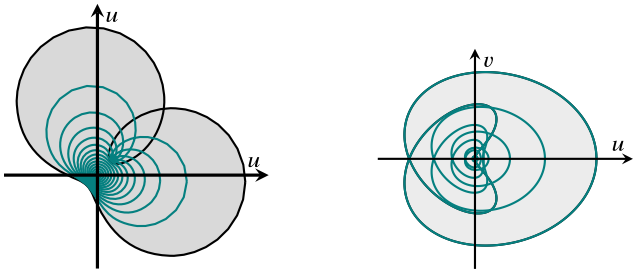
Example 2.1.16 shows that the function f belongs to $\mathcal{S}$. Since $g(z) = \mathrm{i} f(-\mathrm{i}z)$, g is also univalent. Therefore, the functions f and g are in $\mathcal{S}$. The function $h : \mathbb{D} \to \mathbb{C}$ defined by

$$h(z) = \frac{f(z) + g(z)}{2}$$

is analytic in $\mathbb{D}$, $h(0) = 0$ and

$$h'(z) = \frac{f'(z) + g'(z)}{2} = \frac{1 - (1-\mathrm{i})z}{(1-z)^2(1+\mathrm{i}z)^2}.$$

Since $h'((1 + \mathrm{i})/2) = 0$, Theorem 2.1.3 shows that $h \notin \mathcal{S}$. The image of h is shown in Fig. 2.7



(a) $h(z) = (z/(1-z) + z/(1+\mathrm{i}z))/2$ (b) $k(z) = z\,(1+z/2)^2$

Fig. 2.7 Non-univalence of $h = (f+g)/2$ and $k = fg/z$

Example 2.2.2 Let $f, g, k : \mathbb{D} \to \mathbb{C}$ be given by

$$f(z) = g(z) = z + \frac{1}{2}z^2 \quad \text{and} \quad k(z) = \frac{f(z)g(z)}{z} = z\left(1 + \frac{1}{2}z\right)^2.$$

Since

$$k'(z) = \frac{1}{4}(2+z)(2+3z),$$

the derivative k' vanishes at $z = -2/3 \in \mathbb{D}$. It then follows from Theorem 2.1.3 that the function k is not univalent in $\mathbb{D}$. See Fig. 2.7.

The Koebe function κ maps $\mathbb{D}$ onto $\mathbb{C} \setminus (-\infty, -1/4]$ and therefore the composition $\kappa \circ \kappa$ is not defined on $\mathbb{D}$. However, if $f \in \mathcal{S}$ and $g : f(\mathbb{D}) \to \mathbb{C}$ is univalent and normalized by $g(0) = 0$, $g'(0) = 1$, then the composite function $F = g \circ f$ belongs to $\mathcal{S}$. There are other simple transformations T under which the class $\mathcal{S}$ is closed: $f \in \mathcal{S} \Rightarrow T(f) \in \mathcal{S}$.

Theorem 2.2.3 (Rotation) *If $f \in \mathcal{S}$ and $|\alpha| = 1$, then $g : \mathbb{D} \to \mathbb{C}$ defined by $g(z) = \overline{\alpha} f(\alpha z)$ also belongs to $\mathcal{S}$.*

Proof If f is analytic in $\mathbb{D}$, then so is g. Since $g(0) = \overline{\alpha} f(0) = 0$ and $g'(0) = |\alpha|^2 f'(0) = 1$, the function g is normalized. If $g(z_1) = g(z_2)$, then $f(\alpha z_1) = f(\alpha z_2)$ and hence $z_1 = z_2$. Therefore, $g \in \mathcal{S}$. □

Example 2.2.4 The Koebe function

$$\kappa(z) = \frac{z}{(1-z)^2}$$

is in $\mathcal{S}$. See Example 2.1.15. The functions κ_θ defined by

$$\kappa_\theta(z) = \mathrm{e}^{-\mathrm{i}\theta}\kappa(\mathrm{e}^{\mathrm{i}\theta}z) = \frac{z}{(1-\mathrm{e}^{\mathrm{i}\theta}z)^2}, \quad \theta \in [0, 2\pi]$$

are called the rotations of the Koebe function κ and they are all in $\mathcal{S}$ by Theorem 2.2.3.

Theorem 2.2.5 (Square-root Transformation) *If $f \in \mathcal{S}$, then there exists $g \in \mathcal{S}$ such that $g^2(z) = f(z^2)$ for all $z \in \mathbb{D}$. This function g, often denoted as $g(z) = \sqrt{f(z^2)}$, is defined explicitly by*

$$g(z) = z\sqrt{f(z^2)/z^2}.$$

Proof Let $f \in \mathcal{S}$, and define the function φ by $\varphi(z) := f(z)/z$ for $z \neq 0$ and $\varphi(0) = 1$. A computation shows that φ is differentiable at the origin. Thus, φ is analytic in $\mathbb{D}$. Since $f \in \mathcal{S}$, $\varphi(z) \neq 0$ for $z \neq 0$. As $\varphi(0) = 1$, it follows that $\varphi(z) \neq 0$ for all $z \in \mathbb{D}$.

Since every non-vanishing analytic function on a simply connected domain has an analytic square root, there exists an analytic function h such that $h^2(z) = \varphi(z)$ with $h(0) = 1$. Define $g : \mathbb{D} \to \mathbb{C}$ by $g(z) := zh(z^2)$. The function g is also analytic in $\mathbb{D}$, $g(0) = 0$ and $g'(0) = 1$ so that $g \in \mathcal{A}$. It remains to show that $g \in \mathcal{S}$ and $g^2(z) = f(z^2)$.

Since $g(z) := zh(z^2)$, $h^2(z) = \varphi(z)$ and $z\varphi(z) = f(z)$, it follows that $g^2(z) = f(z^2)$. If $g(z_1) = g(z_2)$ for $z_1, z_2 \in \mathbb{D}$, then $f(z_1^2) = f(z_2^2)$. Since f is univalent, this implies $z_1^2 = z_2^2$, that is, $z_1 = \pm z_2$. If $z_1 = -z_2$, then $g(z_1) = z_1 h(z_1^2)$ and

$$g(z_2) = z_2 h(z_2^2) = -z_1 h(z_1^2) = -g(z_1).$$

However, $g(z_1) = g(z_2)$ and so $g(z_1) = 0$. Since $z_1 h(z_1^2) = g(z_1) = 0$ and $h(z_1^2)$ is never zero, it follows that $z_1 = 0$. Thus, in either case, $z_1 = z_2$ and so, the function g is univalent. □

Example 2.2.6 The function

$$f(z) := \frac{z}{1 - z^2}, \quad z \in \mathbb{D},$$

is the square-root transformation of the Koebe function κ, that is,

$$f(z) = \sqrt{\kappa(z^2)}.$$

Hence, by Theorem 2.2.5, f belongs to $\mathcal{S}$.

Definition 2.2.7 Let n be a positive integer. A domain G is said to be n-fold symmetric if a rotation of G about the origin through an angle $2\pi/n$ carries G onto itself. A function $f : \mathbb{D} \to \mathbb{C}$ is said to be *n-fold symmetric* in $\mathbb{D}$ if

$$f(\mathrm{e}^{2\pi \mathrm{i}/n} z) = \mathrm{e}^{2\pi \mathrm{i}/n} f(z), \quad z \in \mathbb{D}.$$

Denote by $\mathcal{S}^{(n)}$ the subclass of functions $f \in \mathcal{S}$ that are n-fold symmetric in $\mathbb{D}$.

A twofold symmetric function is simply an odd function as the condition reduces to $f(-z) = -f(z)$. Functions in the class $\mathcal{S}^{(2)}$ are called *odd univalent functions*. Odd functions in $\mathcal{A}$ have Taylor series of the form $z + a_3 z^3 + a_5 z^5 + \cdots$. More generally, we have the following result.

Theorem 2.2.8 *If $f \in \mathcal{A}$ is n-fold symmetric in $\mathbb{D}$, then it has a Taylor series of the form*

$$f(z) = z + a_{n+1}z^{n+1} + a_{2n+1}z^{2n+1} + \cdots = z + \sum_{k=1}^{\infty} a_{nk+1}z^{nk+1}. \tag{2.2.1}$$

Conversely, if $f \in \mathcal{A}$ has the Taylor series (2.2.1), then the function f is n-fold symmetric in $\mathbb{D}$.

Proof If f is n-fold symmetric in $\mathbb{D}$, then

$$f(\mathrm{e}^{2\pi \mathrm{i}/n}z) = \mathrm{e}^{2\pi \mathrm{i}/n} f(z)$$

and comparing the coefficients of the Taylor series expansions, we see that f has the desired form. The converse is an easy consequence of (2.2.1). □

Notice that the square-root transform of a function f is an odd function. More generally, the following result holds.

Theorem 2.2.9 *If $f \in \mathcal{S}$, then the n-th root transform $g(z) = \sqrt[n]{f(z^n)}$ is a n-fold symmetric function in $\mathcal{S}^{(n)}$. Conversely, if $g \in \mathcal{S}^{(n)}$, then $f(z) = (g(\sqrt[n]{z}))^n$ belongs to $\mathcal{S}$. In both cases, $f(z^n) = (g(z))^n$.*

Proof Let $f \in \mathcal{S}$. Then the function h defined by $h(0) = 1$ and $h(z) = f(z)/z$ for $z \neq 0$ is clearly non-vanishing in $\mathbb{D}$ and therefore $h(z^n)$ is also non-vanishing. Since $\mathbb{D}$ is simply connected, $\sqrt[n]{h(z^n)}$ is analytic and so $g(z) := z\sqrt[n]{h(z^n)}$ is analytic in $\mathbb{D}$. We write this function as $g(z) = \sqrt[n]{f(z^n)}$ (though an analytic branch of $f(z^n)$ cannot be defined in $\mathbb{D}$). Evidently, g is n-fold symmetric.

To check the univalency, let $g(z) = g(w)$ for $z, w \in \mathbb{D}$ so that $f(z^n) = f(w^n)$. The univalence of f implies that $z^n = w^n$ or $z = \mathrm{e}^{2\pi k\mathrm{i}/n}w$. Since g is n-fold symmetric, the assumption $g(z) = g(w)$ becomes $g(z) = g(\mathrm{e}^{2\pi k\mathrm{i}/n}w) = \mathrm{e}^{2\pi k\mathrm{i}/n}g(w) = \mathrm{e}^{2\pi k\mathrm{i}/n}g(z)$. This shows that $g(z) = 0$ or $k = 0$. If $g(z) = 0$, then $f(z^n) = 0$. Thus $z = 0$ and so $z = w$. The case $k = 0$ also clearly implies $z = w$. Thus, g is univalent in $\mathbb{D}$.

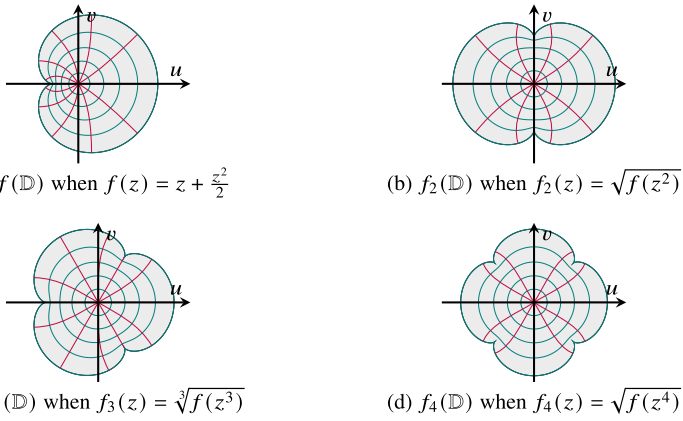
To prove the converse, let $g \in \mathcal{S}^{(n)}$ so that, by Theorem 2.2.8, it has a Taylor series of the form

$$g(z) = z + \sum_{k=1}^{\infty} a_{nk+1}z^{nk+1} = z\big(1 + \sum_{k=1}^{\infty} a_{nk+1}z^{nk}\big).$$

The series

$$1 + \sum_{k=1}^{\infty} a_{nk+1}z^{k}$$

is convergent and non-vanishing in $\mathbb{D}$ as g is univalent. We associate the analytic function



(a) $f(\mathbb{D})$ when $f(z) = z + \frac{z^2}{2}$ (b) $f_2(\mathbb{D})$ when $f_2(z) = \sqrt{f(z^2)}$

(c) $f_3(\mathbb{D})$ when $f_3(z) = \sqrt[3]{f(z^3)}$ (d) $f_4(\mathbb{D})$ when $f_4(z) = \sqrt{f(z^4)}$

Fig. 2.8 The image of $\mathbb{D}$ under $f(z) = z + z^2/2$ and its 2-,3-,fourfold symmetric functions

$$f(z) = z\left(1 + \sum_{k=1}^{\infty} a_{nk+1} z^k\right)^n$$

with g formally by writing $f(z) = (g(\sqrt[n]{z}))^n$. To see the univalence of f, note that $f(z) = f(w)$ becomes $\sqrt[n]{z} = \sqrt[n]{w}$ and so $z = w$. □

The image of the unit disk under $f(z) = z + z^2/2$ and its 2-,3-,fourfold symmetric functions are illustrated in Fig. 2.8.

If $f : A \to B$ and $g : B \to C$ are both one-to-one, then the composite function $g \circ f : A \to C$ is one-to-one. This fact is used in the proof of the following theorem.

Theorem 2.2.10 (Disk Automorphism) *Let $f \in \mathcal{S}$, and $\alpha \in \mathbb{D}$. Then*

$$g(z) = \frac{f\left(\frac{\alpha - z}{1 - \overline{\alpha} z}\right) - f(\alpha)}{(|\alpha|^2 - 1) f'(\alpha)}, \quad z \in \mathbb{D}, \tag{2.2.2}$$

is also in the class $\mathcal{S}$. Further, the mapping $T : \mathcal{S} \to \mathcal{S}$ defined by $T(f) = g$ is invertible.

Proof By Theorem 1.1.2, the Möbius transformation

$$\varphi_\alpha(z) := \frac{\alpha - z}{1 - \overline{\alpha} z}$$

maps $\mathbb{D}$ onto itself. Being a Möbius transformation, φ_α is univalent and

$$\varphi_\alpha'(z) = \frac{|\alpha|^2 - 1}{(1 - \overline{\alpha} z)^2}.$$

The function g in (2.2.2) can be written as

$$g(z) = \frac{f(\varphi_\alpha(z)) - f(\varphi_\alpha(0))}{f'(\varphi_\alpha(0))\varphi'_\alpha(0)}. \tag{2.2.3}$$

Since f and φ_α are univalent, the function $f \circ \varphi_\alpha$ is univalent and hence, by Lemma 2.1.8, the corresponding normalized function g belongs to $\mathcal{S}$.

To prove that T is invertible, we show $T(g) = f$. Since $\varphi_\alpha^{-1} = \varphi_\alpha$, by replacing z with $\varphi_\alpha(z)$ in (2.2.3) yields

$$g(\varphi_\alpha(z)) = \frac{f(\varphi_\alpha(\varphi_\alpha(z))) - f(\varphi_\alpha(0))}{f'(\varphi_\alpha(0))\varphi'_\alpha(0)} = \frac{f(z) - f(\alpha)}{f'(\alpha)\varphi'_\alpha(0)} \tag{2.2.4}$$

and, by a differentiation, we have

$$g'(\alpha)\varphi'_\alpha(0) = g'(\varphi_\alpha(0))\varphi'_\alpha(0) = \frac{f'(0)}{f'(\alpha)\varphi'_\alpha(0)} = \frac{1}{f'(\alpha)\varphi'_\alpha(0)}. \tag{2.2.5}$$

It readily follows from (2.2.4) and (2.2.5) that

$$\begin{aligned} &\frac{g(\varphi_\alpha(z)) - g(\varphi_\alpha(0))}{g'(\varphi_\alpha(0))\varphi'_\alpha(0)} \\ &= \frac{1}{g'(\alpha)\varphi'_\alpha(0)} \left(\frac{f(z) - f(\alpha)}{f'(\alpha)\varphi'_\alpha(0)} - \frac{-f(\alpha)}{f'(\alpha)\varphi'_\alpha(0)} \right) \\ &= f(z). \end{aligned}$$

This proves that $T(g) = f$. □

Remark 2.2.11 We call the function g given by (2.2.2) as the disk automorphism of the function f with parameter α. In the proof of the second part of Theorem 2.2.10, we have shown that if g is the disk automorphism of the function f with parameter α, then f is the disk automorphism of the function g with parameter α.

Example 2.2.12 Let $f(z) = z$ in Theorem 2.2.10. Then

$$g(z) = \frac{z}{1 - \overline{\alpha} z}, \quad \alpha \in \mathbb{D}.$$

Therefore, g belongs to $\mathcal{S}$.

A complex number α is called an *omitted value* of a function f if $\alpha \notin f(\mathbb{D})$. If α is a omitted value of f, then f is said to omit α.

Theorem 2.2.13 (Omitted-value Transformation) *If $f \in \mathcal{S}$ omits α, then*

$$h(z) := \frac{\alpha f(z)}{\alpha - f(z)}, \quad z \in \mathbb{D}$$

belongs to $\mathcal{S}$.

Proof Since $0 = f(0) \in f(\mathbb{D}), \alpha \neq 0$. The function h satisfies $h(0) = 0 = h'(0) - 1$. Note that the function $\varphi : \mathbb{C} \setminus \{\alpha\} \to \mathbb{C}$ defined by

$$\varphi(z) = \frac{\alpha z}{\alpha - z}$$

is a Möbius transformation and $h = \varphi \circ f$. Since Möbius transformations are one-to-one, it follows that the function h is univalent. □

Example 2.2.14 The Koebe function $\kappa(z) = z/(1-z)^2$ omits the interval $(-\infty, -1/4]$. For each $\alpha \leqslant -1/4$, it follows from Theorem 2.2.13 that the function

$$h(z) = \frac{\alpha\kappa(z)}{\alpha - \kappa(z)} = \frac{z}{1 + \beta z + z^2}, \quad z \in \mathbb{D}$$

belongs to $\mathcal{S}$, where $\beta = -2 - 1/\alpha \in (-2, 2]$. In particular, for $\alpha = -1$ and $\alpha = -1/2$, the function h reduces respectively to the functions

$$\frac{z}{1 - z + z^2}, \quad \frac{z}{1 + z^2}$$

and hence these functions are univalent.

Theorem 2.2.15 *If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{S}$, then*

$$g(z) := \overline{f(\overline{z})} = z + \sum_{k=2}^{\infty} \overline{a_k} z^k,$$

and

$$g(z) := f(rz)/r, \quad 0 < r < 1,$$

also belong to the class $\mathcal{S}$.

Proof Each of the function g is analytic in $\mathbb{D}$ and normalized by $g(0) = 0 = g'(0) - 1$. The analyticity of the function g defined by $g(z) = \overline{f(\overline{z})}$ follows from the Taylor series expansion. Univalence is an easy consequence. □

2.3 The Bieberbach Theorem

If Γ is a simple closed curve enclosing a region E, then the area $A(E)$ of E is given by

$$A(E) = \iint_E \mathrm{d}u\mathrm{d}v.$$

From Green's theorem,

$$A(E) = \frac{1}{2}\int_{\Gamma} u\mathrm{d}v - v\mathrm{d}u = \frac{1}{2\mathrm{i}}\int_{\Gamma} \overline{w}\mathrm{d}w, \quad w = u + \mathrm{i}v.$$

If Γ is the image of another curve γ under an analytic univalent function g, then

$$A(E) = \frac{1}{2\mathrm{i}}\int_{\gamma} \overline{g(z)}g'(z)\mathrm{d}z.$$

In the case of a meromorphic function g, γ enclosing a region G, the area $A(E)$ is the area of the omitted region $\mathbb{C} \setminus g(G)$. We illustrate this with an example.

Example 2.3.1 Let $\Delta = \mathbb{C} \setminus \overline{\mathbb{D}}$. The function

$$g(z) = z + \frac{1}{z}$$

is univalent in Δ. To see this, let $g(z) = g(w)$ for two points $z, w \in \Delta$. Then

$$0 = g(z) - g(w) = (z - w)\left(1 - \frac{1}{zw}\right).$$

Since $z, w \in \Delta$, it follows that $|z| > 1$ and $|w| > 1$, and so $|zw| > 1$. This shows that $1/(zw) \neq 1$ so that $z = w$, proving the univalence of g in Δ.

The function g maps the circle γ: $|z| = r > 1$ to an ellipse Γ enclosing a region E_r; the parametric equation of Γ is given by

$$\Gamma(t) = g(r\mathrm{e}^{\mathrm{i}t}) = a\cos t + \mathrm{i}b\sin t \quad (0 \leqslant t \leqslant 2\pi),$$

where $a = r + 1/r$ and $b = r - 1/r$. The area of the region E_r bounded by this ellipse is πab. Here, $\overline{z} = r^2/z$ on $|z| = r$. Using this along with the Cauchy integral formula yield

$$\begin{aligned} A(E_r) &= \frac{1}{2\mathrm{i}}\int_{|z|=r} \overline{g(z)}g'(z)\mathrm{d}z \\ &= \frac{1}{2\mathrm{i}}\int_{|z|=r} \left(\overline{z} + \frac{1}{\overline{z}}\right)\left(1 - \frac{1}{z^2}\right)\mathrm{d}z \\ &= \frac{1}{2\mathrm{i}}\int_{|z|=r} \left(\frac{r^2}{z} + \frac{z}{r^2}\right)\left(1 - \frac{1}{z^2}\right)\mathrm{d}z \\ &= \pi\left(r^2 - \frac{1}{r^2}\right) = \pi ab. \end{aligned}$$

Let $\Delta_r = \{z \in \mathbb{C} : |z| > r\}$. Then $g(\Delta_r)$ omits the interior of the ellipse E_r and therefore $A(E_r)$ is the area of the omitted region $\mathbb{C} \setminus g(\Delta_r)$.

Theorem 2.3.2 (Area Theorem) *Let $g : \mathbb{C} \setminus \overline{\mathbb{D}} \to \mathbb{C}$ be analytic and one-to-one with the Laurent expansion*

$$g(z) = z + \sum_{k=0}^{\infty} \frac{b_k}{z^k}, \quad z \in \mathbb{C} \setminus \overline{\mathbb{D}}.$$

Then

$$\sum_{k=1}^{\infty} k|b_k|^2 \leqslant 1. \tag{2.3.1}$$

As a consequence, $|b_1| \leqslant 1$ with equality if and only if

$$g(z) = z + b_0 + b_1 \frac{1}{z}, \quad |b_1| = 1.$$

Proof Let Γ be the image of the circle γ centered at 0 and radius $r > 1$ under the function g. See Fig. 2.9. The area enclosed by the curve Γ which is the same as the area of the omitted region $\mathbb{C} \setminus g(\Delta_r)$ is given by

$$\begin{aligned} A &= \frac{1}{2\mathrm{i}} \int_{\gamma} \overline{g(z)} g'(z) \mathrm{d}z \\ &= \frac{1}{2} \int_{|z|=r} \overline{g(r\mathrm{e}^{\mathrm{i}\theta})} g'(r\mathrm{e}^{\mathrm{i}\theta}) r\mathrm{e}^{\mathrm{i}\theta} \mathrm{d}\theta \\ &= \frac{1}{2} \int_0^{2\pi} \left(r\mathrm{e}^{-\mathrm{i}\theta} + \sum_{k=0}^{\infty} \overline{b_k} r^{-k} \mathrm{e}^{\mathrm{i}k\theta} \right) \left(r\mathrm{e}^{\mathrm{i}\theta} - \sum_{k=1}^{\infty} k b_k r^{-k} \mathrm{e}^{-\mathrm{i}k\theta} \right) \mathrm{d}\theta \\ &= \pi \left(r^2 - \sum_{k=1}^{\infty} k|b_k|^2 r^{-2k} \right). \end{aligned}$$

The last equality is obtained by using the orthogonality of the set $\{\mathrm{e}^{\mathrm{i}k\theta} : k \in \mathbb{Z}\}$. Since $A \geqslant 0$,

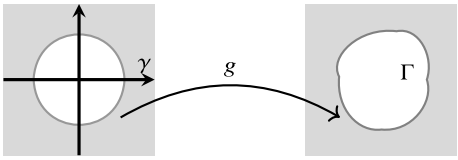


Fig. 2.9 Area theorem

$$r^2 - \sum_{k=1}^{\infty} k|b_k|^2 r^{-2k} \geqslant 0.$$

The inequality (2.3.1) is obtained by letting r decreases to 1.

If the inequality (2.3.1) is satisfied, then

$$|b_1|^2 \leqslant \sum_{k=1}^{\infty} k|b_k|^2 \leqslant 1, \tag{2.3.2}$$

and therefore $|b_1| \leqslant 1$. From the inequality (2.3.2), it follows that $|b_1| = 1$ if and only if $b_n = 0$ for $n \geqslant 2$ and hence

$$g(z) = z + b_0 + b_1 z^{-1}, \qquad |b_1| = 1.$$

□

Theorem 2.3.3 *If $f(z) = \sum_{k=0}^{\infty} a_k z^k$ is analytic in $\mathbb{D}$, then the area of $f(\overline{\mathbb{D}}_r)$ is*

$$A(r) = \pi \sum_{k=1}^{\infty} k\,|a_k|^2\, r^{2k}.$$

Proof Let $f(z) = u + \mathrm{i}v$. Then

$$A(r) = \iint\limits_{f(\overline{\mathbb{D}}_r)} \mathrm{d}u\mathrm{d}v = \iint\limits_{\overline{\mathbb{D}}_r} \frac{\partial(u, v)}{\partial(x, y)} \mathrm{d}x\mathrm{d}y$$

by the usual rule for the transformation of a double integral. The partial derivatives are continuous and satisfy the Cauchy-Riemann equations. Also

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} = \begin{vmatrix} u_x & -v_x \\ v_x & u_x \end{vmatrix} = u_x^2 + v_x^2 = \left|f'(z)\right|^2.$$

Thus,

$$A(r) = \iint\limits_{\overline{\mathbb{D}}_r} \left|f'(z)\right|^2 \mathrm{d}x\mathrm{d}y.$$

Changing $z = x + \mathrm{i}y$ to polar coordinates $z = \varrho \mathrm{e}^{\mathrm{i}\theta}$ gives

$$A(r) = \iint\limits_{\overline{\mathbb{D}}_r} \left|f'(z)\right|^2 \mathrm{d}x\mathrm{d}y = \int\limits_{\varrho=0}^{r} \int\limits_{\theta=0}^{2\pi} \left|f'(\varrho \mathrm{e}^{\mathrm{i}\theta})\right|^2 \varrho \mathrm{d}\theta d\varrho.$$

For the integration on θ, by using the orthogonality of the set $\{e^{in\theta} : n \in \mathbb{Z}\}$ leads to

$$\begin{aligned}
&\int_0^{2\pi} \left|f'(\varrho e^{i\theta})\right|^2 d\theta \\
&= \int_0^{2\pi} f'(\varrho e^{i\theta})\overline{f'(\varrho e^{i\theta})}d\theta \\
&= \int_0^{2\pi} \Big(\sum_{k=1}^{\infty} k a_k \varrho^{k-1} e^{(k-1)i\theta}\Big)\Big(\sum_{m=1}^{\infty} m\overline{a}_m \varrho^{m-1} e^{-(m-1)i\theta}\Big)d\theta \\
&= 2\pi \sum_{k=1}^{\infty} k^2 |a_k|^2 \varrho^{2k-2}.
\end{aligned}$$

Thus,

$$A(r) = \int_{\varrho=0}^{r} 2\pi \sum_{k=1}^{\infty} k^2 |a_k|^2 \varrho^{2k-2} \varrho d\varrho = \pi \sum_{k=1}^{\infty} k |a_k|^2 r^{2k}.$$

□

Theorem 2.3.4 (Bieberbach's Theorem) *If $f(z) = z + a_2 z^2 + \cdots \in \mathcal{S}$, then $|a_2| \leqslant 2$, with equality if and only if f is a rotation of the Koebe function κ.*

Proof From Theorem 2.2.5, there exists $h \in \mathcal{S}$ such that $h^2(z) = f(z^2)$. Then $G : \mathbb{D} \to \mathbb{C}_\infty$ defined by

$$G(z) = \frac{1}{h(z)} = \frac{1}{\sqrt{f(z^2)}} = \frac{1}{z} - \frac{a_2}{2} z + \cdots$$

is also one-to-one in $\mathbb{D}$. Therefore, the function $g : \mathbb{C} \setminus \overline{\mathbb{D}} \to \mathbb{C}$ defined by

$$g(z) = G(1/z) = z - \frac{a_2}{2}\frac{1}{z} + \cdots$$

is analytic and one-to-one. From the area theorem (Theorem 2.3.2), $|a_2| \leqslant 2$. Equality occurs if and only if

$$g(z) = z - \alpha\frac{1}{z}, \quad \left(\alpha = \frac{a_2}{2}\right),$$

or

$$f(z) = \frac{1}{(g(1/\sqrt{z}))^2} = \frac{z}{(1 - \alpha z)^2} \quad \text{with } |\alpha| = 1.$$

□

Corollary 2.3.5 *If $f(z) = z + a_{n+1}z^{n+1} + \cdots \in \mathcal{S}^{(n)}$, then the sharp inequality $|a_{n+1}| \leqslant 2/n$ holds.*

Proof From Theorem 2.2.9, the function

$$g(z) = (f(\sqrt[n]{z}))^n = z\Big(1 + \sum_{k=1}^{\infty} a_{nk+1}z^k\Big)^n = z + na_{n+1}z^2 + \cdots$$

belongs to $\mathcal{S}$. The Bieberbach's theorem (Theorem 2.3.4) immediately shows that $|a_{n+1}| \leqslant 2/n$. Clearly, the equality holds for the function

$$f(z) = \frac{z}{(1 - z^n)^{2/n}} = z + \frac{2}{n}z^{n+1} + \cdots$$

or its rotations. □

An important consequence of Bieberbach's theorem is that the range of any $f \in \mathcal{S}$ contains the disk $\mathbb{D}_{1/4}$. This theorem is also known as the *covering theorem* for the class $\mathcal{S}$.

Theorem 2.3.6 (Koebe One-Quarter Theorem)
If $f \in \mathcal{S}$, then $\mathbb{D}_{1/4} \subset f(\mathbb{D})$. The constant $1/4$ is sharp.

Proof Let $w \notin f(\mathbb{D})$ and define the function

$$h(z) = \frac{wf(z)}{w - f(z)} = f(z)\left(1 - \frac{f(z)}{w}\right)^{-1}.$$

By Theorem 2.2.13, the function h is in $\mathcal{S}$. Let $f(z) = z + a_2z^2 + \cdots$. The Taylor series of h around the origin is given by

$$h(z) = z + \left(a_2 + \frac{1}{w}\right)z^2 + \cdots.$$

Bieberbach's theorem (Theorem 2.3.4) applied to the second coefficients of f and h leads to $|a_2| \leqslant 2$, and

$$\left|a_2 + \frac{1}{w}\right| \leqslant 2.$$

Hence,

$$\left|\frac{1}{w}\right| \leqslant \left|\frac{1}{w} + a_2\right| + |a_2| \leqslant 2 + |a_2| \leqslant 4. \tag{2.3.3}$$

Thus $|w| \geqslant 1/4$ whenever $w \notin f(\mathbb{D})$. Equivalently, $\mathbb{D}_{1/4} \subset f(\mathbb{D})$.

By Example 2.1.15, the image of the Koebe function κ is the entire complex plane minus the slit along the negative real axis from $-1/4$ to infinity. This shows that $1/4$ is sharp. □

The inequality $|1/w| \leqslant 2 + |a_2|$ proved above in (2.3.3) immediately gives the following result.

Corollary 2.3.7 *If $f(z) = z + a_2 z^2 + \cdots \in \mathcal{S}$, then every omitted value w of f satisfies $|w| \geqslant 1/(2 + |a_2|)$.*

Example 2.3.8 The function $f(z) = z + z^2/2$ is in $\mathcal{S}$ and hence $f(\mathbb{D})$ contains the disk $\mathbb{D}_{1/4}$. Since $a_2 = 1/2$, the previous corollary shows that $f(\mathbb{D})$ contains the disk $\mathbb{D}_{2/5}$. We can show that in fact $f(\mathbb{D})$ contains the disk $\mathbb{D}_{1/2}$. To see this, note that

$$||f(z)| - |z|| \leqslant |f(z) - z| \leqslant \frac{1}{2}|z|^2,$$

and hence

$$|z| - \frac{1}{2}|z|^2 \leqslant |f(z)| \leqslant |z| + \frac{1}{2}|z|^2.$$

Letting $|z| \to 1$ shows that $f(\mathbb{D})$ contains the disk $\mathbb{D}_{1/2}$ and is contained in $\mathbb{D}_{3/2}$.

Figure 2.10 shows the containment $\mathbb{D}_{1/4} \subset \mathbb{D}_{2/5} \subset \mathbb{D}_{1/2} \subset f(\mathbb{D}) \subset \mathbb{D}_{3/2}$.

The following theorem, stated without proof, extends the Bieberbach's theorem.

Theorem 2.3.9 (de Branges' Theorem) *If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{S}$, then $|a_k| \leqslant k$ for each $k \geqslant 2$, with equality if and only if f is a rotation of the Koebe function κ.*

Theorem 2.3.9, known as the Bieberbach conjecture, was proposed by Bieberbach in 1916—originally as a footnote of the paper [3] where he proved $|a_2| \leqslant 2$. This conjecture attracted extensive research over the decades until it was ultimately proved by de Branges [7] in 1984. It is known that Bieberbach conjecture is implied by several other conjectures.

For a function $f \in \mathcal{S}$, the coefficients γ_k of

$$\log\left(\frac{f(z)}{z}\right) = 2\sum_{k=1}^{\infty} \gamma_k z^k$$

are known as the *logarithmic coefficients* of the function f. Milin's conjecture states that the logarithmic coefficients γ_k of a function $f \in \mathcal{S}$ satisfies

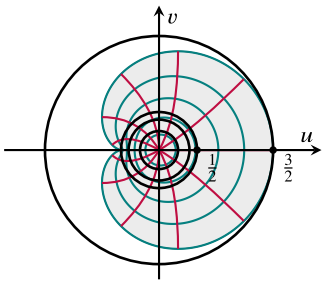


Fig. 2.10 Covering theorem for $f(z) = z + z^2/2$: $\mathbb{D}_{1/4} \subset \mathbb{D}_{2/5} \subset \mathbb{D}_{1/2} \subset f(\mathbb{D}) \subset \mathbb{D}_{3/2}$; the four circles are of radius 1/4, 2/5, 1/2 and 3/2 respectively

$$\sum_{m=1}^{n}\sum_{k=1}^{m}\left(k|\gamma_k|^2-\frac{1}{k}\right)\leqslant 0,$$

and equality holds for the Koebe function κ. The logarithmic coefficients of κ are $\gamma_k = 1/k$ as

$$\log\left(\frac{\kappa(z)}{z}\right) = -2\log(1-z) = 2\sum_{k=1}^{\infty}\frac{1}{k}z^k.$$

Milin's conjecture implies a related conjecture of Robertson which states that

$$\sum_{k=1}^{n}|c_{2k-1}|^2 \leqslant n, \quad n = 1, 2, 3, \ldots$$

for each function $h \in \mathcal{S}^{(2)}$ of the form $h(z) = c_1 z + c_3 z^3 + c_5 z^5 + \cdots$, $c_1 = 1$.

Though the proof of this implication will be omitted, we shall prove that the Robertson's conjecture implies the Bieberbach's conjecture. To see this, let $f \in \mathcal{S}$ and define $h : \mathbb{D} \to \mathbb{C}$ by $h(z) = \sqrt{f(z^2)}$. Then $h \in \mathcal{S}$ and $h(z) = z + c_3 z^3 + c_5 z^5 + \cdots$. Comparing the coefficients of z^{2n} in $f(z^2) = h(z)^2$ yields

$$a_n = c_1 c_{2n-1} + c_3 c_{2n-3} + \cdots + c_{2n-1} c_1.$$

Now the Cauchy-Schwarz inequality shows that

$$|a_n| \leqslant \sqrt{\sum_{k=1}^{n}|c_{2k-1}|^2}\sqrt{\sum_{k=1}^{n}|c_{2n-2k+1}|^2} = \sum_{k=1}^{n}|c_{2k-1}|^2 \leqslant n.$$

This gives the Bieberbach's conjecture. In 1984, Louis de Branges [7] proved the Milin's conjecture on the logarithmic coefficients of functions in $\mathcal{S}$, thereby proving the Bieberbach's conjecture and the Robertson's conjecture.

Example 2.3.10 The nth coefficient of every locally univalent polynomial $p(z) = z + a_2 z^2 + a_3 z^3 + \cdots + a_n z^n$ in $\mathbb{D}$ satisfies

$$|a_n| \leqslant \frac{1}{n}.$$

To see this, note that $p'(0) = 1$ and the local univalence of p implies $p'(z) \neq 0$ in $\mathbb{D}$. Thus all the $(n-1)$ zeros of $p'(z)$ are in $\mathbb{C} \setminus \mathbb{D}$; let these zeros be $c_1, c_2, \ldots, c_{n-1}$. Since $|c_k| \geqslant 1$ for $k = 1, 2, \ldots, n-1$ and

$$p'(z) = n a_n (z - c_1)(z - c_2)\cdots(z - c_{n-1}),$$

it follows that

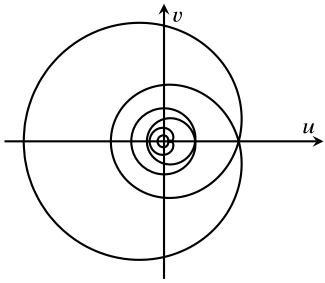


Fig. 2.11 Non-univalence of $f(z) = 2z - z/(1-z)^2$

$$1 = |p'(0)| = |na_n c_1 \dots c_{n-1}| \geqslant n|a_n|.$$

Example 2.3.11 The polynomial $p(z) = z + a_n z^n$ is univalent if and only if

$$|a_n| \leqslant \frac{1}{n}.$$

If p is univalent, then $|a_n| \leqslant 1/n$ follows from Example 2.3.10. For the converse, assume that $a_n \neq 0$. Since $0 < n|a_n| \leqslant 1$, it follows that

$$|p'(z) - 1| = |na_n z^{n-1}| < n|a_n| \leqslant 1,$$

hence $\operatorname{Re} p'(z) > 0$ in $\mathbb{D}$. The Noshiro-Warschawski theorem (Theorem 2.1.6) shows that p is univalent in $\mathbb{D}$.

Example 2.3.12 The function

$$f(z) = 2z - \frac{z}{(1-z)^2} = z - \sum_{k=2}^{\infty} k z^k$$

satisfies the conclusion of the de Branges' theorem but fails to be univalent in $\mathbb{D}$. The non-univalence of f follows as its derivative

$$f'(z) = \frac{1 - 7z + 6z^2 - 2z^3}{(1-z)^3}$$

vanishes at some point $z_0 \approx 0.164878$ belonging to the interval $(0, 1)$ (as the polynomial in the numerator of f' has opposite signs at 0, 1). Figure 2.11 also demonstrate the non-univalence.

As an application of the Bieberbach theorem and the area theorem, we determine the functions in the class $\mathcal{S}$ with integer coefficients.

Theorem 2.3.13 *If all the coefficients a_n of a function $f \in \mathcal{S}$ are integers, then f is one of the following nine functions:*

$$z,\quad \frac{z}{1\pm z},\quad \frac{z}{1\pm z^2},\quad \frac{z}{(1\pm z)^2},\quad \frac{z}{1\pm z+z^2}.$$

Proof Let $z/f(z)=\sum_{n=0}^{\infty} b_n z^n$. Then, the function $g:\mathbb{C}\setminus\overline{\mathbb{D}}\to\mathbb{C}$ defined by $g(z)=1/f(1/z)$ is univalent and

$$g(z)=\frac{1}{f(1/z)}=z+b_1+\frac{b_2}{z}+\frac{b_3}{z^2}+\cdots.$$

From Theorem 2.3.2,

$$|b_2|^2+2|b_3|^2+\cdots\leqslant 1.$$

Clearly, $b_0=1$ and $(n-1)|b_n|^2\leqslant 1$ for $n\geqslant 2$.

First, it is shown that all b_k's are integers. Comparing the coefficients in $f(z)g(1/z)=1$ leads to

$$a_{k+1}+b_k+\sum_{l=1}^{k-1} a_{k-l+1}b_l=0$$

for every $k\geqslant 1$, or equivalently,

$$\begin{pmatrix} 1 & 0 & 0 & \cdots & 0\\ a_2 & 1 & 0 & \cdots & 0\\ a_3 & a_2 & 1 & \cdots & 0\\ \cdots & \cdots & \cdots & \cdots & \cdots\\ a_k & a_{k-1} & a_{k-2} & \cdots & 1 \end{pmatrix}\begin{pmatrix} b_1\\ b_2\\ b_3\\ \vdots\\ b_k \end{pmatrix}=\begin{pmatrix} -a_2\\ -a_3\\ -a_4\\ \vdots\\ -a_{k+1} \end{pmatrix}.$$

Cramer's rule gives

$$b_k=\begin{vmatrix} 1 & 0 & \cdots & 0 & -a_2\\ a_2 & 1 & \cdots & 0 & -a_3\\ a_3 & a_2 & \cdots & 0 & -a_4\\ \cdots & \cdots & \cdots & \cdots & \cdots\\ a_k & a_{k-1} & \cdots & a_2 & -a_{k+1} \end{vmatrix}=(-1)^k\begin{vmatrix} a_2 & 1 & 0 & \cdots & 0\\ a_3 & a_2 & 1 & \cdots & 0\\ a_4 & a_3 & a_2 & \cdots & 0\\ \cdots & \cdots & \cdots & \cdots & \cdots\\ a_{k+1} & a_k & a_{k-1} & \cdots & a_2 \end{vmatrix}.$$

Since all a_k are integers, it follows that each b_k is an integer.

Since $b_1=-a_2$ and $|a_2|\leqslant 2$, it follows that $b_1=0,\pm 1,\pm 2$. Since $|b_2|\leqslant 1$, the possible values are $b_2=0,\pm 1$. Since b_n is an integer and $|b_n|^2\leqslant 1/(n-1)<1$ for $n\geqslant 3$, it follows that b_n must be zero. Combining these values yields a total of 15 functions. Among these, nine are listed in the theorem, while the remaining six are

$$z/(1\pm 2z),\quad z/(1\pm 2z-z^2),\quad z/(1\pm z-z^2).$$

Each of these six functions has a pole in $\mathbb{D}$. □

2.4 Schwarzian Derivative

The *Pre-Schwarzian derivative* of a locally univalent analytic function f is defined by

$$P_f(z) := \frac{f''(z)}{f'(z)}.$$

The *Schwarzian derivative* of a locally univalent analytic function f is defined by

$$\{f, z\} := P_f'(z) - \frac{1}{2}(P_f(z))^2 = \left(\frac{f''(z)}{f'(z)}\right)' - \frac{1}{2}\left(\frac{f''(z)}{f'(z)}\right)^2 = \frac{f'''(z)}{f'(z)} - \frac{3}{2}\left(\frac{f''(z)}{f'(z)}\right)^2.$$

Theorem 2.4.1 *The Schwarzian derivative of a function in $\mathcal{A}$ is zero if and only if it is a Möbius transformation.*

Proof Let

$$S(z) = \frac{az + b}{cz + d} \quad (ad - bc \neq 0)$$

be a Möbius transformation. Then

$$S'(z) = \frac{ad - bc}{(cz + d)^2}, \quad S''(z) = -\frac{2(ad - bc)c}{(cz + d)^3}, \quad S'''(z) = \frac{6(ad - bc)c^2}{(cz + d)^4}.$$

Substituting into the definition of the Schwarzian derivative gives

$$\{S; z\} = \frac{S'''(z)}{S'(z)} - \frac{3}{2}\left(\frac{S''(z)}{S'(z)}\right)^2 = \frac{6c^2}{(cz + d)^2} - \frac{6c^2}{(cz + d)^2} = 0.$$

Conversely, suppose $\{f; z\} = 0$. Setting $w(z) = f''(z)/f'(z)$, the equation $\{f; z\} = 0$ becomes $w'(z) - w^2(z)/2 = 0$. This differential equation has the general solution

$$\frac{f''(z)}{f'(z)} = w(z) = -\frac{2}{z + c}$$

where c is a constant. Integrating,

$$f'(z) = \frac{c^2}{(z + c)^2}$$

with $c \neq 0$. Hence

$$f(z) = \frac{cz}{z + c}$$

is a Möbius transformation. □

In view of Theorem 2.4.1, the Schwarzian derivative quantifies the extent to which a function deviates from being a Möbius transformation. Since every Möbius transformation is univalent, Schwarzian derivative provides a natural tool for formulating necessary conditions for univalence. At the origin, the Schwarzian derivative $\{f; z\}$ satisfies $\{f; 0\} = 6(a_3 - a_2^2)$ when $f(z) = z + \sum_{k=2}^{\infty} a_k z^k$. The following result establishes a sharp bound for this quantity. The general bound for $\{f; z\}$ over the unit disk follows from this special case.

Theorem 2.4.2 *If $f(z) = z + a_2 z^2 + a_3 z^3 + \cdots \in \mathcal{S}$, then*

$$|a_3 - a_2^2| \leqslant 1$$

with equality for the Koebe function κ.

Proof The function

$$g(z) = [f(1/z)]^{-1} = z - a_2 + (a_2^2 - a_3)z^{-1} + \cdots, \quad |z| > 1$$

is univalent as it is a composition of univalent functions. Applying the area theorem (Theorem 2.3.2) to g yields the inequality

$$|a_3 - a_2^2| \leqslant 1.$$

Equality holds for the Koebe function $\kappa(z) = z + 2z^2 + 3z^3 + \cdots$. □

It also follows that

$$|a_3| \leqslant |a_3 - a_2^2| + |a_2^2| \leqslant 5$$

but this inequality is not sharp for $f \in \mathcal{S}$. However, we have the following corollary.

Corollary 2.4.3 *If $f(z) = z + a_3 z^3 + a_4 z^4 + \cdots \in \mathcal{S}$, then the sharp inequality $|a_3| \leqslant 1$ holds.*

Proof Since $a_2 = 0$, the result follows from Theorem 2.4.2. Equality holds for rotations of the function $z/(1 - z^2)$. □

In particular, Corollary 2.4.3 applies to any odd univalent function in $\mathcal{S}$. A function $f \in \mathcal{A}$ is *odd* if $f(z) = -f(-z)$ for all $z \in \mathbb{D}$; such functions are of the form $f(z) = z + a_3 z^3 + a_5 z^5 + \cdots$. The class of all odd univalent functions in $\mathcal{A}$ are precisely the univalent functions in $\mathcal{S}$ with twofold symmetry.

Theorem 2.4.4 (Kraus–Nehari Theorem) *If $f \in \mathcal{S}$, then the Schwarzian derivative $\{f; z\}$ satisfies*

$$|\{f; z\}| \leqslant \frac{6}{(1 - |z|^2)^2}$$

with equality for the Koebe function κ.

Proof Fix $\zeta \in \mathbb{D}$ and consider the disk automorphism F defined by

$$F(z) = \frac{f(\varphi(z)) - f(\varphi(0))}{f'(\varphi(0))\varphi'(0)}, \quad \text{where} \quad \varphi(z) := \frac{\zeta - z}{1 - \overline{\zeta} z}. \tag{2.4.1}$$

Since $f \in \mathcal{S}$, by Theorem 2.2.10, $F \in \mathcal{S}$ and $F(z) = z + A_2 z^2 + A_3 z^3 + \cdots$ where

$$A_2 = \frac{F''(0)}{2!} = -\frac{1}{2}\left\{(1 - |\zeta|^2)\frac{f''(\zeta)}{f'(\zeta)} - 2\overline{\zeta}\right\} \tag{2.4.2}$$

and

$$A_3 = \frac{F'''(0)}{3!} = \frac{1}{6}(1 - |\zeta|^2)^2 \frac{f'''(\zeta)}{f'(\zeta)} - (1 - |\zeta|^2)\overline{\zeta}\frac{f''(\zeta)}{f'(\zeta)} + \overline{\zeta}^2. \tag{2.4.3}$$

The second and third coefficients A_2 and A_3 of F given by (2.4.2) and (2.4.3) readily yield

$$\begin{aligned} A_2^2 - A_3 &= \frac{(1 - |\zeta|^2)^2}{6}\left(\frac{3}{2}\left(\frac{f''(\zeta)}{f'(\zeta)}\right)^2 - \frac{f'''(\zeta)}{f'(\zeta)}\right) \\ &= -\frac{(1 - |\zeta|^2)^2}{6}\{f; \zeta\}. \end{aligned}$$

Theorem 2.4.2 applied to F gives $|A_2^2 - A_3| \leqslant 1$. Hence

$$|\{f; \zeta\}| \leqslant \frac{6}{(1 - |\zeta|^2)^2}.$$

Write ζ as z to get the desired inequality.

For the Koebe function

$$\kappa(z) = \frac{z}{(1 - z)^2},$$

a straightforward calculation shows that

$$\{\kappa; z\} = -\frac{6}{(1 - z^2)^2}.$$

Choosing $z = r \in [0, 1)$, the Koebe function satisfies $|\{\kappa; z\}| = 6/(1 - |z|^2)^2$. □

When the Schwarzian derivative of $f \in \mathcal{A}$ satisfies

$$|\{f; z\}| \leqslant \frac{2}{(1 - |z|^2)^2},$$

Nehari [21] has shown that $f \in \mathcal{S}$. He has also shown that the inequality

$$|\{f; z\}| \leqslant \pi^2/2$$

is sufficient for univalence of $f \in \mathcal{A}$ and the constant $\pi^2/2$ is best possible.

2.5 Growth and Distortion Theorems

The quantities $|f'(z)|$ and $|f'(z)|^2$ represent the infinitesimal magnification factors of the arc length and area, respectively, under the mapping f. A theorem that provides bounds on the distortion of a domain–or of its subsets, boundary, or line elements–at a given point is referred to as a *distortion theorem*. In contrast, a theorem that describes the behavior of $|f(z)|$ is called a *growth theorem*.

The following result, which serves as an example of both types, is derived using Bieberbach's estimate for the second coefficient of disk automorphisms.

Theorem 2.5.1 *If $f \in \mathcal{S}$, then*

$$\left|1 + \frac{zf''(z)}{f'(z)} - \frac{1+r^2}{1-r^2}\right| \leqslant \frac{4r}{1-r^2}, \quad |z| = r < 1. \tag{2.5.1}$$

For each $z \in \mathbb{D}$, the result is sharp for a rotation of the Koebe function κ.

Proof Consider the disk automorphism F given in (2.4.1). Bieberbach's theorem (Theorem 2.3.4) applied to the second coefficient A_2 of F given by (2.4.2) leads to

$$\left|(1-|\zeta|^2)\frac{f''(\zeta)}{f'(\zeta)} - 2\overline{\zeta}\right| \leqslant 4,$$

which, upon a multiplication by $|\zeta|$ and rewriting, yields

$$\left|\frac{\zeta f''(\zeta)}{f'(\zeta)} - \frac{2|\zeta|^2}{1-|\zeta|^2}\right| \leqslant \frac{4|\zeta|}{1-|\zeta|^2}.$$

Now, replace ζ by z and $|\zeta| = r$ to get the desired inequality.

The function

$$f_0(z) = \mathrm{e}^{-\mathrm{i}\alpha}\kappa(\mathrm{e}^{\mathrm{i}\alpha}z) = \frac{z}{(1-\mathrm{e}^{\mathrm{i}\alpha}z)^2}, \quad 0 \leqslant \alpha \leqslant 2\pi,$$

is a rotation of the Koebe function κ,

$$f_0'(z) = \frac{1+\mathrm{e}^{\mathrm{i}\alpha}z}{(1-\mathrm{e}^{\mathrm{i}\alpha}z)^3}$$

and

$$\frac{zf_0''(z)}{f_0'(z)} = \frac{2e^{i\alpha}z(2+e^{i\alpha}z)}{1-e^{2i\alpha}z^2}.$$

For any $z \in \mathbb{D}$ with $|z| = r$, choose $\alpha = -\arg z$ so that $e^{i\alpha}z = r$. Then

$$\left|\frac{zf_0''(z)}{f_0'(z)} - \frac{2r^2}{1-r^2}\right| = \frac{4r}{1-r^2}.$$

Hence, the result is sharp. □

Pommerenke in [22] introduced the notion of a linear-invariant family. A subset $\mathcal{M}$ of locally univalent functions in $\mathcal{A}$ is *linear-invariant* if the disk automorphism F given in (2.4.1) belongs to $\mathcal{M}$ for each $f \in \mathcal{M}$. The *order* of the linear-invariant family $\mathcal{M}$ is defined by

$$\alpha = \sup\left\{\frac{|f''(0)|}{2} : f \in \mathcal{M}\right\}.$$

He proved that the order of any linear-invariant family is always at least one. The class $\mathcal{S}$ is an example of a linear-invariant family of order 2. The class of locally univalent functions in $\mathcal{A}$ is linear-invariant but of infinite order, as exhibited by the function $f(z) = (e^{kz} - 1)/k$, $k > 0$.

For $1 \leqslant \alpha < \infty$, let $\mathcal{U}_\alpha$ denote the universal linear-invariant family consisting of all linear-invariant families of order not greater than α. Evidently, $\mathcal{U}_\alpha$ is also linear-invariant.

The only property of the class $\mathcal{S}$ used in the proof of Theorem 2.5.1 is the linear-invariance. This observation immediately leads to the following theorem.

Theorem 2.5.2 *Let $f \in \mathcal{A}$. Then $f \in \mathcal{U}_\alpha$ if and only if*

$$\left|1 + \frac{zf''(z)}{f'(z)} - \frac{1+r^2}{1-r^2}\right| \leqslant \frac{2\alpha r}{1-r^2}, \quad |z| = r < 1. \tag{2.5.2}$$

Further, equality in (2.5.2) occurs at some point $z \in \mathbb{D}$ for the function

$$f_0(z) = \frac{1}{2\alpha}\left(\left(\frac{1+z}{1-z}\right)^\alpha - 1\right).$$

In this case,

$$1 + \frac{zf_0''(z)}{f_0'(z)} = \frac{1+2\alpha z+z^2}{1-z^2},$$

which at $z = \pm r$ yields

$$\left|1 + \frac{zf_0''(z)}{f_0'(z)} - \frac{1+r^2}{1-r^2}\right| = \frac{2\alpha r}{1-r^2}.$$

Theorem 2.5.3 (Distortion Theorem) *If $f \in \mathcal{S}$, then*

$$\kappa'(-r) = \frac{1-r}{(1+r)^3} \leqslant |f'(z)| \leqslant \frac{1+r}{(1-r)^3} = \kappa'(r), \quad |z| \leqslant r < 1.$$

For each $z \in \mathbb{D} \setminus \{0\}$, equality occurs if and only if f is a suitable rotation of the Koebe function κ.

Proof If $|z - z_0| \leqslant \rho$, then $\operatorname{Re} z_0 - \rho \leqslant \operatorname{Re} z \leqslant \operatorname{Re} z_0 + \rho$. Thus the inequality (2.5.1) of Theorem 2.5.1 yields

$$\frac{2r^2 - 4r}{1-r^2} \leqslant \operatorname{Re} \frac{zf''(z)}{f'(z)} \leqslant \frac{2r^2 + 4r}{1-r^2}. \tag{2.5.3}$$

Since the derivative of a univalent function is non-vanishing, a single-valued analytic branch of $\log f'(z)$ in $\mathbb{D}$ exists that vanishes at the origin. Note that

$$\operatorname{Re}(\log f'(z)) = \log |f'(z)|.$$

For a fixed $z = re^{i\theta}$, $0 \leqslant \theta \leqslant 2\pi$,

$$\begin{aligned} r\frac{\partial}{\partial r}(\log |f'(z)|) &= r\frac{\partial}{\partial r}(\operatorname{Re} \log f'(z)) \\ &= \operatorname{Re}\left(r\frac{\partial}{\partial r}(\log f'(re^{i\theta}))\right) \\ &= \operatorname{Re}\left(\frac{f''(re^{i\theta})}{f'(re^{i\theta})} re^{i\theta}\right) \\ &= \operatorname{Re}\left(\frac{zf''(z)}{f'(z)}\right). \end{aligned} \tag{2.5.4}$$

From (2.5.3) and (2.5.4),

$$\frac{2r-4}{1-r^2} \leqslant \frac{\partial}{\partial r}(\log |f'(re^{i\theta})|) \leqslant \frac{2r+4}{1-r^2}. \tag{2.5.5}$$

By keeping θ fixed and then integrating from $r = 0$ to $r = \rho < 1$ yields

$$\log \frac{(1-\rho)}{(1+\rho)^3} = \int_0^\rho \frac{2r-4}{1-r^2} dr \leqslant \log |f'(\rho e^{i\theta})| \leqslant \int_0^\rho \frac{2r+4}{1-r^2} dr = \log \frac{(1+\rho)}{(1-\rho)^3}.$$

Thus

$$\frac{1-\rho}{(1+\rho)^3} \leqslant |f'(\rho e^{i\theta})| \leqslant \frac{1+\rho}{(1-\rho)^3}.$$

Finally, replacing ρ with r, the desired bounds follow.

For the rotation of the Koebe function κ given by

$$f(z) = e^{-i\alpha}\kappa(e^{i\alpha}z),$$

it follows that $f'(z) = \kappa'(e^{i\alpha}z)$. Let $z \in \mathbb{D}$ and $|z| = r$. Setting $\alpha = -\arg z$ yields $e^{i\alpha}z = r$, and thus $|f'(z)| = |\kappa'(r)|$. Alternatively, setting $\alpha = \pi - \arg z$ yields $e^{i\alpha}z = -r$, resulting in $|f'(z)| = |\kappa'(-r)|$.

If equality occurs in either inequality at $z = \rho e^{i\theta}$, a computation shows that the equality also holds in the corresponding sides of (2.5.5) for all $0 \leqslant r \leqslant \rho$. Applying (2.5.4), this shows that

$$\operatorname{Re}\left(e^{i\theta}\frac{f''(re^{i\theta})}{f'(re^{i\theta})}\right) = \frac{2r \pm 4}{1-r^2}.$$

In particular,

$$2\operatorname{Re}\left(e^{i\theta}a_2\right) = \operatorname{Re}\left(e^{i\theta}\frac{f''(0)}{f'(0)}\right) = \pm 4,$$

which implies

$$2 = |\operatorname{Re}\left(e^{i\theta}a_2\right)| \leqslant |a_2| \leqslant 2,$$

so that $|a_2| = 2$. By the Bieberbach's theorem, this occurs if and only if f is a rotation of the Koebe function. □

The following result, proved by Privalov, applied to the distortion theorem would immediately give the growth theorem.

Lemma 2.5.4 *If $f \in \mathcal{S}$ satisfies*

$$m(r) \leqslant \left|f'(z)\right| \leqslant M(r), \quad |z| = r < 1,$$

where m and M are real-valued integrable functions of r in $[0, 1)$, then

$$\int_0^r m(t)\mathrm{d}t \leqslant |f(z)| \leqslant \int_0^r M(t)\mathrm{d}t, \quad |z| = r < 1.$$

Proof Let $z = re^{i\alpha} \in \mathbb{D}$. The inequality on the right side follows by integrating along a radial line $\zeta(t) = te^{i\alpha}$, $0 \leqslant t \leqslant r$:

$$|f(z)| = \left| \int_0^z f'(\zeta)\mathrm{d}\zeta \right| = \left| \int_0^r f'\left(t\mathrm{e}^{\mathrm{i}\alpha}\right)\mathrm{e}^{\mathrm{i}\alpha}\mathrm{d}t \right|$$
$$\leqslant \int_0^r \left| f'\left(t\mathrm{e}^{\mathrm{i}\alpha}\right)\right| \mathrm{d}t \leqslant \int_0^r M(t)\mathrm{d}t.$$

To prove the left inequality, let Γ be the image under f of the circle $C_r : |z| = r$ and let w_1 be some point of Γ that is nearest to the origin. Since $\mathcal{S}$ is closed under rotation, assume that $w_1 > 0$. Let $\Lambda(t) = tw_1, 0 \leqslant t \leqslant 1$, be the line segment joining the origin to w_1 and suppose that z_1 and the curve L are the preimages of w_1 and Λ, respectively, under f. See Fig. 2.12. Let $\zeta = \zeta(t)$, $0 \leqslant t \leqslant 1$, be the parametric representation of the curve L. Then a computation shows that $|d\zeta(t)| \geqslant d|\zeta(t)|$; the quantity $|d\zeta(t)|$ represents the arc length element of the curve while $d|\zeta(t)|$ measures the rate of change of the modulus of the curve. Hence,

$$w_1 = |w_1| = \int_0^{w_1} |\mathrm{d}w| = \int_L \left|f'(\zeta)\right| |\mathrm{d}\zeta|$$
$$= \int_0^1 \left|f'(\zeta(t))\right| |\mathrm{d}\zeta(t)| \geqslant \int_0^1 \left|f'(\zeta(t))\right| \mathrm{d}|\zeta(t)|$$
$$= \int_0^r m(|\zeta|)\mathrm{d}|\zeta| \geqslant \int_0^r m(t)\mathrm{d}t.$$

□

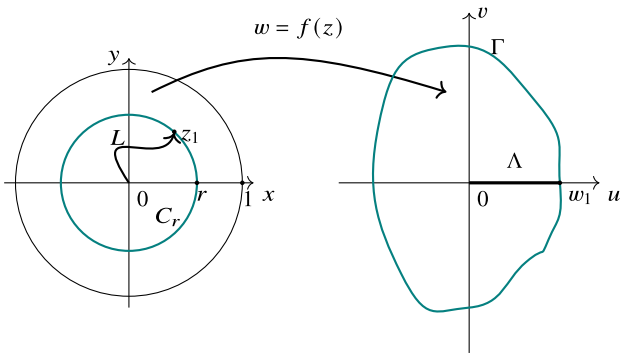


Fig. 2.12 Proof of lemma for distortion

Theorem 2.5.5 (Growth Theorem) *If $f \in \mathcal{S}$, then*

$$-\kappa(-r) = \frac{r}{(1+r)^2} \leqslant |f(z)| \leqslant \frac{r}{(1-r)^2} = \kappa(r), \quad |z| \leqslant r < 1. \tag{2.5.6}$$

For each $z \in \mathbb{D} \setminus \{0\}$, equality occurs if and only if f is a suitable rotation of the Koebe function κ.

Proof The inequalities follow by applying Lemma 2.5.4 to the distortion inequalities in Theorem 2.5.3. Also, if $f(z) = e^{-i\phi}\kappa(e^{i\phi}z)$ is a rotation of the Koebe function, then $|f(z)| = |\kappa(r)|$ when $\phi = -\arg z$ and $|f(z)| = |\kappa(-r)|$ when $\phi = \pi - \arg z$. If the equality occurs for some $z = \rho e^{i\theta}$ in either side of (2.5.6), then equality occurs in the respective side of the distortion inequalities, and hence f is a rotation of the Koebe function. □

The growth inequality (2.5.6) implies that the image of the circle $|z| = r$ lies in annular region $\{w : -\kappa(-r) \leqslant |w| \leqslant \kappa(r)\}$. Since

$$\lim_{r\to 1-} -\kappa(-r) = \lim_{r\to 1-} \frac{r}{(1+r)^2} = \frac{1}{4},$$

it follows that $f(z)$ lies outside a closed disk of radius 1/4 as z approaches any point on $\partial\mathbb{D}$. Thus the boundary of $f(\mathbb{D})$ lies outside the disk $\mathbb{D}_{1/4}$. This proves that $f(\mathbb{D}) \supset \{w : |w| < 1/4\}$ for every $f \in \mathcal{S}$. This result is the Koebe one-quarter theorem (Theorem 2.3.6).

Corollary 2.5.6 *If $g \in \mathcal{S}^{(n)}$, then*

$$\frac{r}{(1+r^n)^{2/n}} \leqslant |g(z)| \leqslant \frac{r}{(1-r^n)^{2/n}}, \quad |z| = r < 1.$$

In addition, $g(\mathbb{D}) \supset \{w : |w| < 4^{-1/n}\}$. For each $z \in \mathbb{D} \setminus \{0\}$, equality occurs if g is a suitable rotation of the function $z/(1-z^n)^{2/n}$.

Proof By Theorem 2.2.9, if g belongs to $\mathcal{S}^{(n)}$, then there exists a function $f \in \mathcal{S}$ such that $f(z^n) = (g(z))^n$. Since

$$\frac{r}{(1+r)^2} \leqslant |f(z)| \leqslant \frac{r}{(1-r)^2},$$

it follows that

$$\frac{r^n}{(1+r^n)^2} \leqslant |f(z^n)| = |g(z)|^n \leqslant \frac{r^n}{(1-r^n)^2}.$$

Taking the nth root yields the desired estimate. The covering result readily follows by letting $r \to 1^-$ in the lower bound. □

Corollary 2.5.7 *The class $\mathcal{S}$ is a compact normal family.*

Proof Since the class $\mathcal{S}$ is a subspace of the space $\mathcal{H}(\mathbb{D})$ of analytic functions, Montel's theorem (Theorem 1.4.6) will be used to prove the normality of the class $\mathcal{S}$. On any closed disk $\overline{\mathbb{D}}_r$, $|f(z)| \leqslant M$ by the growth theorem (Theorem 2.5.6), where $M = r/(1-r)^2$, and so $\mathcal{S}$ is uniformly bounded on $\overline{\mathbb{D}}_r$. Since each compact subset K of $\mathbb{D}$ is contained in some $\overline{\mathbb{D}}_r$, the class $\mathcal{S}$ is uniformly bounded on every compact subset of $\mathbb{D}$. Thus Lemma 1.3.7 shows that it is locally bounded on $\mathbb{D}$. It follows from Montel's theorem that the class $\mathcal{S}$ is a normal family.

To prove compactness, let $\langle f_n \rangle$ be a sequence in $\mathcal{S}$. By normality of the class $\mathcal{S}$, there is a subsequence $\langle f_{n_k} \rangle$ converging to some $f \in \mathcal{H}(\mathbb{D})$. By Corollary 1.4.5, the function $f \in \mathcal{S}$ and this proves the compactness. □

The de Branges' theorem could be used to give an alternate proof of the right-hand inequalities in the growth and distortion theorems. For $f \in \mathcal{S}$, de Branges' theorem gives $|a_n| \leqslant n$, and so, for $|z| \leqslant r < 1$,

$$|f(z)| \leqslant |z| + \sum_{k=2}^{\infty} |a_k|\,|z|^k \leqslant r + \sum_{k=2}^{\infty} k r^k = \frac{r}{(1-r)^2} = \kappa(r),$$

and

$$|f'(z)| \leqslant 1 + \sum_{k=2}^{\infty} k|a_k|\,|z|^{k-1} \leqslant 1 + \sum_{k=2}^{\infty} k^2 r^{k-1} = \frac{1+r}{(1-r)^3} = \kappa'(r).$$

Proceeding in a similar manner for the higher derivatives leads to the following result.

Corollary 2.5.8 *If $f \in \mathcal{S}$, then*

$$\left|f^{(n)}(z)\right| \leqslant n! \frac{n+r}{(1-r)^{n+2}} = \kappa^{(n)}(r), \quad |z| \leqslant r < 1, \quad n = 1, 2, \ldots.$$

For each $z \in \mathbb{D} \setminus \{0\}$, equality occurs if f is the Koebe function κ.

Theorem 2.5.9 *If $f \in \mathcal{S}$, then*

$$-\frac{r\kappa'(-r)}{\kappa(-r)} = \frac{1-r}{1+r} \leqslant \left|\frac{zf'(z)}{f(z)}\right| \leqslant \frac{1+r}{1-r} = \frac{r\kappa'(r)}{\kappa(r)}, \quad |z| \leqslant r < 1.$$

For each $z \in \mathbb{D} \setminus \{0\}$, equality occurs if and only if f is a suitable rotation of the Koebe function κ.

Proof If $f \in \mathcal{S}$, then so is the disk automorphism

$$F(z) = \frac{f\left(\frac{\zeta - z}{1 - \bar{\zeta} z}\right) - f(\zeta)}{(|\zeta|^2 - 1) f'(\zeta)}, \quad \zeta \in \mathbb{D}.$$

The growth theorem (Theorem 2.5.5) applied to F gives

$$\frac{r}{(1+r)^2} \leqslant |F(z)| \leqslant \frac{r}{(1-r)^2} \quad (|z| = r < 1). \tag{2.5.7}$$

Now

$$F(\zeta) = \frac{f(\zeta)}{(1-|\zeta|^2)f'(\zeta)}, \tag{2.5.8}$$

and so, (2.5.7) yields

$$\frac{1-|\zeta|}{1+|\zeta|} \leqslant \left|\frac{f(\zeta)}{\zeta f'(\zeta)}\right| \leqslant \frac{1+|\zeta|}{1-|\zeta|} \tag{2.5.9}$$

which proves the result.

The function

$$f(z) = \mathrm{e}^{-\mathrm{i}\alpha}\kappa(\mathrm{e}^{\mathrm{i}\alpha}z) = \frac{z}{(1-\mathrm{e}^{\mathrm{i}\alpha}z)^2}, \quad 0 \leqslant \alpha \leqslant 2\pi,$$

is a rotation of the Koebe function κ, and for this function

$$\frac{zf'(z)}{f(z)} = \frac{1+\mathrm{e}^{\mathrm{i}\alpha}z}{1-\mathrm{e}^{\mathrm{i}\alpha}z}.$$

Fix $z \in \mathbb{D} \setminus \{0\}$. Choose $\alpha = -\arg z$ so that $\mathrm{e}^{\mathrm{i}\alpha}z = r$. Then

$$\frac{zf'(z)}{f(z)} = \frac{1+r}{1-r}.$$

Choose $\alpha = \pi - \arg z$ so that $\mathrm{e}^{\mathrm{i}\alpha}z = -r$. Then

$$\frac{zf'(z)}{f(z)} = \frac{1-r}{1+r}.$$

Suppose that the equality occurs in the lower bound in (2.5.9) for some $f \in \mathcal{S}$ at a non-zero point $\zeta \in \mathbb{D}$. In view of (2.5.8),

$$|F(\zeta)| = \frac{|\zeta|}{(1-|\zeta|)^2}.$$

Therefore, by the growth theorem, F must be a rotation of the Koebe function

$$F(z) = \frac{z}{(1-\mathrm{e}^{\mathrm{i}\alpha}z)^2}, \quad 0 \leqslant \alpha \leqslant 2\pi.$$

Since the equality happens for the point $z = \zeta$, it follows that $e^{i\alpha}\zeta = |\zeta|$ and

$$F(z) = \frac{z}{\left(1 - \frac{\overline{\zeta}}{|\zeta|}z\right)^2}.$$

Therefore, in view of Remark 2.2.11, the corresponding function f is obtained from F by the disk automorphism with the same parameter ζ. The proof is completed by showing that the disk automorphism of a rotation of the Koebe function given by F above is another rotation of the Koebe function. Write

$$\varphi_\zeta(z) = \frac{\zeta - z}{1 - \overline{\zeta}z}$$

and let $G(z) = F(\varphi_\zeta(z))$ so that

$$f(z) = \frac{G(z) - G(0)}{G'(0)}.$$

Since

$$G(z) = F\left(\frac{\zeta - z}{1 - \overline{\zeta}z}\right) = \frac{(\zeta - z)(1 - \overline{\zeta}z)}{(1 - |\zeta|)^2\left(1 + \frac{\overline{\zeta}}{|\zeta|}z\right)^2}$$

and

$$G(0) = \frac{\zeta}{(1 - |\zeta|)^2},$$

it follows that

$$G(z) - G(0) = -\left(\frac{1 + |\zeta|}{1 - |\zeta|}\right)^2 \frac{z}{\left(1 + \frac{\overline{\zeta}}{|\zeta|}z\right)^2}.$$

Finally, the function

$$f(z) = \frac{G(z) - G(0)}{G'(0)} = \frac{z}{\left(1 + \frac{\overline{\zeta}}{|\zeta|}z\right)^2}$$

is a rotation of the Koebe function. □

The equation (2.5.8) is useful in getting information about $zf'(z)/f(z)$ from $f(z)/z$. For example, Löewner's parametric method (see Duren [9, Theorem 3.5, p. 95]) results in

$$\left|\arg\left(\frac{F(z)}{z}\right)\right| \leqslant \log\left(\frac{1 + |z|}{1 - |z|}\right)$$

for each function $F \in \mathcal{S}$ and this shows by (2.5.8) that

$$\left|\arg\left(\frac{zf'(z)}{f(z)}\right)\right| \leqslant \log\left(\frac{1+|z|}{1-|z|}\right).$$

A generalization of this last inequality is also known [9, Corollary 3, p. 126]

$$\left|\log\left(\frac{zf'(z)}{f(z)}\right)\right| \leqslant \log\left(\frac{1+|z|}{1-|z|}\right). \tag{2.5.10}$$

The image of $|z| \leqslant r$ under the function $zf'(z)/f(z)$ is shown in Fig. 2.13; by Theorem 2.5.9, the region lies inside the annulus $R_1 \leqslant |w| \leqslant R_2$ where $R_1 = (1-r)/(1+r)$ and $R_2 = 1/R_1$.

Corollary 2.5.10 *If $g \in \mathcal{S}^{(n)}$, then*

$$\frac{1-r^n}{(1+r^n)^{\frac{2}{n}+1}} \leqslant |g'(z)| \leqslant \frac{1+r^n}{(1-r^n)^{\frac{2}{n}+1}}, \quad |z| \leqslant r < 1,$$

and

$$\frac{1-r^n}{1+r^n} \leqslant \left|\frac{zg'(z)}{g(z)}\right| \leqslant \frac{1+r^n}{1-r^n}, \quad |z| \leqslant r < 1.$$

For each $z \in \mathbb{D} \setminus \{0\}$, equality occurs if f is a suitable rotation of the function $z/(1-z^n)^{2/n}$.

Proof From Theorem 2.2.9, if g belongs to $\mathcal{S}^{(n)}$, then there is a function $f \in \mathcal{S}$ satisfying $f(z^n) = (g(z))^n$. A calculation shows that

$$g'(z) = \frac{z^n f'(z^n)}{f(z^n)} \frac{g(z)}{z}$$

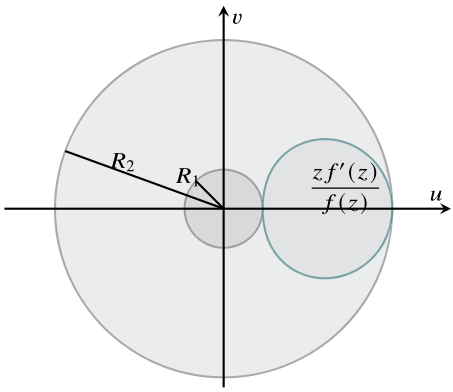


Fig. 2.13 The region given by (2.5.10).

and

$$\frac{zg'(z)}{g(z)} = \frac{z^n f'(z^n)}{f(z^n)}.$$

The result follows by applying Theorem 2.5.9 and Corollary 2.5.6. From Theorem 2.5.9, it follows that

$$\frac{1-r^n}{1+r^n} \leqslant \left|\frac{z^n f'(z^n)}{f(z^n)}\right| \leqslant \frac{1+r^n}{1-r^n}$$

and, Corollary 2.5.6 gives

$$\frac{1}{(1+r^n)^{\frac{2}{n}}} \leqslant \frac{|g(z)|}{|z|} \leqslant \frac{1}{(1-r^n)^{\frac{2}{n}}}.$$

For equality, note that the derivative of the function

$$f(z) = \frac{z}{(1-z^n)^{\frac{2}{n}}}$$

is

$$f'(z) = \frac{1+z^n}{(1-z^n)^{\frac{2}{n}+1}}.$$

Hence, the equalities respectively hold at $z = \mathrm{e}^{\pi \mathrm{i}/n} r$ and $z = r$. □

2.6 Integral Means and Littlewood's Theorem

If the function $f \in \mathcal{S}$ and $f(z) = z + \sum_{k=2}^{\infty} a_k z^k$, then de Branges' Theorem yields the sharp nth coefficient inequality $|a_n| \leqslant n$ for each $n \geqslant 2$, with equality if and only if f is a rotation of the Koebe function κ. The proof of this result requires materials beyond our scope but as an application of the growth theorem (Theorem 2.5.5), non-sharp estimates can be established for the nth coefficient of a univalent function. For $f \in \mathcal{S}$, Cauchy's integral formula yields

$$f^{(n)}(0) = \frac{n!}{2\pi \mathrm{i}} \int\limits_{|z|=r} \frac{f(\zeta)\mathrm{d}\zeta}{\zeta^{n+1}} = \frac{n!}{2\pi} r^{-n} \int\limits_0^{2\pi} f(r\mathrm{e}^{\mathrm{i}\theta}) \mathrm{e}^{-\mathrm{i}n\theta} \mathrm{d}\theta$$

for each $0 \leqslant r < 1$. The upper bound in the growth theorem readily implies that

$$|a_n| = \frac{|f^{(n)}(0)|}{n!} \leqslant \frac{1}{2\pi r^n} \int\limits_0^{2\pi} |f(r\mathrm{e}^{\mathrm{i}\theta})| \mathrm{d}\theta \leqslant \frac{1}{r^{n-1}(1-r)^2} := \varphi(r). \qquad (2.6.1)$$

Since

$$\varphi'(r) = \frac{1 - n + (1 + n)r}{r^n(1 - r)^3},$$

it follows that φ is decreasing for $0 < r < (n-1)/(n+1)$ and increasing for $(n-1)/(n+1) < r < 1$. Therefore, the function φ attains its minimum at $r = (n-1)/(n+1)$. Thus,

$$|a_n| \leqslant \varphi\left(\frac{n-1}{n+1}\right) = \frac{(n+1)^2}{4}\left(\frac{n+1}{n-1}\right)^{n-1}$$
$$= \frac{n^2}{4}\left(\frac{n+1}{n}\right)^2\left(1 + \frac{2}{n-1}\right)^{n-1} < \frac{e^2 n^2}{4}$$

for all n. The last inequality follows as $((n+1)/n)^2\,(1 + 2/(n-1))^{n-1}$ is an increasing function of n. This crude estimate will be improved in Theorem 2.6.2 and in Corollary 2.6.4. The proofs of these results are easier compared to de Branges proof of Milin's conjecture.

For a given function $f \in \mathcal{A}$ and $0 < p < \infty$, the *integral mean* $M_p(r, f)$ of f is defined by

$$M_p(r, f) = \left(\frac{1}{2\pi}\int_0^{2\pi} |f(re^{i\theta})|^p \mathrm{d}\theta\right)^{1/p}, \quad 0 \leqslant r < 1.$$

From (2.6.1), it follows that the coefficient a_n of $f \in \mathcal{S}$ is related to the integral mean $M_1(r, f)$ by the inequality

$$|a_n| \leqslant \frac{1}{r^n} M_1(r, f). \tag{2.6.2}$$

The following lemma on the integral mean will be useful in showing that $|a_n| < en$ for $f \in \mathcal{S}$.

Lemma 2.6.1 *If $f \in \mathcal{S}$, then*

$$M_1(r, f) \leqslant \frac{r}{1 - r}, \quad 0 \leqslant r < 1.$$

Proof The square-root transform of $f \in \mathcal{S}$ is given by

$$h(z) = \sqrt{f\left(z^2\right)} = \sum_{k=1}^{\infty} c_k z^k.$$

Since $|f(z)| \leqslant r/(1-r)^2$ for $|z| = r < 1$ by the growth theorem, it follows that

$$|h(z)| \leqslant \frac{r}{1 - r^2}, \quad |z| = r < 1,$$

and $h(\mathbb{D}_r)$ lies in the disk $\{w \in \mathbb{C} : |w| < r/(1-r^2)\}$. Therefore, the area A_r of $h(\mathbb{D}_r)$ satisfies

$$A_r \leqslant \frac{\pi r^2}{(1-r^2)^2}.$$

By Theorem 2.3.3,

$$A_r = \int_0^{2\pi} \int_0^r |h'(\rho e^{i\theta})|^2 \rho \, d\rho \, d\theta = \pi \sum_{k=1}^{\infty} k|c_k|^2 r^{2k}.$$

Consequently,

$$\sum_{k=1}^{\infty} k\,|c_k|^2\, r^{2k-1} \leqslant \frac{r}{\left(1-r^2\right)^2}, \quad 0 \leqslant r < 1.$$

An integration from 0 to r gives

$$\sum_{k=1}^{\infty} |c_k|^2\, r^{2k} \leqslant \frac{r^2}{1-r^2}. \tag{2.6.3}$$

The orthogonality of $\{e^{ik\theta} : k \in \mathbb{Z}\}$ implies that

$$\begin{aligned} \frac{1}{2\pi} \int_0^{2\pi} |h(re^{i\theta})|^2 d\theta &= \frac{1}{2\pi} \int_0^{2\pi} \left(\sum_{k=1}^{\infty} c_k r^k e^{ik\theta} \right) \left(\sum_{m=1}^{\infty} \overline{c}_m r^m e^{-im\theta} \right) d\theta \\ &= \sum_{k=1}^{\infty} |c_k|^2\, r^{2k}. \end{aligned}$$

Thus, (2.6.3) shows that

$$\frac{1}{2\pi} \int_0^{2\pi} |h(re^{i\theta})|^2 d\theta \leqslant \frac{r^2}{1-r^2}.$$

Since $(h(z))^2 = f(z^2)$, the previous inequality is the same as

$$\frac{1}{2\pi} \int_0^{2\pi} |f(r^2 e^{2i\theta})| d\theta \leqslant \frac{r^2}{1-r^2},$$

or equivalently,

$$\frac{1}{2\pi}\int_0^{2\pi} |f(re^{2i\theta})|d\theta \leqslant \frac{r}{1-r}.$$

Finally,

$$\frac{1}{2\pi}\int_0^{2\pi} |f(re^{2i\theta})|d\theta = \frac{1}{4\pi}\int_0^{4\pi} |f(re^{i\theta})|d\theta = \frac{1}{2\pi}\int_0^{2\pi} |f(re^{i\theta})|d\theta$$

proves the desired inequality. □

Theorem 2.6.2 (Littlewood's Theorem) *If* $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{S}$, *then* $|a_n| < en$ *for* $n = 2, 3, \ldots$.

Proof For each $0 \leqslant r < 1$, (2.6.2) shows that

$$|a_n| \leqslant \frac{M_1(r, f)}{r^n} \leqslant \frac{1}{r^{n-1}(1-r)} := \varphi(r).$$

Since

$$\varphi'(r) = \frac{1 - n + rn}{r^n(1-r)^2},$$

the function φ is decreasing for $0 < r < 1 - n^{-1}$ and increasing for $1 - n^{-1} < r < 1$. Therefore the function φ attains its minimum at $r = 1 - n^{-1}$. Thus

$$|a_n| \leqslant \varphi\left(1 - \frac{1}{n}\right) = n\left(1 + \frac{1}{n-1}\right)^{n-1} < en$$

for all n. □

Recall that a function $f : \mathbb{R} \to \mathbb{R}$ is said to be *convex* if

$$f(tx_1 + (1-t)x_2) \leqslant tf(x_1) + (1-t)f(x_2)$$

for all $x_1, x_2 \in \mathbb{R}$ and t with $0 \leqslant t \leqslant 1$. It is *strictly convex* if

$$f(tx_1 + (1-t)x_2) < tf(x_1) + (1-t)f(x_2)$$

for all $x_1, x_2 \in \mathbb{R}$ with $x_1 \neq x_2$ and for all t with $0 < t < 1$. The estimate $M_1(r, f) \leqslant r/(1-r)$ is not sharp. Baernstein proved the following general result that is sharp.

Theorem 2.6.3 (Baernstein's Theorem) *[2] If $\Phi : \mathbb{R} \to \mathbb{R}$ is a convex non-decreasing function, then*

$$\int_0^{2\pi} \Phi(\log|f(re^{i\theta})|)d\theta \leqslant \int_0^{2\pi} \Phi(\log|\kappa(re^{i\theta})|)d\theta$$

for each $f \in \mathcal{S}$ and $0 < r < 1$. If equality holds for some $r \in (0, 1)$ and for some strictly convex Φ, then f is a rotation of the Koebe function κ.

In particular, for $0 < r < 1$, it follows that

$$\int_0^{2\pi} |f(re^{i\theta})|^p d\theta \leqslant \int_0^{2\pi} |\kappa(re^{i\theta})|^p d\theta \quad (0 < p < \infty),$$

and

$$\int_0^{2\pi} \log^+|f(re^{i\theta})|d\theta \leqslant \int_0^{2\pi} \log^+|\kappa(re^{i\theta})|d\theta.$$

Here, the function $\log^+ : (0, \infty) \to \mathbb{R}$ is defined by

$$\log^+ x = \begin{cases} \log x & \text{if } x > 1 \\ 0 & \text{if } 0 < x \leqslant 1 \end{cases}$$

so that

$$\log x = \log^+ x - \log^+(1/x).$$

The first inequality says that $M_p(r, f) \leqslant M_p(r, \kappa)$ for each $0 < p < \infty$, where κ is the Koebe function. The integral mean $M_p(r, \kappa)$ for the Koebe function κ is given by

$$\begin{aligned} M_p^p(r, \kappa) &= \frac{1}{2\pi} \int_0^{2\pi} |\kappa(re^{i\theta})|^p d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \frac{r^p}{|1 - re^{i\theta}|^{2p}} d\theta \\ &= \frac{1}{2\pi i} \int_{\partial\mathbb{D}} \frac{r^p z^{p-1}}{(1 - rz)^p (z - r)^p} dz. \end{aligned}$$

In particular, Cauchy's integral formula (about the point $z = r$) yields

$$M_1(r,\kappa) = \frac{1}{2\pi \mathrm{i}} \int\limits_{\partial\mathbb{D}} \frac{r}{(1-rz)(z-r)} \mathrm{d}z = \frac{r}{1-r^2}$$

and

$$M_2^2(r,\kappa) = \frac{1}{2\pi \mathrm{i}} \int\limits_{\partial\mathbb{D}} \frac{r^2 z}{(1-rz)^2(z-r)^2} \mathrm{d}z = \frac{r^2\left(1+r^2\right)}{\left(1-r^2\right)^3}.$$

Use of the sharp estimate $M_1(r, f) \leqslant r/(1-r^2)$ instead of $M_1(r, f) \leqslant r/(1-r)$, yields the following result.

Corollary 2.6.4 *If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{S}$, then $|a_n| < en/2$ for $n = 2, 3, \ldots$.*

Proof For each $0 \leqslant r < 1$, by (2.6.2),

$$|a_n| \leqslant \frac{M_1(r, f)}{r^n} \leqslant \frac{1}{r^{n-1}(1-r^2)} := \varphi(r).$$

Since

$$\varphi'(r) = \frac{1 - n + r^2(1+n)}{r^n(1-r^2)^2},$$

the function φ is decreasing for $0 < r < \sqrt{(n-1)/(n+1)}$ and increasing for $\sqrt{(n-1)/(n+1)} < r < 1$. Therefore, the function φ attains its minimum at $r = \sqrt{(n-1)/(n+1)}$ and

$$\begin{aligned} |a_n| &\leqslant \varphi(\sqrt{(n-1)/(n+1)}) \\ &= \frac{n+1}{2}\left(\frac{n+1}{n-1}\right)^{\frac{n-1}{2}} \\ &= \frac{n}{2}\frac{n+1}{n}\left(\frac{n+1}{n-1}\right)^{\frac{n-1}{2}} \\ &< \frac{\mathrm{e}n}{2} \end{aligned}$$

for all n. □

For each rectifiable curve $\gamma : [a, b] \to \mathbb{C}$, its length $V(\gamma)$ is given by

$$V(\gamma) = \int\limits_a^b |\gamma'(t)| \mathrm{d}t.$$

The *arc length* $L_r(f)$ of the image of $|z| = r$ under f is given by

$$L_r(f) = \int_0^{2\pi} |f'(re^{i\theta})ir|d\theta = r\int_0^{2\pi} |f'(re^{i\theta})|d\theta.$$

As another application of the sharp estimate of $M_1(r, f) \leqslant r/(1-r^2)$ yields the following result.

Theorem 2.6.5 *If $f \in \mathcal{S}$, then the arc length $L_r(f)$ of the image of $|z| = r < 1$ under f satisfies*

$$L_r(f) \leqslant \frac{2\pi r}{(1-r)^2}.$$

Proof By applying Theorem 2.5.9 and the estimate $M_1(r, f) \leqslant r/(1-r^2)$, it follows that the arc length $L_r(f)$ of the image of $|z| = r < 1$ under f satisfies

$$\begin{aligned} L_r(f) &= r\int_0^{2\pi} |f'(re^{i\theta})|d\theta \\ &= r\int_0^{2\pi} \frac{|f'(re^{i\theta})|}{|f(re^{i\theta})|}|f(re^{i\theta})|d\theta \\ &\leqslant 2\pi\frac{1+r}{1-r}\frac{r}{1-r^2} = \frac{2\pi r}{(1-r)^2}. \end{aligned}$$

□

Let $C(r, \theta)$ be the image of the ray joining 0 and $z = re^{i\theta}$ under a function $f \in \mathcal{S}$ and let $l(r, \theta)$ be its length given by

$$l(r,\theta) = \int_{C(r,\theta)} |dz| = \int_0^r |f'(\rho e^{i\theta})|d\rho.$$

We note that finding a constant A such that $l(r, \theta) < A|f(re^{i\theta})|$ seems to be a daunting challenge.

Exercises

1. If $f(z) = z/\left(1-z^2\right)$, show that $f \in \mathcal{S}$ and $f(\mathbb{D})$ is the entire plane minus the two half-lines $\frac{1}{2} \leqslant x < \infty$ and $-\infty < x \leq -\frac{1}{2}$.
2. If $f(z) = \frac{1}{2}\log((1+z)/(1-z))$, show that $f \in \mathcal{S}$ and $f(\mathbb{D})$ is the horizontal strip $-\pi/4 < \mathrm{Im}\{w\} < \pi/4$.
3. If $f(z) = z - \frac{1}{2}z^2$, show that $f \in \mathcal{S}$ and $f(\mathbb{D})$ is the interior of a cardioid.
4. If
$$f(z) = z + \frac{8}{9}z^2 + \frac{1}{3}z^3 \quad \text{and} \quad g(z) = z + \frac{3}{5}z^3 + \frac{1}{5}z^5,$$
show that $f, g \in \mathcal{S}$.

5. If $f \in \mathcal{S}$, prove that
$$\left|\frac{zf''(z)}{f'(z)}\right| \leqslant \frac{2|z|(2+|z|)}{1-|z|}.$$
6. For $f \in \mathcal{S}$, prove the non-sharp inequality
$$|\arg f'(z)| \leqslant 2\log\left(\frac{1+|z|}{1-|z|}\right).$$
7. If $f \in \mathcal{U}_\alpha$, show that the following sharp inequality holds:
$$\frac{(1-r)^{\alpha-1}}{(1+r)^{\alpha+1}} \leqslant |f'(z)| \leqslant \frac{(1+r)^{\alpha-1}}{(1-r)^{\alpha+1}}.$$
Use this to prove that the order of any linear-invariant family is always at least one.
8. For $f \in \mathcal{S}$, prove that
$$\left|\arg f'(z)\right| \leqslant \begin{cases} 4\sin^{-1}|z|, & |z| \leqslant 1/\sqrt{2}, \\ \pi + \log\left(\frac{|z|^2}{1-|z|^2}\right), & |z| \geqslant 1/\sqrt{2}. \end{cases}$$
9. For $f \in \mathcal{S}$, prove that
$$\left|\arg\left(\frac{zf'(z)}{f(z)}\right)\right| \leqslant \log\left(\frac{1+|z|}{1-|z|}\right).$$
10. For $f \in \mathcal{S}$, prove that
$$\left|\log\left(\frac{zf'(z)}{f(z)}\right)\right| \leqslant \log\left(\frac{1+|z|}{1-|z|}\right).$$
11. For $f \in \mathcal{S}$, show that the curvature of curve $f(\partial\mathbb{D}_r)$, $0 < r < 1$, at a point $w = f(z)$ is given by
$$K_z = \frac{1}{r\,|f'(z)|}\,\mathrm{Re}\left\{1 + \frac{zf''(z)}{f'(z)}\right\}, \quad |z| = r < 1.$$
and
$$\left(\frac{1-r}{1+r}\right)^2 \frac{1-4r+r^2}{r} \leqslant K_z \leqslant \left(\frac{1+r}{1-r}\right)^2 \frac{1+4r+r^2}{r}, \quad |z| = r < 1$$
12. Show that for each function $f \in \mathcal{S}$ which is not the identity mapping, there are points z_1 and z_2 in $\mathbb{D}$ such that $f(z_1)\,f(z_2) = 1$.
13. For $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{S}$, show that
$$|a_3 - \alpha a_2^2| \leqslant 1 + 2\mathrm{e}^{-\frac{2\alpha}{1-\alpha}}, \quad 0 < \alpha < 1.$$
14. Let a function $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{S}$ be bounded by a number $M > 1$, so that $|f(z)| < M$ for all $z \in \mathbb{D}$. Improve Bieberbach's bound $|a_2| \leqslant 2$ to $|a_2| \leqslant 2(1 - 1/M)$, and show that equality occurs if and only if f maps $\mathbb{D}$ onto the disk $|w| < M$ minus a radial segment.
15. Let g be analytic and univalent in $\mathbb{D}$, with $g(0) = 0$ and $g(z) \neq 1$. Show that the function G defined by
$$G(z) = 2\,g\left(z^2\right) - 2\left(\frac{g\left(z^2\right)}{g\left(z^2\right) - 1}\right)^{1/2}$$
has the same properties.
16. A single slit mapping is a function that maps $\mathbb{D}$ onto $\mathbb{C}$ minus a Jordan arc. Show that the set of all single slit mappings is dense in $\mathcal{S}$.

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Chapter 3
Bounded Functions

Abstract Let $\mathcal{B}$ denote the class of all analytic functions $w : \mathbb{D} \to \overline{\mathbb{D}}$, and let $\mathcal{B}_0$ be its subclass consisting of those functions w satisfying $w(0) = 0$. The functions in the class $\mathcal{B}_0$ are known as bounded or Schur functions. They are also referred to as Schwarz functions, as they satisfy the conditions of the Schwarz's lemma. This chapter is devoted to the study of various properties of these functions, including sharp estimates for their Taylor coefficients.

3.1 Schwarz's Lemma

Let $\mathcal{B}$ be the class of all analytic functions $w : \mathbb{D} \to \overline{\mathbb{D}}$, and let $\mathcal{B}_0$ be its subclass consisting of functions w satisfying $w(0) = 0$. Functions in the class $\mathcal{B}_0$ are known as *bounded* or *Schur functions*. These functions are also referred to as the *Schwarz functions*, as they satisfy the hypothesis of the Schwarz's lemma. For $n \in \mathbb{N}$, the function $w(z) = z^n$ belongs to $\mathcal{B}_0$, while the function $w(z) = (a - z^n)/(1 - \overline{a}z^n)$, $0 < |a| < 1$ belongs to $\mathcal{B} \setminus \mathcal{B}_0$.

Since the functions in $\mathcal{B}$ and $\mathcal{B}_0$ are uniformly bounded by 1, both classes are normal families. In fact, each class is a compact family. To see this, let $w_n \in \mathcal{B}$ and $w_n \to w \in \mathcal{H}(\mathbb{D})$. Since $|w_n(z)| \leqslant 1$, it follows that $|w(z)| \leqslant 1$ and hence $w \in \mathcal{B}$. If, in addition, $w_n(0) = 0$ for all n, then $w(0) = \lim_{n\to\infty} w_n(0) = 0$ and so $w \in \mathcal{B}_0$. Schwarz's lemma plays a central role in studying functions with positive real part, due to a simple relation between these functions and bounded functions. Schwarz's lemma will first be proved, followed by some of its consequences. It implies that a function $w \in \mathcal{B}_0$ maps the disk $\mathbb{D}_r$ into itself for each $0 < r < 1$ (see Fig. 3.1). As a consequence, every function in $\mathcal{B}$ maps $\mathbb{D}$ into $\mathbb{D}$.

Theorem 3.1.1 (Schwarz's Lemma) *If $w \in \mathcal{B}_0$, then*

$$|w(z)| \leqslant |z| \quad \textit{and} \quad |w'(0)| \leqslant 1.$$

Also, $|w(z_0)| = |z_0|$ for some non-zero z_0 in $\mathbb{D}$ or $|w'(0)| = 1$ if and only if $w(z) = e^{it}z$ for some $t \in \mathbb{R}$.

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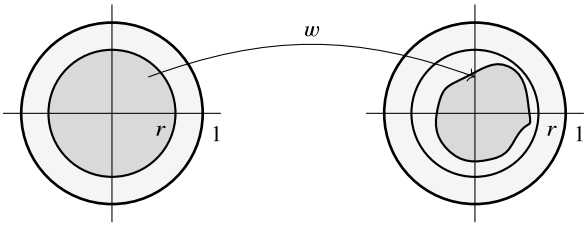


Fig. 3.1 Schwarz's lemma

Proof The function $\widetilde{w} : \mathbb{D} \to \mathbb{C}$ defined by

$$\widetilde{w}(z) = \begin{cases} w(z)/z, & z \neq 0, \\ w'(0), & z = 0, \end{cases} \tag{3.1.1}$$

is analytic. Let $\xi \in \mathbb{D}$ be arbitrary and choose $r < 1$ such that $|\xi| < r$. Since

$$|\widetilde{w}(z)| = \frac{|w(z)|}{|z|} \leqslant \frac{1}{\rho}$$

on $|z| = \rho > r$, the maximum modulus theorem for a bounded domain yields

$$|\widetilde{w}(\xi)| \leqslant \max\{|\widetilde{w}(z)| : |z| \leqslant \rho\} = \max\{|\widetilde{w}(z)| : |z| = \rho\} \leqslant \frac{1}{\rho}.$$

By letting $\rho \to 1^-$, it follows that $|\widetilde{w}(\xi)| \leqslant 1$ and hence $|\widetilde{w}(z)| \leqslant 1$ for all $z \in \mathbb{D}$. This shows that $|w'(0)| \leqslant 1$ and $|w(z)| \leqslant |z|$ for all $z \in \mathbb{D}$.

If $|w'(0)| = 1$, or if $|w(z)| = |z|$ for some $z \neq 0$, then $|\widetilde{w}|$ assumes its maximum inside $\mathbb{D}$. Thus, $\widetilde{w}(z) \equiv c$ with $|c| = 1$, that is, $w(z) = cz = \mathrm{e}^{\mathrm{i}t} z$ for some $t \in \mathbb{R}$. Further, if $w(z) = \mathrm{e}^{\mathrm{i}t} z$ for some $t \in \mathbb{R}$, then $|w(z)| = |z|$ and $|w'(0)| = 1$. □

By Theorem 1.1.2, the mapping $\varphi_a : \mathbb{D} \to \mathbb{D}$ defined by

$$\varphi_a(z) = \frac{a - z}{1 - \overline{a}z}, \quad |a| < 1$$

is a bijection, and the inverse of φ_a is φ_a. Also, the mapping φ_a maps $\partial\mathbb{D}$ onto $\partial\mathbb{D}$, $\varphi_a(a) = 0$, $\varphi_a'(0) = |a|^2 - 1$, and $\varphi_a'(a) = (|a|^2 - 1)^{-1}$.

The next two theorems give an extension of Schwarz's lemma for functions w that do not fix the origin.

Theorem 3.1.2 *If $w \in \mathcal{B}$, then*

$$\frac{|w(0)| - |z|}{1 - |w(0)||z|} \leqslant |w(z)| \leqslant \frac{|w(0)| + |z|}{1 + |w(0)||z|}.$$

Proof Let $\widetilde{w} = \varphi_a \circ w$ with $a = w(0)$. Clearly, $\widetilde{w}$ maps $\mathbb{D}$ into $\mathbb{D}$ and $\widetilde{w}(0) = \varphi_a(w(0)) = \varphi_a(a) = 0$. Then Schwarz's Lemma shows that $|\widetilde{w}(z)| \leqslant |z|$ and hence

$$\left|\frac{a - w(z)}{1 - \overline{a}w(z)}\right| = |\widetilde{w}(z)| \leqslant |z|. \tag{3.1.2}$$

Write $w = w(z)$ and $|z| = r$. Then, the inequality (3.1.2) is equivalent to

$$|w|^2 - 2\,\mathrm{Re}\, w\frac{\overline{a}(1 - r^2)}{1 - r^2|a|^2} \leqslant \frac{r^2 - |a|^2}{1 - r^2|a|^2}.$$

This, by completing the squares, gives

$$\left|w - \frac{a(1 - r^2)}{1 - r^2|a|^2}\right|^2 \leqslant \frac{r^2 - |a|^2}{1 - r^2|a|^2} + \frac{|a|^2(1 - r^2)^2}{(1 - r^2|a|^2)^2} = \frac{r^2(1 - |a|^2)^2}{(1 - r^2|a|^2)^2}$$

and hence,

$$\left|w - \frac{a(1 - r^2)}{1 - r^2|a|^2}\right| \leqslant \frac{r(1 - |a|^2)}{(1 - r^2|a|^2)}.$$

It then follows that

$$|w| \leqslant \frac{|a|(1 - r^2) + r(1 - |a|^2)}{1 - r^2|a|^2} = \frac{|a| + r}{1 + |a|r},$$

and

$$|w| \geqslant \frac{|a|(1 - r^2) - r(1 - |a|^2)}{1 - r^2|a|^2} = \frac{|a| - r}{1 - r|a|}.$$

□

Theorem 3.1.3 (Schwarz-Pick Theorem) *If $w \in \mathcal{B}$, then, for each a in $\mathbb{D}$,*

$$\left|\frac{w(a) - w(z)}{1 - \overline{w}(a)w(z)}\right| \leqslant \left|\frac{a - z}{1 - \overline{a}z}\right|, \quad \text{and} \quad |w'(a)| \leqslant \frac{1 - |w(a)|^2}{1 - |a|^2}. \tag{3.1.3}$$

Equality occurs in the first inequality for some $z \neq a$, or equality occurs in the second inequality if and only if $w(z) = k\varphi_\beta(z)$, where k is a constant with $|k| = 1$ and $\beta \in \mathbb{D}$.

Proof Let $\alpha := w(a)$. The first inequality is the same as $|\varphi_\alpha(w(z))| \leqslant |\varphi_a(z)|$. Let

$$\widetilde{w}(z) = \varphi_\alpha\left(w\left(\varphi_a(z)\right)\right).$$

Then $\widetilde{w}$ maps $\mathbb{D}$ onto $\mathbb{D}$ (see Fig. 3.2) and

$$\widetilde{w}(0) = \varphi_\alpha(w(\varphi_a(0))) = \varphi_\alpha(w(a)) = \varphi_\alpha(\alpha) = 0.$$

By Schwarz's lemma, it follows that $\widetilde{w}$ satisfies $|\widetilde{w}(z)| \leqslant |z|$ for all $z \in \mathbb{D}$ and $|\widetilde{w}'(0)| \leqslant 1$. On using the fact that $\varphi_a(\varphi_a(z)) = z$, the equation $\widetilde{w}(z) = \varphi_\alpha(w(\varphi_a(z)))$, upon replacing z with $\varphi_a(z)$, becomes

$$\widetilde{w}(\varphi_a(z)) = \varphi_\alpha(w(\varphi_a(\varphi_a(z)))) = \varphi_\alpha(w(z)).$$

The inequality $|\widetilde{w}(z)| \leqslant |z|$ shows that

$$|\varphi_\alpha(w(z))| = |\widetilde{w}(\varphi_a(z))| \leqslant |\varphi_a(z)|$$

and this is the same as the first inequality in (3.1.3).

Writing the first inequality (3.1.3) as

$$\left|\frac{w(z) - w(a)}{z - a}\right| \leqslant \left|\frac{1 - \overline{w}(a)w(z)}{1 - \overline{a}z}\right|$$

and taking limit as $z \to a$, the second inequality in (3.1.3) follows. Indeed, the following is a direct proof of it. Since $\varphi_a(a) = 0, \varphi_a'(0) = |a|^2 - 1, \varphi_a'(a) = (|a|^2 - 1)^{-1}$ and

$$\widetilde{w}'(z) = \varphi_\alpha'(w(\varphi_a(z)))w'(\varphi_a(z))\varphi_a'(z),$$

it follows that

$$\begin{aligned}\widetilde{w}'(0) &= \varphi_\alpha'(w(\varphi_a(0)))w'(\varphi_a(0))\varphi_a'(0)\\ &= \varphi_\alpha'(\alpha)w'(a)(|a|^2 - 1)\\ &= \frac{1 - |a|^2}{1 - |\alpha|^2}w'(a).\end{aligned}$$

Since $|\widetilde{w}'(0)| \leqslant 1$, it immediately yields

$$|w'(a)| \leqslant \frac{1 - |\alpha|^2}{1 - |a|^2} = \frac{1 - |w(a)|^2}{1 - |a|^2}.$$

If equality holds in either of the inequalities in (3.1.3), then either $|\widetilde{w}(z)| = |z|$ or $|\widetilde{w}'(0)| = 1$. Thus, $\widetilde{w}(z) = cz$, with $|c| = 1$ and this is equivalent to $\varphi_\alpha(w(\varphi_a(z))) = cz$ or $w(z) = \varphi_\alpha(c\varphi_a(z))$. A simple computation shows that

$$(\varphi_a \circ \varphi_b)(z) = -\frac{a\overline{b} - 1}{\overline{a}b - 1}\varphi_{\varphi_b(a)}(z) \tag{3.1.4}$$

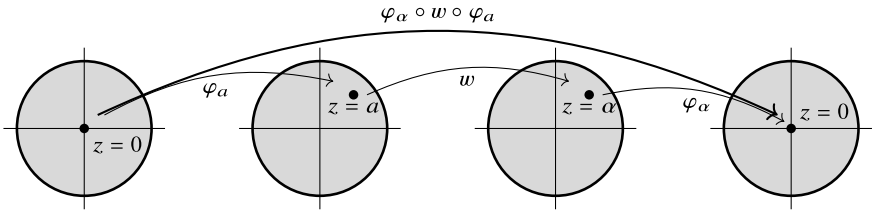


Fig. 3.2 The composition $\varphi_\alpha \circ w \circ \varphi_a$

and, for $|c| = 1$,

$$\varphi_a(cz) = c\varphi_{a\overline{c}}(z). \tag{3.1.5}$$

From (3.1.4) and (3.1.5), it follows that

$$w(z) = \varphi_\alpha(c\varphi_a(z)) = c\varphi_{\alpha\overline{c}}(\varphi_a(z)) = -c\frac{\alpha\overline{a}c - 1}{\overline{\alpha}ac - 1}\varphi_{\varphi_a(\alpha\overline{c})}(z) =: k\varphi_\beta(z).$$

Clearly, $|k| = 1$ and $\beta = \varphi_a(\alpha\overline{c}) \in \mathbb{D}$.

Next is shown that equality holds in (3.1.3) for the function $w = k\varphi_\beta$. The equation (3.1.4) shows that $|(\varphi_a \circ \varphi_b)(z)| = |\varphi_{\varphi_b(a)}(z)|$. Therefore,

$$|\varphi_\alpha(w(z))| = |\varphi_\alpha(k\varphi_\beta(z))| = |k\varphi_{\overline{k}\alpha}(\varphi_\beta(z))| = |\varphi_{\varphi_\beta(\overline{k}\alpha)}(z)| = |\varphi_a(z)|$$

since $\alpha = k\varphi_\beta(a)$ and so $\varphi_\beta(\overline{k}\alpha) = \varphi_\beta(\varphi_\beta(a)) = a$. Alternately, this can be seen directly as a calculation shows that

$$\frac{w(a) - w(z)}{1 - \overline{w}(a)w(z)} = \frac{k\varphi_\beta(a) - k\varphi_\beta(z)}{1 - \overline{k\varphi_\beta(a)}k\varphi_\beta(z)} = -k\frac{1 - \overline{a}\beta}{1 - a\overline{\beta}}\frac{a - z}{1 - \overline{a}z}.$$

□

A function $\varphi : \mathbb{D} \to \mathbb{D}$ is called an automorphism of $\mathbb{D}$ if it is analytic and bijective. Denote by $\mathrm{Aut}(\mathbb{D})$ the set of all automorphisms of $\mathbb{D}$. Clearly, $\mathrm{Aut}(\mathbb{D})$ is a group under composition. By using the Schwarz-Pick theorem, it can be shown that $\mathrm{Aut}(\mathbb{D})$ consists of mappings of the form $c\varphi_\alpha$, where $\alpha \in \mathbb{D}$.

Theorem 3.1.4 *Let $f : \mathbb{D} \to \mathbb{D}$ be analytic and $f(a) = 0$ for some $a \in \mathbb{D}$. Then, the function f is a bijection if and only if there exists $c \in \mathbb{C}$ with $|c| = 1$ such that $f = c\varphi_a$.*

Proof If $f = c\varphi_a$ for some $c \in \mathbb{C}$ with $|c| = 1$, then, by Theorem 1.1.2, the function f has the desired properties. For the converse, let f be a bijection so that there exists an analytic function $g : \mathbb{D} \to \mathbb{D}$ satisfying $g(f(z)) = z$ for all $z \in \mathbb{D}$. Since the function f satisfies $|f(z)| < 1$, the second inequality in (3.1.3) yields

$$|f'(a)| \leqslant \frac{1-|f(a)|^2}{1-|a|^2} = \frac{1}{1-|a|^2}. \tag{3.1.6}$$

On the other hand, since the function g satisfies $|g(z)| < 1$, it follows that

$$|g'(0)| \leqslant 1 - |g(0)|^2 = 1 - |a|^2.$$

Since $g(f(z)) = z$, it follows that $g'(f(a))f'(a) = 1$ or $g'(0)f'(a) = 1$. This shows that $|f'(a)| = 1/|g'(0)| \geqslant 1/(1-|a|^2)$ which together with (3.1.6) implies $|f'(a)| = 1/(1-|a|^2)$.

Since $f(a) = 0$, Theorem 3.1.3 implies that $f(z) = c\varphi_a(z)$ for some c with $|c| = 1$. □

3.2 Sharpened Schwarz's Lemma

In the next lemma, it is shown that, for $p, q \in \mathbb{C}$, the set of points $z \in \mathbb{C}$ satisfying $|(z-p)/(z-q)| \leqslant k$, with $k^2 < 1$, is a closed disk. Also, the center and radius of this disk are computed.

Lemma 3.2.1 *Let $p, q \in \mathbb{C}$, and $k^2 < 1$. The set $\{z \in \mathbb{C} : |(z-p)/(z-q)| \leqslant k\}$ is a closed disk $\overline{\mathbb{D}}(c; r)$ with center and radius given by*

$$c = \frac{p - k^2 q}{1-k^2}, \quad \text{and} \quad r = \frac{|p-q|}{1-k^2}k.$$

Proof The inequality $|z-p| \leqslant k|z-q|$, upon squaring and rewriting, becomes

$$|z|^2 - 2\,\mathrm{Re}(\overline{p}z) + |p|^2 \leqslant k^2\left(|z|^2 - 2\,\mathrm{Re}(\overline{q}z) + |q|^2\right)$$

or

$$\left|z - \frac{p-k^2q}{1-k^2}\right|^2 \leqslant \frac{\left|p-k^2q\right|^2}{\left(1-k^2\right)^2} - \frac{|p|^2 - k^2|q|^2}{1-k^2} = \frac{k^2|p-q|^2}{(1-k^2)^2}.$$

This proves that

$$\left|z - \frac{p-k^2q}{1-k^2}\right| \leqslant \frac{k|p-q|}{1-k^2}.$$

□

Since

$$p - c = \frac{k^2(q-p)}{1-k^2}, \quad \text{and} \quad q - c = \frac{q-p}{1-k^2},$$

we see that $(p-c)\,/\,(q-c)$ is real and positive, and

$$|p-c|\,|q-c| = r^2.$$

Therefore, p and q are inverse points with respect to the circle $|z-p| = k|z-q|$. This circle is known as the *circle of Apollonius*.

The following is a sharpened version of Schwarz's lemma in which the images of two points in $\mathbb{D}$ are known. This version contains Dieudonné's lemma and Rogosinski's lemma as corollaries.

Theorem 3.2.2 (Sharpened version of the Schwarz's lemma) *Let $z_0 \in \mathbb{D}\setminus\{0\}$ be fixed. For all $w \in \mathcal{B}_0$ with $w(z_0) = w_0$, $z_0 \neq 0$, the region of values of $w(\lambda_0)$, $\lambda_0 \in \mathbb{D}$, is the closed disk $\overline{\mathbb{D}}(c;r)$, where the center and radius are given by*

$$c = \frac{1-\left|\varphi_{z_0}(\lambda_0)\right|^2}{1-|w_0/z_0|^2\left|\varphi_{z_0}(\lambda_0)\right|^2}\frac{\lambda_0 w_0}{z_0},$$

and

$$r = \frac{1-|w_0/z_0|^2}{1-|w_0/z_0|^2\left|\varphi_{z_0}(\lambda_0)\right|^2}\,|\lambda_0|\left|\varphi_{z_0}(\lambda_0)\right|.$$

In particular,

$$\frac{|w_0/z_0|-\left|\varphi_{z_0}(\lambda_0)\right|}{1-|w_0/z_0|\left|\varphi_{z_0}(\lambda_0)\right|}\,|\lambda_0| \leqslant |w(\lambda_0)| \leqslant \frac{|w_0/z_0|+\left|\varphi_{z_0}(\lambda_0)\right|}{1+|w_0/z_0|\left|\varphi_{z_0}(\lambda_0)\right|}\,|\lambda_0|,$$

and this inequality is sharp.

Proof Define $\widetilde{w}:\mathbb{D}\to\mathbb{C}$ by $\widetilde{w}(z) = w(z)/z$ for $z\neq 0$ and $\widetilde{w}(0) = w'(0)$. Then the function $\widetilde{w}$ is in the class $\mathcal{H}$ and, by Schwarz lemma, $|\widetilde{w}(z)|<1$ for all $z\in\mathbb{D}$. The Schwarz-Pick theorem then gives

$$\left|\frac{\widetilde{w}(\lambda_0)-\widetilde{w}(z)}{1-\overline{\widetilde{w}(\lambda_0)}\widetilde{w}(z)}\right| \leqslant \left|\frac{\lambda_0 - z}{1-\overline{\lambda_0}z}\right|, \quad z\in\mathbb{D}. \tag{3.2.1}$$

Since $w_0 = w(z_0)$, the inequality (3.2.1) yields

$$\left|\frac{w(\lambda_0)-(\lambda_0 w_0/z_0)}{w(\lambda_0)-\lambda_0\overline{(z_0/w_0)}}\right| \leqslant \left|\frac{w_0}{z_0}\right|\left|\varphi_{z_0}(\lambda_0)\right|. \tag{3.2.2}$$

The inequality (3.2.2) is valid if $w_0\neq 0$ (and the case $w_0 = 0$ is left to the reader). With

$$\sigma := \varphi_{z_0}(\lambda_0), \quad \tau := \frac{w_0}{z_0}, \tag{3.2.3}$$

the inequality (3.2.2) becomes

$$\left|\frac{w(\lambda_0)-\lambda_0\tau}{w(\lambda_0)-(\lambda_0/\overline{\tau})}\right| \leqslant |\tau|\,|\sigma|\,. \tag{3.2.4}$$

Note that $|w_0| = |w(z_0)| \leqslant |z_0|$ and so $|\tau| \leqslant 1$. Let $p = \lambda_0\tau$, $q = \lambda_0/\overline{\tau}$ and $k = |\tau\sigma|$. Since $k \leqslant |\sigma| = |\varphi_{z_0}(\lambda_0)| < 1$, by Lemma 3.2.1, the center and radius of the disk in (3.2.4) are given by

$$c = \frac{p-k^2q}{1-k^2} = \frac{1-|\sigma|^2}{1-|\tau\sigma|^2}\lambda_0\tau, \quad \text{and} \quad r = \frac{|p-q|}{1-k^2}k = \frac{1-|\tau|^2}{1-|\tau\sigma|^2}|\lambda_0\sigma|.$$

An application of Lemma 3.2.1 gives the desired result.

To show that $\overline{\mathbb{D}}(c;r)$ is covered, let $\alpha \in \mathbb{D}$ and set $w(z) = z\varphi_{w_0/z_0}\left(\alpha\varphi_{z_0}(z)\right)$. Then $w(0) = 0$, $w(z_0) = w_0$ and, by using (3.2.3),

$$w(\lambda_0) = \lambda_0\frac{\tau-\alpha\sigma}{1-\overline{\tau}\alpha\sigma}.$$

It is now shown that

$$w(\lambda_0) = \frac{1-|\sigma|^2}{1-|\tau\sigma|^2}\lambda_0\tau + \varphi_{\tau\overline{\sigma}}(\alpha)\frac{1-|\tau|^2}{1-|\tau\sigma|^2}\lambda_0\sigma.$$

Since

$$\varphi_{\tau\overline{\sigma}}(\alpha) = \frac{\tau\overline{\sigma}-\alpha}{1-\sigma\alpha\overline{\tau}}$$

and

$$\tau(1-|\sigma|^2)(1-\sigma\alpha\overline{\tau}) + (\tau\overline{\sigma}-\alpha)(1-|\tau|^2)\sigma = (\tau-\sigma\alpha)(1-|\tau\sigma|^2),$$

it follows that

$$\begin{aligned}
&\frac{1-|\sigma|^2}{1-|\tau\sigma|^2}\lambda_0\tau + \varphi_{\tau\overline{\sigma}}(\alpha)\frac{1-|\tau|^2}{1-|\tau\sigma|^2}\lambda_0\sigma \\
&= \lambda_0\frac{\tau(1-|\sigma|^2)(1-\sigma\alpha\overline{\tau}) + (\tau\overline{\sigma}-\alpha)(1-|\tau|^2)\sigma}{(1-\alpha\sigma\overline{\tau})(1-|\tau\sigma|^2)} \\
&= w(\lambda_0)\,.
\end{aligned}$$

Thus, $w(\lambda_0) = c + re^{i\beta}\varphi_{\tau\overline{\sigma}}(\alpha)$, where β is an argument of the number

$$\frac{1-|\tau|^2}{1-|\tau|^2|\sigma|^2}\lambda_0\sigma.$$

Since $\alpha \in \mathbb{D}$ is arbitrary and $\varphi_{\tau\overline{\sigma}}$ is a self-map of $\mathbb{D}$, the disk $\overline{\mathbb{D}}(c;r)$ is covered.

The last inequality in the theorem follows from

$$|c| \pm r = \frac{1-|\sigma|^2}{1-|\tau\sigma|^2}|\lambda_0\tau| \pm \frac{1-|\tau|^2}{1-|\tau\sigma|^2}|\lambda_0\sigma| = \frac{|\tau| \pm |\sigma|}{1 \pm |\tau||\sigma|}\,|\lambda_0|\,.$$

Sharpness follows as the entire disk $\overline{\mathbb{D}}(c;r)$ is covered. □

Corollary 3.2.3 (Dieudonné's Lemma) *For all $w \in \mathcal{B}_0$ with $w(z_0) = w_0$, $z_0 \neq 0$, the region of values of $w'(z_0)$ is the closed disk $\overline{\mathbb{D}}(c_1; r_1)$, where*

$$c_1 = \frac{w_0}{z_0}, \quad \textit{and} \quad r_1 = \frac{|z_0|^2 - |w_0|^2}{|z_0|\left(1-|z_0|^2\right)}.$$

Proof Let $\lambda_0 \in \mathbb{D}$. Theorem 3.2.2 applied to w shows that, by using translation and dilation, the quotient $(w(\lambda_0) - w_0)/(\lambda_0 - z_0)$ lies in $\mathbb{D}(\hat{c}; \hat{r})$, where

$$\hat{c} = \frac{c - w_0}{\lambda_0 - z_0}, \quad \text{and} \quad \hat{r} = \frac{r}{|\lambda_0 - z_0|}.$$

With σ and τ as in the proof of Theorem 3.2.2, it follows that

$$\begin{aligned}\hat{c} &= \frac{(\lambda_0 w_0/z_0)\left(1-|\sigma|^2\right) - w_0\left(1-|\tau|^2|\sigma|^2\right)}{(\lambda_0 - z_0)\left(1-|\tau|^2|\sigma|^2\right)} \\ &= \frac{w_0\left((\lambda_0 - z_0)/z_0 + |\sigma|^2\left(|\tau|^2 - \lambda_0/z_0\right)\right)}{(\lambda_0 - z_0)\left(1-|\tau|^2|\sigma|^2\right)}\end{aligned}$$

and

$$\hat{r} = \frac{|\lambda_0|\,|\sigma|\left(1-|\tau|^2\right)}{|\lambda_0 - z_0|\left(1-|\tau|^2|\sigma|^2\right)} = \frac{|\lambda_0|\left(1-|\tau|^2\right)}{|1-\overline{z}_0\lambda_0|\left(1-|\tau|^2|\sigma|^2\right)}.$$

Now $\lim_{\lambda_0 \to z_0} \hat{c} = c_1$, $\lim_{\lambda_0 \to z_0} \hat{r} = r_1$ and

$$\frac{w(\lambda_0) - w_0}{\lambda_0 - z_0} = \frac{w(\lambda_0) - w(z_0)}{\lambda_0 - z_0} \to w'(z_0)$$

as $\lambda_0 \to z_0$. Thus, the desired result is established. □

Corollary 3.2.4 (Distortion Theorem) *If $w \in \mathcal{B}_0$, then*

$$\left|w'(z)\right| \leqslant \begin{cases} 1, & r = |z| \leqslant \sqrt{2} - 1, \\ \frac{\left(1+r^2\right)^2}{4r\left(1-r^2\right)}, & r \geqslant \sqrt{2} - 1. \end{cases}$$

This bound is sharp for each $r < 1$.

Proof For fixed z, let $r = |z|$ and $R = |w(z)|$. Then, by Schwarz's lemma, $R = |w(z)| \leqslant |z| = r$, and Corollary 3.2.3 gives

$$\left|w'(z)\right| \leqslant \frac{R}{r} + \frac{r^2 - R^2}{r\left(1 - r^2\right)} = \frac{\psi(R)}{r\left(1 - r^2\right)}$$

where $\psi : [0, r] \to \mathbb{R}$ is defined by

$$\psi(R) := \left(1 - r^2\right) R + r^2 - R^2 = (1 - R)(r^2 + R).$$

Since $\psi'(R) = 1 - r^2 - 2R$ and $\psi''(R) < 0$, ψ has a (local) maximum at $R_0 = \frac{1}{2}\left(1 - r^2\right)$. Note that R_0 is less than r if and only if $r > \sqrt{2} - 1$. In the case $r > \sqrt{2} - 1$, the sharp upper bound for $\left|w'(z)\right|$ is

$$|w'(z)| \leqslant \frac{\psi\left(\frac{1}{2}\left(1 - r^2\right)\right)}{r\left(1 - r^2\right)} = \frac{\left(1 + r^2\right)^2}{4r\left(1 - r^2\right)}.$$

For $r \leqslant \sqrt{2} - 1$, it follows that $\psi(R) \leqslant \psi(r) = r\left(1 - r^2\right)$ and therefore $\left|w'(z)\right| \leqslant 1$.

Equality holds in the first case for some z_0 if and only if $w(z) = \mathrm{e}^{\mathrm{i}\theta} z$ for some real constant θ. Equality holds in the latter case for some z_0 if and only if

$$w(z) = \mathrm{e}^{\mathrm{i}\theta} z \frac{z - a}{1 - \overline{a} z},$$

where $r_0 = |z_0|$,

$$a = \frac{3r_0^2 - 1}{r_0^2\left(1 + r_0^2\right)} z_0$$

and $\theta \in \mathbb{R}$ is arbitrary. It is easy to check that $|a| < 1$ for $r_0 > \sqrt{2} - 1$ and so $w \in \mathcal{B}_0$. The result is sharp since

$$|w'(z_0)| = \left|\frac{z_0 - a}{(1 - \overline{a} z_0)^2} + \frac{z_0}{1 - \overline{a} z_0}\right| = \frac{(1 + r_0^2)^2}{4r_0(1 - r_0^2)}.$$

□

Corollary 3.2.5 (Rogosinski's Lemma) *Let $\lambda_0 \in \mathbb{D}$. Then, for all functions $w \in \mathcal{B}_0$, the region of values of $w(\lambda_0)$ is $\mathbb{D}(c_2; r_2)$, where*

$$c_2 = \frac{\lambda_0 w'(0)\left(1 - |\lambda_0|^2\right)}{1 - |\lambda_0|^2 |w'(0)|^2}, \quad \textit{and} \quad r_2 = \frac{|\lambda_0|^2\left(1 - \left|w'(0)\right|^2\right)}{1 - |\lambda_0|^2 |w'(0)|^2}.$$

Proof Since

$$\lim_{z_0 \to 0} \frac{w_0}{z_0} = \lim_{z_0 \to 0} \frac{w(z_0)}{z_0} = w'(0),$$

the corollary is inferred by letting $z_0 \to 0$ in Theorem 3.2.2. □

3.3 Hardy-Hilbert Space

Recall that $\mathcal{H} = \mathcal{H}(\mathbb{D})$ is the class of all functions analytic in $\mathbb{D}$. For $f \in \mathcal{H}$, $0 < p < \infty$, and $0 \leqslant r < 1$, define

$$M_p(r, f) := \left(\frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})|^p d\theta \right)^{\frac{1}{p}}$$

and, for $p = \infty$,

$$M_\infty(r, f) := \sup\{|f(re^{i\theta})| : 0 \leqslant \theta < 2\pi\}.$$

The *Hardy space* $\mathcal{H}^p$ consists of all analytic functions $f \in \mathcal{H}$ for which

$$\|f\|_p := \lim_{r \uparrow 1} M_p(r, f) < \infty.$$

We shall show that the Hardy space $\mathcal{H}^2$ is indeed a Hilbert space. In view of this fact, the space $\mathcal{H}^2$ is called the Hardy-Hilbert space.

The following lemma expresses the norm of a function in terms of its Taylor coefficients.

Lemma 3.3.1 *If* $f(z) = \sum_{k=0}^{\infty} a_k z^k \in \mathcal{H}^2$, *then*

$$\|f\|_2^2 = \sum_{k=0}^{\infty} |a_k|^2 . \tag{3.3.1}$$

Proof Since

$$\int_0^{2\pi} e^{i(n-m)\theta} d\theta = \begin{cases} 2\pi, & \text{for } n = m, \\ 0, & \text{for } n \neq m, \end{cases}$$

the identity

$$|f(re^{i\theta})|^2 = \left(\sum_{n=0}^{\infty} a_n r^n e^{in\theta}\right)\left(\sum_{m=0}^{\infty} \overline{a}_m r^m e^{-im\theta}\right)$$
$$= \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} a_n \overline{a}_m r^{n+m} e^{i(n-m)\theta} \tag{3.3.2}$$

for $0 < r < 1$ yields

$$(M_2(r, f))^2 = \frac{1}{2\pi}\int_0^{2\pi} |f(re^{i\theta})|^2 d\theta = \sum_{k=0}^{\infty} |a_k|^2 r^{2k}. \tag{3.3.3}$$

From (3.3.3), it is evident that $(M_2(r, f))^2$ is an increasing function of r and therefore,

$$\|f\|_2^2 = \lim_{r\uparrow 1}(M_2(r, f))^2 = \sum_{k=0}^{\infty} |a_k|^2.$$

□

For two functions $f(z) = \sum_{k=0}^{\infty} a_k z^k$ and $g(z) = \sum_{k=0}^{\infty} b_k z^k$ in $\mathcal{H}^2$, define the inner product $\langle f, g\rangle$ by

$$\langle f, g\rangle = \sum_{k=0}^{\infty} a_k \overline{b_k}. \tag{3.3.4}$$

The Cauchy-Schwarz inequality yields

$$\left|\sum_{k=0}^{n} a_k \overline{b_k}\right|^2 \leqslant \left(\sum_{k=0}^{n} |a_k|^2\right)\left(\sum_{k=0}^{n} |b_k|^2\right)$$
$$\leqslant \left(\sum_{k=0}^{\infty} |a_k|^2\right)\left(\sum_{k=0}^{\infty} |b_k|^2\right)$$
$$= \|f\|_2^2 \|g\|_2^2.$$

Taking limit as $n \to \infty$ shows that

$$\left|\sum_{k=0}^{\infty} a_k \overline{b_k}\right| \leqslant \|f\|_2 \|g\|_2 < \infty.$$

Therefore, $\sum_{k=0}^{\infty} a_k \overline{b_k}$ converges, and hence the inner product is well-defined. One can verify that $\langle f, g\rangle$ defined by (3.3.4) satisfies the properties of an inner product, so that the space $\mathcal{H}^2$ becomes a Hilbert space with this inner product.

Example 3.3.2 The space $\mathcal{H}^2$ contains all polynomials. For $\alpha > 1/2$, the functions $\sum_{k=1}^{\infty} z^k/k^\alpha$ is in the class $\mathcal{H}^2$ because the sum

$$\sum_{k=1}^{\infty} \left(\frac{1}{k^\alpha}\right)^2 = \sum_{k=1}^{\infty} \frac{1}{k^{2\alpha}} < \infty.$$

Example 3.3.3 For $\alpha \in (3/4, 1)$, the *generalized Koebe function* $\kappa_\alpha : \mathbb{D} \to \mathbb{C}$ given by

$$\kappa_\alpha(z) = \frac{z}{(1-z)^{2(1-\alpha)}}$$

belongs to the class $\mathcal{H}^2$. For a proof, see [6, Example 1.1.14, p. 8].

Example 3.3.4 Every function in $\mathcal{H}^\infty$ is in the class $\mathcal{H}^2$. To see this, note that $|f(z)| \leqslant \|f\|_\infty$ for a function $f \in \mathcal{H}^\infty$ and so

$$\|f\|_2^2 = \sup_{0<r<1} \frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})|^2 d\theta \leqslant \|f\|_\infty^2 < \infty.$$

Hence, the function $f \in \mathcal{H}^2$.

In particular, a function $f \in \mathcal{H}$ satisfying $|f(z)| \leqslant 1$ belongs to the Hardy-Hilbert space $\mathcal{H}^2$. Thus, $\mathcal{B} \subset \mathcal{H}^2$.

For $f \in \mathcal{H}$, let $M_f : \mathcal{H} \to \mathcal{H}$ be the *multiplication operator* defined by $M_f(g) = fg$. The following theorem gives a characterization of a function f in the class $\mathcal{B}$ in terms of the multiplication operator M_f.

Theorem 3.3.5 *A function $f \in \mathcal{B}$ if and only if $M_f\left(\mathcal{H}^2\right) \subset \mathcal{H}^2$ and $\left\|M_f\right\|_2 \leqslant 1$.*

Proof Let $f \in \mathcal{B}$. For any $g \in \mathcal{H}^2$,

$$\|M_f(g)\|_2^2 = \frac{1}{2\pi} \lim_{r\uparrow 1} \int_0^{2\pi} |f(re^{i\theta})g(re^{i\theta})|^2 d\theta \leqslant \|f\|_\infty^2 \|g\|_2^2 < \infty.$$

Therefore,

$$\|M_f(g)\|_2 \leqslant \|f\|_\infty \|g\|_2.$$

Since $f \in \mathcal{B}$, it follows that $\|f\|_\infty \leqslant 1$, and hence $\|M_f(g)\|_2 \leqslant \|g\|_2$. This shows that $M_f g \in \mathcal{H}^2$ whenever $g \in \mathcal{H}^2$. Moreover, the operator norm of M_f satisfies

$$\|M_f\|_2 = \sup\{\|M_f g\|_2 : \|g\|_2 = 1\} \leqslant 1.$$

For the converse, we first note that $1 \in \mathcal{H}^2$. Since $M_f\left(\mathcal{H}^2\right) \subset \mathcal{H}^2$, it follows that $f = M_f(1) \in \mathcal{H}^2$, and hence $f^2 = M_f(f) \in \mathcal{H}^2$. By induction, $f^n = M_f(f^{n-1}) \in \mathcal{H}^2$ for all n, and hence $\|f^n\|_2 \leqslant 1$ for all n. Therefore,

$$\|f\|_{2n}^{2n} = \lim_{r\uparrow 1}\left(\frac{1}{2\pi}\int_0^{2\pi}\left|f\left(re^{i\theta}\right)\right|^{2n}\mathrm{d}\theta\right) = \|f^n\|_2^2 \leqslant 1,$$

implying $\|f\|_{2n} \leqslant 1$. Taking the limit as $n \to \infty$ yields $\|f\|_\infty \leqslant 1$. Thus, $|f(z)| \leqslant 1$ for all z, and hence $f \in \mathcal{B}$. □

3.4 Coefficient Inequalities

Let $w(z) = \sum_{k=0}^{\infty} w_k z^k \in \mathcal{B}$. By Cauchy's integral formula, it follows that

$$w_n = \frac{1}{2\pi i}\oint_{|z|=r}\frac{w(z)}{z^{n+1}}\mathrm{d}z, \quad 0 < r < 1,$$

and so

$$|w_n| \leqslant \frac{1}{2\pi}\oint_{|z|=r}\frac{|w(z)|}{|z|^{n+1}}|\mathrm{d}z| \leqslant \frac{1}{r^n}.$$

By letting $r \to 1^-$ gives $|w_n| \leqslant 1$. This is clearly sharp for each n for the function $w(z) = z^n$. Since $\mathcal{B} \subset \mathcal{H}^2$, Lemma 3.3.1 gives

$$\sum_{k=0}^{\infty}|w_k|^2 = \|w\|_2^2 \leqslant 1.$$

In particular, the following *Gutzmer's inequality* holds:

$$\sum_{k=0}^{n}|w_k|^2 \leqslant 1, \quad n \geqslant 0.$$

This implies that $|w_n|^2 \leqslant 1 - |w_0|^2$ for $n \geqslant 1$. However, an even sharper result can be established.

Theorem 3.4.1 *If* $w(z) = \sum_{k=0}^{\infty} w_k z^k \in \mathcal{B}$, *then*

$$|w_n| \leqslant 1 - |w_0|^2, \quad n \geqslant 1.$$

Proof Case (i). If $n = 1$, let $a = 0$ in (3.1.3) to obtain $|w_1| \leqslant 1 - |w_0|^2$.

Case (ii). For $n > 1$, consider

$$w_n(z) = \frac{1}{n} \sum_{j=0}^{n-1} w\left(\eta^j z\right),$$

where $\eta = \mathrm{e}^{2\pi \mathrm{i}/n}$. Since

$$1 + \eta^k + \eta^{2k} + \cdots + \eta^{(n-1)k} = \begin{cases} 0, & \text{if } k \not\equiv 0 \bmod n, \\ n, & \text{if } k \equiv 0 \bmod n, \end{cases}$$

it follows that

$$\begin{aligned} w_n(z) &= \frac{1}{n} \sum_{j=0}^{n-1} \sum_{k=0}^{\infty} w_k \left(\eta^j z\right)^k \\ &= \frac{1}{n} \sum_{k=0}^{\infty} \left(\sum_{j=0}^{n-1} \eta^{jk} \right) w_k z^k \\ &= w_0 + w_n z^n + w_{2n} z^{2n} + \cdots . \end{aligned}$$

Now $|w(z)| \leqslant 1$ implies that $|w(\eta^j z)| \leqslant 1$ and $|w_n(z)| \leqslant 1$. Thus

$$\widetilde{w}(z) := w_0 + w_n z + w_{2n} z^2 + \cdots$$

satisfies $|\widetilde{w}(z^n)| \leqslant 1$ and therefore $|\widetilde{w}(z)| \leqslant 1$. Therefore, by case (i),

$$|w_n| \leqslant 1 - |w_0|^2.$$

□

Corollary 3.4.2 (Bohr's inequality) *If* $w(z) = \sum_{n=0}^{\infty} w_n z^n \in \mathcal{B}$, *then*

$$\sum_{n=0}^{\infty} |w_n z^n| \leq 1$$

for all $z \in \mathbb{D}_{1/3}$. *The constant* $1/3$ *is sharp.*

Proof Since $|w(z)| < 1$, it follows by Theorem 3.4.1 that $|w_n| \leqslant 1 - |w_0|^2$ and hence, for $|z| = r \leq 1/3$,

$$\begin{aligned}\sum_{n=0}^{\infty}|w_n z^n| &\leqslant |w_0| + (1-|w_0|^2)\sum_{n=1}^{\infty} r^n \\ &= |w_0| + (1-|w_0|^2)\frac{r}{1-r} \\ &\leqslant |w_0| + (1-|w_0|^2)\frac{1}{2} \\ &= 1 - \frac{1}{2}(1-|w_0|)^2 \\ &\leqslant 1.\end{aligned}$$

For the sharpness, consider the function

$$\widetilde{w}(z) = \frac{a-z}{1-az} = \sum_{n=0}^{\infty} w_n z^n, \quad 0 < a < 1$$

where $w_0 = a$ and $w_n = (a^2-1)a^{n-1}$ for $n \geqslant 1$. Clearly, $|\widetilde{w}(z)| < 1$ and, for $|z| = r = 1/3$,

$$\sum_{n=0}^{\infty}|w_n z^n| = a + (1-a^2)\frac{1}{3-a} \to 1$$

as $a \to 1$. □

For a generalization of Corollary 3.4.2, see Corollary 4.5.2.

Corollary 3.4.3 *If* $w(z) = \sum_{k=1}^{\infty} w_k z^k \in \mathcal{B}_0$, *then*

$$|w_n| \leqslant 1 - |w_1|^2, \quad n \geqslant 2.$$

Proof The function $\widetilde{w}$ defined by $\widetilde{w}(z) = w(z)/z$ for $z \neq 0$ and $\widetilde{w}(0) = w'(0)$ is in $\mathcal{B}$. Then by Theorem 3.4.1, the result follows. □

Lemma 3.4.4 *If* $w(z) = \sum_{k=1}^{\infty} w_k z^k \in \mathcal{B}_0$, *then the coefficients* w_1, w_2 *satisfy*

$$\left|w_2 - s w_1^2\right| \leqslant \max\{1, |s|\}, \quad s \in \mathbb{C}, \tag{3.4.1}$$

$$\left|w_2 - (s-1)w_1^2\right| + s|w_1|^2 \leqslant 1, \quad 0 \leqslant s \leqslant 1, \tag{3.4.2}$$

$$\left|w_2 - (s-1)w_1^2\right| + (2-s)|w_1|^2 \leqslant 1, \quad 1 \leqslant s \leqslant 2. \tag{3.4.3}$$

Equality holds in (3.4.1) *for the function* $w(z) = z$ *when* $|s| \geqslant 1$ *and for* $w(z) = z^2$ *for* $|s| < 1$. *Equality holds in* (3.4.2) *and* (3.4.3) *for* $w(z) = z^2$.

Proof It follows by Corollary 3.4.3 that $|w_2| \leqslant 1 - |w_1|^2$. Therefore,

$$\left|w_2 - s w_1^2\right| \leqslant |w_2| + |s|\,|w_1|^2 \leqslant 1 + (|s|-1)\,|w_1|^2 \leqslant \max\{1, |s|\}.$$

Moreover, the functions $w(z) = z$ and $w(z) = z^2$, respectively, show that the result is sharp for $|s| \geqslant 1$ and for $|s| < 1$.

For $0 \leqslant s \leqslant 1$, it follows from $|w_2| \leqslant 1 - |w_1|^2$ that

$$\begin{aligned}\left|w_2 - (s-1)w_1^2\right| + s|w_1|^2 &\leqslant |w_2| + (1-s)|w_1|^2 + s|w_1|^2 \\ &= |w_2| + |w_1|^2 \leqslant 1.\end{aligned}$$

For $1 \leqslant s \leqslant 2$, it follows that

$$\begin{aligned}\left|w_2 - (s-1)w_1^2\right| + (2-s)|w_1|^2 &\leqslant |w_2| + (s-1)|w_1|^2 + (2-s)|w_1|^2 \\ &= |w_2| + |w_1|^2 \leqslant 1.\end{aligned}$$

Equality holds in (3.4.2) and (3.4.3) for $w(z) = z^2$. □

The next theorem gives a characterization of bounded analytic functions which will be useful in getting coefficient bounds.

Theorem 3.4.5 (Schur Theorem) *Let $w(z) = \sum_{k=0}^{\infty} w_k z^k$ be analytic in $\mathbb{D}$. The function $w \in \mathcal{B}$ if and only if*

$$\sum_{k=0}^{n} \left| \sum_{j=k}^{n} w_{j-k} \lambda_j \right|^2 \leqslant \sum_{k=0}^{n} |\lambda_k|^2 \tag{3.4.4}$$

for all $n \in \mathbb{N}$ and for all $\lambda_0, \ldots, \lambda_n \in \mathbb{C}$.

Proof Let $w \in \mathcal{B}$. For every sequence $\mu_0, \mu_1, \ldots \in \mathbb{C}$ with $\mu_{n+1} = \mu_{n+2} = \cdots = 0$, the polynomial g defined by $g(z) = \sum_{k=0}^{n} \mu_k z^k$ belongs to $\mathcal{H}^2$ and

$$\|g\|_2^2 = \sum_{k=0}^{n} |\mu_k|^2 .$$

Since

$$M_w(g)(z) = \sum_{m=0}^{\infty} \left(\sum_{l=0}^{m} w_{m-l} \mu_l \right) z^m ,$$

Theorem 3.3.5 shows that

$$\sum_{m=0}^{n} \left| \sum_{l=0}^{m} w_{m-l} \mu_l \right|^2 \leqslant \sum_{m=0}^{\infty} \left| \sum_{l=0}^{m} w_{m-l} \mu_l \right|^2 \leqslant \sum_{k=0}^{n} |\mu_k|^2 .$$

For $k = 0, \ldots, n$, take $\mu_k = \lambda_{n-k}$ to obtain

$$\sum_{m=0}^{n} \left| \sum_{l=0}^{m} w_{m-l} \lambda_{n-l} \right|^2 \leqslant \sum_{k=0}^{n} |\lambda_k|^2 .$$

The change of variables $m = n - k, l = n - j$ leads to

$$\sum_{k=0}^{n} \left| \sum_{j=k}^{n} w_{j-k} \lambda_j \right|^2 \leqslant \sum_{k=0}^{n} |\lambda_k|^2 .$$

To prove the converse, let $z \in \mathbb{D}$. For $\lambda_j = z^j$,

$$\sum_{k=0}^{n} \left| \sum_{j=k}^{n} w_{j-k} \lambda_j \right|^2 = \sum_{k=0}^{n} |z|^{2k} \left| \sum_{j=0}^{n-k} w_j z^j \right|^2 ,$$

and so (3.4.4) reduces to

$$\sum_{k=0}^{n} |z|^{2k} \left| \sum_{j=0}^{n-k} w_j z^j \right|^2 \leqslant \sum_{k=0}^{n} |z|^{2k} .$$

Letting $n \to \infty$ yields

$$\frac{1}{1-|z|^2} |w(z)|^2 \leqslant \frac{1}{1-|z|^2}.$$

This shows that $|w(z)| \leqslant 1$. □

When $\lambda_0 = \cdots = \lambda_{n-1} = 0$ and $\lambda_n = 1$, Theorem 3.4.5 gives

$$\sum_{k=0}^{n} |w_{n-k}|^2 \leqslant 1, \quad n \geqslant 0,$$

which is the Gutzmer's inequality

$$\sum_{k=0}^{n} |w_k|^2 \leqslant 1, \quad n \geqslant 0.$$

Theorem 3.4.6 *If* $w(z) = \sum_{k=0}^{\infty} w_k z^k \in \mathcal{B}$, *then*

$$|w_n|^2 \leqslant \left(1 - |w_0|^2\right) \left(1 - \sum_{k=0}^{n-1} |w_k|^2\right), \quad n \geqslant 1. \tag{3.4.5}$$

Proof The inequality (3.4.5) trivially holds in the case when $|w_0| = 1$; indeed in this case, $w_j = 0$ for all $j \geqslant 1$. Otherwise, by applying Theorem 3.4.5 with $\lambda_0 = x$, $\lambda_i = 0$ for $i = 1, 2, \ldots, n-1$ and $\lambda_n = \mathrm{e}^{\mathrm{i}\theta}$, where $x \geqslant 0$ and $\theta \in \mathbb{R}$, it follows that

$$\left|x w_0 + w_n \mathrm{e}^{\mathrm{i}\theta}\right|^2 + \sum_{k=0}^{n-1} \left|w_k \mathrm{e}^{\mathrm{i}\theta}\right|^2 \leqslant x^2 + 1.$$

Choosing θ so that $\left|x w_0 + w_n \mathrm{e}^{\mathrm{i}\theta}\right| = |x w_0| + \left|w_n \mathrm{e}^{\mathrm{i}\theta}\right|$, the above inequality takes the form

$$\left(x\,|w_0| + \left|w_n \mathrm{e}^{\mathrm{i}\theta}\right|\right)^2 + \sum_{k=0}^{n-1} \left|w_k \mathrm{e}^{\mathrm{i}\theta}\right|^2 \leqslant x^2 + 1$$

which can be rewritten as

$$a x^2 - 2bx + c \geqslant 0$$

where $a = 1 - |w_0|^2 > 0$, $b = |w_0|\,|w_n| \geqslant 0$ and $c = 1 - \sum_{k=0}^{n} |w_k|^2$. Let $x = b/a \geqslant 0$ For this choice of x, the inequality reduces to $b^2 \leqslant ac$. Thus, $|w_0| < 1$, and

$$\begin{aligned} |w_0|^2\,|w_n|^2 &\leqslant \left(1 - |w_0|^2\right)\left(1 - \sum_{k=0}^{n} |w_k|^2\right) \\ &= \left(1 - |w_0|^2\right)\left(1 - \sum_{k=0}^{n-1} |w_k|^2\right) - \left(1 - |w_0|^2\right)|w_n|^2 \end{aligned}$$

and this gives the desired result. □

Evidently, $1 - \sum_{k=0}^{n-1} |w_k|^2 \leqslant 1 - |w_0|^2$, and so (3.4.5) yields

$$|w_n| \leqslant 1 - |w_0|^2, \quad n \geqslant 1,$$

for each function $w \in \mathcal{B}$.

Corollary 3.4.7 *If $w(z) = \sum_{k=0}^{\infty} w_k z^k \in \mathcal{B}$, then*

$$\sum_{k=1}^{n} |w_k|^2 \leqslant \left(1 - |w_0|^2\right)\left(1 - |w_0|^{2n}\right)$$

for all $n \geqslant 1$. In particular,

$$|w_n|^2 \leqslant \left(1 - |w_0|^2\right)\left(1 - |w_0|^{2n}\right).$$

Proof The second inequality trivially follows from the first. The first inequality can be established by induction from (3.4.5), which may be written as

$$\sum_{k=1}^{n} |w_k|^2 \leqslant (1-|w_0|^2)^2 + |w_0|^2 \sum_{k=1}^{n-1} |w_k|^2 \,. \tag{3.4.6}$$

Assuming the inequality for $n = m-1$, (3.4.6) yields

$$\begin{aligned}\sum_{k=1}^{m} |w_k|^2 &\leqslant (1-|w_0|^2)^2 + |w_0|^2(1-|w_0|^2)(1-|w_0|^{2m-2}) \\ &= (1-|w_0|^2)(1-|w_0|^{2m}).\end{aligned}$$

This proves the result for $n = m$. □

Corollary 3.4.8 *If* $w(z) = \sum_{k=1}^{\infty} w_k z^k \in \mathcal{B}_0$, *then*

$$\sum_{k=2}^{n+1} |w_k|^2 \leqslant \left(1-|w_1|^2\right)\left(1-|w_1|^{2n}\right)$$

for all $n \geqslant 1$, *and, in particular,*

$$|w_{n+1}|^2 \leqslant \left(1-|w_1|^2\right)\left(1-|w_1|^{2n}\right).$$

Proof The result follows by applying Corollary 3.4.7 to the function $\widetilde{w}$ defined by $\widetilde{w}(z) = w(z)/z$ for $z \neq 0$ and $\widetilde{w}(0) = w'(0)$. □

Exercises

1. If $w \in \mathcal{B}_0$, show that
$$p(z) = \frac{1+w(z)}{1-w(z)}$$
has positive real part in $\mathbb{D}$ and conversely. Use this to prove that the function p satisfy
$$\frac{1-|z|}{1+|z|} \leqslant |p(z)| \leqslant \frac{1+|z|}{1-|z|}.$$
2. Let w be a function in $\mathcal{B}_0$. Let $w_0 \in \mathcal{B}_0$ be given by $w_0(z) = z$. Show that $w(\overline{\mathbb{D}}_r) \subset w_0(\overline{\mathbb{D}}_r)$ for each $0 < r < 1$.
3. Let $0 < s < 2$. Assume that $w(z) = \sum_{k=0}^{\infty} w_k z^k \in \mathcal{B}$ satisfies the inequalities $|f(z)+1| \geqslant s$ in $\mathbb{D}$. Show that $\sum_{k=0}^{\infty} |w_k|^2 \leqslant 1 - s^2\,\mathrm{Re}((1-w_0)/(1+w_0))$ and the inequality is sharp for every admissible choice of s and $w_0 = w(0)$.
4. If $w \in \mathcal{B}$, show that, for each $n \geqslant 1$,
$$|w^{(n)}(z)| \leqslant n! \frac{1-|w(z)|^2}{(1-|z|)^n(1+|z|)}, \quad z \in \mathbb{D}.$$
For $n > 1$, $z \neq 0$, equality holds for $w = \varepsilon$, $|\varepsilon| = 1$.

5. If $w \in \mathcal{B}$, show that, for every $z \in \mathbb{D}$, there exist functions $w_j \in \mathcal{B}$, $j \in \mathbf{N}$, such that

$$\lim_{j\to\infty} \frac{|w_j^{(n)}(z)|}{1-|w_j(z)|^2} = \frac{n!}{(1-|z|)^n(1+|z|)}.$$

6. A finite Blaschke product is a function of the form

$$e^{i\varphi} \prod_{k=1}^{n} \frac{z-z_k}{1-\overline{z}_k z}$$

where $|z_k| < 1$. For a function $w \in \mathcal{B}$, show that there exists a sequence $\langle w_k \rangle$ of finite Blaschke products that converges to w.

7. If $w \in \mathcal{B}$ has two distinct fixed points, then $w(z) = z$.
8. For $\alpha \in (3/4, 1)$, prove that the generalized Koebe function

$$\kappa_\alpha(z) = \frac{z}{(1-z)^{2(1-\alpha)}}, \quad z \in \mathbb{D},$$

is in the class $\mathcal{H}^2$.

9. For each real number t, show that the function $w : \mathbb{D} \to \mathbb{C}$ defined by

$$w(z) = \left(\frac{1+z}{1-z}\right)^{it}$$

is in the class $\mathcal{H}^\infty$.

10. Show that the function $w : \mathbb{D} \to \mathbb{C}$ defined by

$$w(z) = \left(\frac{1}{1-z}\right)^{1/2}$$

is not in $\mathcal{H}^2$.

11. If $w(z) = \sum_{k=1}^{\infty} w_k z^k \in \mathcal{B}_0$, show that

$$\left|\sum_{k=1}^{n-1} (n-k) w_k\right| \leqslant n-1$$

for each fixed $n \in \mathbb{N}$.

12. If $w(z) = \sum_{k=1}^{\infty} w_k z^k \in \mathcal{B}_0$, show that

$$\left|\sum_{k=1}^{n-1} (n-k)\lambda_{n-k} w_k\right| \leqslant (n-1)\lambda_{n-1}$$

for each fixed $n \in \mathbb{N}$ and any numbers $\lambda_k \in \mathbb{R}$ with $\lambda_2 \geqslant \lambda_1 \geqslant 0$ and $k\lambda_k + (k-2)\lambda_{k-2} \geqslant 2(k-1)\lambda_{k-1}$ for $3 \leqslant k \leqslant n-1$.

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[1] The idea of using $\varphi_a(z) = (a - z)/(1 - \overline{a}z)$ instead of $(z - a)/(1 - \overline{a}z)$ is in [1]. For Schwarz's lemma and its extensions to various other spaces, see Dineen [2]. Most of the material on the sharpened forms of the Schwarz's lemma was derived from Mercer [7]. For an elementary introduction to Hardy-Hilbert space, see Martínez-Avendaño and Rosenthal [6]. For general H^p spaces, see Duren [3] or Koosis [5]. Theorem 3.3.5 and the proof of Schur's theorem (Theorem 3.4.5) are contained in Kortram [4]. Theorem 3.4.6 and its corollaries as well as two of the exercises are drawn from Hotta and Michalski [8]. Some exercises are selected from Ruscheweyh [9].

Chapter 4
Functions with Positive Real Part



Abstract This chapter studies the class $\mathcal{P}$ of analytic functions $p : \mathbb{D} \to \mathbb{C}$ satisfying $p(0) = 1$ and $\mathrm{Re}\, p(z) > 0$ for all $z \in \mathbb{D}$. A function in $\mathcal{P}$ is referred to as a function with positive real part or a Carathéodory function. Several important classes of univalent functions are closely related to this class. For instance, by the Noshiro-Warschawski theorem, a function $f \in \mathcal{A}$ is univalent if its derivative $f' \in \mathcal{P}$. The class $\mathcal{P}$ is also related to $\mathcal{B}_0$ (see Theorem 4.4.5). Moreover, the classes of starlike and convex functions, as well as several others to be discussed later, are linked to $\mathcal{P}$. Distortion, growth and coefficient estimates for functions in $\mathcal{P}$ often lead to the corresponding estimates for these related classes. In view of these connections, we examine various properties of the functions with positive real part, including growth estimates and coefficient inequalities.

4.1 Carathéodory Functions

This chapter focuses on a fundamental class of analytic functions known as the Carathéodory functions.

Definition 4.1.1 Let $p : \mathbb{D} \to \mathbb{C}$ be analytic with $p(0) = 1$. The function p is said to have positive real part if $\mathrm{Re}\, p(z) > 0$ for all $z \in \mathbb{D}$. The class of such functions is denoted by $\mathcal{P}$, and is known as the class of *Carathéodory functions*, or the class of *functions with positive real part*.

The classes of starlike, convex, and close-to-convex functions to be introduced later are all closely connected to the class $\mathcal{P}$. We shall show that for a function $f \in \mathcal{A}$, the image $f(\mathbb{D})$ is starlike with respect to the origin if and only if $zf'(z)/f(z) \in \mathcal{P}$ and f is convex if and only if $1 + zf''(z)/f'(z) \in \mathcal{P}$. Thus, properties of starlike and convex functions can be deduced from the corresponding properties of functions in the class $\mathcal{P}$. This approach applies more generally to any class of functions that can be analytically associated with $\mathcal{P}$.

For example, if f belongs to the class $\mathcal{R} = \{f \in \mathcal{A} : \mathrm{Re}\, f'(z) > 0\}$, then its derivative $f' \in \mathcal{P}$. Suppose $f(z) = z + \sum_{n=2}^{\infty} a_n z^n \in \mathcal{R}$ and $p(z) = 1 + \sum_{n=1}^{\infty} p_n z^n \in \mathcal{P}$.

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Then the coefficients are related by $na_n = p_{n-1}$. Hence, once bounds for the coefficients of Carathéodory functions are known, corresponding bounds for the estimate on $|a_n|$ can be readily obtained.

By the following theorem, the class $\mathcal{P}$ is non-empty.

Theorem 4.1.2 *If $p : \mathbb{D} \to \mathbb{C}$ is analytic, $p(z) = 1 + \sum_{k=1}^{\infty} c_k z^k$ and $\sum_{k=1}^{\infty} |c_k| \leqslant 1$, then $p \in \mathcal{P}$. The condition is necessary if the coefficients c_k are negative.*

Proof If $\sum_{k=1}^{\infty} |c_k| \leqslant 1$, then

$$|\mathrm{Re}\, p(z) - 1| \leqslant |p(z) - 1| \leqslant \sum_{k=1}^{\infty} |c_k|\, |z|^k \leqslant \sum_{k=1}^{\infty} |c_k| \leqslant 1,$$

and therefore $0 \leqslant \mathrm{Re}\, p(z) \leqslant 2$. By Theorem 1.2.12, it follows that $\mathrm{Re}\, p(z) > 0$ and hence $p \in \mathcal{P}$.

If the coefficients c_k are negative, by letting $z \to 1^-$ in

$$\mathrm{Re}\left(1 + \sum_{k=1}^{\infty} c_k z^k\right) > 0$$

yields

$$1 + \sum_{k=1}^{\infty} c_k \geqslant 0.$$

Hence,

$$\sum_{k=1}^{\infty} |c_k| = -\sum_{k=1}^{\infty} c_k \leqslant 1,$$

which establishes the necessity condition. □

Example 4.1.3 The function $p(z) = 1 + cz^n$, defined in $\mathbb{D}$, satisfies

$$\mathrm{Re}\, p(z) \geqslant 1 - |c|\, |z|^n > 1 - |c|$$

and therefore, $p \in \mathcal{P}$ if $|c| \leqslant 1$. The function

$$p(z) = 1 + \frac{1}{n} z^n + \frac{1}{m} z^m$$

belongs to $\mathcal{P}$ when the positive integers n and m satisfy

$$\frac{1}{n} + \frac{1}{m} \leqslant 1.$$

Similarly, Theorem 4.1.2 shows that the function

$$p(z) = \frac{2}{2-z} = 1 + \sum_{k=1}^{\infty} \frac{z^k}{2^k}, \quad z \in \mathbb{D},$$

belongs to $\mathcal{P}$ since $\sum_{k=1}^{\infty} 2^{-k} = 1$.

The image of the function p is given by (see Fig. 4.1)

$$\left| p(z) - \frac{4}{3} \right| = \frac{2}{3} \left| \frac{2z-1}{2-z} \right| < \frac{2}{3}.$$

A function in the class $\mathcal{P}$ need not be univalent as the following example shows.

Example 4.1.4 By Example 4.1.3, the function $p(z) = 1 + z^n$ belongs to $\mathcal{P}$. However, if $n \geqslant 2$, then $p'(z) = nz^{n-1} = 0$ at $z = 0$ and hence the function p is not univalent. Thus, a function from the class $\mathcal{P}$ need not be univalent.

Example 4.1.5 The function

$$p(z) = \frac{1+\gamma z}{1-\gamma z}, \quad |\gamma| \leqslant 1, z \in \mathbb{D},$$

satisfies

$$\operatorname{Re} p(z) = \operatorname{Re} \frac{(1+\gamma z)(1-\overline{\gamma z})}{|1-\gamma z|^2} = \frac{1-|\gamma z|^2}{|1-\gamma z|^2} = \frac{1-|z|^2}{|1-\gamma z|^2} > 0.$$

This shows that $p \in \mathcal{P}$.

The class $\mathcal{H}(\mathbb{D})$ is a vector space with the usual addition of functions and scalar multiplication. A subset $D \subset \mathcal{H}(\mathbb{D})$ is convex if $\alpha_1 p_1 + \alpha_2 p_2 \in D$ whenever $p_1, p_2 \in D, 0 \leqslant \alpha_1, \alpha_2 \leqslant 1$ and $\alpha_1 + \alpha_2 = 1$. We show that the class $\mathcal{P}$ is convex.

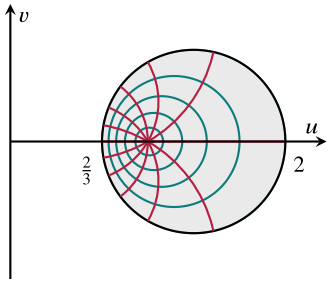


Fig. 4.1 $p(\mathbb{D}) = \mathbb{D}(4/3; 2/3)$ when $p(z) = 2/(2-z)$

Theorem 4.1.6 *The class $\mathcal{P}$ is convex. More generally, if $0 \leqslant \alpha_k \leqslant 1$, $\sum_{k=1}^{n} \alpha_k = 1$ and $p_k \in \mathcal{P}$ for $k = 1, 2, 3, \ldots, n$, then the function $p(z) = \sum_{k=1}^{n} \alpha_k p_k(z)$ belongs to $\mathcal{P}$. In particular, for these α_k's and γ_k's with $|\gamma_k| \leqslant 1$, the function*

$$p(z) = \sum_{k=1}^{n} \alpha_k \frac{1 + \gamma_k z}{1 - \gamma_k z}, \quad z \in \mathbb{D},$$

belongs to $\mathcal{P}$.

Proof Clearly, $p(0) = 1$. Also,

$$\operatorname{Re} p(z) = \sum_{k=1}^{n} \alpha_k \operatorname{Re} p_k(z) > 0.$$

Thus, $p \in \mathcal{P}$. By Example 4.1.5,

$$p_k(z) = \frac{1 + \gamma_k z}{1 - \gamma_k z}$$

belongs to $\mathcal{P}$, and hence

$$p(z) = \sum_{k=1}^{n} \alpha_k p_k(z)$$

belongs to $\mathcal{P}$. □

Example 4.1.7 By Example 4.1.5, the functions

$$p_1(z) = \frac{1 + \gamma z}{1 - \gamma z} \quad \text{and} \quad p_2(z) = \frac{1 + \overline{\gamma} z}{1 - \overline{\gamma} z}$$

are in $\mathcal{P}$ for each γ with $|\gamma| = 1$. Hence, by Theorem 4.1.6,

$$p(z) = \frac{1}{2} \frac{1 + \gamma z}{1 - \gamma z} + \frac{1}{2} \frac{1 + \overline{\gamma} z}{1 - \overline{\gamma} z} = \frac{1 - z^2}{1 - (2 \cos \theta) z + z^2}, \quad \theta = \arg \gamma,$$

belongs to $\mathcal{P}$. The function p can be written as

$$p(z) = 1 + \sum_{n=1}^{\infty} (2 \cos n\theta) z^n.$$

The following theorem gives a number of transformations under which the class $\mathcal{P}$ is preserved.

Theorem 4.1.8 *If $h, h_1, h_2 \in \mathcal{P}$, then each of the function $H : \mathbb{D} \to \mathbb{C}$ defined below belongs to $\mathcal{P}$:*

(i) $H(z) = h(e^{it}z), \quad t \in \mathbb{R}$,
(ii) $H(z) = h(tz), \quad t \in [-1, 1)$,
(iii) $H(z) = h(z^n), \quad n = 1, 2, \ldots$,
(iv) $H(z) = (1/a)\left(h\left((z+\lambda)/(1+\bar{\lambda}z)\right) - ib\right), \quad h(\lambda) = a + ib, \ \lambda \in \mathbb{D}$,
(v) $H(z) = h(z)^t, \quad t \in [-1, 1]$,
(vi) $H(z) = 1/h(z)$,
(vii) $H(z) = (h(z) + ib)/(1 + ibh(z)), \quad b \in \mathbb{R}$,
(viii) $H(z) = (h_1(z))^{t_1}(h_2(z))^{t_2}, \quad t_1, \ t_2, \ t_1 + t_2 \in [0, 1]$.

Proof Let $p \in \mathcal{P}$. The results follow by observing that $p \circ g$ has positive real part when g maps $\mathbb{D}$ into itself, and similarly, $g \circ p$ has positive real part when g maps the right half-plane $H^+ = \{z \in \mathbb{C} : \mathrm{Re} z > 0\}$ into itself. The mappings $z \to \mathrm{e}^{\mathrm{i}t}z$, $z \to tz$, $z \to z^n$ and $z \to (z+\lambda)/(1+\bar{\lambda}z)$ all map $\mathbb{D}$ into itself. This establishes parts (i)–(iv).

Note that $\mathrm{Re} w > 0$ if and only if $|\arg(w)| < \pi/2$. The mappings $z \to z^t$ with $t \in [-1, 1]$ and $z \to 1/z$ map the right half-plane H^+ into itself, since

$$|\arg(z^t)| = |t|\,|\arg(z)| < \frac{\pi}{2}$$

and

$$|\arg(1/z)| = |-\arg(z)| = |\arg(z)| < \frac{\pi}{2}.$$

The mapping $z \to (z + \mathrm{i}b)/(1 + \mathrm{i}bz)$ also maps H^+ into itself. This can be seen from

$$\mathrm{Re}\frac{z + \mathrm{i}b}{1 + \mathrm{i}bz} = \frac{(1 + b^2)\mathrm{Re} z}{|1 + \mathrm{i}bz|^2}.$$

This completes the proof of parts (v)–(vii).

For $t_1, \ t_2, \ t_1 + t_2 \in [0, 1]$, the function H given by $H(z) = (h_1(z))^{t_1}(h_2(z))^{t_2}$ satisfies

$$|\arg H(z)| \leqslant t_1|\arg h_1(z)| + t_2|\arg h_2(z)| < (t_1 + t_2)\frac{\pi}{2} < \frac{\pi}{2}.$$

Hence, $\mathrm{Re} H(z) > 0$, which shows that $H \in \mathcal{P}$. □

There are several other transformations under which $\mathcal{P}$ is preserved, see [13].

Example 4.1.9 As shown in Example 4.1.3, the function $p(z) = 2/(2 - z) = 1/(1 - z/2)$ belongs to $\mathcal{P}$. More generally, the function $p(z) = 1/(1 + cz)$ belongs to $\mathcal{P}$ for $|c| \leqslant 1$.

Indeed, by Example 4.1.3, the function $q(z) = 1 + cz$ belongs to $\mathcal{P}$ whenever $|c| \leqslant 1$. By Theorem 4.1.8 (vi), the function $p(z) = 1/q(z) = 1/(1 + cz)$ belongs to $\mathcal{P}$. In particular, for $c = -1/2$, the function $p(z) = 1/(1 - z/2) = 2/(2 - z)$ belongs to $\mathcal{P}$.

4.2 Herglotz's Representation Theorem

Let G be an open subset of $\mathbb{C}$. A function $u : G \to \mathbb{R}$ is called *harmonic* if it has continuous second-order partial derivatives and satisfies Laplace's equation

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

for all $z = x + \mathrm{i}y \in G$. A function $f : G \to \mathbb{C}$ is analytic if and only if its real and imaginary parts $u := \mathrm{Re}\, f$ and $v := \mathrm{Im} f$ are harmonic functions that satisfy the Cauchy-Riemann equations. As a consequence, every harmonic function is infinitely differentiable.

Let $u : G \to \mathbb{R}$ be a harmonic function, and let $\overline{\mathbb{D}}(a; r)$ be a closed disk contained in G. By applying the Cauchy integral formula to an analytic function f whose real part is u, and using the parametric representation of circle, one obtains the mean value formula:

$$u(a) = \frac{1}{2\pi} \int_0^{2\pi} u(a + r\mathrm{e}^{\mathrm{i}t})\mathrm{d}t. \tag{4.2.1}$$

A continuous function $u : G \to \mathbb{R}$ is said to satisfy the *Mean Value Property* (MVP), if Eq. (4.2.1) holds whenever $\overline{\mathbb{D}}(a; r) \subset G$. For such a function, if there exists a point $a \in G$ such that $u(z) \leqslant u(a)$ for all $z \in G$, then the function u is a constant. This is a version of the maximum principle for harmonic functions.

The function

$$P_r(\theta) = \sum_{n=-\infty}^{\infty} r^{|n|}\mathrm{e}^{\mathrm{i}n\theta} = \frac{1 - r^2}{1 + r^2 - 2r\cos\theta} = \mathrm{Re}\left(\frac{1 + r\mathrm{e}^{\mathrm{i}\theta}}{1 - r\mathrm{e}^{\mathrm{i}\theta}}\right),$$

defined for $0 \leqslant r < 1$, $-\infty < \theta < \infty$, is called the *Poisson kernel*. Every harmonic function on the closed unit disk $\overline{\mathbb{D}}$ admits an integral representation in terms of Poisson kernel.

Theorem 4.2.1 *If $u : \overline{\mathbb{D}} \to \mathbb{R}$ is a continuous harmonic function in $\mathbb{D}$, then*

$$u(re^{i\theta}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P_r(\theta - t)u(e^{it})dt$$

for $0 \leqslant r < 1$ and $\theta \in \mathbb{R}$. Moreover, u is the real part of the analytic function $f : \mathbb{D} \to \mathbb{R}$ given by

$$f(z) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{e^{it} + z}{e^{it} - z} u(e^{it})dt.$$

A proof of Theorem 4.2.1 can be found in Conway [5]. The Herglotz's representation theorem (Theorem 4.2.3) below gives a necessary and sufficient condition for a function to be in the class $\mathcal{P}$. For a proof of this theorem, we require the following theorem.

Theorem 4.2.2 (Helly Selection Theorem) *Let $\langle \alpha_n \rangle$ be a sequence of non-decreasing functions on the bounded interval $[a, b]$, with $\alpha_n(a) = 0$ and $\alpha_n(b) = 1$. Then there exists a subsequence $\langle \alpha_{n_k} \rangle$ that converges everywhere in $[a, b]$ to a non-decreasing function α, and for each function φ continuous on $[a, b]$,*

$$\lim_{k\to\infty} \int_a^b \varphi(t) d\alpha_{n_k}(t) = \int_a^b \varphi(t) d\alpha(t).$$

Theorem 4.2.3 (Herglotz's Representation Theorem) *A function $p \in \mathcal{P}$ if and only if there is a non-decreasing function μ on $[0, 2\pi]$ with $\int_0^{2\pi} d\mu(t) = 1$ satisfying*

$$p(z) = \int_0^{2\pi} \frac{1 + ze^{-it}}{1 - ze^{-it}} d\mu(t). \tag{4.2.2}$$

Proof If a function p is given by (4.2.2), then clearly $p(0) = 1$. Moreover, by Example 4.1.5,

$$\mathrm{Re}\left(\frac{1 + ze^{-it}}{1 - ze^{-it}}\right) > 0, \quad z \in \mathbb{D}.$$

Therefore,

$$\mathrm{Re}\, p(z) = \int_0^{2\pi} \mathrm{Re}\left(\frac{1 + ze^{-it}}{1 - ze^{-it}}\right) \mathrm{d}\mu(t) > 0,$$

which proves that $p \in \mathcal{P}$.

Conversely, suppose $p \in \mathcal{P}$. For $0 \le r < 1$, define the harmonic function $u_r(\zeta) = \mathrm{Re}\, p(r\zeta)$ on $\mathbb{D}$. Since p is analytic, u_r is continuous on $\overline{\mathbb{D}}$ and harmonic in $\mathbb{D}$. By Theorem 4.2.1, there exists an analytic function F_r in $\mathbb{D}$ such that

$$F_r(w) = \frac{1}{2\pi} \int_0^{2\pi} \frac{e^{it} + w}{e^{it} - w} u_r(e^{it})\, \mathrm{d}t, \qquad |w| < 1,$$

and $\mathrm{Re}\, F_r(w) = u_r(w) = \mathrm{Re}\, p(rw)$. Since two analytic functions with the same real part differ by a purely imaginary constant, there exists a constant ic_r such that $F_r(w) = p(rw) + ic_r$. Evaluating at $w = 0$ gives

$$F_r(0) = \frac{1}{2\pi} \int_0^{2\pi} u_r(e^{it})\, \mathrm{d}t = \frac{1}{2\pi} \int_0^{2\pi} \mathrm{Re}\, p(re^{it})\, \mathrm{d}t = \mathrm{Re}\, p(0) = 1,$$

so $c_r = 0$. Hence

$$p(rw) = \frac{1}{2\pi} \int_0^{2\pi} \frac{\mathrm{e}^{\mathrm{i}t} + w}{\mathrm{e}^{\mathrm{i}t} - w} \operatorname{Re} p(r\mathrm{e}^{\mathrm{i}t})\, \mathrm{d}t, \qquad |w| < 1.$$

Letting $z = rw$, so that $|z| < r$, gives

$$p(z) = \frac{1}{2\pi} \int_0^{2\pi} \frac{r\mathrm{e}^{\mathrm{i}t} + z}{r\mathrm{e}^{\mathrm{i}t} - z} \operatorname{Re} p\big(r\mathrm{e}^{\mathrm{i}t}\big)\mathrm{d}t, \quad |z| < r < 1.$$

Define the function $\mu : [0, 2\pi] \times [0, 1) \to \mathbb{R}$ by

$$\mu(t, r) = \frac{1}{2\pi} \int_0^t \operatorname{Re} p\big(r\mathrm{e}^{\mathrm{i}\tau}\big)\mathrm{d}\tau.$$

Then $\mu(0, r) = 0$. From the series expansion $p(z) = 1 + \sum_{k=1}^{\infty} c_k z^k$,

$$\mu(2\pi, r) = \frac{1}{2\pi} \operatorname{Re} \left(\int_0^{2\pi} \Big(1 + \sum_{k=1}^{\infty} c_k r^k \mathrm{e}^{\mathrm{i}k\tau}\Big)\mathrm{d}\tau \right) = 1$$

since all integrals with the exponential terms vanishes. Moreover, since

$$\frac{\mathrm{d}}{\mathrm{d}t}\mu(t, r) = \frac{1}{2\pi} \operatorname{Re} p\big(r\mathrm{e}^{\mathrm{i}t}\big) > 0,$$

it follows that $\mu(\cdot, r)$ is non-decreasing on $[0, 2\pi]$. With this notation, the above integral representation becomes

$$p(z) = \int_0^{2\pi} \frac{r\mathrm{e}^{\mathrm{i}t} + z}{r\mathrm{e}^{\mathrm{i}t} - z}\, \mathrm{d}\mu(t, r). \tag{4.2.3}$$

By the Helly selection theorem, there exists a sequence $\langle r_n \rangle$ with $r_n \to 1^-$ such that $\mu(t, r_n)$ converges pointwise to a non-decreasing function $\mu(t)$ and, for each continuous φ on $[0, 2\pi]$,

$$\lim_{n\to\infty} \int_0^{2\pi} \varphi(t)\, \mathrm{d}\mu(t, r_n) = \int_0^{2\pi} \varphi(t)\, \mathrm{d}\mu(t).$$

Now fix z with $|z| < 1$. Then $|z| < r_n$ for sufficiently large n, and hence

$$p(z) = \int_0^{2\pi} \frac{r_n e^{it} + z}{r_n e^{it} - z} \, d\mu(t, r_n).$$

Since the function $(r_n e^{it} + z)/(r_n e^{it} - z)$ converges uniformly in t to $(e^{it} + z)/(e^{it} - z)$ as $r_n \to 1^-$, passing to the limit under the integral sign and applying the Helly's convergence result gives

$$p(z) = \int_0^{2\pi} \frac{e^{it} + z}{e^{it} - z} \, d\mu(t) = \int_0^{2\pi} \frac{1 + ze^{-it}}{1 - ze^{-it}} \, d\mu(t).$$

Since $\mu(0) = 0$ and $\mu(2\pi) = 1$, this proves the required representation. □

An immediate consequence of Theorem 4.2.3 is the following result.

Corollary 4.2.4 *If $p(z) = 1 + \sum_{k=1}^{\infty} c_k z^k \in \mathcal{P}$, then $|c_k| \leqslant 2$ for each $k \geqslant 1$. Equality holds for functions of the form*

$$p_0(z) = \sum_{j=1}^{n} \alpha_j \frac{1 + \gamma^j z}{1 - \gamma^j z}$$

where $\gamma = e^{2\pi i/k}$, $0 \leqslant \alpha_j \leqslant 1$ with $\sum_{j=1}^{n} \alpha_j = 1$.

Proof It follows from the Herglotz's representation theorem that

$$c_k = 2 \int_0^{2\pi} e^{-ik\theta} d\mu(\theta), \quad k \geqslant 1.$$

Therefore,

$$|c_k| \leqslant 2 \int_0^{2\pi} d\mu(\theta) = 2, \quad k \geqslant 1.$$

The kth coefficient of p_0 is

$$c_k = 2 \sum_{j=1}^{n} \alpha_j (\gamma^j)^k = 2 \sum_{j=1}^{n} \alpha_j (\gamma^k)^j = 2 \sum_{j=1}^{n} \alpha_j = 2.$$

□

Example 4.2.5 Recall that the class $\mathcal{R}$ is defined by

$$\mathcal{R} := \{f \in \mathcal{A} : \operatorname{Re} f'(z) > 0\}.$$

If $f \in \mathcal{R}$, then $p := f' \in \mathcal{P}$. If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k$ and $p(z) = 1 + \sum_{k=1}^{\infty} c_k z^k$, then $a_k = c_{k-1}/k$. An application of Corollary 4.2.4 shows that

$$|a_k| = \frac{|c_{k-1}|}{k} \leqslant \frac{2}{k}$$

for each $f \in \mathcal{R}$. Equality holds for the function

$$f(z) = -z - 2\log(1-z) = z + 2\sum_{k=2}^{\infty} \frac{z^k}{k}, \quad z \in \mathbb{D}.$$

Definition 4.2.6 Let $p : \mathbb{D} \to \mathbb{C}$ be analytic with $p(0) = 1$. Then p is said to have real part greater than α, $0 \leqslant \alpha < 1$, if $\operatorname{Re} p(z) > \alpha$ for all $z \in \mathbb{D}$. The class of such functions is denoted by $\mathcal{P}(\alpha)$ and is known as the class of *Carathéodory functions of order* α or the class of *functions with real part greater than* α.

Corollary 4.2.7 *If* $p(z) = 1 + \sum_{k=1}^{\infty} c_k z^k \in \mathcal{P}(\alpha)$, *then* $|c_k| \leqslant 2(1-\alpha)$ *for each* k. *Equality holds for functions of the form*

$$p(z) = \alpha + (1-\alpha)\sum_{j=1}^{k} \alpha_j \frac{1+\gamma^j z}{1-\gamma^j z}$$

where $\gamma = e^{2\pi i/k}$, $0 \leqslant \alpha_j \leqslant 1$ *with* $\sum_{j=1}^{k} \alpha_j = 1$.

Proof The results follows from the observation that if $p \in \mathcal{P}(\alpha)$, then the function

$$q(z) = \frac{p(z) - \alpha}{1 - \alpha}$$

belongs to $\mathcal{P}$, and conversely, every function q in $\mathcal{P}$ yields a function p in $\mathcal{P}(\alpha)$ via this transformation. □

Corollary 4.2.7 can also be derived using the following Herglotz's representation theorem for functions in the class $\mathcal{P}(\alpha)$.

Corollary 4.2.8 *A function* $p \in \mathcal{P}(\alpha)$ *if and only if there is a non-decreasing function* μ *on* $[0, 2\pi]$ *with* $\int_0^{2\pi} d\mu(t) = 1$ *satisfying*

$$p(z) = \int_0^{2\pi} \frac{1 + (1-2\alpha)ze^{-it}}{1 - ze^{-it}} d\mu(t).$$

Proof The function $p \in \mathcal{P}(\alpha)$ if and only if the function

$$q(z) = \frac{p(z) - \alpha}{1 - \alpha}$$

belongs to $\mathcal{P}$. By the Herglotz's representation theorem, $q \in \mathcal{P}$ if and only if there is a non-decreasing function μ on $[0, 2\pi]$ with $\int_0^{2\pi} \mathrm{d}\mu(t) = 1$ satisfying

$$q(z) = \int_0^{2\pi} \frac{1 + z\mathrm{e}^{-\mathrm{i}t}}{1 - z\mathrm{e}^{-\mathrm{i}t}} \mathrm{d}\mu(t).$$

Since

$$\begin{aligned} p(z) &= (1 - \alpha)q(z) + \alpha \\ &= (1 - \alpha) \int_0^{2\pi} \frac{1 + z\mathrm{e}^{-\mathrm{i}t}}{1 - z\mathrm{e}^{-\mathrm{i}t}} \mathrm{d}\mu(t) + \alpha \int_0^{2\pi} \mathrm{d}\mu(t) \\ &= \int_0^{2\pi} \frac{1 + (1 - 2\alpha)z\mathrm{e}^{-\mathrm{i}t}}{1 - z\mathrm{e}^{-\mathrm{i}t}} \mathrm{d}\mu(t), \end{aligned}$$

the result follows. □

For $\alpha > 1$, let $\tilde{P}(\alpha)$ be the class of functions p with $p(0) = 1$ satisfying $\mathrm{Re}\, p(z) < \alpha$. If $p \in \tilde{P}(\alpha)$, then the function q defined by $q(z) = (p(z) - \alpha)/(1 - \alpha)$ belongs to $\mathcal{P}$. Therefore, Corollary 4.2.8 holds even when $\mathcal{P}(\alpha)$ is replaced by $\tilde{P}(\alpha)$.

4.3 Growth and Distortion Estimates

The coefficient estimates $|c_k| \leqslant 2$ is first used to establish certain inequalities satisfied by functions in the class $\mathcal{P}$.

Theorem 4.3.1 *If $p \in \mathcal{P}$, then, for $|z| = r < 1$,*

$$\frac{1 - r}{1 + r} \leqslant |p(z)| \leqslant \frac{1 + r}{1 - r},$$

$$\left|p^{(n)}(z)\right| \leqslant \frac{2(n!)}{(1 - r)^{n+1}},$$

$$|p(z) - 1| \leqslant \frac{2r}{1 - r},$$

and

$$|p(z) - zp'(z) - 1| \leqslant \frac{2r^2}{(1-r)^2}.$$

These inequalities are sharp. Also, for $|\alpha| \leqslant 1, |\beta| \leqslant 1$,

$$|p(\alpha z) - p(\beta z)| \leqslant \frac{2|\alpha - \beta| r}{(1-r)^2}.$$

Proof For $p(z) = 1 + \sum_{k=1}^{\infty} c_k z^k \in \mathcal{P}$, the coefficient estimate $|c_k| \leqslant 2$ yields

$$|p(z)| \leqslant 1 + \sum_{k=1}^{\infty} |c_k||z|^k \leqslant 1 + 2\sum_{k=1}^{\infty} r^k = \frac{1+r}{1-r}.$$

Since

$$\operatorname{Re}\left(\frac{1}{p(z)}\right) = \operatorname{Re}\left(\frac{\overline{p(z)}}{|p(z)|^2}\right) = \frac{\operatorname{Re} p(z)}{|p(z)|^2}$$

and

$$\left(\frac{1}{p}\right)(0) = \frac{1}{p(0)} = 1,$$

it follows that $p \in \mathcal{P}$ if and only if $1/p \in \mathcal{P}$ (This also follows by Theorem 4.1.8 (vi)). Applying the inequality $|p(z)| \leqslant (1+r)/(1-r)$ to the function $1/p$ yields

$$\frac{1-r}{1+r} \leqslant |p(z)|.$$

The coefficient estimate $|c_n| \leqslant 2$ yields

$$\begin{aligned} |p^{(n)}(z)| &= \left| \sum_{k=n}^{\infty} k(k-1)\dots(k-n+1) c_k z^{k-n} \right| \\ &\leqslant 2\sum_{k=n}^{\infty} k(k-1)\dots(k-n+1) r^{k-n} \\ &= \frac{2(n!)}{(1-r)^{n+1}}. \end{aligned}$$

Similarly,

$$|p(z) - 1| \leqslant \sum_{k=1}^{\infty} |c_k| r^k \leqslant 2\sum_{k=1}^{\infty} r^k = \frac{2r}{1-r},$$

and

$$|p(z) - zp'(z) - 1| = \left| \sum_{k=2}^{\infty} c_k (k-1) z^k \right| \leqslant 2 \sum_{k=2}^{\infty} (k-1) r^k = \frac{2r^2}{(1-r)^2}.$$

All these inequalities become equalities if $c_k = 2$ for all $k \geqslant 1$ and this happens for the function $p_0(z) = (1+z)/(1-z)$, $z \in \mathbb{D}$.

Since

$$p(\alpha z) - p(\beta z) = \sum_{k=1}^{\infty} (\alpha^k - \beta^k) c_k z^k = (\alpha - \beta) \sum_{k=1}^{\infty} (\alpha^{k-1} + \alpha^{k-2}\beta + \cdots + \beta^{k-1}) c_k z^k,$$

it follows that

$$|p(\alpha z) - p(\beta z)| \leqslant 2|\alpha - \beta| \sum_{k=1}^{\infty} k r^k = \frac{2|\alpha - \beta| r}{(1-r)^2}$$

for $|\alpha| \leqslant 1$, $|\beta| \leqslant 1$, and $|z| = r < 1$. □

Theorem 4.3.2 (Growth Theorem) *If $p \in \mathcal{P}$, then*

$$\left| p(z) - \frac{1+r^2}{1-r^2} \right| \leqslant \frac{2r}{1-r^2}, \quad |z| = r < 1, \tag{4.3.1}$$

and

$$\frac{1-r}{1+r} \leqslant Re\, p(z) \leqslant |p(z)| \leqslant \frac{1+r}{1-r}, \quad |z| = r < 1. \tag{4.3.2}$$

Equality holds for $p_0(z) = (1+z)/(1-z)$, $z \in \mathbb{D}$.

Proof The regions defined by (4.3.1) and (4.3.2) are shown in Fig. 4.2. The function $p_0(z) = (1+z)/(1-z)$ maps $\partial \mathbb{D}_r$ onto the circle

$$\left\{ w \in \mathbb{C} : \left| w - \frac{1+r^2}{1-r^2} \right| = \frac{2r}{1-r^2} \right\}.$$

To see this, write $w = (1+z)/(1-z)$ so that $z = (w-1)/(w+1)$. If $|z| = r$, then $|w-1| = |w+1|r$ and hence

$$(1-r^2)|w|^2 - 2(1+r^2)\mathrm{Re}\, w = r^2 - 1$$

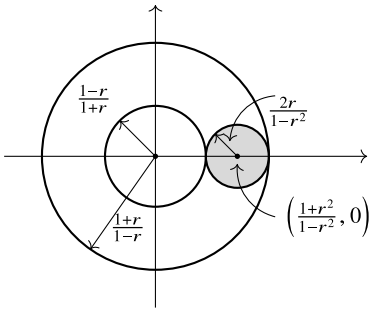


Fig. 4.2 $p_0(\mathbb{D}_r)$ where $p_0(z) = (1+z)/(1-z)$

This is equivalent to

$$\left| w - \frac{1+r^2}{1-r^2} \right|^2 = \frac{4r^2}{(1-r^2)^2}.$$

Therefore,

$$\left| p_0(z) - \frac{1+r^2}{1-r^2} \right| = \frac{2r}{1-r^2}, \quad |z| = r < 1.$$

The function $p_t(z) = (1+\mathrm{e}^{-\mathrm{i}t}z)/(1-\mathrm{e}^{-\mathrm{i}t}z)$, $z \in \mathbb{D}$, satisfies $p_t(z) = p_0(\mathrm{e}^{-\mathrm{i}t}z)$, and thus

$$\left| p_t(z) - \frac{1+r^2}{1-r^2} \right| = \frac{2r}{1-r^2}. \tag{4.3.3}$$

This proves that the inequality (4.3.1) becomes equality for the function p_0 (as well as for p_t).

Now, if $p \in \mathcal{P}$, it follows from the Herglotz's representation theorem that there is a non-decreasing function μ on $[0, 2\pi]$ with $\int_0^{2\pi} \mathrm{d}\mu(t) = 1$ satisfying

$$p(z) = \int_0^{2\pi} p_t(z) \mathrm{d}\mu(t), \quad z \in \mathbb{D}.$$

Hence, for $|z| = r < 1$, the inequality (4.3.3) yields

$$\begin{aligned} \left| p(z) - \frac{1+r^2}{1-r^2} \right| &= \left| \int_0^{2\pi} \left(p_t(z) - \frac{1+r^2}{1-r^2} \right) \mathrm{d}\mu(t) \right| \\ &\leqslant \int_0^{2\pi} \left| p_t(z) - \frac{1+r^2}{1-r^2} \right| \mathrm{d}\mu(t) \\ &= \frac{2r}{1-r^2}. \end{aligned}$$

This establishes (4.3.1). The other inequality (4.3.2) follows as $|w - c| \leqslant d$ with $c \in \mathbb{R}$ implies $c - d \leqslant \mathrm{Re} w \leqslant |w| \leqslant c + d$. Clearly, the equality holds in (4.3.2) for the function p_0. □

Corollary 4.3.3 *The class $\mathcal{P}$ is a compact normal family.*

Proof Following the same argument as in the proof of Corollary 2.5.7, it follows that $\mathcal{P} \subset \mathcal{H}(\mathbb{D})$ is locally bounded and therefore a normal family. To establish compactness, let $p_n \in \mathcal{P}$ with $p_n \to p$, where $p \in \mathcal{H}(\mathbb{D})$. Since $\mathrm{Re} p_n(z) > 0$, then $\mathrm{Re} p(z) \geqslant 0$. However, by Theorem 1.2.12, the condition $\mathrm{Re} p(z) > 0$ must hold, and thus $p \in \mathcal{P}$, establishing the compactness of $\mathcal{P}$. □

Theorem 4.3.4 *(Distortion Theorem) If $p \in \mathcal{P}$, then*

$$|p'(z)| \leqslant \frac{2Re\, p(z)}{1 - r^2} \leqslant \frac{2}{(1 - r)^2}, \quad |z| = r < 1 \tag{4.3.4}$$

and

$$\left| \frac{z p'(z)}{p(z)} \right| \leqslant \frac{2r}{1 - r^2}, \quad |z| = r < 1. \tag{4.3.5}$$

Equality holds for $p_0(z) = (1 + z)/(1 - z)$, $z \in \mathbb{D}$.

Proof The function defined by

$$\varphi_a(z) = \frac{a - z}{1 - \overline{a} z}$$

maps $\mathbb{D}$ onto itself and so the function $h = p \circ \varphi_a$ satisfies $\mathrm{Re}\, h(z) > 0$ and

$$h(z) = h(0) + h'(0) z + \cdots = p(a) - (1 - |a|^2) p'(a) z + \cdots .$$

Then

$$\tilde{h}(z) := (h(z) - i \mathrm{Im} p(a))/\mathrm{Re} p(a) = 1 - (1 - |a|^2) \frac{p'(a)}{\mathrm{Re} p(a)} z + \cdots$$

belongs to $\mathcal{P}$. Since every coefficient of a function in the class $\mathcal{P}$ is bounded by 2, it follows that

$$\left| \frac{(1 - |a|^2) p'(a)}{\mathrm{Re} p(a)} \right| \leqslant 2.$$

Replacing a by z yields the first inequality in (4.3.4). The second follows by using the growth inequality (4.3.2). Since $\mathrm{Re} p(z) \leqslant |p(z)|$ (4.3.5) follows by (4.3.4). A direct calculation shows that equality holds in (4.3.4) for p_0 at $z = r$. □

By Corollary 4.2.8, $p \in \mathcal{P}(\alpha)$ if and only if there is a non-decreasing function μ on $[0, 2\pi]$ with $\int_0^{2\pi} \mathrm{d}\mu(t) = 1$ satisfying

$$p(z) = \int_0^{2\pi} \frac{1 + (1-2\alpha)z\mathrm{e}^{-\mathrm{i}t}}{1 - z\mathrm{e}^{-\mathrm{i}t}} \mathrm{d}\mu(t) = \int_0^{2\pi} p_\alpha(z\mathrm{e}^{-\mathrm{i}t}) \mathrm{d}\mu(t), \quad z \in \mathbb{D} \tag{4.3.6}$$

where $p_\alpha(z) := (1 + (1-2\alpha)z)/(1-z)$.

Corollary 4.3.5 *If $p \in \mathcal{P}(\alpha)$, then, for $|z| = r$,*

$$\left| p(z) - \frac{1 + (1-2\alpha)r^2}{1 - r^2} \right| \leqslant \frac{2(1-\alpha)r}{1 - r^2}, \tag{4.3.7}$$

$$\frac{1 - (1-2\alpha)r}{1 + r} \leqslant Re\, p(z) \leqslant |p(z)| \leqslant \frac{1 + (1-2\alpha)r}{1 - r}, \tag{4.3.8}$$

$$|p'(z)| \leqslant \frac{2(Re\, p(z) - \alpha)}{1 - r^2} \tag{4.3.9}$$

$$\left| \frac{zp'(z)}{p(z)} \right| \leqslant \frac{2(1-\alpha)r}{(1-r)(1 + (1-2\alpha)r)}. \tag{4.3.10}$$

Equality holds for

$$p_\alpha(z) := \frac{1 + (1-2\alpha)z}{1 - z}, \quad z \in \mathbb{D}.$$

Proof If $p \in \mathcal{P}(\alpha)$, then the function $q(z) = (p(z) - \alpha)/(1-\alpha)$ belongs to $\mathcal{P}$. The inequality (4.3.1) applied to q gives

$$\left| q(z) - \frac{1 + r^2}{1 - r^2} \right| \leqslant \frac{2r}{1 - r^2},$$

which leads to the inequality (4.3.7). The inequality (4.3.8). follows readily from (4.3.7) because $|w - c| \leqslant d$ with $c \in \mathbb{R}$ implies $c - d \leqslant \mathrm{Re} w \leqslant |w| \leqslant c + d$.

The inequality (4.3.4) applied to q gives the inequality (4.3.9). Further, the inequality (4.3.9) yields

$$\begin{aligned} \left| \frac{zp'(z)}{p(z)} \right| &\leqslant \frac{|zp'(z)|}{\mathrm{Re} p(z)} \\ &\leqslant \frac{r}{\mathrm{Re} p(z)} \frac{2(\mathrm{Re} p(z) - \alpha)}{1 - r^2} \\ &= \frac{2r}{1 - r^2} \left(1 - \frac{\alpha}{\mathrm{Re} p(z)} \right). \end{aligned} \tag{4.3.11}$$

By using (4.3.8), it follows that

$$1 - \frac{\alpha}{\mathrm{Re}\, p(z)} \leqslant 1 - \alpha \frac{1-r}{1+(1-2\alpha)r} = \frac{(1-\alpha)(1+r)}{1+(1-2\alpha)r},$$

and thus (4.3.11) yields (4.3.10).

The left hand side of the inequality in (4.3.8) becomes an equality at $z = -r$ and all other inequalities in (4.3.7)–(4.3.10) become equality at $z = r$ for p_α. □

4.4 Subordination

Though complex numbers can be ordered in many ways, there is no ordering that makes $\mathbb{C}$ an ordered field. The inequality $a < x < y < b$ is the same as the inclusion $(x, y) \subset (a, b)$ and the inclusion of sets could be used in the complex plane as a natural analogue of inequalities on the real line. The concept of subordination is defined by using the containment of the image of two functions. Recall that $\mathcal{B}_0$ denotes the class of all analytic functions $w : \mathbb{D} \to \mathbb{D}$ with $w(0) = 0$.

Definition 4.4.1 Let $p, q : \mathbb{D} \to \mathbb{C}$ be analytic functions. The function p is said to be *subordinate* to q, written $p \prec q$, if there is a function $w \in \mathcal{B}_0$ such that $p = q \circ w$. The functions p and q are, respectively, called subordinate function and superordinate function.

Example 4.4.2 If $p(z) = z^2$ and $q(z) = z$, then $p \prec q$. This follows because $p(z) = q(w(z))$ where $w(z) = z^2$ is a function in $\mathcal{B}_0$. However, q is not subordinate to p. For, if $q \prec p$, then there is a $w \in \mathcal{B}_0$ such that $z = q(z) = p(w(z)) = w(z)^2$. Thus, w is an analytic square-root of z in $\mathbb{D}$, which however does not exist.

The following theorem gives an equivalent definition for subordination when the superordinate function q is univalent.

Theorem 4.4.3 *Let $p, q : \mathbb{D} \to \mathbb{C}$ be analytic functions and let q be univalent. The function p is subordinate to q if and only if $p(0) = q(0)$ and $p(\mathbb{D}) \subset q(\mathbb{D})$.*

Proof Suppose $p \prec q$, so that $p(z) = q(w(z))$ for some $w \in \mathcal{B}_0$. Since $w(0) = 0$, it follows that $p(0) = q(w(0)) = q(0)$. As w maps $\mathbb{D}$ into $\mathbb{D}$, the composite function $p = q \circ w$ maps $\mathbb{D}$ into $q(\mathbb{D})$, and hence $p(\mathbb{D}) \subset q(\mathbb{D})$.

Conversely, since $p(0) = q(0)$, $p(\mathbb{D}) \subset q(\mathbb{D})$, and q is univalent, the inverse function q^{-1} maps $p(\mathbb{D}) \subset q(\mathbb{D})$ into $\mathbb{D}$. Let $w(z) = q^{-1}(p(z))$ in $\mathbb{D}$; then w maps $\mathbb{D}$ into $\mathbb{D}$ and satisfies $w(0) = q^{-1}(p(0)) = q^{-1}(q(0)) = 0$. Thus, $p(z) = q(w(z))$ with $w \in \mathcal{B}_0$, showing that $p \prec q$. □

Theorem 4.4.4 *If $p \prec q$, then*

(i) $p + c \prec q + c$ *for each constant c;*
(ii) $p^n \prec q^n$ *for each integer n; and more generally,*
(iii) $h \circ p \prec h \circ q$ *for each analytic function $h : q(\mathbb{D}) \to \mathbb{C}$.*

Proof (i) and (ii) are special cases of (iii). To prove (iii), since $p \prec q$, there exists $w \in \mathcal{B}_0$ such that $p = q \circ w$. Thus, $h \circ p = (h \circ q) \circ w$, or $h \circ p \prec h \circ q$. □

The class $\mathcal{P}$ is related to the class $\mathcal{B}_0$ in a natural way as shown in the following theorem.

Theorem 4.4.5 *If $p \in \mathcal{P}$ and*

$$w(z) = \frac{p(z) - 1}{p(z) + 1}, \quad z \in \mathbb{D},$$

then $w \in \mathcal{B}_0$. Conversely, if $w \in \mathcal{B}_0$ and

$$p(z) = \frac{1 + w(z)}{1 - w(z)}, \quad z \in \mathbb{D},$$

then $p \in \mathcal{P}$.

Proof For $p \in \mathcal{P}$, it follows that $\mathrm{Re}\, p(z) + 1 > 1$ and

$$w(z) = \frac{p(z) - 1}{p(z) + 1}$$

is analytic with $w(0) = 0$. Further,

$$1 - |w(z)|^2 = 1 - \left|\frac{p(z) - 1}{p(z) + 1}\right|^2 = \frac{4\mathrm{Re}\, p(z)}{|p(z) + 1|^2} > 0.$$

Thus, $|w(z)| < 1$ and hence $w \in \mathcal{B}_0$.

To prove the converse, let $w \in \mathcal{B}_0$ so that $|w(z)| < 1$. The function

$$p(z) = \frac{1 + w(z)}{1 - w(z)}$$

satisfies $p(0) = 1$ and

$$\mathrm{Re}\, p(z) = \mathrm{Re}\frac{\big(1 + w(z)\big)\big(1 - \overline{w(z)}\big)}{|1 - w(z)|^2} = \frac{1 - |w(z)|^2}{|1 - w(z)|^2} > 0. \tag{4.4.1}$$

Thus, $p \in \mathcal{P}$. □

By Theorem 4.4.5, it follows that $p \in \mathcal{P}$ if and only if $p(z) \prec (1 + z)/(1 - z)$. This connection between the class $\mathcal{P}$ and $\mathcal{B}_0$ is used to give another proof of the growth inequality (4.3.2) and the distortion inequality (4.3.4) in the following corollary.

Corollary 4.4.6 *If $p \in \mathcal{P}$, then, for each $z \in \mathbb{D}$,*

$$\frac{1-r}{1+r} \leqslant Rep(z) \leqslant |p(z)| \leqslant \frac{1+r}{1-r}, \quad |z| = r < 1,$$

and

$$|p'(z)| \leqslant \frac{2Rep(z)}{1-r^2}, \quad |z| = r < 1.$$

Proof The function

$$w(z) = \frac{p(z)-1}{p(z)+1}, \quad z \in \mathbb{D}$$

belongs to $\mathcal{B}_0$. By Schwarz's lemma, $|w(z)| \leqslant |z| = r$. Since

$$p(z) = \frac{1+w(z)}{1-w(z)},$$

it follows that

$$|p(z)| \leqslant \frac{1+|w(z)|}{1-|w(z)|} \leqslant \frac{1+r}{1-r}.$$

By (4.4.1),

$$\mathrm{Re}\, p(z) = \frac{1-|w(z)|^2}{|1-w(z)|^2} \geqslant \frac{1-r^2}{(1+r)^2} = \frac{1-r}{1+r},$$

proving the growth inequality.

To prove the distortion inequality, first note that $w \in \mathcal{B}_0$ and hence, by the Schwarz–Pick theorem (Theorem 3.1.3),

$$|w'(z)| \leqslant \frac{1-|w(z)|^2}{1-|z|^2}. \tag{4.4.2}$$

Since

$$p'(z) = -\frac{2w'(z)}{(1-w(z))^2},$$

(4.4.2) and (4.4.1) yield

$$|p'(z)| = \frac{2|w'(z)|}{|1-w(z)|^2} \leqslant \frac{2}{1-|z|^2}\frac{1-|w(z)|^2}{|1-w(z)|^2} = \frac{2\mathrm{Re}\, p(z)}{1-r^2}.$$

□

Let w be a function in the class $\mathcal{B}_0$, and let $w_0(z) = z$. Then $w(z) \prec w_0(z)$. By Schwarz's lemma, the inequality $|w(z)| \leqslant |z| \leqslant r$ holds whenever $|z| \leqslant r < 1$.

Consequently, $w(\overline{\mathbb{D}}_r) \subset \overline{\mathbb{D}}_r = w_0(\overline{\mathbb{D}}_r)$ for each $0 < r < 1$. In other words, if w maps $\mathbb{D}$ into the image of $\mathbb{D}$ under w_0, then w maps $\mathbb{D}_r$ into the image of $\mathbb{D}_r$ under w_0 for each $0 < r < 1$. The general case is addressed in the following theorem (see Fig. 4.3).

Theorem 4.4.7 (Lindelöf subordination principle) *If $p \prec q$, then for each r in $[0, 1)$, $p(\overline{\mathbb{D}}_r) \subset q(\overline{\mathbb{D}}_r)$. Further, if $p(re^{i\theta})$ is on the boundary of $q(\overline{\mathbb{D}}_r)$ at a point $z_0 = re^{i\theta_0}$, then there is a real α such that $p(z) = q(e^{i\alpha}z)$, and $p(re^{i\theta})$ is on the boundary of $q(\overline{\mathbb{D}}_r)$ for every point $z = re^{i\theta}$ in $\mathbb{D}$.*

Proof Since $p \prec q$, there is a function $w \in \mathcal{B}_0$ such that $p = q \circ w$, and so $p(\overline{\mathbb{D}}_r) = q(w(\overline{\mathbb{D}}_r))$. Since $w(\overline{\mathbb{D}}_r) \subset \overline{\mathbb{D}}_r$, it follows that $p(\overline{\mathbb{D}}_r) = q(w(\overline{\mathbb{D}}_r)) \subset q(\overline{\mathbb{D}}_r)$.

If there is a z_0 such that $0 < r = |z_0| < 1$ and $p(z_0)$ is a boundary point of $q(\overline{\mathbb{D}}_r)$, then $|w(z_0)| = r$. Hence, $w(z) = \mathrm{e}^{\mathrm{i}\alpha}z$ for some real α, and $p(z) = q(\mathrm{e}^{\mathrm{i}\alpha}z)$. □

The Lindelöf subordination principle gives another proof of the inequality (4.3.1). If $p \in \mathcal{P}$, then $p \prec p_0$ where $p_0(z) = (1 + z)/(1 - z)$. From the proof of Theorem 4.3.2, this function p_0 maps $\overline{\mathbb{D}}_r$ onto the disk

$$p_0(\overline{\mathbb{D}}_r) = \left\{ w \in \mathbb{C} : \left| w - \frac{1 + r^2}{1 - r^2} \right| \leqslant \frac{2r}{1 - r^2} \right\}.$$

By Lindelöf subordination principle, it follows that $p(\overline{\mathbb{D}}_r) \subset p_0(\overline{\mathbb{D}}_r)$. This proves (4.3.1).

Definition 4.4.8 For $A \neq B$, $|B| \leqslant 1$, the Janowski class $\mathcal{P}[A, B]$ consists of all analytic functions p with $p(0) = 1$ satisfying the subordination

$$p(z) \prec \frac{1 + Az}{1 + Bz}.$$

The condition $|B| \leqslant 1$ ensures that the function $(1 + Az)/(1 + Bz)$ is defined and analytic in $\mathbb{D}$, and the condition $A \neq B$ guarantees that the mapping does not reduce to

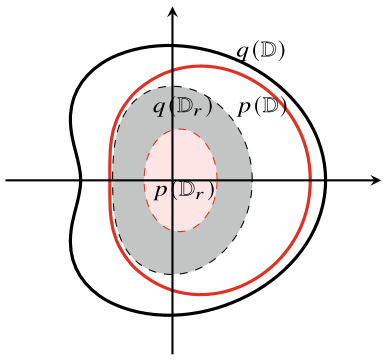


Fig. 4.3 Lindelöf subordination principle

a constant. With the additional restriction $-1 \leqslant B < A \leqslant 1$, this class—introduced and studied by Janowski [11, p. 297] in 1973—is a subclass of $\mathcal{P}$; in this case, the class $\mathcal{P}[A, B]$ is referred to as the Janowski class of functions with positive real part. In particular, $\mathcal{P}[1-2\alpha, -1] =: \mathcal{P}(\alpha)$ $(0 \leqslant \alpha < 1)$, and $\mathcal{P}[1, -1] = \mathcal{P}$. The mapping $(1 + Az)/(1 + Bz)$ is a Möbius transformation. If $|B| = 1$, then the image of $(1 + Az)/(1 + Bz)$ is a half-plane; whereas for $|B| < 1$ the image is a disk.

Theorem 4.4.9 *The function $p \in \mathcal{P}[A, B]$ if and only if*

$$p(z) = \frac{(1 + A)q(z) + 1 - A}{(1 + B)q(z) + 1 - B} \tag{4.4.3}$$

for some function $q \in \mathcal{P}$.

Proof Since $p \in \mathcal{P}[A, B]$, there exists a function $w \in \mathcal{B}_0$ such that

$$p(z) = \frac{1 + Aw(z)}{1 + Bw(z)}. \tag{4.4.4}$$

The function

$$q(z) = \frac{1 + w(z)}{1 - w(z)}, \quad z \in \mathbb{D} \tag{4.4.5}$$

belongs to $\mathcal{P}$. Since $w(z) = (q(z) - 1)/(q(z) + 1)$, (4.4.4) yields

$$p(z) = \frac{1 + Aw(z)}{1 + Bw(z)} = \frac{(q(z) + 1) + A(q(z) - 1)}{(q(z) + 1) + B(q(z) - 1)} = \frac{(1 + A)q(z) + 1 - A}{(1 + B)q(z) + 1 - B}.$$

Conversely, suppose that (4.4.3) holds for some $q \in \mathcal{P}$. Then (4.4.5) holds for some $w \in \mathcal{B}_0$. Using (4.4.5) in (4.4.3), it follows that (4.4.4) holds, proving $p \in \mathcal{P}[A, B]$. □

Theorem 4.4.10 *If $p \in \mathcal{P}[A, B]$, then*

$$\left| p(z) - \frac{1 - A\overline{B}r^2}{1 - |B|^2 r^2} \right| \leqslant \frac{|B - A|r}{1 - |B|^2 r^2}, \quad |z| = r < 1. \tag{4.4.6}$$

Proof The inequality (4.4.6) follows from the Lindelöf subordination principle once it is established that $w = (1 + Az)/(1 + Bz)$ maps $\overline{\mathbb{D}}_r$ onto the disk

$$\left\{ w : \left| w - \frac{1 - A\overline{B}r^2}{1 - |B|^2 r^2} \right| \leqslant \frac{|B - A|r}{1 - |B|^2 r^2} \right\}.$$

To show this, note that $z = (w - 1)/(A - Bw)$. Then the condition $|z| \leqslant r$ becomes $|w - 1| \leqslant |A - Bw|r$. Squaring both sides and simplifying yields

$$|w|^2 \left(1 - |B|^2 r^2\right) - 2\mathrm{Re}\left(w(1 - \overline{A}Br^2)\right) \leqslant |A|^2 r^2 - 1,$$

or equivalently,

$$|w|^2 - 2\mathrm{Re}\left(w\frac{(1-\overline{A}Br^2)}{1-|B|^2r^2}\right) \leqslant -\frac{1-|A|^2r^2}{1-|B|^2r^2}.$$

This leads to the inequality

$$\left|w - \frac{1-A\overline{B}r^2}{1-|B|^2r^2}\right|^2 \leqslant \frac{|1-A\overline{B}r^2|^2}{(1-|B|^2r^2)^2} - \frac{1-|A|^2r^2}{1-|B|^2r^2} = \frac{|B-A|^2r^2}{(1-|B|^2r^2)^2}.$$

Hence, the mapping $w = (1+Az)/(1+Bz)$ maps $\overline{\mathbb{D}}_r$ onto the disk

$$\left|w - \frac{1-\overline{A}Br^2}{1-|B|^2r^2}\right| \leqslant \frac{|B-A|r}{1-|B|^2r^2}.$$

□

If $p \in \mathcal{P}[A, B]$, $-1 \leqslant B < A \leqslant 1$, then the inequality (4.4.6) reduces to

$$\left|p(z) - \frac{1-ABr^2}{1-B^2r^2}\right| \leqslant \frac{(A-B)r}{1-B^2r^2}. \tag{4.4.7}$$

Equation (4.4.7) readily yields

$$\frac{1-ABr^2}{1-B^2r^2} - \frac{(A-B)r}{1-B^2r^2} \leqslant \mathrm{Re}\,p(z) \leqslant \frac{1-ABr^2}{1-B^2r^2} + \frac{(A-B)r}{1-B^2r^2}.$$

Thus,

$$\frac{1-Ar}{1-Br} \leqslant \mathrm{Re}\,p(z) \leqslant \frac{1+Ar}{1+Br}.$$

See Fig. 4.4. If $B = -1$, then $p \in \mathcal{P}[A, B]$ if and only if $\mathrm{Re}\,p(z) > (1-A)/2$. If $B > -1$, then $p \in \mathcal{P}[A, B]$ if and only if $p(z)$ lies in a disk whose diametric end points are $(1-A)/(1-B)$ and $(1+A)/(1+B)$.

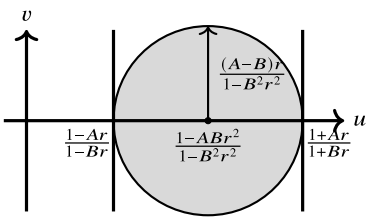


Fig. 4.4 $p(\mathbb{D}_r)$ when $p(z) = (1+Az)/(1+Bz)$

In the next result, the Schwarz's lemma is applied to derive coefficient bounds for functions in the class $\mathcal{P}[A, B]$.

Theorem 4.4.11 *If* $p(z) = 1 + \sum_{k=1}^{\infty} c_k z^k \in \mathcal{P}[A, B]$, *then the sharp inequality* $|c_k| \leqslant |A - B|$ *holds for each* $k \geqslant 1$.

Proof Let $q(z) = (1 + Az)/(1 + Bz)$. Since $p \prec q$, there is a function $w \in \mathcal{B}_0$ with $p(z) = q(w(z))$ and so, by Schwarz's lemma,

$$|p'(0)| = |q'(w(0))w'(0)| = |q'(0)||w'(0)| \leqslant |q'(0)|.$$

Hence,

$$|c_1| = |p'(0)| \leqslant |q'(0)| = |A - B|.$$

This bound of c_1 gives the bound for c_k for any $k \geqslant 2$. Let $\eta = \mathrm{e}^{2\pi \mathrm{i}/k}$ and

$$p_k(z) = \frac{1}{k} \sum_{j=0}^{k-1} p\left(\eta^j z\right).$$

Since

$$1 + \eta^l + \eta^{2l} + \cdots + \eta^{(k-1)l} = \begin{cases} 0, & \text{if } l \not\equiv 0 \bmod k \\ k, & \text{if } l \equiv 0 \bmod k \end{cases}$$

it follows that

$$\begin{aligned} p_k(z) &= \frac{1}{k} \sum_{j=0}^{k-1} \left(1 + \sum_{l=1}^{\infty} c_l \left(\eta^j z\right)^l\right) \\ &= 1 + \frac{1}{k} \sum_{l=1}^{\infty} \left(\sum_{j=0}^{k-1} \eta^{jl}\right) c_l z^l = 1 + c_k z^k + c_{2k} z^{2k} + \cdots. \end{aligned}$$

Since $p \prec q$, the image of $p(\eta^j z)$ lie in $q(\mathbb{D})$. The convexity of $q(\mathbb{D})$ implies that $p_k(z) \in q(\mathbb{D})$. The function $\tilde{p}_k$ defined by

$$\tilde{p}_k(z) = 1 + c_k z + c_{2k} z^2 + \cdots \tag{4.4.8}$$

belongs to $\mathcal{P}[A, B]$ and so $|c_k| \leqslant |A - B|$.

For each k, the inequality is sharp for the function

$$p_0(z) = \frac{1 + Az^k}{1 + Bz^k}, \quad z \in \mathbb{D}.$$

□

4.5 Coefficient Estimates

By Corollary 4.2.7, the coefficients of functions in $\mathcal{P}(\alpha)$ are bounded by $2(1-\alpha)$. The following result extends the coefficient bound to functions that are not necessarily normalized.

Theorem 4.5.1 *If $p : \mathbb{D} \to \mathbb{C}$ is analytic, $p(z) = \sum_{k=0}^{\infty} c_k z^k$ and $\operatorname{Re} p(z) > \alpha$, then $|c_k| \leqslant 2(\operatorname{Re} c_0 - \alpha)$ for $k \geqslant 1$ with equality for the function p defined by*

$$p(z) = \alpha + (\operatorname{Re} c_0 - \alpha)\frac{1+z}{1-z} + i\operatorname{Im} c_0.$$

Proof Since $\operatorname{Re} p(z) > \alpha$, then $\operatorname{Re} c_0 > \alpha$, and

$$q(z) := \frac{p(z) - \alpha - i\operatorname{Im} c_0}{\operatorname{Re} c_0 - \alpha} = 1 + \sum_{k=1}^{\infty} \frac{c_k}{\operatorname{Re} c_0 - \alpha} z^k$$

belongs to $\mathcal{P}$. The result is deduced from the fact that the modulus of each coefficient of functions in $\mathcal{P}$ is bounded above by 2. □

Corollary 3.4.2 states that $\sum_{n=0}^{\infty} |w_n z^n| \leq 1$ for all $z \in \mathbb{D}_{1/3}$ for each analytic function $w(z) = \sum_{n=0}^{\infty} w_n z^n$ satisfying $|w(z)| < 1$. If the condition $|w(z)| < 1$ is replaced by the weaker assumption $\operatorname{Re}(w(z)) < 1$, the result remains valid.

Corollary 4.5.2 (Bohr's inequality) *If $f(z) = \sum_{k=0}^{\infty} a_k z^k$ is analytic in $\mathbb{D}$, and $\operatorname{Re} f(z) < 1$, then $\sum_{k=0}^{\infty} |a_k z^k| \leq 1$ for all $z \in \mathbb{D}_{1/3}$. The constant $1/3$ is sharp.*

Proof There is no loss of generality in assuming $a_0 \geqslant 0$. Since $\operatorname{Re}(1 - f(z)) > 0$, by Theorem 4.5.1,

$$|a_k| \leqslant 2(1 - a_0), \quad k \geqslant 1.$$

Therefore, for $|z| = r \leq 1/3$,

$$\sum_{k=0}^{\infty} |a_k z^k| \leqslant a_0 + 2(1-a_0)\sum_{k=1}^{\infty} r^k \leqslant a_0 + 2(1-a_0)\frac{r}{1-r} \leqslant 1.$$

To show sharpness, consider

$$f_0(z) = 1 - (1-a_0)\frac{1+z}{1-z} = a_0 - 2(1-a_0)(z + z^2 + z^3 + \cdots), \quad 0 < a_0 < 1.$$

Then $\operatorname{Re} f_0(z) < 1$ and

$$\sum_{k=0}^{\infty} |a_k z^k| = a_0 + 2(1-a_0)\frac{r}{1-r} = 1$$

for $|z| = r = 1/3$. □

Since the classes $\mathcal{B}_0$ and $\mathcal{P}$ are related, coefficient estimates for functions in one class can often be derived from the corresponding estimates in the other. This is achieved by expressing the coefficients of the function $w \in \mathcal{B}_0$ in terms of those of $p \in \mathcal{P}$, and vice versa.

Theorem 4.5.3 *Let $w \in \mathcal{B}_0$ be related to $p \in \mathcal{P}$ by*

$$p(z) = \frac{1 + w(z)}{1 - w(z)}.$$

If $w(z) = \sum_{k=1}^{\infty} w_k z^k$ and $p(z) = 1 + \sum_{k=1}^{\infty} c_k z^k$, then the coefficients c_k and w_k are given by

$$c_k = 2 \begin{vmatrix} w_1 & -1 & 0 & \cdots & 0 \\ w_2 & w_1 & -1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ w_k & w_{k-1} & w_{k-2} & \cdots & w_1 \end{vmatrix} \tag{4.5.1}$$

and

$$w_k = \frac{(-1)^{k+1}}{2^k} \begin{vmatrix} c_1 & 2 & 0 & \cdots & 0 \\ c_2 & c_1 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_k & c_{k-1} & c_{k-2} & \cdots & c_1 \end{vmatrix}. \tag{4.5.2}$$

Proof Since

$$p(z)(1 - w(z)) = 1 + w(z),$$

it follows that

$$(1 + c_1 z + c_2 z^2 + \cdots)(1 - w_1 z - w_2 z^2 - \cdots) = 1 + w_1 z + w_2 z^2 + \cdots .$$

Comparison of the coefficients of z^n yields

$$c_n - c_{n-1} w_1 - c_{n-2} w_2 - \cdots - c_1 w_{n-1} = 2w_n, \quad n = 1, 2, \ldots, k. \tag{4.5.3}$$

The resulting system of equations can be written in matrix form as

$$\begin{pmatrix} 1 & 0 & \cdots & 0 & 0 \\ -w_1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ -w_{k-1} & -w_{k-2} & \cdots & -w_1 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_k \end{pmatrix} = \begin{pmatrix} 2w_1 \\ 2w_2 \\ \vdots \\ 2w_k \end{pmatrix}.$$

By Cramer's rule, c_k is given by

$$c_k = \begin{vmatrix} 1 & 0 & \cdots & 0 & 2w_1 \\ -w_1 & 1 & \cdots & 0 & 2w_2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ -w_{k-1} & -w_{k-2} & \cdots & -w_1 & 2w_k \end{vmatrix}$$

and this gives (4.5.1).

From (4.5.3), it follows that

$$c_{n-1}w_1 + c_{n-2}w_2 + \cdots + c_1 w_{n-1} + 2w_n = c_n.$$

The system of equations obtained by taking $n = 1, 2, \ldots, k$ can be written in matrix form as

$$\begin{pmatrix} 2 & 0 & \cdots & 0 & 0 \\ c_1 & 2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ c_{k-1} & c_{k-2} & \cdots & c_1 & 2 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_k \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_k \end{pmatrix}.$$

By Cramer's rule, w_k is given by (4.5.2). □

In particular, by using (4.5.1),

$$\begin{aligned} &c_1 = 2w_1, \quad c_2 = 2w_2 + 2w_1^2, \quad c_3 = 2w_3 + 4w_1w_2 + 2w_1^3, \\ &c_4 = 2w_4 + 4w_1w_3 + 2w_2^2 + 6w_1^2w_2 + 2w_1^4 \end{aligned} \tag{4.5.4}$$

and, by using (4.5.2),

$$\begin{aligned} &2w_1 = c_1, \quad 4w_2 = 2c_2 - c_1^2, \quad 8w_3 = 4c_3 - 4c_1c_2 + c_1^3 \\ &16w_4 = 8c_4 - 8c_1c_3 - 4c_2^2 + 6c_1^2c_2 - c_1^4. \end{aligned} \tag{4.5.5}$$

Estimates for the coefficients of Schwarz functions and Carathéodory functions can be obtained from each other whenever bounds for either class are known. For example, the estimate $|c_n| \leqslant 2$ for $p \in \mathcal{P}$, when applied to (4.5.4) yields the following coefficient estimates for $w \in \mathcal{B}_0$:

$$|w_1| \leqslant 1, \quad |w_2 + w_1^2| \leqslant 1, \quad |w_3 + 2w_1w_2 + w_1^3| \leqslant 1,$$

and

$$|w_4 + 2w_1w_3 + w_2^2 + 3w_1^2w_2 + w_1^4| \leqslant 1.$$

The estimate $|w_i| \leqslant 1$ for $w \in \mathcal{B}_0$ applied to (4.5.5) yields the following coefficient estimates for $p \in \mathcal{P}$:

$$|c_1| \leqslant 2, \quad |2c_2 - c_1^2| \leqslant 4, \quad |4c_3 - 4c_1c_2 + c_1^3| \leqslant 8,$$

and

$$|8c_4 - 8c_1c_3 - 4c_2^2 + 6c_1^2c_2 - c_1^4| \leqslant 16.$$

Theorem 4.5.4 *If $p(z) = 1 + c_1z + c_2z^2 + \cdots \in \mathcal{P}$, then*

$$|c_2 - \mu c_1^2| \leqslant 2\max\{1, |2\mu - 1|\}, \quad \mu \in \mathbb{C}. \tag{4.5.6}$$

For $0 < \mu < 1$, the following improved estimates hold:

$$|c_2 - \mu c_1^2| + \mu|c_1|^2 \leqslant 2, \quad 0 < \mu \leqslant 1/2, \tag{4.5.7}$$

and

$$|c_2 - \mu c_1^2| + (1 - \mu)|c_1|^2 \leqslant 2, \quad 1/2 \leqslant \mu < 1. \tag{4.5.8}$$

Let $p_1, p_2 \in \mathcal{P}$ be given by

$$p_1(z) = \frac{1+z}{1-z}, \quad \textit{and} \quad p_2(z) = \frac{1+z^2}{1-z^2}.$$

Inequality (4.5.6) is sharp for the function p_1 when $|2\mu - 1| > 1$ and for function p_2 when $|2\mu - 1| \leqslant 1$. Inequalities (4.5.7) and (4.5.8) are sharp for both p_1 and p_2.

Proof Since $p \in \mathcal{P}$, the function w defined by

$$p(z) = \frac{1 + w(z)}{1 - w(z)}$$

belongs to $\mathcal{B}_0$. If $w(z) = \sum_{k=1}^{\infty} w_k z^k$, then, by (4.5.4),

$$c_1 = 2w_1, \quad c_2 = 2w_2 + 2w_1^2$$

and therefore

$$c_2 - \mu c_1^2 = 2\big(w_2 - (2\mu - 1)w_1^2\big).$$

The results now follow by applying Lemma 3.4.4. The sharpness of the estimates are also readily established. □

Corollary 4.5.5 *If $p(z) = 1 + c_1z + c_2z^2 + \cdots \in \mathcal{P}$, then*

$$|c_{2n} - \mu c_n^2| \leqslant 2\max\{1, |2\mu - 1|\}, \quad \mu \in \mathbb{C}.$$

For $0 < \mu < 1$, the following improved estimates hold:

$$|c_{2n} - \mu c_n^2| + \mu|c_n|^2 \leqslant 2, \quad 0 < \mu \leqslant 1/2$$

and

$$|c_{2n} - \mu c_n^2| + (1-\mu)|c_n|^2 \leqslant 2, \quad 1/2 \leqslant \mu < 1.$$

The results are sharp for one of the functions

$$p(z) = \frac{1+z^n}{1-z^n}, \quad \textit{and} \quad p(z) = \frac{1+z^{2n}}{1-z^{2n}}.$$

Proof Given $p \in \mathcal{P}$, the function $\tilde{p}_n$ defined by (4.4.8) belongs to $\mathcal{P}$. Since $\tilde{p}_n(z) = 1 + c_n z + c_{2n} z^2 + \cdots$, Theorem 4.5.4 applied to $\tilde{p}_n$ gives the desired results. □

In Theorem 4.1.8, several simple transformations are shown to preserve the class $\mathcal{P}$ and two of them were used to prove the growth and distortion theorems. In the next theorem, for a function $p \in \mathcal{P}$, it is shown that the function q defined by the transformation

$$q(z) = \frac{1-z^2}{z} - \frac{(1-z)^2}{z} p(z)$$

satisfies $\mathrm{Re}\, q(z) \geqslant 0$ for all $z \in \mathbb{D}$. By using this fact, estimates are obtained on the difference of successive coefficients for functions with positive real part.

Theorem 4.5.6 *If $p \in \mathcal{P}$, then*

$$q(z) = \frac{1-z^2}{z} - \frac{(1-z)^2}{z} p(z) \tag{4.5.9}$$

is analytic in $\mathbb{D}$ and satisfies $\mathrm{Re}\, q(z) \geqslant 0$ for all $z \in \mathbb{D}$. Equality $\mathrm{Re}\, q(z) = 0$ holds for some $z \in \mathbb{D}$ if and only if the corresponding function $p : \mathbb{D} \to \mathbb{C}$ is given by

$$p(z) = \frac{1+z}{1-z}.$$

Proof Choose ρ in the interval $(0, 1)$, and define $q_\rho : \overline{\mathbb{D}} \to \mathbb{C}$ by

$$z q_\rho(z) = \left(1 - z^2\right) - (1-z)^2 p(\rho z).$$

The function q_ρ is analytic in $\overline{\mathbb{D}}$ and $q_\rho \to q$ as $\rho \to 1^-$. For $z = \mathrm{e}^{\mathrm{i}\theta}$,

$$\mathrm{Re} q_\rho(z) = 2(1 - \cos\theta)\mathrm{Re}\, p(\rho z) \geqslant 0.$$

Since the minimum of a harmonic function occurs on the boundary, it follows that $\mathrm{Re} q_\rho(z) \geqslant 0$ for all $z \in \overline{\mathbb{D}}$, and hence, $\mathrm{Re} q(z) \geqslant 0$ for all $z \in \mathbb{D}$. In the special case $p(z) = (1+z)/(1-z)$, the expression simplifies to $q(z) = 0$. If equality holds in $\mathrm{Re} q(z) \geqslant 0$, then $q(z)$ is identically zero, and hence $p(z) = (1+z)/(1-z)$. □

Theorem 4.5.7 is useful in getting the sharp estimate on the difference of successive coefficients for the starlike and convex functions.

Theorem 4.5.7 *If* $p(z) = 1 + \sum_{k=1}^{\infty} c_k z^k \in \mathcal{P}$, *then*

$$|c_{n+1} - c_n| \leqslant (2n+1)|2 - c_1|, \quad n \geqslant 1. \tag{4.5.10}$$

The factor $2n + 1$ *cannot be replaced by a smaller factor.*

Proof By Theorem 4.5.6, the function

$$q(z) = \frac{1 - z^2}{z} - \frac{(1-z)^2}{z} p(z) = (2 - c_1) + \sum_{n=1}^{\infty} d_n z^n$$

is analytic in $\mathbb{D}$ and $\operatorname{Re} q(z) \geqslant 0$ for all $z \in \mathbb{D}$. If $\operatorname{Re} q(z) = 0$ for some $z \in \mathbb{D}$, then $p(z) = (1+z)/(1-z)$ and the results hold.

Assume now that $\operatorname{Re} q(z) > 0$ for all $z \in \mathbb{D}$. Then, by Theorem 4.5.1,

$$|d_n| \leqslant 2\operatorname{Re}(2 - c_1), \quad n = 1, 2, \ldots . \tag{4.5.11}$$

From the identity

$$2 - \frac{1-z}{z}(p(z) - 1) = \frac{q(z)}{1-z},$$

it follows that

$$2 - c_1 + \sum_{k=1}^{\infty} (c_k - c_{k+1}) z^k = \left(1 + \sum_{k=1}^{\infty} z^k\right)\left(2 - c_1 + \sum_{k=1}^{\infty} d_k z^k\right).$$

Equating the coefficient of z^n yields

$$c_n - c_{n+1} = 2 - c_1 + d_1 + \cdots + d_n$$

and hence, using (4.5.11),

$$\begin{aligned} |c_{n+1} - c_n| &\leqslant |2 - c_1| + 2n\operatorname{Re}(2 - c_1) \\ &\leqslant (2n+1)\,|2 - c_1|\,. \end{aligned}$$

Consider the function $p_\theta \in \mathcal{P}$ given by

$$p_\theta(z) = \frac{1 - z^2}{1 - 2z\cos\theta + z^2}. \tag{4.5.12}$$

By Example 4.1.7, the function p_θ defined by (4.5.12) belongs to $\mathcal{P}$ and equality holds in (4.5.10) for the function p_0. Also, for this function p_θ, $c_n = 2\cos n\theta$ and hence

$$\lim_{\theta\to 0}\frac{c_{n+1}-c_n}{2-c_1} = \lim_{\theta\to 0}\frac{\cos(n+1)\theta - \cos n\theta}{1-\cos\theta} = -(2n+1),$$

proving that the factor $(2n+1)$ in (4.5.10) cannot be replaced by a smaller factor. □

A characterization of a function with positive real part in terms of its coefficients is particularly useful. The condition $\sum_{k=1}^{\infty}|c_k| \leqslant 1$, given in Theorem 4.1.2, is sufficient for the function $p(z) = 1 + \sum_{k=1}^{\infty} c_k z^k$ to belong to $\mathcal{P}$. However, this condition is not necessary in general. It becomes necessary when all the coefficients c_k are non-positive. The following theorem provides a necessary and sufficient condition for $p \in \mathcal{P}$ in terms of its Taylor coefficients.

Theorem 4.5.8 (Carathéodory's positivity criterion) *Let $p(z) = 1 + \sum_{k=1}^{\infty} c_k z^k$ be analytic in $\mathbb{D}$. Then, $p \in \mathcal{P}$ if and only if*

$$\sum_{m=0}^{N}\sum_{n=0}^{N} c_{m-n}\lambda_m\overline{\lambda_n} \geqslant 0 \tag{4.5.13}$$

for each complex numbers $\lambda_0, \lambda_1, \ldots, \lambda_N$, where $c_0 = 2$ and $c_{-m} = \overline{c_m}$ for $m > 0$.

Proof From the Herglotz's representation theorem, there is a non-decreasing function μ on $[0, 2\pi]$ with $\int_0^{2\pi} \mathrm{d}\mu(t) = 1$ satisfying

$$p(z) = \int_0^{2\pi} \frac{1+z\mathrm{e}^{-\mathrm{i}t}}{1-z\mathrm{e}^{-\mathrm{i}t}}\mathrm{d}\mu(t).$$

Thus,

$$c_n = 2\int_0^{2\pi} \mathrm{e}^{-\mathrm{i}n\theta}\mathrm{d}\mu(\theta), \quad n \geqslant 1. \tag{4.5.14}$$

For $n \leqslant 0$, c_n satisfies (4.5.14). Using (4.5.14), it follows that

$$\begin{aligned} 0 &\leqslant \int_0^{2\pi}\left|\sum_{n=0}^{N}\lambda_n\mathrm{e}^{-\mathrm{i}n\theta}\right|^2 \mathrm{d}\mu(\theta) \\ &= \sum_{m=0}^{N}\sum_{n=0}^{N}\int_0^{2\pi}\lambda_m\overline{\lambda_n}\mathrm{e}^{-\mathrm{i}(n-m)\theta}\mathrm{d}\mu(\theta) \\ &= \frac{1}{2}\sum_{m=0}^{N}\sum_{n=0}^{N} c_{m-n}\lambda_m\overline{\lambda_n}. \end{aligned}$$

Conversely, setting $\lambda_n = z^n$, it follows that

$$\begin{aligned}
0 &\leqslant \sum_{m=0}^{\infty}\sum_{n=0}^{\infty} c_{m-n}\lambda_m\overline{\lambda_n} \\
&= 2\sum_{m=0}^{\infty}|z|^{2m} + 2\,\mathrm{Re}\sum_{n=0}^{\infty}\sum_{m=n+1}^{\infty} c_{m-n}z^{m-n}|z|^{2n} \\
&= \frac{2}{1-|z|^2} + 2\mathrm{Re}\left(\sum_{n=0}^{\infty}|z|^{2n}\left(\sum_{m=n+1}^{\infty} c_{m-n}z^{m-n}\right)\right) \\
&= \frac{2}{1-|z|^2} + 2\mathrm{Re}\left(\sum_{n=0}^{\infty}|z|^{2n}(p(z)-1)\right) \\
&= \frac{2}{1-|z|^2}\mathrm{Re}\,p(z).
\end{aligned}$$

This proves that $\mathrm{Re}\,p(z) > 0$. □

Theorem 4.5.8 provides an alternative proof for the estimate $|c_N| \leqslant 2$ when the function $p(z) = 1 + \sum_{k=1}^{\infty} c_k z^k$ belongs to $\mathcal{P}$. Since the class $\mathcal{P}$ is rotationally invariant, assume that $c_N \leqslant 0$ by applying a suitable rotation. Choosing $\lambda_0 = \lambda_N = 1$ and $\lambda_i = 0$ for $i = 1, 2, 3, \ldots, N-1$, the expression in (4.5.13) simplifies to

$$c_0 + c_{-N} + c_N + c_0 \geqslant 0,$$

which is equivalent to $\mathrm{Re}\,c_N \geqslant -2$.

The quadratic form given by (4.5.13) is a Hermitian form, since its associated matrix is Hermitian. A *principal submatrix* of a matrix is obtained by selecting a subset of indices $\{i_1, i_2, \ldots, i_k\}$ and retaining only the rows and columns corresponding to these indices. The *principal minor* is the determinant of such a submatrix. For example, the submatrix formed by keeping rows and columns numbered 1, 3, and 5 is a principal submatrix, and its determinant is a principal minor. A *leading principal submatrix* is a special case where the first k rows and columns (with consecutive indices) are retained.

Another important consequence of Theorem 4.5.8 is the following result, which follows from the characterization of positive semi-definite Hermitian forms.

Theorem 4.5.9 *Let $p(z) = 1 + \sum_{k=1}^{\infty} c_k z^k$ be analytic in $\mathbb{D}$. Then, $p \in \mathcal{P}$ if and only if, for each $k \in \mathbb{N}$, the Hermitian Toeplitz matrix*

$$T_k := \begin{pmatrix}
2 & c_1 & c_2 & \cdots & c_k \\
\overline{c_1} & 2 & c_1 & \cdots & c_{k-1} \\
\overline{c_2} & \overline{c_1} & 2 & \cdots & c_{k-2} \\
\vdots & \vdots & \vdots & \ddots & \vdots \\
\overline{c_k} & \overline{c_{k-1}} & \overline{c_{k-2}} & \cdots & 2
\end{pmatrix}$$

is positive semi-definite. That is, all of its principal minors are non-negative.

Proof The result follows from the fact that $p \in \mathcal{P}$ if and only if the Hermitian form associated with (4.5.13) is non-negative for all choices of complex scalars. By Sylvester's criterion, this is equivalent to the condition that all principal minors of the associated Hermitian Toeplitz matrices are non-negative. □

Corollary 4.5.10 *If* $p(z) = 1 + \sum_{k=1}^{\infty} c_k z^k \in \mathcal{P}$, *then*

$$\left| c_{2k} - \frac{1}{2} c_k^2 \right| \leqslant 2 - \frac{1}{2}|c_k|^2. \tag{4.5.15}$$

In particular,

$$\left| c_2 - \frac{1}{2} c_1^2 \right| \leqslant 2 - \frac{1}{2}|c_1|^2. \tag{4.5.16}$$

Proof By the non-negativity of the special minor obtained by choosing rows with indices $1, r, s$, it follows that

$$\begin{vmatrix} 2 & c_r & c_s \\ \bar{c}_r & 2 & c_{s-r} \\ \bar{c}_s & \bar{c}_{s-r} & 2 \end{vmatrix} \geqslant 0.$$

This simplifies to

$$|c_r|^2 + |c_s|^2 + |c_{s-r}|^2 \leqslant 4 + \mathrm{Re}(c_s \bar{c}_r \bar{c}_{s-r}),$$

which can be rewritten as

$$\left| c_s - \frac{1}{2} c_r c_{s-r} \right|^2 \leqslant \frac{1}{4}(4 - |c_r|^2).(4 - |c_{s-r}|^2).$$

The latter inequality yields (4.5.15) with $s = 2k$, $r = k$. The inequality (4.5.16) is a special case of (4.5.15) when $k = 1$. □

The following result gives a representation for the coefficients c_2 and c_3 in terms of c_1, assuming $c_1 \in \mathbb{R}$. The formula for c_2 follows directly from (4.5.16).

Theorem 4.5.11 *If* $p(z) = 1 + c_1 z + c_2 z^2 + \cdots \in \mathcal{P}$ *with* $-2 \leqslant c_1 \leqslant 2$, *then*

$$2c_2 = c_1^2 + x(4 - c_1^2) \tag{4.5.17}$$

and

$$4c_3 = c_1^3 + 2c_1(4 - c_1^2)x - c_1(4 - c_1^2)x^2 + 2(4 - c_1^2)(1 - |x|^2)y \tag{4.5.18}$$

for some $x, y \in \mathbb{C}$ *with* $|x| \leqslant 1$ *and* $|y| \leqslant 1$.

Proof Assume $c_1 \in \mathbb{R}$. By Theorem 4.5.9, the determinant

$$\det(T_2) = \begin{vmatrix} 2 & c_1 & c_2 \\ c_1 & 2 & c_1 \\ \bar{c}_2 & c_1 & 2 \end{vmatrix}$$

is non-negative. A direct computation gives:

$$\det(T_2) = \frac{1}{2}\left((4 - c_1^2)^2 - |2c_2 - c_1^2|^2\right) \geqslant 0,$$

which implies

$$|2c_2 - c_1^2| \leqslant 4 - c_1^2.$$

Therefore,

$$2c_2 = c_1^2 + x(4 - c_1^2)$$

for some $x \in \mathbb{C}$ with $|x| \leqslant 1$, proving (4.5.17).

To obtain (4.5.18), consider $\det(T_3) \geqslant 0$, where T_3 is the Toeplitz matrix of order 4:

$$T_3 = \begin{pmatrix} 2 & c_1 & c_2 & c_3 \\ c_1 & 2 & c_1 & c_2 \\ \bar{c}_2 & c_1 & 2 & c_1 \\ \bar{c}_3 & \bar{c}_2 & c_1 & 2 \end{pmatrix}.$$

Since

$$\begin{aligned} 16(4 - c_1^2)\det(T_3) = &\left(2\left(4 - c_1^2\right)^2 - 2\left|2c_2 - c_1^2\right|^2\right)^2 \\ &- \left|\left(4c_3 - 4c_1c_2 + c_1^3\right)\left(4 - c_1^2\right) + c_1\left(2c_2 - c_1^2\right)^2\right|^2, \end{aligned}$$

the inequality $\det(T_3) \geqslant 0$ is equivalent to the inequality

$$\left|\left(4c_3 - 4c_1c_2 + c_1^3\right)(4 - c_1^2) + c_1(2c_2 - c_1^2)^2\right| \leqslant 2(4 - c_1^2)^2 - 2|2c_2 - c_1^2|^2.$$

Substituting (4.5.17), this becomes

$$\left|\left(4c_3 - 4c_1c_2 + c_1^3\right)(4 - c_1^2) + c_1(4 - c_1^2)^2x^2\right| \leqslant 2(4 - c_1^2)^2(1 - |x|^2).$$

If $c_1 \neq \pm 2$, solving for $4c_3$ and substituting the expression for c_2 from (4.5.17) yields

$$4c_3 = c_1^3 + 2c_1(4 - c_1^2)x - c_1(4 - c_1^2)x^2 + 2(4 - c_1^2)(1 - |x|^2)y,$$

for some $y \in \mathbb{C}$ with $|y| \leqslant 1$, completing the proof.

In the case $c_1 = \pm 2$, we have $4 - c_1^2 = 0$, and the above expressions degenerate. From Exercise 4, it follows that $c_3 = \pm 2$ in this case, and the formula holds trivially. □

Theorem 4.5.12 *If $p(z) = 1 + c_1 z + c_2 z^2 + \cdots \in \mathcal{P}$, then the following sharp bounds hold:*

(i) $|c_1^2 - c_2| \leqslant 2$,
(ii) $|c_1^3 - 2c_1 c_2 + c_3| \leqslant 2$,
(iii) $|c_1^4 + 2c_1 c_3 + c_2^2 - 3c_1^2 c_2 - c_4| \leqslant 2$,
(iv) $|c_1^5 + 3c_1 c_2^2 + 3c_1^2 c_3 - 4c_1^3 c_2 - 2c_1 c_4 - 2c_2 c_3 + c_5| \leqslant 2$,
(v) $|c_1^6 + 6c_1^2 c_2^2 + 4c_1^3 c_3 + 2c_1 c_5 + 2c_2 c_4 + c_3^2 - c_2^3 - 5c_1^4 c_2 - 3c_1^2 c_4 - 6c_1 c_2 c_3 - c_6| \leqslant 2$.

Proof The expressions within the modulus are the coefficients of the function $q := 1/p$. To see this, consider the series expansion

$$q(z) = \frac{1}{p(z)} = 1 + d_1 z + d_2 z^2 + \cdots .$$

Since

$$1 = p(z)q(z) = (1 + c_1 z + c_2 z^2 + \cdots)(1 + d_1 z + d_2 z^2 + \cdots),$$

equating the coefficients of z^k on both sides yields

$$c_k + c_{k-1} d_1 + c_{k-2} d_2 + \cdots + c_1 d_{k-1} + d_k = 0.$$

In particular, choosing $k = 1, 2, 3$ gives

$$c_1 + d_1 = 0, \quad d_2 + c_1 d_1 + c_2 = 0, \quad c_3 + c_2 d_1 + c_1 d_2 + d_3 = 0.$$

The first two of these equations imply that $d_1 = -c_1$, and

$$d_2 = -c_2 - c_1 d_1 = c_1^2 - c_2.$$

Substituting these into the third equation gives

$$d_3 = -c_3 - c_2 d_1 - c_1 d_2 = -c_3 + 2c_1 c_2 - c_1^3.$$

Continuing in this way yields expressions for d_4, d_5, d_6 and, therefore,

$$\begin{aligned}\frac{1}{p(z)} &= 1 - c_1 z + \left(c_1^2 - c_2\right) z^2 + \left(-c_1^3 + 2c_2c_1 - c_3\right) z^3 \\ &\quad + \left(c_1^4 - 3c_2c_1^2 + 2c_3c_1 + c_2^2 - c_4\right) z^4 \\ &\quad + \left(-c_1^5 + 4c_2c_1^3 - 3c_3c_1^2 - 3c_2^2c_1 + 2c_4c_1 + 2c_2c_3 - c_5\right) z^5 \\ &\quad + \Big(c_1^6 - 5c_2c_1^4 + 4c_3c_1^3 + 6c_2^2c_1^2 - 3c_4c_1^2 - 6c_2c_3c_1 \\ &\quad + 2c_5c_1 - c_2^3 + c_3^2 + 2c_2c_4 - c_6\Big) z^6 + \cdots .\end{aligned}$$

Since $1/p \in \mathcal{P}$ whenever $p \in \mathcal{P}$, it follows that each coefficient in the expansion of $1/p$ is bounded by 2. □

Exercises

1. Using Schwarz's lemma, prove that $|p(z) - 1| < 2r/(1 - r)$ for $p \in \mathcal{P}$.
2. If $p : \mathbb{D} \to \mathbb{C}$ is analytic, show that the function

$$P(z) = \frac{1}{z^\gamma} \int_0^z p(\tau)\tau^{\gamma-1} d\tau, \quad z \in \mathbb{D},$$

   is analytic and

$$P(z) = \int_0^1 p(tz)t^{\gamma-1} \mathrm{d}t.$$

   Hence, prove that $P \in \mathcal{P}$ whenever $p \in \mathcal{P}$.
3. If $p \in \mathcal{P}(\alpha)$, show that the following sharp estimate holds:

$$|p(z) - zp'(z) - 1| \leqslant 2(1 - \alpha)\frac{|z|^2}{(1 - |z|)^2}.$$

   What is the corresponding estimate for the functions in the class $\mathcal{P}[A, B]$?
4. If $p(z) = 1 + c_1 z + c_2 z^2 + \cdots \in \mathcal{P}$, show that $|c_N| = 2$ for some N if and only if $p(z) = (1 + \alpha z^N)/(1 - \alpha z^N)$ where $|\alpha| = 1$.
5. If $p(z) = 1 + c_1 z + c_2 z^2 + \cdots \in \mathcal{P}$, prove the following inequalities:

   (i) $|2c_1^2 - c_2| \leqslant 6$,
   (ii) $|-6c_1^3 + 7c_1c_2 - 2c_3| \leqslant 24$,
   (iii) $|24c_1^4 - 46c_1^2c_2 + 22c_1c_3 + 7c_2^2 - 6c_4| \leqslant 120$,
   (iv) $|-120c_1^5 + 96c_4c_1 + 50c_2c_3 + 326c_1^3c_2 - 202c_1^2c_3 - 127c_1c_2^2 - 24c_5| \leqslant 720$.

6. Find the Taylor series of the function q given by (4.5.9) and use it prove that

$$|c_{n-1} - 2c_n + c_{n+1}| \leqslant 2\text{Re}(2 - c_1) \leqslant 2|2 - c_1|, \quad n \geqslant 2$$

for $p(z) = 1 + c_1 z + c_2 z^2 + \cdots \in \mathcal{P}$.

7. If $f \in \mathcal{A}$ satisfies $\text{Re}(f'(z)) > \alpha$ for $\alpha < 1$, find a sharp estimate for its kth coefficient and obtain the growth and distortion estimates for the function f.

8. If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{A}$ satisfies $\text{Re}(f'(z) + zf''(z)) > \alpha$ for $\alpha < 1$, show that the coefficient a_k of f satisfies the sharp estimate

$$|a_k| \leqslant \frac{2(1-\alpha)}{k^2}.$$

Also, show that

$$|f(z)| \leqslant (1-\alpha)\left(\frac{\pi^2}{3} - 1\right) + \alpha.$$

9. If $p \in \mathcal{P}$ and $z = re^{it}$, show that

$$\frac{1}{2\pi}\int_0^{2\pi} |p(z)|^2 \mathrm{d}t \leqslant \frac{1+3r^2}{1-r^2}.$$

10. Let $p \in \mathcal{P}$ and $z = re^{it}$. Show that

$$\frac{1}{2\pi}\int_0^{2\pi} |p'(z)| \mathrm{d}t \leqslant \frac{2}{1-r^2}.$$

11. Let $p \in \mathcal{P}$, $p(z) = 1 + c_1 z + c_2 z^2 + \cdots$, and $\lambda_n \geqslant 0$, for $n \in \mathbb{N}$. If $q(z) = \sum_{n=1}^{\infty} \lambda_n p_n z^n$ is analytic in $\mathbb{D}$, and $\text{Re}\, q(z) \leqslant M$ for some $M > 0$ and all $z \in \mathbb{D}$, show that

$$\sum_{n=1}^{\infty} \lambda_n |p_n|^2 \leqslant 2M.$$

12. For every $p(z) = 1 + c_1 z + c_2 z^2 + \cdots \in \mathcal{P}$, and every positive integer n, show that there exists ζ, with $|\zeta| = 1$, such that

$$\sum_{k=1}^{n} \frac{1}{k}\left|p_k - \zeta^k\right|^2 \leqslant \sum_{k=1}^{n} \frac{1}{k}.$$

13. Let $p(z) = 1 + c_1 z + c_2 z^2 + \cdots \in \mathcal{P}$, and $c_1 \geqslant 0$. Show that

$$\begin{aligned} 8c_4 =& c_1^4 + \left(4 - c_1^2\right)\zeta\left(c_1^2\left(\zeta^2 - 3\zeta + 3\right) + 4\zeta\right) \\ & - 4\left(4 - c_1^2\right)\left(1 - |\zeta|^2\right)\left(c_1(\zeta - 1)\eta + \bar{\zeta}\eta^2 - \left(1 - |\eta|^2\right)\xi\right) \end{aligned}$$

for some $\zeta, \eta, \xi \in \overline{\mathbb{D}}$.

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[1] Most of the results in this chapter can be found in [7]. Functions with positive real part were first studied by Carathéodory in [1, 2]. The Herglotz's representation theorem (Theorem 4.2.3) and the distortion theorem (Theorem 4.3.4) are due respectively to Herglotz [9] and Goluzin [6]. The Janowski functions were introduced in [10, 11]. Robertson in [15] proved Theorem 4.5.6 and Theorem 4.5.7. Carathéodory's positivity criterion (Theorem 4.5.8) and Theorem 4.5.9 are treated in Chap. 9 of the book by Grenander and Szegö [8]. In [14], Libera and Zlotkiewicz gave a proof of Theorem 4.5.11. For a survey on coefficient estimates, the work of Cho et al. [4] is suggested. We have selected Exercise 13 from Kwon et al. [12] (see also [3]). Finally, Lecko in [13] obtained several coefficient inequalities for the class $\mathcal{P}(\alpha)$.

Chapter 5
Differential Subordinations

Abstract Differential subordination generalizes the concept of differential inequalities from the real line to the complex plane. This chapter focuses on the theory of first and second-order differential subordinations and their applications in deriving sufficient conditions for functions p to belong to certain subclasses of the class $\mathcal{P}$.

5.1 Dominants of Differential Subordinations

In Sect. 4.1, it was noted that a function $f \in \mathcal{A}$ is starlike if and only if $zf'(z)/f(z) \in \mathcal{P}$, and it is convex whenever $1 + zf''(z)/f'(z) \in \mathcal{P}$. Geometrically, it is evident that every convex function is starlike. Analytically, this means the following implication holds:

$$\operatorname{Re}\left(1 + \frac{zf''(z)}{f'(z)}\right) > 0 \quad \Longrightarrow \quad \operatorname{Re}\left(\frac{zf'(z)}{f(z)}\right) > 0.$$

In fact, it will be shown in this chapter that the stronger implication

$$\operatorname{Re}\left(1 + \frac{zf''(z)}{f'(z)}\right) > 0 \quad \Longrightarrow \quad \operatorname{Re}\left(\frac{zf'(z)}{f(z)}\right) > \frac{1}{2} \tag{5.1.1}$$

also holds.

Let $p : \mathbb{D} \to \mathbb{C}$ be defined by $p(z) = zf'(z)/f(z)$, and consider the sets

$$\Omega = \left\{w : \operatorname{Re} w > 0\right\} \quad \text{and} \quad \Delta = \left\{w : \operatorname{Re} w > \frac{1}{2}\right\}.$$

Then the implication (5.1.1) can be restated as

$$p(z) + \frac{zp'(z)}{p(z)} \in \Omega \quad \Longrightarrow \quad p(z) \in \Delta,$$

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or, in terms of subordination,

$$p(z) + \frac{zp'(z)}{p(z)} \prec h(z) \quad \Longrightarrow \quad p(z) \prec q(z)$$

where the functions $h, q : \mathbb{D} \to \mathbb{C}$ are given by

$$h(z) = \frac{1+z}{1-z} \quad \text{and} \quad q(z) = \frac{1}{1-z}.$$

This chapter focuses on such differential implications.

Let p be analytic in the unit disk $\mathbb{D}$ and let $\psi : \mathbb{C}^3 \times \mathbb{D} \to \mathbb{C}$. For given domains Ω and Δ in $\mathbb{C}$, consider the implication:

$$\{\psi(p(z), zp'(z), z^2p''(z); z) : z \in \mathbb{D}\} \subset \Omega \Longrightarrow \{p(z) : z \in \mathbb{D}\} \subset \Delta. \tag{5.1.2}$$

If Ω and Δ are simply connected domains that are not equal to the entire complex plane $\mathbb{C}$, then there exist univalent functions h and q mapping $\mathbb{D}$, respectively, onto Ω and Δ, such that $h(0) = \psi(p(0), 0, 0; 0)$ and $q(0) = p(0)$. Moreover, if $\psi(p(z), zp'(z), z^2p''(z); z)$ is analytic, then (5.1.2) can be expressed in terms of subordination as

$$\psi(p(z), zp'(z), z^2p''(z); z) \prec h(z) \Longrightarrow p(z) \prec q(z). \tag{5.1.3}$$

The inclusion

$$\{\psi(p(z), zp'(z), z^2p''(z); z) : z \in \mathbb{D}\} \subset \Omega$$

is referred to as a second-order differential inclusion, while the subordination

$$\psi(p(z), zp'(z), z^2p''(z); z) \prec h(z)$$

is called as a second-order differential subordination.

Definition 5.1.1 An analytic function p that satisfies the second-order differential subordination

$$\psi(p(z), zp'(z), z^2p''(z); z) \prec h(z), \tag{5.1.4}$$

is called a *solution* of the differential subordination (5.1.4). A univalent function q is a *dominant* (of the solutions) of the differential subordination (5.1.4) if $p \prec q$ for all p satisfying (5.1.4). If $\widetilde{q}$ is a dominant of (5.1.4) and $\widetilde{q} \prec q$ for all dominants q of (5.1.4), then $\widetilde{q}$ is the *best dominant* of (5.1.4).

A dominant of (5.1.4) need not be a solution of (5.1.4). However, if a dominant q of (5.1.4) is a solution, then q is the best dominant. Indeed, if q_* is any dominant of (5.1.4), then $p \prec q_*$ for every function p satisfying (5.1.4). In particular, since q satisfies (5.1.4), it follows that $q \prec q_*$. This holds for all dominants q_* of (5.1.4), and hence q is the best dominant.

Let $\mathcal{H}[a, n]$ denote the class of all analytic functions of the form $f(z) = a + \sum_{k=n}^{\infty} a_k z^k$. With the additional condition $p \in \mathcal{H}[a, n]$, then p is referred to as an *(a,n)-solution*, q an *(a,n)-dominant*, and $\widetilde{q}$ the *best (a,n)-dominant*. The best dominant of (5.1.4), if it exists, is unique up to a rotation of the unit disk. That is, if q_1 and q_2 are two best dominants, then $q_1(z) = q_2(\mathrm{e}^{\mathrm{i}\theta} z)$ for some real θ. This fact follows from the next result.

Theorem 5.1.2 *If $q_1, q_2 : \mathbb{D} \to \mathbb{C}$ are univalent and subordinate to each other, then $q_1(z) = q_2(e^{i\theta} z)$.*

Proof If $q_1 \prec q_2$ and $q_2 \prec q_1$, then, from the definition of subordination, there are analytic functions w_1 and w_2 in $\mathbb{D}$ satisfying $q_1(z) = q_2(w_1(z))$ and $q_2(z) = q_1(w_2(z))$ with $w_1(0) = w_2(0) = 0$ and $|w_1(z)| < 1$, $|w_2(z)| < 1$ for $z \in \mathbb{D}$. Hence,

$$q_1(z) = q_1(w_2(w_1(z))) = q_1((w_2 \circ w_1)(z)).$$

Univalence of q_1 implies that $w_2 \circ w_1(z) = z$ for $z \in \mathbb{D}$. Similarly, it can be shown that $w_1 \circ w_2(z) = z$ for $z \in \mathbb{D}$. Therefore, $w_1 = w_2^{-1}$ is a conformal mapping of the unit disk $\mathbb{D}$ onto itself. This implies that w_1 must be of the form

$$w_1(z) = \mathrm{e}^{\mathrm{i}\theta} \frac{z - \alpha}{1 - \overline{\alpha} z}, \quad |\alpha| < 1.$$

Since $w_1(0) = 0$, then $w_1(z) = \mathrm{e}^{\mathrm{i}\theta} z$. Thus, $q_1(z) = q_2(\mathrm{e}^{\mathrm{i}\theta} z)$. □

The subordination implication (5.1.2) depends on the function ψ, and the sets Ω and Δ while the differential subordination (5.1.3) involve the functions ψ, h, and q. Given any two among these three components, determining the third constitutes a fundamental problem in the theory of differential subordinations:

1. Given Ω and Δ, determine conditions on ψ so that (5.1.2) holds. Such a ψ is called an *admissible function*. Equivalently, given univalent functions h and q, find a class of admissible functions $\Psi(h, q)$ for which the differential subordination (5.1.3) holds.
2. Given ψ and Ω, determine Δ such that (5.1.2) holds. Furthermore, find the "smallest" such Δ. For the differential subordination in (5.1.3), this becomes the problem of finding a dominant q and the best dominant.
3. Given ψ and Δ, determine Ω such that (5.1.2) holds. Furthermore, find the "largest" such Ω. For the differential subordination in (5.1.3), given ψ and dominant q, the problem is to find the largest class of univalent functions h such that (5.1.3) holds.

5.2 Miller-Mocanu Theorems

The development of the theory of differential subordination relies on the following fundamental lemma.

Lemma 5.2.1 *(Clunie-Jack-Miller-Mocanu Lemma) Let $z_0 = r_0 e^{i\theta_0}$ be a point in $\mathbb{D}$. Let $g(z) = g_n z^n + g_{n+1} z^{n+1} + \cdots$ be continuous in $\overline{\mathbb{D}}_{r_0}$ and analytic in $\mathbb{D}_{r_0} \cup \{z_0\}$ with $g(z) \not\equiv 0$ and $n \geqslant 1$. If*

$$|g(z_0)| = \max_{|z| \leqslant r_0} |g(z)|, \tag{5.2.1}$$

then there exists an $m \geqslant n$ such that

$$\frac{z_0 g'(z_0)}{g(z_0)} = m \quad \text{and} \quad Re\left(\frac{z_0 g''(z_0)}{g'(z_0)} + 1\right) \geqslant m.$$

Proof For $z = r_0 \mathrm{e}^{\mathrm{i}\theta}$, let $g(z) = g(r_0 \mathrm{e}^{\mathrm{i}\theta}) = R(r_0, \theta)\mathrm{e}^{\mathrm{i}\Phi(r_0,\theta)}$. By differentiating with respect to θ, then

$$\begin{aligned} \mathrm{i}z g'(z) &= \mathrm{e}^{\mathrm{i}\Phi} \frac{\partial R}{\partial \theta} + \mathrm{i} R \mathrm{e}^{\mathrm{i}\Phi} \frac{\partial \Phi}{\partial \theta} \\ &= \frac{g(z)}{R} \frac{\partial R}{\partial \theta} + \mathrm{i} g(z) \frac{\partial \Phi}{\partial \theta} \end{aligned}$$

so that

$$\frac{z g'(z)}{g(z)} = \frac{\partial \Phi}{\partial \theta} - \frac{\mathrm{i}}{R} \frac{\partial R}{\partial \theta}. \tag{5.2.2}$$

Since z_0 is a point of maximum of $R(r_0, \theta)$, it follows that

$$\frac{\partial R}{\partial \theta}(z_0) = 0.$$

Consequently, by (5.2.2),

$$\frac{z_0 g'(z_0)}{g(z_0)} = \frac{\partial \Phi}{\partial \theta}(z_0) = m,$$

where m is a real number.

To show $m \geqslant n$, consider the function defined by

$$h(z) := \frac{g(z_0 z)}{g(z_0) z^{n-1}} = \frac{1}{g(z_0)} \left(g_n z_0^n z + g_{n+1} z_0^{n+1} z^2 + \cdots\right)$$

in $\overline{\mathbb{D}}$. The function h is continuous on $\overline{\mathbb{D}}$ and analytic on $\mathbb{D} \cup \{1\}$. For $|z| \leqslant r < 1$, the maximum principle implies that

$$|h(z)| \leqslant \max_{|z|=r} |h(z)| = \max_{|z|=r} \left|\frac{g(z_0 z)}{g(z_0) z^{n-1}}\right| = \frac{1}{|g(z_0)| r^{n-1}} \max_{|z|=r} |g(z_0 z)|.$$

Using (5.2.1), it follows that $|h(z)| \leqslant 1/r^{n-1}$. Letting $r \to 1$, this yields $|h(z)| \leqslant 1$. Since $h(0) = 0$, the Schwarz Lemma implies that $|h(z)| \leqslant |z|$ for $z \in \mathbb{D}$, or equivalently,

$$\left|\frac{g(z_0 z)}{g(z_0) z^{n-1}}\right| \leqslant |z|,$$

and therefore

$$\left|\frac{g(z_0 z)}{g(z_0)}\right| \leqslant |z|^n.$$

In particular, for $z = r$, with $0 \leqslant r < 1$, this gives

$$\operatorname{Re}\frac{g(z_0 r)}{g(z_0)} \leqslant \left|\frac{g(z_0 r)}{g(z_0)}\right| \leqslant r^n. \tag{5.2.3}$$

Since $m = z_0 g'(z_0)/g(z_0)$, it follows that

$$\begin{aligned} m &= \frac{d}{dr}\left[\frac{g(z_0 r)}{g(z_0)}\right]\Bigg|_{r=1} \\ &= \lim_{r\to 1} \frac{g(z_0 r) - g(z_0)}{(r-1)z_0} \frac{z_0}{g(z_0)} \\ &= \lim_{r\to 1}\left[1 - \frac{g(z_0 r)}{g(z_0)}\right]\frac{1}{1-r}. \end{aligned}$$

Taking real parts and using (5.2.3), it follows that

$$m = \lim_{r\to 1}\left[1 - \operatorname{Re}\frac{g(z_0 r)}{g(z_0)}\right]\frac{1}{1-r} \geqslant \lim_{r\to 1}\frac{1-r^n}{1-r} = n.$$

This proves $m \geqslant n$.

Differentiating equation (5.2.2) yields

$$\frac{g(z)[zg''(z) + g'(z)] - z[g'(z)]^2}{g^2(z)} = \frac{1}{\mathrm{i}z}\left[\frac{\partial^2 \Phi}{\partial \theta^2} - \mathrm{i}\frac{1}{R}\frac{\partial^2 R}{\partial \theta^2} + \mathrm{i}\frac{1}{R^2}\left(\frac{\partial R}{\partial \theta}\right)^2\right],$$

and therefore

$$\mathrm{i}\left[\frac{zg'(z)}{g(z)}\left(\frac{zg''(z)}{g'(z)} + 1\right) - \left(\frac{zg'(z)}{g(z)}\right)^2\right] = \frac{\partial^2 \Phi}{\partial \theta^2} - \mathrm{i}\left[\frac{1}{R}\frac{\partial^2 R}{\partial \theta^2} - \left(\frac{1}{R}\frac{\partial R}{\partial \theta}\right)^2\right]. \tag{5.2.4}$$

Since z_0 is a maximum point of $R(r_0, \theta)$, it follows that

$$\frac{\partial R}{\partial \theta}(z_0) = 0 \quad \text{and} \quad \frac{\partial^2 R}{\partial \theta^2}(z_0) \leqslant 0.$$

Since $m = z_0 g'(z_0)/g(z_0)$, (5.2.4) yields

$$\mathrm{i}\left[m\left(\frac{z_0 g''(z_0)}{g'(z_0)}+1\right)-m^2\right]=\frac{\partial^2\Phi}{\partial\theta^2}(z_0)-\mathrm{i}\frac{1}{R(z_0)}\frac{\partial^2 R}{\partial\theta^2}(z_0).$$

Taking imaginary parts on both sides yields

$$\operatorname{Re} m\left(\frac{z_0 g''(z_0)}{g'(z_0)}+1\right)-m^2=-\frac{1}{R(z_0)}\frac{\partial^2 R}{\partial\theta^2}(z_0)\geqslant 0,$$

and so

$$\operatorname{Re}\left(\frac{z_0 g''(z_0)}{g'(z_0)}+1\right)\geqslant m.$$

□

Lemma 5.2.1 is based on the fact that the domain $g(\mathbb{D}_{r_0})$ lies inside the disk $|w|\leqslant R=|g(z_0)|$ and that the boundary of $g(\mathbb{D}_{r_0})$ is tangent to the circle $|w|=R$. The first part of Lemma 5.2.1, that is, $m = z_0 g'(z_0)/g(z_0)$, is known as Jack's lemma.

Let Q denote the class of all functions q that are analytic and one-to-one on $\overline{\mathbb{D}}\setminus E(q)$, where $E(q)$ consists of all points $\zeta\in\partial\mathbb{D}$ for which $q(z)\to\infty$ as $z\to\zeta$. Every function $q\in Q$ has a nice boundary. If $q\in Q$, then $\Delta=q(\mathbb{D})$ is a simply connected domain whose boundary is either a simple closed regular curve or a (possibly infinite) union of pairwise disjoint simple regular arcs, each of which converges to ∞ in both directions. (An arc γ is regular if its derivative $\gamma'(t)$ exists and is continuous.) The functions $q_1(z)=z$ and $q_2(z)=(1+z)/(1-z)$ illustrate these two cases, respectively.

Lemma 5.2.2 *Let $p(z)=a+p_n z^n+p_{n+1}z^{n+1}+\cdots$ be analytic in $\mathbb{D}$ with $p(z)\not\equiv a$ and $n\geqslant 1$. Further, let $q\in Q$ with $q(0)=a$. If there exists a point $z_0\in\mathbb{D}$ such that $p(z_0)\in q(\partial\mathbb{D})$ and $p(\mathbb{D}_{r_0})\subset q(\mathbb{D})$, $r_0=|z_0|$, then*

$$z_0 p'(z_0)=m\zeta_0 q'(\zeta_0) \tag{5.2.5}$$

and

$$Re\left(\frac{z_0 p''(z_0)}{p'(z_0)}+1\right)\geqslant mRe\left(\frac{\zeta_0 q''(\zeta_0)}{q'(\zeta_0)}+1\right), \tag{5.2.6}$$

where $q^{-1}(p(z_0))=\zeta_0=e^{i\theta_0}$ and $m\geqslant n\geqslant 1$.

Proof Since p is analytic in $\mathbb{D}$, the set $p(\overline{\mathbb{D}}_{r_0})$ is bounded and

$$p(\overline{\mathbb{D}}_{r_0})\subset q(\overline{\mathbb{D}})\setminus E(q)$$

where $E(q)=\{\zeta\in\partial\mathbb{D}:\lim_{z\to\zeta}q(z)=\infty\}$. Consider the function g defined by

$$g(z)=q^{-1}(p(z))\quad(z\in\overline{\mathbb{D}}_{r_0}). \tag{5.2.7}$$

The lemma is proved by applying Lemma 5.2.1 to the function g. This function g is analytic in $\overline{\mathbb{D}}_{r_0}$, satisfies $g(0) = q^{-1}(p(0)) = q^{-1}(a) = 0$ and $|g(z_0)| = |q^{-1}(p(z_0))| = |\zeta_0| = 1$. Since $p(\overline{\mathbb{D}}_{r_0}) \subset q(\overline{\mathbb{D}}) \setminus E(q)$, it follows that $|g(z)| \leqslant 1$ for $z \in \overline{\mathbb{D}}_{r_0}$. Thus, $\max_{|z|\leqslant r_0} |g(z)| = 1 = |g(z_0)|$. From (5.2.7),

$$q(g(z)) = p(z), \quad z \in \overline{\mathbb{D}}_{r_0} \tag{5.2.8}$$

so that

$$q'(g(z))g'(z) = p'(z).$$

Since $g(0) = 0, g(z) = g_n z^n + g_{n+1} z^{n+1} + \cdots$ for $n = 1$. For $n > 1$, since $g(0) = 0$ and $p(z) = a + p_n z^n + p_{n+1} z^{n+1} + \cdots$, it follows that

$$q'(0)g'(0) = q'(g(0))g'(0) = p'(0) = 0.$$

Since q is univalent, $q'(z) \neq 0$ in $\mathbb{D}$. Thus, $g'(0) = p'(0) = 0$. A similar argument shows that

$$g^{(k)}(0) = p^{(k)}(0) \quad \text{for} \quad k = 1, 2, \ldots n-1.$$

Since $p^{(k)}(0) = 0$ for $k = 1, 2, \ldots n-1$, it follows that $g^{(k)}(0) = 0$ for $k = 1, 2, \ldots n-1$ and so $g(z) = g_n z^n + g_{n+1} z^{n+1} + \cdots$. Therefore, the function g satisfies the conditions of Lemma 5.2.1.

Hence, there exists an $m \geqslant n \geqslant 1$ such that

$$\frac{z_0 g'(z_0)}{g(z_0)} = m \quad \text{and} \quad \operatorname{Re}\left(\frac{z_0 g''(z_0)}{g'(z_0)} + 1\right) \geqslant m. \tag{5.2.9}$$

From (5.2.8),

$$zp'(z) = zq'(g(z))g'(z) = \zeta q'(\zeta)\frac{zg'(z)}{g(z)},$$

where $\zeta = g(z)$. Since $g(z_0) = \zeta_0$, the first equation in (5.2.9) shows that

$$z_0 p'(z_0) = \zeta_0 q'(\zeta_0)\frac{z_0 g'(z_0)}{g(z_0)} = m\zeta_0 q'(\zeta_0).$$

This proves (5.2.5).

From $zp'(z) = zq'(g(z))g'(z)$, it is easily shown that

$$\frac{zp''(z)}{p'(z)} + 1 = \zeta\frac{q''(\zeta)}{q'(\zeta)}\frac{zg'(z)}{g(z)} + \frac{zg''(z)}{g'(z)} + 1$$

where $\zeta = g(z)$. From (5.2.9),

$$\operatorname{Re}\left(\frac{z_0 p''(z_0)}{p'(z_0)}+1\right) \geqslant m \operatorname{Re}\left(\frac{\zeta_0 q''(\zeta_0)}{q'(\zeta_0)}+1\right)$$

which establishes (5.2.6). □

Note that (5.2.5) is a relation between the outer normals to the curves $p(\partial \mathbb{D}_{r_0})$ and $q(\partial \mathbb{D})$ at their point of tangency $p(z_0) = q(\zeta_0)$, and (5.2.6) is a relation between the curvature of these curves at their point of tangency.

Lemma 5.2.3 *Let $p(z) = a + p_n z^n + p_{n+1} z^{n+1} + \cdots$ be analytic in $\mathbb{D}$ with $p(z) \not\equiv a$ and $n \geqslant 1$. Further, let $q \in Q$ with $q(0) = a$. If p is not subordinate to q, then there exist points $z_0 \in \mathbb{D}$ and $\zeta_0 \in \partial \mathbb{D} \setminus E(q)$ and an $m \geqslant n \geqslant 1$ for which $p(\mathbb{D}_{r_0}) \subset q(\mathbb{D})$, $r_0 = |z_0|$, $p(z_0) = q(\zeta_0)$, $z_0 p'(z_0) = m \zeta_0 q'(\zeta_0)$ and*

$$Re\left(\frac{z_0 p''(z_0)}{p'(z_0)}+1\right) \geqslant m Re\left(\frac{\zeta_0 q''(\zeta_0)}{q'(\zeta_0)}+1\right).$$

Proof Let $r_0 := \sup\{r : p(\mathbb{D}_r) \subset q(\mathbb{D})\}$. We shall show that $0 < r_0 < 1$. By the Koebe 1/4-theorem, the univalence of q shows that the disk

$$\mathbb{D}(q(0); |q'(0)|/4) \subset q(\mathbb{D}).$$

Since p is continuous, $p^{-1}(\mathbb{D}(q(0); |q'(0)|/4))$ is open and contains the origin. So there is some $r > 0$ such that

$$\mathbb{D}_r \subset p^{-1}(\mathbb{D}(q(0); |q'(0)|/4)),$$

and therefore

$$p(\mathbb{D}_r) \subset \mathbb{D}(q(0), |q'(0)|/4) \subset q(\mathbb{D}).$$

This shows that $r_0 \geqslant r > 0$. If $r_0 = 1$, then $p(\mathbb{D}_r) \subset q(\mathbb{D})$ for every r with $0 < r < 1$ and so

$$p(\mathbb{D}) = p\left(\bigcup_{0<r<1} \mathbb{D}_r\right) = \bigcup_{0<r<1} p(\mathbb{D}_r) \subset q(\mathbb{D}).$$

Since p is not subordinate to q, then $p(\mathbb{D}) \not\subset q(\mathbb{D})$ and so $r_0 < 1$.

Clearly

$$p(\mathbb{D}_{r_0}) = \bigcup_{0<r<r_0} p(\mathbb{D}_r) \subset q(\mathbb{D}).$$

However, $p(\overline{\mathbb{D}}_{r_0}) \not\subset q(\mathbb{D})$. Otherwise, if $p(\overline{\mathbb{D}}_{r_0}) \subset q(\mathbb{D})$, then $\overline{\mathbb{D}}_{r_0} \subset p^{-1}(q(\mathbb{D}))$. Since $\overline{\mathbb{D}}_{r_0}$ is compact, and $\partial p^{-1}(q(\mathbb{D}))$ is closed, it follows that the distance $2\epsilon := d(\overline{\mathbb{D}}_{r_0}, \partial p^{-1}(q(\mathbb{D}))) > 0$. Thus, the disk $\mathbb{D}r_0 + \epsilon \subset p^{-1}(q(\mathbb{D}))$ or $p(\mathbb{D}r_0 + \epsilon) \subset p(p^{-1}(q(\mathbb{D}))) \subset q(\mathbb{D})$, a contradiction to the definition of r_0.

Now $p(\overline{\mathbb{D}}_{r_0}) \subset q(\overline{\mathbb{D}})$. Choose $z_0 \in q(\overline{\mathbb{D}}) \setminus q(\mathbb{D}_{r_0})$ satisfying $|z_0| = r_0$, and $p(z_0) \in \partial q(\mathbb{D}) = q(\partial \mathbb{D})$. Thus, there is $\zeta_0 \in \partial \mathbb{D} \setminus E(q)$ such that $p(z_0) = q(\zeta_0)$. The other parts of the lemma immediately follow from Lemma 5.2.2. □

Definition 5.2.4 Let Ω be a subset of $\mathbb{C}$ and $q \in Q$. The class of *admissible functions* $\Psi_n(\Omega, q)$ consists of all functions $\psi : \mathbb{C}^3 \times \mathbb{D} \to \mathbb{C}$ that satisfy the admissibility condition $\psi(r, s, t; z) \notin \Omega$ when $r = q(\zeta)$, $s = m\zeta q'(\zeta)$,

$$\operatorname{Re}\left(\frac{t}{s} + 1\right) \geqslant m\operatorname{Re}\left(\frac{\zeta q''(\zeta)}{q'(\zeta)} + 1\right),$$

$\zeta \in \partial\mathbb{D} \setminus E(q)$, and $m \geqslant n$. In the case $n = 1$, $\Psi(\Omega, q)$ denotes $\Psi_1(\Omega, q)$.

In the proceeding definition, the domain Ω is not required to be simply connected or to have a particularly nice boundary, in contrast to the image $q(\mathbb{D})$. In the next definition, however, $\Omega \neq \mathbb{C}$ is a simply connected domain.

Definition 5.2.5 Let $\Omega \neq \mathbb{C}$ be a simply connected domain in $\mathbb{C}$ and let $q \in Q$. Suppose that h is a conformal mapping of $\mathbb{D}$ onto Ω. The class of admissible functions $\Psi_n(h, q)$ consists of functions $\psi \in \Psi_n(\Omega, q) = \Psi_n(h(\mathbb{D}), q)$. The symbol $\Psi(h, q)$ denotes $\Psi_1(h, q)$.

Remark 5.2.6 Definition 5.2.4 shows that if $\Omega \subset \widetilde{\Omega}$ then $\Psi_n(\widetilde{\Omega}, q) \subset \Psi_n(\Omega, q)$, that is, enlarging Ω decreases the class of admissible functions. Also, $\Psi_n(\Omega, q) \subset \Psi_{n+1}(\Omega, q)$.

The next theorem is a foundation result in the theory of first and second-order differential subordinations.

Theorem 5.2.7 (Miller-Mocanu Theorem) *Let $q(0) = a$ and $\psi \in \Psi_n(\Omega, q)$. If $p \in \mathcal{H}[a, n]$ satisfies*

$$\psi(p(z), zp'(z), z^2p''(z); z) \in \Omega, \quad z \in \mathbb{D}, \tag{5.2.10}$$

then $p \prec q$.

Proof If p is not subordinate to q, then, by Lemma 5.2.3, there exists a point z_0 such that

$$p(z_0) = q(\zeta_0), \quad z_0p'(z_0) = m\zeta_0q'(z_0)$$

and

$$\operatorname{Re}\left(\frac{z_0p''(z_0)}{p'(z_0)} + 1\right) \geqslant m\operatorname{Re}\left(\frac{\zeta_0q''(\zeta_0)}{q'(\zeta_0)} + 1\right),$$

where $|\zeta_0| = 1$, $q(\zeta_0)$ is finite, and $m \geqslant n$. Set $r_0 = p(z_0)$, $s_0 = z_0p'(z_0)$, and $t_0 = z_0^2p''(z_0)$. Then, the admissibility condition in Definition 5.2.4 implies

$$\psi(p(z_0), z_0p'(z_0), z_0^2p''(z_0); z_0) \notin \Omega$$

which contradicts (5.2.10). This contradiction shows that $p \prec q$. □

In view of the definitions of Q and $\Psi_n(\Omega, q)$, the hypothesis of Theorem 5.2.7 requires that q exhibits nice boundary behavior on $\partial\mathbb{D}$. If this is not the case or if the behavior of q on $\partial\mathbb{D}$ is not known, it may still be possible to prove that $p \prec q$ by a limiting argument.

Theorem 5.2.8 *Let q be univalent in $\mathbb{D}$ with $q(0) = a$, and let $q_\rho(z) = q(\rho z)$ for $0 < \rho < 1$. Let $\psi \in \Psi_n(\Omega, q_\rho)$ for some $\rho \in (0, 1)$. If $p \in \mathcal{H}[a, n]$ satisfies*

$$\psi(p(z), zp'(z), z^2p''(z); z) \in \Omega, \quad z \in \mathbb{D},$$

then $p \prec q$.

Proof Clearly, the function q_ρ is univalent in $\overline{\mathbb{D}}$ for $0 < \rho < 1$. Hence, $q_\rho \in Q$ and $q_\rho(0) = q(0) = a$, so that $\psi \in \Psi_n(\Omega, q_\rho)$ is a well defined admissibility class. By Theorem 5.2.7, it follows that $p \prec q_\rho$, $0 < \rho < 1$. Since q is univalent, $q_\rho(0) = q(0)$ and $q_\rho(\mathbb{D}) \subset q(\mathbb{D})$, the subordination $q_\rho \prec q$ holds for each $0 < \rho < 1$. By the transitivity of subordination, this gives $p \prec q$. □

For a simply connected domain $\Omega \neq \mathbb{C}$, Theorem 5.2.7 reduces to the following result.

Theorem 5.2.9 (Miller-Mocanu Theorem) *Let $q(0) = a$ and $\psi \in \Psi_n(h, q)$. If $p \in \mathcal{H}[a, n]$ and $\psi(p(z), zp'(z), z^2p''(z); z)$ is analytic in $\mathbb{D}$, then*

$$\psi(p(z), zp'(z), z^2p''(z); z) \prec h(z) \quad \Rightarrow \quad p(z) \prec q(z). \tag{5.2.11}$$

Remark 5.2.10 The set of functions p satisfying (5.2.10) is non-empty when Ω is a domain, $\psi(a, 0, 0; 0) \in \Omega$ and ψ continuous at $(a, 0, 0; 0)$. If $p(z) = a + p_n z^n$, then for a sufficiently small $|p_n|$,

$$\psi(p(z), zp'(z), z^2p''(z); z) = \psi(a + p_n z^n, np_n z^n, n(n-1)p_n z^n) \in \Omega, \quad z \in \mathbb{D}.$$

Theorem 5.2.9 can be extended to those cases in which the behavior of q on the boundary of $\mathbb{D}$ is unknown.

Theorem 5.2.11 *Let h and q be univalent in $\mathbb{D}$ with $q(0) = a$, and let $h_\rho(z) = h(\rho z)$, $q_\rho(z) = q(\rho z)$. Let $\psi : \mathbb{C}^3 \times \mathbb{D} \to \mathbb{C}$ satisfies one of the following conditions:*

(i) $\psi \in \Psi_n(h, q_\rho)$ for some $\rho \in (0, 1)$, or
(ii) there exists $\rho_0 \in (0, 1)$ such that $\psi \in \Psi_n(h_\rho, q_\rho)$ for all $\rho \in (\rho_0, 1)$.

If $p \in \mathcal{H}[a, n]$, $\psi(p(z), zp'(z), z^2p''(z); z)$ is analytic in $\mathbb{D}$, and

$$\psi(p(z), zp'(z), z^2p''(z); z) \prec h(z),$$

then $p \prec q$.

Proof (i) By Theorem 5.2.8, $p \prec q_\rho$, and since $q_\rho \prec q$, it follows that $p \prec q$.

(ii) Clearly, the function q_ρ is univalent in $\overline{\mathbb{D}}$ for $0 < \rho < 1$. Hence, $q_\rho \in \mathcal{Q}$ and $q_\rho(0) = q(0) = a$. Therefore, the class $\psi \in \Psi_n(h_\rho, q_\rho)$ is a well-defined admissibility class. Define $p_\rho(z) = p(\rho z)$, $0 < \rho < 1$. Then

$$\psi(p_\rho(z), zp'_\rho(z), z^2 p''_\rho(z); \rho z) = \psi(p(\rho z), \rho z p'(\rho z), \rho^2 z^2 p''(\rho z); \rho z) \in h_\rho(\mathbb{D}).$$

By Theorem 5.2.9, this implies $p_\rho(z) \prec q_\rho(z)$, $0 < \rho < 1$. Hence, $p(\rho z) \prec q(\rho z)$ so that $p(\rho\mathbb{D}) \subset q(\rho\mathbb{D}) \subset q(\mathbb{D})$. Taking the limit as $\rho \to 1^-$ yields $p(\mathbb{D}) \subset q(\mathbb{D})$. Since q is univalent and $p(0) = q(0)$, the subordination $p \prec q$ is deduced. □

The following theorems prove existence of the best dominant of (5.1.2) for certain ψ; they also provide a method for obtaining the best dominant. First, the result is given for the case $n = 1$.

Theorem 5.2.12 *Let h be univalent in $\mathbb{D}$ and $\psi : \mathbb{C}^3 \times \mathbb{D} \to \mathbb{C}$. Suppose that the differential equation*

$$\psi(q(z), zq'(z), z^2 q''(z); z) = h(z) \tag{5.2.12}$$

has a solution q with $q(0) = a$, and one of the following conditions is satisfied:

(i) $q \in \mathcal{Q}$ *and* $\psi \in \Psi(h, q)$,
(ii) q *is univalent in* $\mathbb{D}$ *and* $\psi \in \Psi(h, q_\rho)$ *for some* $\rho \in (0, 1)$, *or*
(iii) q *is univalent in* $\mathbb{D}$ *and there exists* $\rho_0 \in (0, 1)$ *such that* $\psi \in \Psi(h_\rho, q_\rho)$ *for all* $\rho \in (\rho_0, 1)$.

If $p \in \mathcal{H}[a, 1]$ satisfies $\psi(p(z), zp'(z), z^2 p''(z); z) \prec h(z)$, then $p \prec q$ and q is the best dominant.

Proof By Theorems 5.2.9 and 5.2.11, the function q is a dominant of the subordination $\psi(p(z), zp'(z), z^2 p''(z); z) \prec h(z)$. Since q satisfies (5.2.12), it is a solution of $\psi(p(z), zp'(z), z^2 p''(z); z) \prec h(z)$ and so it is the best dominant. □

Theorem 5.2.13 *Let h be univalent in $\mathbb{D}$ and $\psi : \mathbb{C}^3 \times \mathbb{D} \to \mathbb{C}$. Suppose that the differential equation*

$$\psi(q(z), nzq'(z), n(n-1)zq'(z) + n^2 z^2 q''(z); z) = h(z) \tag{5.2.13}$$

has a solution q with $q(0) = a$, and one of the following conditions is satisfied:

(i) $q \in \mathcal{Q}$ *and* $\psi \in \Psi_n(h, q)$,
(ii) q *is univalent in* $\mathbb{D}$ *and* $\psi \in \Psi_n(h, q_\rho)$ *for some* $\rho \in (0, 1)$, *or*
(iii) q *is univalent in* $\mathbb{D}$ *and there exists* $\rho_0 \in (0, 1)$ *such that* $\psi \in \Psi_n(h_\rho, q_\rho)$ *for all* $\rho \in (\rho_0, 1)$.

If $p \in \mathcal{H}[a, n]$ satisfies $\psi(p(z), zp'(z), z^2 p''(z); z) \prec h(z)$, then $p \prec q$ and q is the best (a, n)-dominant.

Proof By Theorems 5.2.9 and 5.2.11, q is a dominant of the subordination $\psi(p(z), zp'(z), z^2p''(z); z) \prec h(z)$. Let $\hat{q}(z) = q(z^n)$. Then

$$z\hat{q}'(z) = nz^n q'(z^n), \quad z^2\hat{q}''(z) = n(n-1)z^n q'(z^n) + n^2 z^{2n} q''(z^n).$$

Then, the equation (5.2.13) shows that

$$\psi(\hat{q}(z), z\hat{q}'(z), z^2\hat{q}''(z); z) = h(z^n) \prec h(z).$$

Since $\hat{q}(\mathbb{D}) = q(\mathbb{D})$, it follows that q is the best dominant. □

By Theorems 5.2.12 and 5.2.13, the problem of finding the best dominant to the differential subordination (5.1.2) reduces to solving the differential equation (5.2.12) and verifying whether $\psi \in \Psi(h, q)$ or $\psi \in \Psi(h(\rho z), q(\rho z))$ for some $\rho \in (0, 1)$. The conclusion of these theorems can be written in the symmetric form

$$\psi(p(z), zp'(z), z^2p''(z); z) \prec \psi(q(z), zq'(z), z^2q''(z); z) \quad \Rightarrow \quad p(z) \prec q(z).$$

Theorem 5.2.7 can be applied to show that the solution of certain second-order differential equations lie within a specific domain. For simplicity, the case $n = 1$ is considered again.

Theorem 5.2.14 *Let $\psi \in \Psi(\Omega, q)$ and let w be an analytic function in $\mathbb{D}$ satisfying $w(\mathbb{D}) \subset \Omega$. If the differential equation*

$$\psi(p(z), zp'(z), z^2p''(z); z) = w(z)$$

has a solution p analytic in $\mathbb{D}$ with $p(0) = q(0)$, then $p \prec q$.

Proof Since $w(\mathbb{D}) \subset \Omega$, it follows that $\psi(p(z), zp'(z), z^2p''(z); z) \in \Omega$ for $z \in \mathbb{D}$. The result follows directly from Theorem 5.2.7. □

5.3 Bounded Analytic Functions

The function q defined by $q(z) = z$ belongs to $\mathcal{B}_0$ and maps $\mathbb{D}$ onto itself. Then $E(q) = \emptyset$ and $|q(z)| = 1$ for $z \in \partial\mathbb{D}$. Moreover,

$$zq'(z) = z \quad \text{and} \quad \frac{zq''(z)}{q'(z)} + 1 = 1. \tag{5.3.1}$$

If $\zeta \in \partial\mathbb{D}$, then $r = q(\zeta) = e^{i\theta}$, where $\theta \in \mathbb{R}$. It follows from (5.3.1) that

$$s = m\zeta q'(\zeta) = me^{i\theta} \quad \text{and} \quad \operatorname{Re}\left(\frac{\zeta q''(\zeta)}{q'(\zeta)} + 1\right) = 1.$$

Hence, the admissibility condition in Definition 5.2.4 reduces to

$$\psi(e^{i\theta}, Ke^{i\theta}, L; z) \notin \Omega$$

whenever

$$K \geqslant n,\ \mathrm{Re}(Le^{-i\theta}) \geqslant (n-1)K,\ \theta \in \mathbb{R} \text{ and } z \in \mathbb{D}.$$

Definition 5.3.1 Let Ω be a subset of $\mathbb{C}$. The class $\mathcal{B}_n(\Omega)$ of *admissible functions* consists of all functions $\psi : \mathbb{C}^3 \times \mathbb{D} \to \mathbb{C}$ that satisfy the admissibility condition

$$\psi(e^{i\theta}, Ke^{i\theta}, L; z) \notin \Omega \quad \text{when } K \geqslant n, \mathrm{Re}(Le^{-i\theta}) \geqslant (n-1)K, \theta \in \mathbb{R} \text{ and } z \in \mathbb{D}.$$

With $q(z) = z$, Theorem 5.2.7 becomes

Theorem 5.3.2 *Let $\psi \in \mathcal{B}_n(\Omega)$. If $p \in \mathcal{H}[0, n]$ satisfies*

$$\psi(p(z), zp'(z), z^2p''(z); z) \in \Omega, \quad z \in \mathbb{D},$$

then $p \in \mathcal{B}_0$.

For first-order differential subordination, the class $\mathcal{B}_n(\Omega)$ of admissible functions consists of all functions $\psi : \mathbb{C}^2 \times \mathbb{D} \to \mathbb{C}$ that satisfy the admissibility condition

$$\psi(e^{i\theta}, Ke^{i\theta}; z) \notin \Omega \quad \text{when } K \geqslant n, \theta \in \mathbb{R} \text{ and } z \in \mathbb{D}.$$

In addition, if there is no function of z appearing in our subordination other than q and q', the condition is simplified as

$$\psi(e^{i\theta}, Ke^{i\theta}) \notin \Omega \quad \text{when } K \geqslant n \text{ and } \theta \in \mathbb{R}.$$

Theorem 5.3.3 *Let $Re(\alpha\overline{\beta}) \geqslant 0$. If $p \in \mathcal{H}[0, n]$ satisfies*

$$|\alpha p(z) + \beta zp'(z)| < |\alpha + n\beta|, \quad z \in \mathbb{D},$$

then $p \in \mathcal{B}_0$.

Proof Let $\psi(r, s) = \alpha r + \beta s$, and $\Omega = \{w \in \mathbb{C} : |w| < |\alpha + n\beta|\}$. The function ψ satisfies

$$|\psi(e^{i\theta}, Ke^{i\theta})| = |\alpha e^{i\theta} + K\beta e^{i\theta}| = |\alpha + K\beta| \geqslant |\alpha + n\beta|$$

whenever $K \geqslant n$, and $\theta \in \mathbb{R}$. The inequality $|\alpha + K\beta| \geqslant |\alpha + n\beta|$ can be verified by just squaring both sides. Thus, $\psi \in \mathcal{B}_n(\Omega)$. Hence, the result follows by Theorem 5.3.2. □

With $\beta = 1, n = 1$, the above theorem gives the following result.

Corollary 5.3.4 *Let* $\operatorname{Re}\alpha \geqslant 0$. *If* $p \in \mathcal{H}[0,1]$ *satisfies*

$$|\alpha p(z) + zp'(z)| < |1+\alpha|, \quad z \in \mathbb{D},$$

then $p \in \mathcal{B}_0$.

Theorem 5.3.5 *If* F *is analytic in* $\mathbb{D}$, $F(0) = 1$ *and* $zF'(z) \prec kz$ *for some* $k > 0$, *then*

$$F(z) \prec 1 + kz.$$

Proof Let

$$p(z) = \frac{F(z) - 1}{k}, \quad z \in \mathbb{D}.$$

Then the theorem reduces to showing[1] that

$$|zp'(z)| < 1 \Rightarrow |p(z)| < 1.$$

This follows from Corollary 5.3.4 with $\alpha = 0$. □

Theorem 5.3.6 *If* F *is analytic in* $\mathbb{D}$ *and* $F(z) \prec 1 + kz$ *for some* $k > 0$, *then*

$$\frac{1}{z}\int_0^z F(t)dt \prec 1 + \frac{k}{2}z.$$

Proof Let

$$p(z) = \frac{2}{k}\left(\frac{1}{z}\int_0^z F(t)\mathrm{d}t - 1\right), \quad z \in \mathbb{D}.$$

Then,

$$zp'(z) + p(z) = \frac{2}{k}(F(z) - 1) \prec 2z.$$

Therefore, the theorem reduces to showing

$$|zp'(z) + p(z)| < 2 \Rightarrow |p(z)| < 1,$$

which follows from Corollary 5.3.4 with $\alpha = 1$. □

Theorem 5.3.7 *If* $p \in \mathcal{H}[0,n]$ *satisfies*

$$|p^2(z) + p(z) + zp'(z)| < n, \quad z \in \mathbb{D},$$

then $p \in \mathcal{B}_0$.

[1] This can be proved directly by an integration.

Proof Let $\psi(r,s) = r + r^2 + s$, and $\Omega = \{w \in \mathbb{C} : |w| < n\}$. The function ψ satisfies

$$\begin{aligned}|\psi(e^{i\theta}, Ke^{i\theta})| &= |e^{2i\theta} + e^{i\theta} + Ke^{i\theta}| = |e^{i\theta} + 1 + K| \\ &\geqslant 1 + K - 1 = K \geqslant n,\end{aligned}$$

whenever $K \geqslant n$ and $\theta \in \mathbb{R}$. Thus, $\psi \in \mathcal{B}_n(\Omega)$, and the result follows by Theorem 5.3.2. □

The following theorem provides a simple condition for the univalence of the solution of a second-order differential equation.

Theorem 5.3.8 *Let $zC(z)$ be analytic in $\mathbb{D}$ with $|zC(z)| < 1$. Let $f \in \mathcal{A}$ be the unique solution of*

$$f''(z) + C(z)f(z) = 0. \tag{5.3.2}$$

Then f is starlike, and satisfies

$$\left|\frac{zf'(z)}{f(z)} - 1\right| < 1. \tag{5.3.3}$$

Proof Let

$$p(z) := \frac{zf'(z)}{f(z)} - 1, \quad z \in \mathbb{D}. \tag{5.3.4}$$

By Exercise 8, $f(z) \neq 0$ for $z \neq 0$, the function p is analytic in $\mathbb{D}$ and $p(0) = 0$. Using (5.3.2) and (5.3.4),

$$p^2(z) + p(z) + zp'(z) = -z^2C(z). \tag{5.3.5}$$

Since $|-z^2C(z)| < 1$, Theorem 5.3.7 with $n = 1$ implies that the solution p of the differential equation (5.3.5) belongs to $\mathcal{B}_0$. Since $|p(z)| < 1$, (5.3.4) yields (5.3.3), from which

$$\operatorname{Re}\frac{zf'(z)}{f(z)} > 0$$

follows. Therefore, the function f is univalent and starlike in $\mathbb{D}$. □

Example 5.3.9 The function $f \in \mathcal{A}$ given by $f(z) = \sin z$ satisfies $f''(z) + C(z)f(z) = 0$ with $C(z) = 1$. Theorem 5.3.8 shows that f is starlike.

Theorem 5.3.10 *If $p \in \mathcal{H}[0,n]$ satisfies*

$$|p(z) + zp'(z) + z^2p''(z)| < n^2 + 1, \quad z \in \mathbb{D},$$

then $p \in \mathcal{B}_0$.

Proof Let $\psi(r, s, t) = r + s + t$, and $\Omega = \{w \in \mathbb{C} : |w| < n^2 + 1\}$. The function ψ satisfies

$$\begin{aligned} |\psi(\mathrm{e}^{\mathrm{i}\theta}, K\mathrm{e}^{\mathrm{i}\theta}, L)| &= |\mathrm{e}^{\mathrm{i}\theta} + K\mathrm{e}^{\mathrm{i}\theta} + L| = |1 + K + L\mathrm{e}^{-\mathrm{i}\theta}| \\ &\geqslant 1 + K + \mathrm{Re}(L\mathrm{e}^{-\mathrm{i}\theta}) \geqslant 1 + K + (n-1)K \\ &= 1 + nK \geqslant n^2 + 1, \end{aligned}$$

whenever $K \geqslant n$, $\mathrm{Re}(L\mathrm{e}^{-\mathrm{i}\theta}) \geqslant (n-1)K$ and $\theta \in \mathbb{R}$. Thus, $\psi \in \mathcal{B}_n(\Omega)$, and the result follows by Theorem 5.3.2. □

Corollary 5.3.11 *Let $n \geqslant 2$. If $p \in \mathcal{H}[1, n]$ satisfies*

$$|z^2 p''(z) + 3zp'(z) - p^2(z) + 1| < (n-1)(n+3), \quad z \in \mathbb{D},$$

then

$$|p(z) - 1| < 1, \quad z \in \mathbb{D}.$$

Proof Let $P(z) = p(z) - 1$ in $\mathbb{D}$. Then the problem becomes

$$|z^2 P''(z) + 3zP'(z) - P^2(z) - 2P(z)| < (n-1)(n+3) \Longrightarrow P \in \mathcal{B}_0.$$

Let $\psi(r, s, t) = t + 3s - r^2 - 2r$, and $\Omega = \{w \in \mathbb{C} : |w| < (n-1)(n+3)\}$. The function ψ satisfies

$$\begin{aligned} |\psi(\mathrm{e}^{\mathrm{i}\theta}, K\mathrm{e}^{\mathrm{i}\theta}, L)| &= |L + 3K\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{2\mathrm{i}\theta} - 2\mathrm{e}^{\mathrm{i}\theta}| \\ &= |L\mathrm{e}^{-\mathrm{i}\theta} + 3K - \mathrm{e}^{\mathrm{i}\theta} - 2| \\ &\geqslant \mathrm{Re}(L\mathrm{e}^{-\mathrm{i}\theta} + 3K - \mathrm{e}^{\mathrm{i}\theta} - 2) \\ &\geqslant 3n - 3 + n(n-1) \\ &= (n-1)(n+3) \end{aligned}$$

whenever $K \geqslant n$, $\mathrm{Re}(L\mathrm{e}^{-\mathrm{i}\theta}) \geqslant (n-1)K$ and $\theta \in \mathbb{R}$. Thus, $\psi \in \mathcal{B}_n(\Omega)$, and the result follows by Theorem 5.3.2. □

5.4 Carathéodory Functions

Consider the function $q \in \mathcal{P}$ defined by

$$q(z) = \frac{1+z}{1-z}.$$

This function q maps $\mathbb{D}$ onto the right half-plane $\{w \in \mathbb{C} : \mathrm{Re}\, w > 0\}$ with $E(q) = \{1\}$. For $\zeta \neq 1$ with $|\zeta| = 1$,

$$\operatorname{Re} q(\zeta) = \operatorname{Re}\frac{(1+\zeta)(1-\overline{\zeta})}{|1-\zeta|^2} = \frac{1-|\zeta|^2}{|1-\zeta|^2} = 0,$$

so that $q(\zeta)$ is purely imaginary on $\partial\mathbb{D} \setminus \{1\}$. Thus, $r = q(\zeta) = \mathrm{i}\rho$, where $\zeta \in \partial\mathbb{D} \setminus \{1\}$ and ρ is a real number. Evidently,

$$zq'(z) = \frac{2z}{(1-z)^2} = \frac{1}{2}((q(z))^2 - 1) \tag{5.4.1}$$

and

$$1 + \frac{zq''(z)}{q'(z)} = \frac{1+z}{1-z} = q(z). \tag{5.4.2}$$

From (5.4.1) and (5.4.2), it follows that

$$s = m\zeta q'(\zeta) = -\frac{m}{2}(1+\rho^2) \quad \text{and} \quad \operatorname{Re}\left(1 + \frac{\zeta q''(\zeta)}{q'(\zeta)}\right) = 0.$$

Thus, the admissibility condition in Definition 5.2.4 reduces to

$$\psi(\mathrm{i}\rho, \sigma, \mu + \mathrm{i}\nu; z) \notin \Omega$$

when

$$\rho \in \mathbb{R},\ \sigma \leq -\frac{n}{2}(1+\rho^2),\ \sigma + \mu \leqslant 0,\ \text{and } z \in \mathbb{D}.$$

Definition 5.4.1 Let Ω be a subset of $\mathbb{C}$. The class $\mathcal{P}_n(\Omega)$ of *admissible functions* consists of all functions $\psi : \mathbb{C}^3 \times \mathbb{D} \to \mathbb{C}$ that satisfy the admissibility condition

$$\psi(\mathrm{i}\rho, \sigma, \mu + \mathrm{i}\nu; z) \notin \Omega \text{ whenever } \rho \in \mathbb{R},\ \sigma \leq -\frac{n}{2}(1+\rho^2),\ \sigma + \mu \leqslant 0,\ \text{and } z \in \mathbb{D}.$$

Let $\Omega \neq \mathbb{C}$ be a simply connected domain in $\mathbb{C}$ and let h be a conformal mapping of $\mathbb{D}$ onto Ω. In this case, $\mathcal{P}_n(h) := \mathcal{P}_n(\Omega) = \mathcal{P}_n(h(\mathbb{D}))$. The notations $\mathcal{P}(\Omega) := \mathcal{P}_1(\Omega)$ and $\mathcal{P}(h) := \mathcal{P}_1(h)$ are used when $n = 1$.

If ψ involves only the quantities $p(z)$, $zp'(z)$, and $z^2 p''(z)$, then the class $\mathcal{P}_n(\Omega)$ of admissible functions is defined as the class of all functions $\psi : \mathbb{C}^3 \to \mathbb{C}$ that satisfy the admissibility condition

$$\psi(\mathrm{i}\rho, \sigma, \mu + \mathrm{i}\nu) \notin \Omega \text{ when } \rho \in \mathbb{R},\ \sigma \leq -\frac{n}{2}(1+\rho^2),\ \text{and}\ \sigma + \mu \leqslant 0.$$

For first-order differential subordinations, the class $\mathcal{P}_n(\Omega)$ of admissible functions consists of all functions $\psi : \mathbb{C}^2 \times \mathbb{D} \to \mathbb{C}$ that satisfy the admissibility condition

$$\psi(i\rho, \sigma; z) \notin \Omega \text{ when } \rho \in \mathbb{R},\ \sigma \leq -\frac{n}{2}(1+\rho^2),\ \text{and } z \in \mathbb{D}.$$

If ψ involves only the quantities $p(z)$, and $zp'(z)$, then the class $\mathcal{P}_n(\Omega)$ of admissible functions is defined as the class of all functions $\psi : \mathbb{C}^2 \to \mathbb{C}$ that satisfy the admissibility condition

$$\psi(i\rho, \sigma) \notin \Omega \text{ when } \rho \in \mathbb{R},\ \text{and } \sigma \leq -\frac{n}{2}(1+\rho^2).$$

While dealing with subordination, the class $\mathcal{P}_n(\Omega)$ of admissible functions consists of all functions $\psi : \mathbb{C} \times \mathbb{D} \to \mathbb{C}$ that satisfy the admissibility condition

$$\psi(i\rho; z) \notin \Omega \text{ when } \rho \in \mathbb{R},\ \text{and } z \in \mathbb{D}.$$

With $q(z) = (1+z)/(1-z)$, Theorem 5.2.7 reduces to the following theorem.

Theorem 5.4.2 *Let $\psi \in \mathcal{P}_n(\Omega)$. If $p \in \mathcal{H}[1, n]$ satisfies*

$$\psi(p(z), zp'(z), z^2p''(z); z) \in \Omega, \quad z \in \mathbb{D},$$

then $p \in \mathcal{P}$.

Theorem 5.4.2 is a simple elegantly stated theorem. Yet it yields profound ramification as illustrated in the theorems of this remaining section.

Theorem 5.4.3 *Let $Re\beta > 0$ and $\alpha > -nRe\beta/2$. If $p \in \mathcal{H}[1, n]$ satisfies*

$$Re(\alpha p(z) + \beta zp'(z)) > -\frac{nRe\beta}{2}, \quad z \in \mathbb{D},$$

then $p \in \mathcal{P}$.

Proof Let $\Omega = \{w \in \mathbb{C} : \text{Re}w > -n\text{Re}\beta/2\}$ and $\psi : \mathbb{C}^2 \to \mathbb{C}$ be given by $\psi(r, s) = \alpha r + \beta s$. For $\rho \in \mathbb{R}$, and $\sigma \leqslant -n(1+\rho^2)/2 \leqslant -n/2$, the function ψ satisfies

$$\text{Re}\psi(i\rho, \sigma) = \text{Re}(i\alpha\rho + \beta\sigma) = (\text{Re}\beta)\sigma \leqslant -n\text{Re}\beta/2.$$

Thus, $\psi(i\rho, \sigma) \notin \Omega$ when $\rho \in \mathbb{R}$, and $\sigma \leqslant -n(1+\rho^2)/2$ and so $\psi \in \mathcal{P}_n(\Omega)$. Since $\psi(p(z), zp'(z)) \in \Omega$, the result now follows from Theorem 5.4.2. □

Corollary 5.4.4 *Let $\alpha \in \mathbb{R}$, $|\alpha| < n|Re\beta|/2$. If $p \in \mathcal{H}[1, n]$ satisfies*

$$|\alpha p(z) + \beta zp'(z)| < \frac{n|Re\beta|}{2}, \quad z \in \mathbb{D},$$

then $p \in \mathcal{P}$.

Theorem 5.4.5 *Let $P, R : \mathbb{D} \to \mathbb{C}$ satisfy*

$$|ImR(z)| \leqslant nReP(z), \quad z \in \mathbb{D}.$$

If $p \in \mathcal{H}[1, n]$ satisfies

$$Re(P(z)zp'(z) + R(z)p(z)) > 0, \quad z \in \mathbb{D}, \tag{5.4.3}$$

then $p \in \mathcal{P}$.

Proof Let $\Omega = \{w \in \mathbb{C} : \mathrm{Re}w > 0\}$ and $\psi : \mathbb{C}^2 \times \mathbb{D} \to \mathbb{C}$ be $\psi(r, s; z) := P(z)s + R(z)r$. For $\rho \in \mathbb{R}$, $\sigma \leqslant -n(1 + \rho^2)/2$, and $z \in \mathbb{D}$, the function ψ satisfies

$$\begin{aligned}
\mathrm{Re}\psi(\mathrm{i}\rho, \sigma; z) &= \mathrm{Re}(\sigma P(z) + \mathrm{i}\rho R(z)) = \sigma \mathrm{Re}P(z) - \rho \mathrm{Im}R(z) \\
&\leqslant -\frac{n}{2}(1 + \rho^2)\mathrm{Re}P(z) + \rho|\mathrm{Im}R(z)| \\
&\leqslant -\frac{n}{2}(1 + \rho^2)\mathrm{Re}P(z) + n\rho \mathrm{Re}P(z) \\
&= -\frac{n}{2}(1 - \rho)^2 \mathrm{Re}P(z) \leqslant 0,
\end{aligned}$$

or $\psi(\mathrm{i}\rho, \sigma; z) \notin \Omega$. Hence $\psi \in \mathcal{P}_n(\Omega)$. Since $\psi(p(z), zp'(z); z) \in \Omega$ by (5.4.3), it follows from Theorem 5.4.2 that $p \in \mathcal{P}$. □

Corollary 5.4.6 *Let $R : \mathbb{D} \to \mathbb{C}$ satisfy $|ImR(z)| \leqslant n$. If $p \in \mathcal{H}[1, n]$ satisfies*

$$Re(zp'(z) + R(z)p(z)) > 0, \quad z \in \mathbb{D},$$

then $p \in \mathcal{P}$.

Proof Take $P(z) \equiv 1$ in Theorem 5.4.5. □

Corollary 5.4.7 *Let $P : \mathbb{D} \to \mathbb{C}$ satisfy $ReP(z) \geqslant 0$. If $p \in \mathcal{H}[1, n]$ satisfies*

$$Re(p(z) + P(z)zp'(z)) > 0, \quad z \in \mathbb{D},$$

then $p \in \mathcal{P}$.

Proof Take $R(z) \equiv 1$ in Theorem 5.4.5. □

The following result is an application of the above corollary.

Theorem 5.4.8 *Let $\gamma \in \mathbb{R}$. If the functions M, N are analytic, $M(0) = N(0) = 0$ and $Re(zN'(z)/N(z)) > 0$, then*

$$\mathrm{Re}\,\frac{M'(z)}{N'(z)} > \gamma \quad \Rightarrow \quad \mathrm{Re}\,\frac{M(z)}{N(z)} > \gamma.$$

Proof Define the function $p : \mathbb{D} \to \mathbb{C}$ by

$$p(z) = \frac{1}{1-\gamma}\left(\frac{M(z)}{N(z)} - \gamma\right).$$

Then p is analytic in $\mathbb{D}$,

$$\mathrm{Re}\, p(0) = \frac{1}{1-\gamma}\mathrm{Re}\left(\frac{M(0)}{N(0)} - \gamma\right) = \frac{1}{1-\gamma}\mathrm{Re}\left(\frac{M'(0)}{N'(0)} - \gamma\right) = 1$$

and

$$p(z) + \frac{N(z)}{zN'(z)}zp'(z) = \frac{1}{1-\gamma}\left(\frac{M'(z)}{N'(z)} - \gamma\right). \tag{5.4.4}$$

Since $\mathrm{Re}(zN'(z)/N(z)) > 0$, the function

$$P(z) = \frac{N(z)}{zN'(z)}$$

satisfies $\mathrm{Re}\, P(z) = \mathrm{Re}(|P(z)|^2/P(z)) = |P(z)|^2\mathrm{Re}(zN'(z)/N(z)) > 0$. Using (5.4.4), the hypothesis $\mathrm{Re}(M'(z)/N'(z)) > \gamma$ becomes

$$\mathrm{Re}\left(p(z) + P(z)\cdot zp'(z)\right) > 0.$$

Since $p \in \mathcal{H}[p(0), n]$ and $\mathrm{Re}\, P(z) > 0$, Corollary 5.4.7 shows that $\mathrm{Re}\, p(z) > 0$, and so $\mathrm{Re}(M(z)/N(z)) > \gamma$. □

Theorem 5.4.9 *Let $\alpha \in \mathbb{R}$. If $p \in \mathcal{H}[1, n]$ satisfies*

$$Re\left(p(z) + \alpha\frac{zp'(z)}{p(z)}\right) > 0, \quad z \in \mathbb{D}, \tag{5.4.5}$$

then $p \in \mathcal{P}$.

Proof Let $\psi : \mathbb{C}^2 \to \mathbb{C}$ be defined by $\psi(r, s) = r + \alpha s/r$ and $\Omega = \{w \in \mathbb{C} : \mathrm{Re}\, w > 0\}$. The function ψ satisfies

$$\mathrm{Re}\,\psi(\mathrm{i}\rho, \sigma) = \mathrm{Re}\left(\mathrm{i}\rho + \frac{\alpha\sigma}{\mathrm{i}\rho}\right) = 0.$$

Thus, $\psi(\mathrm{i}\rho, \sigma) \notin \Omega$ for $\rho \in \mathbb{R}$, $\sigma \leqslant -n(1+\rho^2)/2$ and so $\psi \in \mathcal{P}_n(\Omega)$. Since $\psi(p(z), zp'(z)) \in \Omega$ by (5.4.5), it follows from Theorem 5.4.2 that $p \in \mathcal{P}$. □

When $\alpha > 0$, the following result holds.

Theorem 5.4.10 *Let $\alpha > 0$ and $\Omega = \mathbb{C} \setminus \{iy : y \in \mathbb{R},\ |y| \geqslant \sqrt{\alpha n(2+\alpha n)}\}$. If $p \in \mathcal{H}[1, n]$ satisfies*

$$p(z) + \alpha \frac{zp'(z)}{p(z)} \in \Omega, \quad z \in \mathbb{D}, \tag{5.4.6}$$

then $p \in \mathcal{P}$.

Proof The domain Ω is shown in Fig. 5.1. Here, ψ given by $\psi(r, s) = r + \alpha s/r$ satisfies

$$\psi(\mathrm{i}\rho, \sigma) = \mathrm{i}\tau \quad \text{where} \quad \tau = \frac{\rho^2 - \alpha\sigma}{\rho}.$$

For $\rho \in \mathbb{R}$, $\sigma \leqslant -n(1 + \rho^2)/2$, the quantity τ satisfies

$$|\tau| \geqslant \frac{(2 + \alpha n)\rho^2 + \alpha n}{2|\rho|} \geqslant \sqrt{\alpha n(2 + \alpha n)}.$$

(The last inequality follows as $ax^2 + b \geqslant 2\sqrt{ab}|x|$ for all $x \in \mathbb{R}$ when $a, b > 0$.) This shows that $\psi \in \mathcal{P}_n(\Omega)$. Since $\psi(p(z), zp'(z)) \in \Omega$ by (5.4.6), it follows from Theorem 5.4.2 that $p \in \mathcal{P}$. □

Theorem 5.4.11 *Let* $\alpha \leqslant 1$ *and* $\beta \geqslant 0$. *If* $p \in \mathcal{H}[1, n]$ *satisfies*

$$Re(\alpha p(z) + (1 - \alpha)p^2(z) + \beta zp'(z)) > -\frac{n\beta}{2}, \quad z \in \mathbb{D}, \tag{5.4.7}$$

then $p \in \mathcal{P}$.

Proof Let $\psi : \mathbb{C}^2 \to \mathbb{C}$ be defined by $\psi(r, s) = \alpha r + (1 - \alpha)r^2 + \beta s$ and $\Omega = \{w \in \mathbb{C} : \mathrm{Re}w > -\beta n/2\}$. The function ψ satisfies

$$\begin{aligned} \mathrm{Re}\psi(\mathrm{i}\rho, \sigma) &= \mathrm{Re}(i\alpha\rho - (1 - \alpha)\rho^2 + \beta\sigma) = -(1 - \alpha)\rho^2 + \beta\sigma \\ &\leqslant -(1 - \alpha)\rho^2 - \frac{n\beta}{2}(1 + \rho^2) \leqslant -\frac{n\beta}{2}, \end{aligned}$$

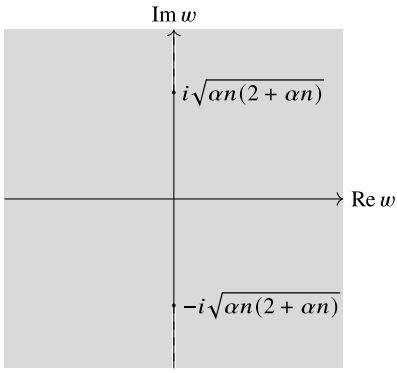


Fig. 5.1 Set Ω with two slits

or $\psi(\mathrm{i}\rho, \sigma) \notin \Omega$ when $\rho \in \mathbb{R}, \sigma \leqslant -n(1+\rho^2)/2$. Thus, $\psi \in \mathcal{P}_n(\Omega)$. Since (5.4.7) shows that $\psi(p(z), zp'(z)) \in \Omega$, it follows from Theorem 5.4.2 that $p \in \mathcal{P}$. □

Theorem 5.4.12 *Let $n \geqslant 2$. If $p \in \mathcal{H}[1, n]$ satisfies*

$$Re(p(z) + zp'(z) + 1 - p^2(z)) > 0, \quad z \in \mathbb{D}, \tag{5.4.8}$$

then $p \in \mathcal{P}$.

Proof Let $\psi(r, s) = r + s + 1 - r^2$ and $\Omega = \{w \in \mathbb{C} : \mathrm{Re}w > 0\}$. It is first shown that $\psi \in \mathcal{P}_n(\Omega)$, for $n \geqslant 2$. The function ψ satisfies

$$\begin{aligned}\mathrm{Re}\psi(\mathrm{i}\rho, \sigma) &= \mathrm{Re}(\mathrm{i}\rho + \sigma + 1 + \rho^2) = \sigma + 1 + \rho^2 \\ &\leqslant -\frac{n}{2}(1+\rho^2) + 1 + \rho^2 = \frac{2-n}{2}(1+\rho^2) \leqslant 0,\end{aligned}$$

or $\psi(\mathrm{i}\rho, \sigma) \notin \Omega$ when $\rho \in \mathbb{R}, \sigma \leqslant -n(1+\rho^2)/2$ and $n \geqslant 2$. Thus, $\psi \in \mathcal{P}_n(\Omega)$ for $n \geqslant 2$. Since $\psi(p(z), zp'(z)) \in \Omega$ by (5.4.8), it follows from Theorem 5.4.2 that $p \in \mathcal{P}$. □

Note that the condition $n \geqslant 2$ is needed since $\psi \notin \mathcal{P}_1(\Omega)$.

Theorem 5.4.13 *If $p \in \mathcal{H}[1, n]$ satisfies*

$$Re\left(2p(z) - \frac{zp''(z)}{p'(z)}\right) > 1, \quad z \in \mathbb{D}, \tag{5.4.9}$$

then $p \in \mathcal{P}$.

Proof Let $\psi(r, s, t) = 2r - t/s - 1$ and $\Omega = \{w \in \mathbb{C} : \mathrm{Re}w > 0\}$. The function ψ satisfies

$$\mathrm{Re}\psi(\mathrm{i}\rho, \sigma, \mu + \mathrm{i}\nu) = \mathrm{Re}\left(2\mathrm{i}\rho - \frac{\mu + \mathrm{i}\nu}{\sigma} - 1\right) = -\frac{\mu + \sigma}{\sigma} \leqslant 0,$$

or $\psi(\mathrm{i}\rho, \sigma, \mu + \mathrm{i}\nu) \notin \Omega$ when $\rho \in \mathbb{R}, \sigma \leqslant -n(1+\rho^2)/2$ and $\sigma + \mu \leqslant 0$. Since $\psi \in \mathcal{P}_n(\Omega)$, and $\psi(p(z), zp'(z), z^2p''(z)) \in \Omega$ by (5.4.9), evidently $p \in \mathcal{P}$ by Theorem 5.4.2. □

Theorem 5.4.14 *If $p \in \mathcal{H}[1, n]$ satisfies*

$$|z^2p''(z) + 5zp'(z) - p^2(z) + 1| < 2n - 1, \quad z \in \mathbb{D}, \tag{5.4.10}$$

then $p \in \mathcal{P}$.

Proof Let $\Omega = \{w \in \mathbb{C} : |w| < 2n - 1\}$. The function $\psi : \mathbb{C}^3 \to \mathbb{C}$ given by $\psi(r, s, t) = 1 - r^2 + 5s + t$ satisfies

$$\begin{aligned} |\psi(\mathrm{i}\rho, \sigma, \mu + \mathrm{i}\nu)| &= |1 + \rho^2 + 5\sigma + \mu + \mathrm{i}\nu| \geqslant -(1 + \rho^2 + 5\sigma + \mu) \\ &= -(1 + \rho^2) - 4\sigma - (\sigma + \mu) \geqslant -(1 + \rho^2) - 4\sigma \\ &\geqslant -(1 + \rho^2) + 2n(1 + \rho^2) = (2n - 1)(1 + \rho^2) \geqslant 2n - 1, \end{aligned}$$

or $\psi(\mathrm{i}\rho, \sigma, \mu + \mathrm{i}\nu) \notin \Omega$ when $\rho \in \mathbb{R}$, $\sigma \leqslant -n(1 + \rho^2)/2$ and $\sigma + \mu \leqslant 0$. Since $\psi \in \mathcal{P}_n(\Omega)$, and $\psi(p(z), zp'(z), z^2p''(z)) \in \Omega$ by (5.4.10), it follows from Theorem 5.4.2 that $p \in \mathcal{P}$. □

Theorem 5.4.15 *Let $\alpha > 0$ and $\beta > 0$. If $p \in \mathcal{H}[1, n]$ satisfies*

$$Re\left(\alpha p(z) + \beta(zp'(z) + z^2p''(z))\right) > 0, \quad z \in \mathbb{D}, \tag{5.4.11}$$

then $p \in \mathcal{P}$.

Proof Let $\Omega = \{w \in \mathbb{C} : \mathrm{Re}w > 0\}$. The function $\psi(r, s, t) = \alpha r + \beta(s + t)$ satisfies

$$\mathrm{Re}\psi(\mathrm{i}\rho, \sigma, \mu + \mathrm{i}\nu) = \mathrm{Re}(\mathrm{i}\alpha\rho + \beta(\sigma + \mu + \mathrm{i}\nu)) = \beta(\sigma + \mu) \leqslant 0,$$

or $\psi(\mathrm{i}\rho, \sigma, \mu + \mathrm{i}\nu) \notin \Omega$ when $\rho \in \mathbb{R}$, $\sigma \leqslant -n(1 + \rho^2)/2$ and $\sigma + \mu \leqslant 0$. Thus, $\psi \in \mathcal{P}_n(\Omega)$. Since $\psi(p(z), zp'(z), z^2p''(z)) \in \Omega$ by (5.4.11), it follows from Theorem 5.4.2 that $p \in \mathcal{P}$. □

Theorem 5.4.16 *If $p \in \mathcal{H}[1, n]$ satisfies*

$$Re(z^2p''(z) + 3zp'(z) - p^2(z)) > -n, \quad z \in \mathbb{D}, \tag{5.4.12}$$

then $p \in \mathcal{P}$.

Proof Let $\Omega = \{w \in \mathbb{C} : \mathrm{Re}w > -n\}$. The function $\psi(r, s, t) = t - r^2 + 3s$ satisfies

$$\begin{aligned} \mathrm{Re}\psi(\mathrm{i}\rho, \sigma, \mu + \mathrm{i}\nu) &= \mathrm{Re}(\rho^2 + 3\sigma + \mu + \mathrm{i}\nu) = \rho^2 + 3\sigma + \mu \\ &= \rho^2 + 2\sigma + (\sigma + \mu) \leqslant \rho^2 + 2\sigma \leqslant \rho^2 - n(1 + \rho^2) \\ &= -n + (1 - n)\rho^2 \leqslant -n. \end{aligned}$$

Thus, $\psi(\mathrm{i}\rho, \sigma, \mu + \mathrm{i}\nu) \notin \Omega$ when $\rho \in \mathbb{R}, \sigma \leqslant -n(1 + \rho^2)/2$ and $\sigma + \mu \leqslant 0$, which shows that $\psi \in \mathcal{P}_n(\Omega)$. Since $\psi(p(z), zp'(z), z^2p''(z)) \in \Omega$ by (5.4.12), it follows from Theorem 5.4.2 that $p \in \mathcal{P}$. □

5.5 Carathéodory Functions with Image in Sectors

For $0 < \alpha \leqslant 2$, the class $\mathcal{P}[\alpha]$ consists of analytic functions p with $p(0) = 1$ satisfying the subordination $p(z) \prec q(z) = ((1+z)/(1-z))^{\alpha}$. For $0 < \alpha \leqslant 1$, the class $\mathcal{P}[\alpha]$ is a subclass of $\mathcal{P}$. This class is associated with the class of strongly starlike functions that will be studied later. Since $\mathrm{Re}\, w > 0$ is the same as $|\arg w| < \pi/2$, it follows that $|\arg((1+z)/(1-z))| < \pi/2$ and so the function

$$q(z) = \left(\frac{1+z}{1-z}\right)^{\alpha}$$

in $\mathcal{P}$ satisfies

$$|\arg(q(z))| = \alpha \left|\arg\left(\frac{1+z}{1-z}\right)\right| < \frac{\alpha\pi}{2}.$$

Thus the function q maps $\mathbb{D}$ onto the sector $\{w \in \mathbb{C} : |\arg w| < \alpha\pi/2\}$ and $E(q) = \{1\}$ (see Fig. 5.2).

For $\zeta = \mathrm{e}^{\mathrm{i}\theta} \neq 1$,

$$\frac{1+z}{1-z} = \mathrm{i}\cot(\theta/2) = \pm\mathrm{i}\rho,$$

where $\rho \geqslant 0$. Thus, for $\zeta \neq 1$ with $|\zeta| = 1$,

$$\arg q(\zeta) = \alpha \arg(\pm\mathrm{i}\rho) = \pm\frac{\alpha\pi}{2}$$

and so

$$r = q(\zeta) = (\pm\mathrm{i}\rho)^{\alpha} = \rho^{\alpha}\mathrm{e}^{\pm\frac{\mathrm{i}\alpha\pi}{2}} \quad \text{where} \quad \rho \geqslant 0. \tag{5.5.1}$$

A computation shows that

$$zq'(z) = \alpha\left(\frac{1+z}{1-z}\right)^{\alpha-1}\frac{2z}{(1-z)^2} = \frac{\alpha}{2}q(z)^{1-\frac{1}{\alpha}}((q(z))^{\frac{2}{\alpha}} - 1) \tag{5.5.2}$$

and

$$\frac{zq''(z)}{q'(z)} + 1 = \frac{1+2\alpha z+z^2}{1-z^2}. \tag{5.5.3}$$

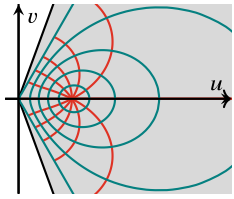


Fig. 5.2 $q(\mathbb{D})$ when $q(z) = ((1+z)/(1-z))^{\alpha}$, $\alpha = 0.7$

If $\zeta \in \partial\mathbb{D} \setminus \{1\}$, then, by using (5.5.1) in (5.5.2),

$$s = m\zeta q'(\zeta) = \frac{m\alpha}{2}(1+\rho^2)\rho^{\alpha-1}\mathrm{e}^{\pm\frac{i(\alpha+1)\pi}{2}}.$$

Also, for $\zeta \in \partial\mathbb{D} \setminus \{1\}$, (5.5.3) yields

$$\mathrm{Re}\left(\frac{\zeta q''(\zeta)}{q'(\zeta)} + 1\right) = 0.$$

Thus, Definition 5.2.4 of the class of admissible functions is equivalent to the following:

Definition 5.5.1 Let Ω be a subset of $\mathbb{C}$. The class $\mathcal{P}_n(\Omega, \alpha)$ of *admissible functions* consists of all functions $\psi : \mathbb{C}^3 \times \mathbb{D} \to \mathbb{C}$ that satisfy the admissibility condition $\psi(r, s, t; z) \notin \Omega$ when $\rho > 0$, $z \in \mathbb{D}$, $m \geqslant n$,

$$r = \rho^\alpha \mathrm{e}^{\frac{\mathrm{i}\alpha\pi}{2}}, \quad s = \frac{m\alpha}{2}(1+\rho^2)\rho^{\alpha-1}\mathrm{e}^{\frac{\mathrm{i}(\alpha+1)\pi}{2}}, \quad \mathrm{Re}\frac{t}{s} \geqslant -1,$$

and

$$r = \rho^\alpha \mathrm{e}^{-\frac{\mathrm{i}\alpha\pi}{2}}, \quad s = \frac{m\alpha}{2}(1+\rho^2)\rho^{\alpha-1}\mathrm{e}^{-\frac{\mathrm{i}(\alpha+1)\pi}{2}}, \quad \mathrm{Re}\frac{t}{s} \geqslant -1.$$

If $r^{1/\alpha} = \mathrm{i}\rho$, then

$$\frac{s}{r} = \frac{m\alpha}{2\rho}(1+\rho^2)\mathrm{e}^{\frac{\mathrm{i}\pi}{2}} = \mathrm{i}k\alpha \quad \text{where} \quad k \geqslant \frac{n}{2}\left(\rho + \frac{1}{\rho}\right) \geqslant n. \tag{5.5.4}$$

Similarly, if $r^{1/\alpha} = -\mathrm{i}\rho$, then

$$\frac{s}{r} = \frac{m\alpha}{2\rho}(1+\rho^2)\mathrm{e}^{-\frac{\mathrm{i}\pi}{2}} = \mathrm{i}k\alpha \quad \text{where} \quad k \leqslant -\frac{n}{2}\left(\rho + \frac{1}{\rho}\right) \leqslant -n. \tag{5.5.5}$$

For the function q defined by $q(z) = ((1+z)/(1-z))^\alpha$, Theorem 5.2.7 becomes

Theorem 5.5.2 *Let $\psi \in \mathcal{P}_n(\Omega, \alpha)$. If $p \in \mathcal{H}[1, n]$ satisfies*

$$\psi(p(z), zp'(z), z^2p''(z); z) \in \Omega, \quad z \in \mathbb{D},$$

then $p \in \mathcal{P}[\alpha]$.

Theorem 5.5.3 *Let $0 < \alpha \leqslant 2$ and β be given by*

$$\beta = \frac{2}{\pi}\tan^{-1}(n\alpha).$$

If $p \in \mathcal{H}[1, n]$ satisfies

$$\left|\arg\left(1+\frac{zp'(z)}{p(z)}\right)\right| < \frac{\beta\pi}{2}, \quad z \in \mathbb{D}, \tag{5.5.6}$$

then $p \in \mathcal{P}[\alpha]$.

Proof Let $\Omega = \{w \in \mathbb{C} : |\arg w| < \beta\pi/2\}$, and $\psi(r,s) = 1 + s/r$. Then, the inequality (5.5.6) shows that $\psi(p(z), zp'(z)) \in \Omega$. Note that $\psi(r,s) \notin \Omega$ if either $\beta\pi/2 \leqslant \arg\psi(r,s) \leqslant 2\pi - \beta\pi/2$ or $-2\pi + \beta\pi/2 \leqslant \arg\psi(r,s) \leqslant -\beta\pi/2$. First, consider the case when

$$r = \rho^{\alpha} e^{\frac{i\alpha\pi}{2}}, \quad s = \frac{m\alpha}{2}(1+\rho^2)\rho^{\alpha-1} e^{\frac{i(\alpha+1)\pi}{2}}$$

where $m \geqslant n$ and $k \geqslant \frac{n}{2}(\rho + \frac{1}{\rho}) \geqslant n$. In this case, using (5.5.4),

$$\arg\psi(r,s) = \arg(1 + ik\alpha) = \tan^{-1}(k\alpha) \geqslant \tan^{-1}(n\alpha) = \frac{\beta\pi}{2}.$$

Since $\beta \leqslant 1$,

$$\arg\psi(r,s) = \tan^{-1}(k\alpha) \leqslant \frac{\pi}{2} \leqslant 2\pi - \frac{\beta\pi}{2}.$$

Similarly, when

$$r = \rho^{\alpha} e^{-\frac{i\alpha\pi}{2}}, \quad s = \frac{m\alpha}{2}(1+\rho^2)\rho^{\alpha-1} e^{-\frac{i(\alpha+1)\pi}{2}}$$

where $m \geqslant n$ and $k \leqslant -\frac{n}{2}(\rho + \frac{1}{\rho}) \leqslant -n$, using (5.5.5),

$$\arg\psi(r,s) = \tan^{-1}(k\alpha) \leqslant -\tan^{-1}(n\alpha) = -\frac{\beta\pi}{2},$$

and since $\beta \leqslant 1$,

$$\arg\psi(r,s) = \tan^{-1}(k\alpha) \geqslant -\frac{\pi}{2} \geqslant -2\pi + \frac{\beta\pi}{2}.$$

Thus, $\psi(r,s) \notin \Omega$, where r and s are as in the Definition 5.5.1. This proves $\psi \in \mathcal{P}_n(\Omega, \alpha)$ and, by Theorem 5.5.2, $p \in \mathcal{P}[\alpha]$. □

Theorem 5.5.4 *Let* $\gamma \geqslant 0$ *and* $\alpha_0 = \alpha_0(\gamma, n)$ *be the root of the equation*

$$2\alpha = 3 - \frac{2}{\pi}\tan^{-1}(n\gamma\alpha). \tag{5.5.7}$$

For $0 < \alpha \leqslant \alpha_0$, *let* β *be given by*

$$\beta = \alpha + \frac{2}{\pi}\tan^{-1}(n\gamma\alpha). \tag{5.5.8}$$

If $p \in \mathcal{H}[1, n]$ *satisfies*

$$|\arg(p(z) + \gamma z p'(z))| < \frac{\beta\pi}{2}, \quad z \in \mathbb{D}, \tag{5.5.9}$$

then $p \in \mathcal{P}[\alpha]$.

Proof Let $\Omega = \{w \in \mathbb{C} : |\arg w| < \beta\pi/2\}$, and $\psi(r, s) = r + \gamma s$. Then, (5.5.9) shows that $\psi(p(z), zp'(z)) \in \Omega$. By Theorem 5.5.2, $p \in \mathcal{P}[\alpha]$ if $\psi \in \mathcal{P}_n(\Omega, \alpha)$. Note that $\psi(r, s) \notin \Omega$ if either $\beta\pi/2 \leqslant \arg\psi(r, s) \leqslant 2\pi - \beta\pi/2$ or $-2\pi + \beta\pi/2 \leqslant \arg\psi(r, s) \leqslant -\beta\pi/2$. Since α_0 is the root of Eq. (5.5.7), it follows that $2\alpha \leqslant 3 - \frac{2}{\pi}\tan^{-1}(n\gamma\alpha)$ for $0 < \alpha \leqslant \alpha_0$, and by using (5.5.8) this gives $\alpha + \beta \leqslant 3$.

First consider the case when

$$r = \rho^{\alpha} e^{\frac{i\alpha\pi}{2}}, \quad s = \frac{m\alpha}{2}(1 + \rho^2)\rho^{\alpha-1} e^{\frac{i(\alpha+1)\pi}{2}},$$

where $m \geqslant n$ and $\rho + 1/\rho \geqslant 2$. Then

$$\begin{aligned}
\arg\psi(r, s) &= \arg\left(\rho^{\alpha} e^{\frac{i\alpha\pi}{2}}\left(1 + \frac{m\alpha\gamma}{2}\left(\rho + \frac{1}{\rho}\right)i\right)\right) \\
&= \frac{\alpha\pi}{2} + \tan^{-1}\left(\frac{m\alpha\gamma}{2}\left(\rho + \frac{1}{\rho}\right)\right) \\
&\geqslant \frac{\alpha\pi}{2} + \tan^{-1}(n\alpha\gamma) = \frac{\beta\pi}{2}.
\end{aligned}$$

It is also evident from $\alpha + \beta \leqslant 3$ that

$$\begin{aligned}
\arg\psi(r, s) &= \frac{\alpha\pi}{2} + \tan^{-1}\left(\frac{m\alpha\gamma}{2}\left(\rho + \frac{1}{\rho}\right)\right) \\
&\leqslant \frac{\alpha\pi}{2} + \frac{\pi}{2} \leqslant 2\pi - \frac{\beta\pi}{2}.
\end{aligned}$$

Next, consider the case when

$$r = \rho^{\alpha} e^{-\frac{i\alpha\pi}{2}}, \quad s = \frac{m\alpha}{2}(1 + \rho^2)\rho^{\alpha-1} e^{-\frac{i(\alpha+1)\pi}{2}},$$

where $m \geqslant n$ and $\rho + 1/\rho \geqslant 2$. Here,

$$\begin{aligned}
\arg\psi(r, s) &= -\frac{\alpha\pi}{2} - \tan^{-1}\left(\frac{m\alpha\gamma}{2}\left(\rho + \frac{1}{\rho}\right)\right) \\
&\leqslant -\frac{\alpha\pi}{2} - \tan^{-1}(n\alpha\gamma) = -\frac{\beta\pi}{2},
\end{aligned}$$

and,

$$\arg \psi(r,s) \geqslant -\frac{\alpha\pi}{2} - \frac{\pi}{2} \geqslant -2\pi + \frac{\beta\pi}{2}.$$

Thus, $\psi(r,s) \notin \Omega$, and this shows that $\psi \in \mathcal{P}_n(\Omega,\alpha)$. □

Theorem 5.5.5 *Let* $0 < \alpha \leqslant 1$. *If* $p \in \mathcal{H}[1,1]$ *satisfies*

$$\left|\arg\left(p(z) + \frac{zp'(z)}{p(z)}\right)\right| < \frac{\beta\pi}{2}, \quad z \in \mathbb{D}, \tag{5.5.10}$$

where

$$\beta = \alpha + \frac{2}{\pi}\tan^{-1}\frac{\alpha(1-\alpha)^{\frac{\alpha-1}{2}}\cos\frac{\alpha\pi}{2}}{(1+\alpha)^{\frac{1+\alpha}{2}} + \alpha(1-\alpha)^{\frac{\alpha-1}{2}}\sin\frac{\alpha\pi}{2}},$$

then $p \in \mathcal{P}[\alpha]$.

Proof Let $\Omega = \{w \in \mathbb{C} : |\arg w| < \beta\pi/2\}$ and $\psi(r,s) = r + s/r$. Then (5.5.10), yields $\psi(p(z), zp'(z)) \in \Omega$. Thus, $p \in \mathcal{P}[\alpha]$ follows from Theorem 5.5.2 if $\psi \in \mathcal{P}_1(\Omega,\alpha)$. As noted previously, $\psi(r,s) \notin \Omega$ whenever $\beta\pi/2 \leqslant \arg\psi(r,s) \leqslant 2\pi - \beta\pi/2$ or $-2\pi + \beta\pi/2 \leqslant \arg\psi(r,s) \leqslant -\beta\pi/2$. Let

$$r = \rho^{\alpha} e^{\frac{i\alpha\pi}{2}}, \quad s = \frac{m\alpha}{2}(1+\rho^2)\rho^{\alpha-1}e^{\frac{i(\alpha+1)\pi}{2}},$$

where $m \geqslant 1$ and $k \geqslant (\rho + 1/\rho)/2$. Using (5.5.4),

$$\begin{aligned}\psi(r,s) &= r + \frac{s}{r} = r\left(1 + \frac{s}{r}\frac{1}{r}\right)\\ &= (i\rho)^{\alpha}\left(1 + i\alpha k\frac{1}{(i\rho)^{\alpha}}\right)\\ &= \rho^{\alpha}e^{i\frac{\pi\alpha}{2}}\left(1 + e^{i\frac{\pi}{2}(1-\alpha)}\alpha k\frac{1}{\rho^{\alpha}}\right).\end{aligned}$$

Thus,

$$\begin{aligned}\arg\psi(r,s) &= \frac{\pi\alpha}{2} + \arg\left(1 + e^{i\frac{\pi}{2}(1-\alpha)}\frac{\alpha k}{\rho^{\alpha}}\right)\\ &= \frac{\pi\alpha}{2} + \tan^{-1}\frac{h(\rho)\sin(1-\alpha)\frac{\pi}{2}}{1 + h(\rho)\cos(1-\alpha)\frac{\pi}{2}}\end{aligned} \tag{5.5.11}$$

where

$$h(\rho) := \alpha k\frac{1}{\rho^{\alpha}} \geqslant \frac{\alpha}{2}\left(\rho^{1-\alpha} + \rho^{-1-\alpha}\right) =: g(\rho).$$

A straightforward computation yields

$$g'(\rho) = \frac{\alpha}{2}\left((1-\alpha)\rho^{-\alpha} - (1+\alpha)\rho^{-2-\alpha}\right),$$

with the minimum attained at $\rho_0 = \sqrt{(1+\alpha)/(1-\alpha)}$. Thus,

$$h(\rho) \geqslant g(\rho_0) = \frac{\alpha}{2}\left(\left(\frac{1+\alpha}{1-\alpha}\right)^{\frac{1-\alpha}{2}} + \left(\frac{1+\alpha}{1-\alpha}\right)^{\frac{-1-\alpha}{2}}\right) = \alpha\frac{(1-\alpha)^{\frac{\alpha-1}{2}}}{(1+\alpha)^{\frac{1+\alpha}{2}}}.$$

Since the function $\tan^{-1}(a\,h(\rho)/(1+b\,h(\rho))$ is an increasing function of ρ,

$$\begin{aligned}\arg\psi(r,s) &\geqslant \frac{\pi\alpha}{2} + \tan^{-1}\frac{g(\rho_0)\sin(1-\alpha)\frac{\pi}{2}}{1+g(\rho_0)\cos(1-\alpha)\frac{\pi}{2}}\\ &\geqslant \frac{\pi\alpha}{2} + \tan^{-1}\frac{\alpha(1-\alpha)^{\frac{\alpha-1}{2}}\cos\frac{\alpha\pi}{2}}{(1+\alpha)^{\frac{1+\alpha}{2}} + \alpha(1-\alpha)^{\frac{\alpha-1}{2}}\sin\frac{\alpha\pi}{2}} = \frac{\beta\pi}{2}.\end{aligned}$$

Since $\beta \leqslant \alpha + 1$, it follows that $\alpha + \beta \leqslant 2\alpha + 1 \leqslant 3$. By using (5.5.11),

$$\arg\psi(r,s) \leqslant \frac{\pi\alpha}{2} + \frac{\pi}{2} \leqslant 2\pi - \frac{\beta\pi}{2}.$$

Similarly, for

$$r = \rho^{\alpha}\mathrm{e}^{-\frac{\mathrm{i}\alpha\pi}{2}}, \quad s = \frac{m\alpha}{2}(1+\rho^2)\rho^{\alpha-1}\mathrm{e}^{-\frac{i(\alpha+1)\pi}{2}},$$

it follows that

$$\arg\psi(r,s) \leqslant -\frac{\pi\alpha}{2} - \tan^{-1}\frac{\alpha(1-\alpha)^{\frac{\alpha-1}{2}}\cos\frac{\alpha\pi}{2}}{(1+\alpha)^{\frac{1+\alpha}{2}} + \alpha(1-\alpha)^{\frac{\alpha-1}{2}}\sin\frac{\alpha\pi}{2}} = -\frac{\beta\pi}{2}$$

and

$$\arg\psi(r,s) \geqslant -\frac{\pi\alpha}{2} - \frac{\pi}{2} \geqslant -2\pi + \frac{\beta\pi}{2}.$$

Hence, $\psi(r,s) \notin \Omega$ for all r and s as in the Definition 5.5.1. This proves that $\psi \in \mathcal{P}_1(\Omega,\alpha)$. □

Proceeding as in the proof of the preceding theorem, we derive the following corollary.

Corollary 5.5.6 *Let* $0 < \alpha \leqslant 1$. *If* $p \in \mathcal{H}[1,1]$ *satisfies*

$$\left|\arg\left(1 + \frac{zp'(z)}{p^2(z)}\right)\right| < \frac{\beta\pi}{2}, \quad z \in \mathbb{D},$$

where

$$\beta = \frac{2}{\pi}\tan^{-1}\frac{\alpha(1-\alpha)^{\frac{\alpha-1}{2}}\cos\frac{\alpha\pi}{2}}{(1+\alpha)^{\frac{1+\alpha}{2}}+\alpha(1-\alpha)^{\frac{\alpha-1}{2}}\sin\frac{\alpha\pi}{2}},$$

then $p \in \mathcal{P}[\alpha]$.

5.6 Functions Subordinate to e^z

This section focuses on the class $\mathcal{P}_e$ consisting of all functions $p \in \mathcal{P}$ satisfying the subordination $p(z) \prec e^z$. Consider the function $q(z) = e^z$ in $\mathbb{D}$. The function q is univalent in $\overline{\mathbb{D}}$ and its image is $q(\mathbb{D}) = \{w \in \mathbb{C} : |\log w| < 1\}$ (see Fig. 5.3). Hence, $q \in \mathcal{Q}$ with $E(q) = \emptyset$.

For $|\zeta| = 1$, $q(\zeta) \in q(\partial\mathbb{D}) = \partial q(\mathbb{D}) = \{w \in \mathbb{C} : |\log w| = 1\}$. This gives $|\log q(\zeta)| = 1$ so that $\log q(\zeta) = e^{i\theta}$, where $\theta \in [0, 2\pi)$ and hence $q(\zeta) = e^{e^{i\theta}}$. Since $q(\zeta) = e^{\zeta}$, this implies that $\zeta = e^{i\theta}$. Also $\zeta q'(\zeta) = e^{i\theta}e^{e^{i\theta}}$ and

$$\operatorname{Re}\left(1 + \frac{\zeta q''(\zeta)}{q'(\zeta)}\right) = \operatorname{Re}(1 + e^{i\theta}) = 1 + \cos\theta.$$

For the second-order differential subordination $\psi(p(z), zp'(z), z^2p''(z); z) \in \Omega$, Definition 5.2.4 of the class of admissible functions is equivalent to the following:

Definition 5.6.1 The class of *admissible functions* $\mathcal{E}_n(\Omega)$ consists of all functions $\psi : \mathbb{C}^3 \times \mathbb{D} \to \mathbb{C}$ satisfying $\psi(r, s, t; z) \notin \Omega$ whenever

$$r = e^{e^{i\theta}}, \quad s = me^{i\theta}r, \quad \text{and} \quad \operatorname{Re}(1 + t/s) \geqslant m(1 + \cos\theta) \tag{5.6.1}$$

where $z \in \mathbb{D}, \theta \in [0, 2\pi)$ and $m \geqslant n \geqslant 1$. The case $n = 1$ is simply written $\mathcal{E}(\Omega) := \mathcal{E}_1(\Omega)$.

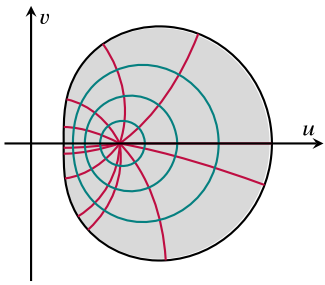


Fig. 5.3 $q(\mathbb{D})$ when $q(z) = e^z$

If $\psi : \mathbb{C}^2 \times \mathbb{D} \to \mathbb{C}$, then the admissibility condition (5.6.1) reduces to

$$\psi(e^{e^{i\theta}}, me^{i\theta}e^{e^{i\theta}}; z) \notin \Omega$$

where $z \in \mathbb{D}$, $\theta \in [0, 2\pi)$ and $m \geqslant n \geqslant 1$. We shall drop z if $\psi : \mathbb{C}^2 \to \mathbb{C}$.

The following theorem is a particular case of Theorem 5.2.7.

Theorem 5.6.2 *Let $\psi \in \mathcal{E}_n(\Omega)$. If $p \in \mathcal{H}[1, n]$ satisfies*

$$\psi(p(z), zp'(z), z^2p''(z); z) \in \Omega,$$

then $p \in \mathcal{P}_e$.

The examples below serve to illustrate Theorem 5.6.2. The conclusion $p \in \mathcal{P}_e$ or equivalently $p(z) \prec e^z$ is inferred from Theorem 5.6.2 once the function ψ is shown to be in the class $\mathcal{E}(\Omega)$.

Example 5.6.3 If $p \in \mathcal{H}[1, 1]$ satisfies

$$|p(z) + (1 + 2e)zp'(z)| < 2, \quad z \in \mathbb{D},$$

then $p \in \mathcal{P}_e$.

Let $\psi(r, s) = r + (1 + 2e)s$, and $\Omega = \{w \in \mathbb{C} : |w| < 2\}$. Let $r = e^{e^{i\theta}}$, and $s = me^{i\theta}r$ where $\theta \in [0, 2\pi)$ and $m \geqslant 1$. Then

$$\begin{aligned} |\psi(r, s)| &= e^{\cos\theta}|1 + (1 + 2e)me^{i\theta}| \\ &\geqslant e^{-1}((1 + 2e)m - 1) \geqslant 2. \end{aligned}$$

Hence, $\psi(r, s) \notin \Omega$ or $\psi \in \mathcal{E}(\Omega)$.

Example 5.6.4 If $p \in \mathcal{H}[1, 1]$ satisfies

$$1 + (1 + \sqrt{2})e\, zp'(z) \prec \sqrt{1 + z},$$

then $p \in \mathcal{P}_e$.

Let $\psi(r, s) = 1 + (1 + \sqrt{2})es$ and $h(z) = \sqrt{1 + z}$. Then $\Omega = h(\mathbb{D}) = \{w \in \mathbb{C} : |w^2 - 1| < 1\}$. Let $r = e^{e^{i\theta}}$ and $s = me^{i\theta}r$ where $\theta \in [0, 2\pi)$ and $m \geqslant 1$. Then

$$\begin{aligned} |(\psi(r, s))^2 - 1| &= (1 + \sqrt{2})e|s||(1 + \sqrt{2})es + 2| \\ &\geqslant (1 + \sqrt{2})e|s|((1 + \sqrt{2})e|s| - 2) \\ &= (1 + \sqrt{2})me^{1+\cos\theta}((1 + \sqrt{2})me^{1+\cos\theta} - 2) \\ &\geqslant (1 + \sqrt{2})(\sqrt{2} - 1) = 1 \end{aligned}$$

Thus, $\psi(r, s) \notin \Omega$ and therefore $\psi \in \mathcal{E}(\Omega)$.

Example 5.6.5 If $p \in \mathcal{H}[1, 1]$ satisfies

$$|2zp'(z) + z^2 p''(z)| < \mathrm{e}^{-1}, \quad z \in \mathbb{D},$$

then $p \in \mathcal{P}_e$.

Let $\psi(r, s, t) = 1 + 2s + t$ and let $\Omega = \{w \in \mathbb{C}: |w - 1| < \mathrm{e}^{-1}\}$. Then, $\psi(p(z), zp'(z), z^2 p''(z)) \in \Omega$. To verify the admissibility condition, let $r = \mathrm{e}^{\mathrm{e}^{i\theta}}$, $s = m\mathrm{e}^{i\theta} r$ and $\mathrm{Re}\,(1 + t/s) \geq m(1 + \cos\theta)$, where $z \in \mathbb{D}$, $\theta \in [0, 2\pi)$ and $m \geqslant 1$. Now,

$$\begin{aligned} |\psi(r, s, t; z) - 1| = |2s + t| &= |s| \left|2 + \frac{t}{s}\right| \\ &\geqslant m\mathrm{e}^{\cos\theta} \mathrm{Re}\left(2 + \frac{t}{s}\right) \\ &\geqslant m\mathrm{e}^{\cos\theta}(1 + m(1 + \cos\theta)) \\ &\geqslant m\mathrm{e}^{\cos\theta} \geqslant \mathrm{e}^{-1}. \end{aligned}$$

This shows that $\psi \in \mathcal{E}(\Omega)$.

Example 5.6.6 If $p \in \mathcal{H}[1, 1]$, then each of the following conditions is sufficient for $p \in \mathcal{P}_e$:

1. $\left|zp'(z)\right| < \mathrm{e}^{-1}$,
2. $|p^2(z) - p(z) + (2 + \mathrm{e})zp'(z)| < \mathrm{e}^{-1}$,
3. $\left|zp'(z)/p^2(z)\right| < \mathrm{e}^{-1}$,

for $z \in \mathbb{D}$.

Define the following functions: $\psi_1(r, s) = 1 + s$, $\psi_2(r, s) = r^2 - r + (2 + \mathrm{e})s + 1$, and $\psi_3(r, s) = 1 + s/r^2$ and let $\Omega = \{w \in \mathbb{C}: |w - 1| < \mathrm{e}^{-1}\}$. Then, under each of the corresponding hypotheses, $\psi_i(p(z), zp'(z)) \in \Omega$. The desired conclusion follows once it is verified that $\psi_i \in \mathcal{E}(\Omega)$.

To verify the admissibility condition, let $r = \mathrm{e}^{\mathrm{e}^{i\theta}}$, and $s = m\mathrm{e}^{i\theta} r$, where $\theta \in [0, 2\pi)$ and $m \geqslant 1$. Then

$$\begin{aligned} |\psi_1(r, s) - 1| &= |s| = |m\mathrm{e}^{i\theta}\mathrm{e}^{\mathrm{e}^{i\theta}}| \\ &= m\mathrm{e}^{\cos\theta} \geqslant m\mathrm{e}^{-1} \geqslant \mathrm{e}^{-1}, \\ |\psi_2(r, s) - 1| &= |r^2 - r + (2 + \mathrm{e})s| \\ &= \mathrm{e}^{\cos\theta}|\mathrm{e}^{\mathrm{e}^{i\theta}} - 1 + (2 + \mathrm{e})m\mathrm{e}^{i\theta}| \\ &\geqslant \mathrm{e}^{-1}((2 + \mathrm{e})m - |\mathrm{e}^{\mathrm{e}^{i\theta}} - 1|) \\ &\geqslant \mathrm{e}^{-1}((2 + \mathrm{e})m - 1 - \mathrm{e}^{\cos\theta}) \\ &\geqslant \mathrm{e}^{-1}((2 + \mathrm{e})m - 1 - \mathrm{e}) \geqslant \mathrm{e}^{-1}, \end{aligned}$$

and

$$|\psi_3(r,s)-1| = |s/r^2| = |me^{i\theta}e^{-e^{i\theta}}| \\ = me^{-\cos\theta} \geqslant me^{-1} \geqslant e^{-1}.$$

Thus, $\psi_i(r,s) \notin \Omega$ and $\psi_i \in \mathcal{E}(\Omega)$ for each $i = 1, 2, 3$.

Example 5.6.7 Let $\psi(r,s) = 1 + s/r$ and $\Omega = h(\mathbb{D}) = \{w \in \mathbb{C}: |w-1| < 1\}$, where $h(z) = 1 + z$. Consider

$$|\psi(r,s)-1| = |s/r| = |me^{i\theta}| = m \geqslant 1$$

whenever $r = e^{e^{i\theta}}$, and $s = me^{i\theta}r$, where $\theta \in [0, 2\pi)$ and $m \geqslant 1$. Therefore, $\psi(r,s) \notin \Omega$ and hence $\psi \in \mathcal{E}(\Omega)$. Using Theorem 5.6.2, in terms of subordination, the result can be written in the form

$$1 + \frac{zp'(z)}{p(z)} \prec 1 + z \quad \Rightarrow \quad p(z) \prec e^z$$

where $p \in \mathcal{H}[1,1]$. With $q(z) = e^z$, then $\psi(q(z), zq'(z)) = 1 + z = h(z)$ and $\psi \in \mathcal{E}(\Omega)$. Thus, e^z is the best dominant by Theorem 5.2.12.

Theorem 5.6.8 *If n is a non-negative integer, $0 \leqslant \alpha < 1$ and $p \in \mathcal{H}[1,1]$ satisfies the subordination*

$$1 + \beta\frac{zp'(z)}{p^n(z)} \prec 1 + (1-\alpha)z, \quad \textit{where } \beta \geqslant \begin{cases} e(1-\alpha), & n = 0 \\ e^{n-1}(1-\alpha), & n \neq 0 \end{cases},$$

then $p \in \mathcal{P}_e$.

Proof Let $h(z) = 1 + (1-\alpha)z$, where $z \in \mathbb{D}$, $0 \leqslant \alpha < 1$ and $\Omega = h(\mathbb{D}) = \{w \in \mathbb{C}: |w-1| < 1-\alpha\}$.

Case (i). If $n = 0$, let $\psi(r,s) = 1 + \beta s$. Then

$$\psi(p(z), zp'(z)) \prec 1 + (1-\alpha)z.$$

Let $r = e^{e^{i\theta}}$, and $s = me^{i\theta}r$, where $\theta \in [0, 2\pi)$ and $m \geqslant 1$. A simple calculation yields

$$|\psi(r,s)-1| = \beta m e^{\cos\theta} \geqslant \beta e^{-1} \geqslant 1 - \alpha.$$

Hence, $\psi(r,s) \notin \Omega$ which gives $\psi \in \mathcal{E}(\Omega)$. Theorem 5.6.2 shows that $p(z) \prec e^z$.

Case (ii). When $n \neq 0$, the function $\psi(r,s) = 1 + \beta s/r^n$ satisfies

$$|\psi(r,s)-1| = \beta m e^{-(n-1)\cos\theta} \geqslant \beta e^{-(n-1)} \geq 1 - \alpha$$

whenever $r = e^{e^{i\theta}}$ and $s = me^{i\theta}r$, where $\theta \in [0, 2\pi)$ and $m \geqslant 1$. Therefore, as argued in Case (i), $\psi(r, s) \notin \Omega$ which implies $\psi \in \mathcal{E}(\Omega)$. The desired result follows by an application of Theorem 5.6.2. □

Theorem 5.6.9 *If n is any non-negative integer and $p \in \mathcal{H}[1, 1]$ satisfies the subordination*

$$1 + \beta \frac{zp'(z)}{p^{n+1}(z)} \prec \frac{2+z}{2-z}, \quad \text{where } \beta \geqslant 2e^n,$$

then $p \in \mathcal{P}_e$.

Proof By considering the function $\psi(r, s) = 1 + \beta s/r^{n+1}$ and $\Omega = \{w \in \mathbb{C}: |(2w-2)/(w+1)| < 1\}$, it suffices to show that $\psi \in \Psi(\Omega, e^z)$. For this, note that

$$\left| \frac{2\psi(r, s) - 2}{\psi(r, s, t; z) + 1} \right| \geqslant \frac{2\beta m e^{-n\cos\theta}}{2 + \beta m e^{-n\cos\theta}}.$$

Since the real-valued function $g(x) = 2x/(2+x)$ is increasing for $x \geqslant 0$ and $\beta m e^{-n\cos\theta} \geqslant 2$, it is evident that

$$\left| \frac{2\psi(r, s) - 2}{\psi(r, s, t; z) + 1} \right| \geqslant 1$$

whenever $r = e^{e^{i\theta}}$, and $s = me^{i\theta}r$ where $\theta \in [0, 2\pi)$ and $m \geqslant 1$. Thus, Theorem 5.6.2 implies the required result. □

Exercises

1. Let q be univalent in $\mathbb{D}$ and ϑ and φ be analytic in a domain D containing $q(\mathbb{D})$ with $\varphi(w) \neq 0$ when $w \in q(\mathbb{D})$. Set $Q(z) := zq'(z)\varphi(q(z))$, $h(z) := \vartheta(q(z)) + Q(z)$. Suppose that either (i) $h(z)$ is convex, or (ii) $Q(z)$ is starlike univalent in $\mathbb{D}$. In addition, assume that

$$\operatorname{Re} \frac{zh'(z)}{Q(z)} > 0, \quad z \in \mathbb{D}.$$

   If p is analytic in $\mathbb{D}$, with $p(0) = q(0)$, $p(\mathbb{D}) \subset D$ and

$$\vartheta(p(z)) + zp'(z)\varphi(p(z)) \prec \vartheta(q(z)) + zq'(z)\varphi(q(z)) = h(z),$$

   show that $p \prec q$ and that q is the best dominant.
2. Let α, β be any complex numbers, $\beta \neq 0$. Let $q \in \mathcal{A}$ be univalent in $\mathbb{D}$ and satisfy the following conditions for $z \in \mathbb{D}$:

   (i) $\operatorname{Re}\left(\frac{2\alpha}{\beta} q(z) + (zq''(z)/q'(z) + 1)\right) > 0$;

   (ii) $zq'(z)$ is starlike.

   If $p \in \mathcal{A}$ satisfies

$$\alpha p(z)^2 + \beta z p'(z) \prec \alpha q(z)^2 + \beta z q'(z),$$

   show that $p \prec q$ and q is the best dominant.

3. If $p \in \mathcal{H}[1,1]$ satisfies $|\arg(p(z)+zp'(z))| < 3\pi/4$, show that $p \in \mathcal{P}$.
4. If $p \in \mathcal{H}[1,1]$ satisfies $|1+p(z)+zp'(z)| > |\mathrm{Re}(p(z)+zp'(z))|$, show that $p \in \mathcal{P}$.
5. Let α and β be positive numbers. If $p \in \mathcal{H}[1,1]$ satisfies

$$\alpha p(z) - \beta \frac{zp'(z)}{p(z)^2} \prec \alpha \frac{1+z}{1-z} - \frac{2\beta z}{(1+z)^2} = \alpha \frac{1+z}{1-z} + \frac{\beta}{2}\left(\frac{1-z}{1+z}\right)^2 - \frac{\beta}{2},$$

show that $p \in \mathcal{P}$.

6. Let α and β be complex numbers and γ be a real number such that either

$$\mathrm{Re}\beta \geqslant 0, \gamma \geqslant 0 \quad \text{and} \quad \mathrm{Re}(\alpha+n\beta)+n(n-1)\gamma > 0, \quad \text{or}$$

$$\mathrm{Re}\beta \leqslant 0, \gamma \leqslant 0 \quad \text{and} \quad \mathrm{Re}(\alpha+n\beta)+n(n-1)\gamma < 0.$$

If a function $w \in \mathcal{H}[0,n]$ satisfies the inequality

$$\left|\alpha w(z) + \beta z w'(z) + \gamma z^2 w''(z)\right| < |\mathrm{Re}(\alpha+n\beta)+n(n-1)\gamma|, \quad z \in \mathbb{D},$$

show that $w \in \mathcal{B}_0$.

7. Let α and β be complex numbers such that either

$$\mathrm{Re}\beta \geqslant 0 \quad \text{and} \quad \mathrm{Re}(\alpha+n^2\beta) > 0 \quad \text{or}$$

$$\mathrm{Re}\beta \leqslant 0 \quad \text{and} \quad \mathrm{Re}(\alpha+n^2\beta) < 0.$$

If $w \in \mathcal{H}[0,n]$ satisfies the inequality

$$\left|\alpha w^2(z) + \beta z^2 (w'(z))^2\right| < \left|\mathrm{Re}(\alpha+n^2\beta)\right|, \quad z \in \mathbb{D},$$

show that $w \in \mathcal{B}_0$.

8. Let $zC(z)$ be analytic in $\mathbb{D}$ with $|zC(z)| < 1$. Let $f \in \mathcal{A}$ be the unique solution of $f''(z) + C(z)f(z) = 0$. Show that $f(z) \neq 0$ for $z \neq 0$. (Hint: Consider $p(z) = f(z)/z$ and show that $\mathrm{Re}\,p(z) > 0$.)
9. If $p \in \mathcal{H}[1,n]$ satisfies

$$|z^2 p''(z) + 5zp'(z) - p^2(z) + 1| < 1, \quad z \in \mathbb{D},$$

show that $p \in \mathcal{P}$.

10. If $p \in \mathcal{H}[1,n]$ satisfies

$$\mathrm{Re}(A(z)zp'(z) + B(z)p^2(z) + C(z)p(z) + D(z) > 0, \quad z \in \mathbb{D},$$

where A, B, C, D are functions that satisfy $\mathrm{Re}A(z) > 0$ and $(\mathrm{Im}C(z))^2 \leqslant \mathrm{Re}(A(z)+2B(z))\mathrm{Re}(A(z)-2D(z))$ for all $z \in \mathbb{D}$, show that $p \in \mathcal{P}$.

11. Let $P \in \mathcal{H}[1,n]$ satisfy

$$P(z) \prec \frac{1+z}{1-z} + \frac{2nz}{1-z^2}.$$

If $p \in \mathcal{H}[1,n]$ satisfies the differential equation

$$zp'(z) + P(z)p(z) = 1,$$

show that $p \in \mathcal{P}$.

12. Let h be convex in $\mathbb{D}$ with $\mathrm{Re}(\beta h(z)+\gamma) > 0$. If p is analytic in $\mathbb{D}$ with $p(0) = h(0)$, show that

$$p(z) + \frac{zp'(z)}{\beta p(z)+\gamma} \prec h(z) \Rightarrow p(z) \prec h(z).$$

13. Let n be a positive integer and $A(z) = A \geqslant 0$. Suppose that the functions $B, C, D : \mathbb{D} \to \mathrm{C}$ satisfy $\mathrm{Re}B(z) \geqslant A$ and

$$\mathrm{Im}(C(z))^2 \leqslant n(\mathrm{Re}B(z) - A)\mathrm{Re}(nB(z) - nA - 2D(z)).$$

If $p \in \mathcal{H}[1, n]$ and if

$$\mathrm{Re}\left(Az^2 p''(z) + B(z)zp'(z) + C(z)p(z) + D(z)\right) > 0, \quad z \in \mathbb{D},$$

show that $p \in \mathcal{P}$.

14. Let h be convex in $\mathbb{D}$ and let M and N be analytic in $\mathbb{D}$ with $M(0) = N(0)$. If $\mathrm{Re}(zN'(z)/N(z)) > 0$, show that

$$\frac{M'(z)}{N'(z)} \prec h(z) \Rightarrow \frac{M(z)}{N(z)} \prec h(z).$$

15. If n is any positive integer and $p \in \mathcal{H}[1, 1]$ satisfies the subordination

$$1 + \beta(zp'(z))^n \prec \mathrm{e}^z, \quad \text{where } \beta \geqslant \begin{cases} \mathrm{e}^{n+1} + \mathrm{e}^n & \text{when } n \text{ is odd,} \\ \mathrm{e}^{n+1} - \mathrm{e}^n & \text{when } n \text{ is even,} \end{cases}$$

show that $p \in \mathcal{P}_e$.

16. If $p \in \mathcal{H}[1, 1]$ satisfies the subordination

$$1 + \beta \frac{zp'(z)}{p^{n+1}(z)} \prec \mathrm{e}^z,$$

where $\beta \geqslant \mathrm{e}^{n+1} - \mathrm{e}^n$ and n is any non-negative integer, show that $p \in \mathcal{P}_e$.

17. Let $p \in \mathcal{H}[1, 1]$. Show that each of the following subordinations is sufficient for $p \in \mathcal{P}_e$:

    a. $p(z) + \beta zp'(z) \prec \mathrm{e}^z$ for $\beta \geqslant \mathrm{e}^2 + 1 \approx 8.38906$.
    b. $p(z) + \beta zp'(z)/p(z) \prec \mathrm{e}^z$ for $\beta \geqslant \mathrm{e} + \mathrm{e}^{-1} \approx 3.08616$.

18. Let n be any positive integer and β_n a positive root of the equation

$$\left(\mathrm{e}^{1+n} - x(-1+n)\right)^2 - \left(\mathrm{e}^{2+2n}n - x^2 n + x\mathrm{e}^{1+n}(1 - n + n^2)\right)\ln(\mathrm{e} + \mathrm{e}^{-n}x) = 0.$$

If $p \in \mathcal{H}[1, 1]$ satisfies the subordination

$$p(z) + \beta \frac{zp'(z)}{p^{n+1}(z)} \prec \mathrm{e}^z, \quad \text{for } \beta > \beta_n,$$

show that $p \in \mathcal{P}_e$.

19. Let $p \in \mathcal{H}[1, 1]$. Show that each of the following subordinations are sufficient for $p \in \mathcal{P}_e$:

    a. $p^2(z) + \beta zp'(z) \prec \mathrm{e}^z$ for $\beta \geqslant \mathrm{e}^2 + \mathrm{e}^{-1} \approx 7.75694$.
    b. $p^2(z) + \beta zp'(z)/p(z) \prec \mathrm{e}^z$ for $\beta \geqslant \mathrm{e} + \mathrm{e}^{-2} \approx 2.85362$.
    c. $p^2(z) + \beta zp'(z)/p^2(z) \prec \mathrm{e}^z$ for $\beta > \beta^* \approx 104.122$, where β^* is a positive root of the equation

$$6\mathrm{e}^6 + 5x\mathrm{e}^3 - x^2 + (-2\mathrm{e}^6 - 5x\mathrm{e}^3 + x^2)\ln(\mathrm{e}^3 + x) = 0.$$

20. Let $p \in \mathcal{H}[1, 1]$. Show that each of the following subordinations are sufficient for $p \in \mathcal{P}_e$:

    a. $p(z) + \beta zp(z)p'(z) \prec \mathrm{e}^z$ for $\beta \geqslant \mathrm{e}^3 + \mathrm{e} \approx 22.8038$.
    b. $p^2(z) + \beta zp(z)p'(z) \prec \mathrm{e}^z$ for $\beta \geqslant \mathrm{e}^3 + 1 \approx 21.0855$.
    c. $p^3(z) + \beta zp(z)p'(z) \prec \mathrm{e}^z$ for $\beta \geqslant \mathrm{e}^3 + \mathrm{e}^{-1} \approx 20.4534$.

21. If $p \in \mathcal{H}[1, 1]$ satisfies the subordination

$$p^2(z) + p(z) - 1 + \beta z p'(z) \prec \mathrm{e}^z \quad \text{for } \beta \geqslant \mathrm{e}^2 + \mathrm{e}^{-1} - \mathrm{e} + 1 \approx 6.03865,$$

show that $p \in \mathcal{P}_e$.

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[2] The theory of differential subordination originated with the works of Miller [5, 6] and Miller and Mocanu [7–9]. The general theory appears in [10, 11]. The monograph by Miller and Mocanu [12], which form the basis of Sects. 5.1–5.4, contains significant results and applications of differential subordination. The earliest result of the Clunie-Jack-Miller-Mocanu Lemma is due to Jack [2]. Section 5.5 is based on a lemma by Nunokawa [14] (see also Marjono and Thomas [4]), while the final section draws on results from Naz, Nagpal and Ravichandran [13]. Some of the exercises on bounded functions are taken from [3]. For further developments in subordination and the corresponding theory of superordination, we referred to the monograph by Bulboacă [1].

Chapter 6
Starlike Functions

Abstract An analytic function $f : \mathbb{D} \to \mathbb{C}$ is called a starlike function if its image $f(\mathbb{D})$ is starlike with respect to the origin. The Koebe function is a prominent example of a starlike function. The class of starlike functions shares several properties with the class of univalent functions. This chapter is devoted to the study of starlike functions. An analytic characterization of starlikeness is established and used to derive various properties of functions in this class.

6.1 Analytic Characterization

Recall that the line segment joining the points z_1 and z_2 is the set

$$[z_1, z_2] := \{tz_1 + (1-t)z_2 : 0 \leqslant t \leqslant 1\}.$$

A domain G is said to be *starlike* with respect to a point $z_0 \in G$ if the line segment $[z, z_0]$ lies entirely within G for every $z \in G$. Domains that are starlike with respect to a are shown in Fig. 6.1 while domains that are not starlike with respect to b are shown in Fig. 6.2. A function f in $\mathcal{A}$ is called *starlike* if its image $f(\mathbb{D})$ is a starlike domain with respect to the origin. The subclass of $\mathcal{S}$ consisting of all starlike functions is denoted by $\mathcal{ST}$. Since the points on the line segment from z to the origin are of the form tz for $0 \leqslant t \leqslant 1$, a function $f \in \mathcal{ST}$ if and only if $f(tz) \in f(\mathbb{D})$ for all $z \in \mathbb{D}$ and all $t \in [0, 1]$.

Example 6.1.1 (Koebe function) In Example 2.1.15, it was already shown that the Koebe function $\kappa : \mathbb{D} \to \mathbb{C}$ given by

$$\kappa(z) = \frac{z}{(1-z)^2}$$

belongs to $\mathcal{S}$. Since $\kappa(\mathbb{D}) = \mathbb{C} \setminus (-\infty, -1/4]$, which is the entire complex plane except for a slit along the negative real axis from $w = -\infty$ to $w = -\frac{1}{4}$, is clearly starlike with respect to the origin, the Koebe function $\kappa \in \mathcal{ST}$.

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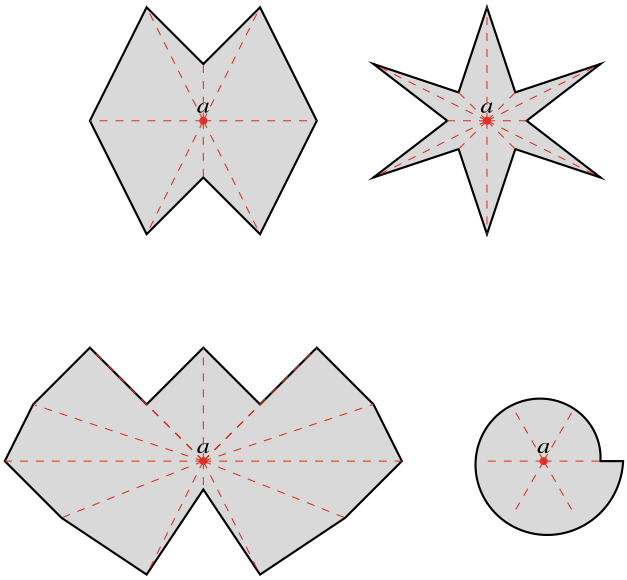
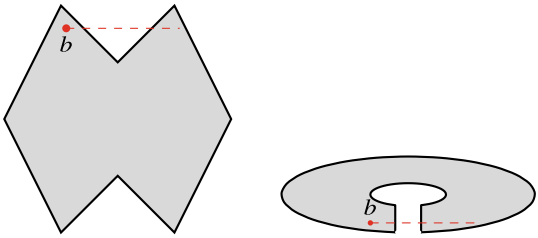


Fig. 6.1 Starlike domains

Fig. 6.2 Non-starlike domains



The following theorem provides growth, distortion and covering estimates for starlike functions.

Theorem 6.1.2 *If $f \in \mathcal{ST}$, then*

$$-\kappa(-r) = \frac{r}{(1+r)^2} \leqslant |f(z)| \leqslant \frac{r}{(1-r)^2} = \kappa(r), \quad |z| = r < 1,$$

$$\kappa'(-r) = \frac{1-r}{(1+r)^3} \leqslant |f'(z)| \leqslant \frac{1+r}{(1-r)^3} = \kappa'(r), \quad |z| = r < 1$$

and for each $k \geqslant 2$

$$|f^{(k)}(z)| \leqslant \frac{k!(k+r)}{(1-r)^{k+2}} = \kappa^{(k)}(r), \quad |z| = r < 1.$$

In addition, $\mathbb{D}_{1/4} \subset f(\mathbb{D})$. All of these results are sharp, with equality if and only if f is a rotation of the Koebe function κ.

Proof These inequalities have been shown to hold for all functions in the class $\mathcal{S}$ with κ as an extremal function. Thus, these results remain valid for the subclass $\mathcal{ST}$, since $\mathcal{ST} \subset \mathcal{S}$. Sharpness is attained by the Koebe function $\kappa \in \mathcal{ST}$. □

Since $f(z) \equiv z$ belongs to $\mathcal{ST}$ and $f^{(k)}(z) \equiv 0$ for $k = 2, 3, \ldots$, the sharp lower bound for $|f^{(k)}(z)|$ is 0. The upper bound for $|f^{(k)}(z)|$ gives the sharp coefficient estimate $|a_n| \leqslant n$ for $f \in \mathcal{ST}$. This will be proved again later.

Recall that a region G is *convex* if the line segment $[z_1, z_2]$ joining any two points z_1 and z_2 in G also lies entirely within G. Examples of convex domains are shown in Fig. 6.3. A function $f \in \mathcal{S}$ is *convex* if the image $f(\mathbb{D})$ is a convex domain, and the class of all such functions is denoted by $\mathcal{CV}$. By definition, both $\mathcal{CV}$ and $\mathcal{ST}$ are subclasses of $\mathcal{S}$. Geometrically, it is evident that every convex function is starlike, and hence the inclusion $\mathcal{CV} \subset \mathcal{ST} \subset \mathcal{S}$ holds. Further properties of starlike and convex functions can be derived by characterizing these functions in terms of functions with positive real part. To obtain an analytic characterization of starlike functions, the following theorem is required.

Theorem 6.1.3 *(Hereditary Property of Starlike Functions) Let $f \in \mathcal{A}$. A function $f \in \mathcal{ST}$ if and only if $f(\mathbb{D}_r)$ is starlike with respect to the origin for every r with $0 < r < 1$. Equivalently, $f \in \mathcal{ST}$ if and only if the function $f_r : \mathbb{D} \to \mathbb{C}$ defined by $f_r(z) = r^{-1} f(rz)$ belongs to $\mathcal{ST}$ for each $r \in (0, 1)$.*

Proof Suppose that f is a starlike function. Fix $t \in (0, 1)$, and let $z \in \mathbb{D}$. Since $f(\mathbb{D})$ is starlike with respect to the origin, then $tf(z) \in f(\mathbb{D})$ and hence $f^{-1}(tf(z)) \in \mathbb{D}$.

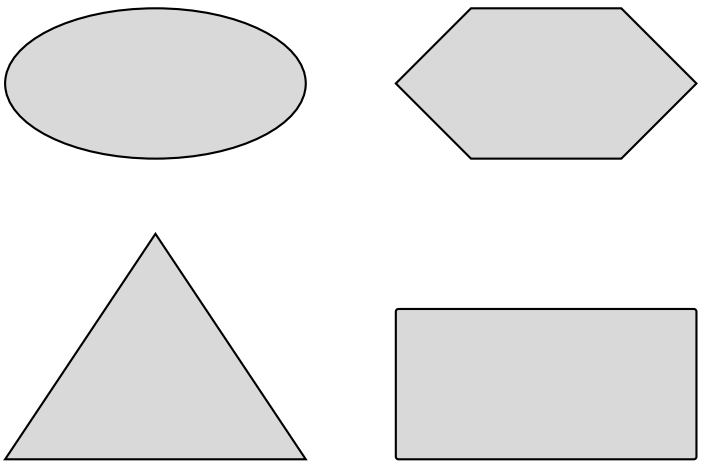


Fig. 6.3 Convex domains

Define the function $w : \mathbb{D} \to \mathbb{C}$ by $w(z) = f^{-1}(tf(z))$ so that $tf(z) = f(w(z))$. Using this identity, it follows that the function $f_r : \mathbb{D} \to \mathbb{C}$ given by $f_r(z) = r^{-1} f(rz)$ satisfies

$$tf_r(z) = r^{-1} tf(rz) = r^{-1} f(w(rz)) = r^{-1} f(rw_1(z)) = f_r(w_1(z))$$

where the function $w_1 : \mathbb{D} \to \mathbb{C}$ is defined by $w_1(z) = w(rz)/r$. Since f is univalent, the function w is analytic in $\mathbb{D}$, maps $\mathbb{D}$ into itself and satisfies

$$w(0) = f^{-1}(tf(0)) = f^{-1}(0) = 0.$$

By the Schwarz lemma, $|w(z)| \leqslant |z|$ for all $z \in \mathbb{D}$. The function w_1 satisfies

$$|w_1(z)| = \frac{|w(rz)|}{r} \leqslant \frac{|rz|}{r} = |z|.$$

Hence, $w_1(\mathbb{D}) \subset \mathbb{D}$, and therefore $tf_r(z) = f_r(w_1(z)) \in f_r(\mathbb{D})$, which shows that $f_r \in \mathcal{ST}$.

For the converse, assume that f_r is starlike for all $0 < r < 1$. Then, the function rf_r is starlike. Since

$$f(\mathbb{D}) = \bigcup_{0<r<1} rf_r(\mathbb{D})$$

and the union of domains starlike with respect to the origin is again a domain starlike with respect to the origin, the result now follows. □

Hereditary property of starlike functions is illustrated for the function f when $f(z) = z + z^2/2$ and $f(z) = (2/\pi)\sin(\pi z/2)$ in Fig. 6.4.

Theorem 6.1.4 *A function f in $\mathcal{A}$ belongs to $\mathcal{ST}$ if and only if*

$$\operatorname{Re} \frac{zf'(z)}{f(z)} > 0, \quad z \in \mathbb{D}. \tag{6.1.1}$$

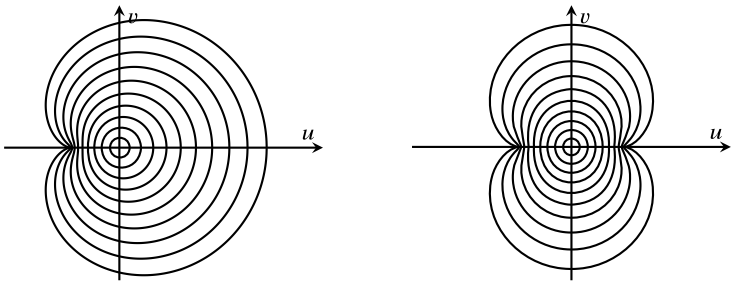


Fig. 6.4 $f(\mathbb{D}_r)$ when $f(z) = z + z^2/2$ and $f(z) = (2/\pi)\sin(\pi z/2)$

Equivalently, a function $f \in \mathcal{ST}$ if and only if the function $p : \mathbb{D} \to \mathbb{C}$ given by

$$p(z) = \frac{zf'(z)}{f(z)} \tag{6.1.2}$$

belongs to the class $\mathcal{P}$ of Carathéodory functions.

Proof Let $f \in \mathcal{ST}$. By the hereditary property of starlike functions, the function f maps $\mathbb{D}_r$ into a domain starlike with respect to the origin. Therefore, the image curve C_r of the circle $|z| = r$ under the function f bounds a starlike domain. Geometrically, this means that $\arg(f(z))$ increases as z moves around the circle $|z| = r$ (see Fig. 6.5). Thus,

$$\frac{\partial}{\partial \theta} \arg(f(re^{i\theta})) \geqslant 0.$$

Now $\log z = \log|z| + i \arg z$, and so $\arg z = \operatorname{Im} \log z$, $z \neq 0$. By using this identity with $z = re^{i\theta}$,

$$\begin{aligned} 0 &\leqslant \frac{\partial}{\partial \theta} \arg(f(re^{i\theta})) \\ &= \frac{\partial}{\partial \theta} \operatorname{Im} \log(f(re^{i\theta})) \\ &= \operatorname{Im} \frac{\partial}{\partial \theta} \log(f(re^{i\theta})) \\ &= \operatorname{Im} \frac{ire^{i\theta} f'(re^{i\theta})}{f(re^{i\theta})} \\ &= \operatorname{Re} \frac{zf'(z)}{f(z)}. \end{aligned}$$

The function p in (6.1.2) is well-defined since the univalence of f ensures that $f(z) \neq 0$ for $z \neq 0$. At the origin, the function p has a removable singularity, which is removed by defining $p(0) = 1$. It is readily shown that p is analytic. By Theorem 1.2.12, it follows that $\operatorname{Re} p(z) > 0$ for all $z \in \mathbb{D}$, and consequently, (6.1.1) also holds. This establishes the function p in (6.1.2) belongs to $\mathcal{P}$.

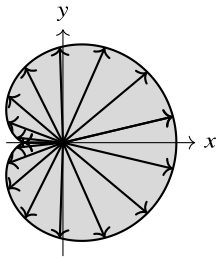


Fig. 6.5 Argument of $f(re^{i\theta})$ is increasing on the boundary of a starlike domain

For the converse, assume that $f \in \mathcal{A}$ satisfies the inequality (6.1.1). If $f(z_0) = 0$ for some $z_0 \neq 0$, then z_0 would be a pole of the function p defined by (6.1.2), contradicting the analyticity of p. Thus, $f(z) \neq 0$ for all $z \neq 0$, and hence f has only one zero in $\mathbb{D}$, namely at the origin.

Let C_r be the image of the circle $|z| = r$ under f, for fixed $0 < r < 1$. By the argument principle, the winding number of the circle C_r around the origin is given by

$$n(C_r; 0) = \frac{1}{2\pi \mathrm{i}} \int_{C_r} \frac{\mathrm{d}w}{w} = \frac{1}{2\pi \mathrm{i}} \int_{\partial \mathbb{D}_r} \frac{f'(z)}{f(z)} \mathrm{d}z = 1.$$

Hence, the curve C_r winds around the origin exactly once. Furthermore, by condition (6.1.1), the function $\arg(f(z))$ increases as z traces the circle $|z| = r$, implying that the curve C_r does not intersect itself. Hence, the function f is one-to-one on $\partial \mathbb{D}_r$ for each $r \in (0, 1)$. Therefore, by Darboux Theorem (Theorem 2.1.7), f is univalent in $\mathbb{D}_r$, and hence univalent in $\mathbb{D}$.

Since $\arg(f(z))$ increases as z traverses the circle $|z| = r$, it follows that the image $f(\mathbb{D}_r)$ is starlike with respect to the origin. As this holds for each $r \in (0, 1)$, we deduced that $f(\mathbb{D})$ is starlike with respect to the origin. □

Corollary 6.1.5 *The class $\mathcal{ST}$ is a compact normal family.*

Proof Since $\mathcal{ST} \subset \mathcal{S} \subset \mathcal{H}(\mathbb{D})$, it follows that $\mathcal{ST}$ is normal, being a subset of the normal family $\mathcal{S}$. To prove compactness, let $\langle f_n \rangle$ be a sequence of functions in $\mathcal{ST}$. By the compactness of $\mathcal{S}$, there exists a subsequence $\langle f_{n_k} \rangle$ that converges to a function $f \in \mathcal{S}$.

Since $f_{n_k} \in \mathcal{ST}$, then

$$\mathrm{Re}\left(\frac{z f'_{n_k}(z)}{f_{n_k}(z)}\right) > 0.$$

By Weierstrass' Theorem (Theorem 1.4.1), the sequence $f'_{n_k} \to f'$ as $k \to \infty$, and hence

$$\mathrm{Re}\left(\frac{z f'(z)}{f(z)}\right) \geqslant 0.$$

Applying the minimum principle for harmonic functions (Theorem 1.2.12), it follows that $\mathrm{Re}(zf'(z)/f(z)) > 0$. Thus, $f \in \mathcal{ST}$, proving that the class $\mathcal{ST}$ is compact. □

Theorem 6.1.4 is very useful in generating starlike functions.

Example 6.1.6 The Koebe function $\kappa(z) = z/(1 - z)^2$ satisfies

$$\mathrm{Re}\,\frac{z\kappa'(z)}{\kappa(z)} = 1 + \mathrm{Re}\,\frac{2z}{1 - z} = \mathrm{Re}\,\frac{1 + z}{1 - z} > 0,$$

and therefore, by Theorem 6.1.4, the Koebe function $\kappa \in \mathcal{ST}$. Note that the inequality (6.1.1) can be written as the subordination

$$\frac{zf'(z)}{f(z)} \prec \frac{z\kappa'(z)}{\kappa(z)}.$$

This subordination is necessary and sufficient for a function $f \in \mathcal{A}$ to be starlike.

Example 6.1.7 If $f \in \mathcal{A}$ satisfies the inequality

$$\left|\frac{zf'(z)}{f(z)} - 1\right| < 1, \quad z \in \mathbb{D},$$

then $f \in \mathcal{ST}$. The converse does not hold in general, as demonstrated by the Koebe function.

The given condition asserts that $w = zf'(z)/f(z)$ lies in the disk $\{w : |w - 1| < 1\}$, which is clearly contained in the right half-plane. Since $\operatorname{Re} w \geqslant -|w|$, then

$$\operatorname{Re} \frac{zf'(z)}{f(z)} = 1 + \operatorname{Re}\left(\frac{zf'(z)}{f(z)} - 1\right) \geqslant 1 - \left|\frac{zf'(z)}{f(z)} - 1\right| > 0.$$

Therefore, by Theorem 6.1.4, the function $f \in \mathcal{ST}$. In particular, the function $f_0(z) = ze^z$ satisfies

$$\frac{zf_0'(z)}{f_0(z)} = 1 + z,$$

and hence

$$\left|\frac{zf_0'(z)}{f_0(z)} - 1\right| = |z| < 1.$$

Thus, the function $f_0 \in \mathcal{ST}$. This could be seen directly from the estimate

$$\operatorname{Re} \frac{zf_0'(z)}{f_0(z)} = 1 + \operatorname{Re} z \geqslant 1 - |z| > 0.$$

Example 6.1.8 If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{A}$ satisfies

$$\sum_{k=2}^{\infty} k|a_k| \leqslant 1,$$

then

$$\left|\frac{zf'(z)}{f(z)} - 1\right| < 1, \quad z \in \mathbb{D},$$

and $f \in \mathcal{ST}$.

By Theorem 2.1.13, $f \in \mathcal{S}$ and so $f(z) \neq 0$ for $z \neq 0$. Also

$$\begin{aligned} |zf'(z) - f(z)| - |f(z)| &= \left| \sum_{k=2}^{\infty} (k-1) a_k z^k \right| - \left| z + \sum_{k=2}^{\infty} a_k z^k \right| \\ &\leqslant \sum_{k=2}^{\infty} (k-1)|a_k||z|^k - \left(|z| - \sum_{k=2}^{\infty} |a_k||z|^k \right) \\ &\leqslant |z| \left(\sum_{k=2}^{\infty} k|a_k| - 1 \right) \\ &\leqslant 0. \end{aligned}$$

Thus,

$$\left| \frac{zf'(z)}{f(z)} - 1 \right| \leqslant 1$$

and the Maximum Modulus Theorem (Theorem 1.2.8) shows that

$$\left| \frac{zf'(z)}{f(z)} - 1 \right| < 1.$$

By Example 6.1.7, $f \in \mathcal{ST}$.

The closure of the class $\mathcal{ST}$ under the kth root transformation (Theorem 2.2.9) and the transformations in Theorem 2.2.15 is established in the next theorem.

Theorem 6.1.9 *Let $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{ST}$. Then, the following holds:*

(i) The k-th root transform

$$g(z) = z \sqrt[k]{f(z^k)/z^k}, \quad z \in \mathbb{D}$$

belongs to $\mathcal{ST}^{(k)} := \mathcal{ST} \cap \mathcal{S}^{(k)}$.

(ii) The function

$$g(z) := \overline{f(\overline{z})} = z + \sum_{n=2}^{\infty} \overline{a}_n z^n, \quad z \in \mathbb{D}$$

belongs to $\mathcal{ST}$.

(iii) The function

$$g(z) := f(rz)/r, \quad 0 < r < 1, \quad z \in \mathbb{D}$$

belongs to $\mathcal{ST}$.

Proof

(i) The univalence of the function g was shown in Theorem 2.2.9. The result readily follows from

$$\frac{zg'(z)}{g(z)} = \frac{z^k f'(z^k)}{f(z^k)}.$$

(ii) The univalence of g was shown in Theorem 2.2.15. Note that

$$\frac{zg'(z)}{g(z)} = \overline{p}(\overline{z})$$

where $p(z) = zf'(z)/f(z)$.

(iii) The univalence of g was shown in Theorem 2.2.15 and the result follows from

$$\frac{zg'(z)}{g(z)} = \frac{rzf'(rz)}{f(rz)}.$$

□

6.2 Coefficients Estimates

As an application of Theorem 6.1.4, bounds for the Taylor coefficients of starlike functions are obtained. The approach involves relating the coefficients of the given function $f \in \mathcal{ST}$ to the coefficients c_k of a function $p \in \mathcal{P}$, where $p(z) = 1 + \sum_{k=1}^{\infty} c_k z^k$, and using the known estimate $|c_k| \leqslant 2$. These bounds are also derived using the estimates for the logarithmic coefficients (see Corollary 6.5.2).

Theorem 6.2.1 *If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{ST}$, then $|a_k| \leqslant k$ for each $k \geqslant 2$ with equality if f is a rotation of the Koebe function κ.*

Proof If $f \in \mathcal{ST}$, then, by Theorem 6.1.4, the function

$$p(z) := \frac{zf'(z)}{f(z)}$$

belongs to $\mathcal{P}$. Let $p(z) = 1 + \sum_{k=1}^{\infty} c_k z^k$. Then

$$z + \sum_{k=2}^{\infty} k a_k z^k = \left(z + \sum_{k=2}^{\infty} a_k z^k\right)\left(1 + \sum_{k=1}^{\infty} c_k z^k\right).$$

Equating coefficients of z^k in the above equation gives

$$k a_k = a_k + \sum_{n=1}^{k-1} a_{k-n} c_n, \quad k \geqslant 2$$

where $a_1 = 1$, or

$$(k-1)a_k = \sum_{n=1}^{k-1} a_{k-n}c_n, \quad k \geqslant 2. \tag{6.2.1}$$

For $k = 2$, the equation (6.2.1) reduces to $a_2 = c_1$, and hence $|a_2| = |c_1| \leqslant 2$. Assume that $|a_n| \leqslant n$ for $2 \leqslant n \leqslant k-1$. Then

$$|(k-1)a_k| \leqslant \sum_{n=1}^{k-1} |a_{k-n}|\,|c_n| \leqslant 2\sum_{n=1}^{k-1}(k-n) = (k-1)k.$$

Hence, by the principle of mathematical induction, $|a_k| \leqslant k$. Equality clearly holds for the rotations κ_θ of the Koebe function κ given by

$$\kappa_\theta(z) = \mathrm{e}^{-\mathrm{i}\theta}\kappa(\mathrm{e}^{\mathrm{i}\theta}z) = \frac{z}{(1-\mathrm{e}^{\mathrm{i}\theta}z)^2} = z + \sum_{k=2}^{\infty} k\mathrm{e}^{\mathrm{i}(k-1)\theta}z^k, \quad \theta \in [0, 2\pi]. \qquad \square$$

Theorem 6.2.2 *(Fekete-Szegö Inequality) If* $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{ST}$, *then*

$$|a_3 - \mu a_2^2| \leqslant \max\{1, |4\mu - 3|\}, \quad \mu \in \mathbb{C},$$

and

$$|a_3 - \mu a_2^2| \leqslant \begin{cases} 1 - (\mu - 1/2)|a_2|^2 & 1/2 \leqslant \mu \leqslant 3/4, \\ 1 - (1-\mu)|a_2|^2 & 3/4 \leqslant \mu \leqslant 1. \end{cases}$$

Equality holds in the first inequality for the Koebe function κ *or its square root transformation. Equality holds in the second inequality for the Koebe function.*

Proof From (6.2.1),

$$a_2 = c_1, \quad a_3 = \frac{1}{2}(c_2 + c_1^2),$$

and hence

$$a_3 - \mu a_2^2 = \frac{1}{2}\left(c_2 - (2\mu - 1)\,c_1^2\right).$$

The results follows by an application of Theorem 4.5.4.

Equality in the first result holds for the Koebe function when $|4\mu - 3| \geqslant 1$ and for the square root transformation of the Koebe function in the other case. Equality in the second result holds for the Koebe function. $\square$

For $q \geqslant 1$, and $n \geqslant 1$, the *q-th Hankel determinant* of an analytic function $f(z) = \sum_{k=1}^{\infty} a_k z^k$ is defined by

$$H_q(n) = \begin{vmatrix} a_n & a_{n+1} & \dots & a_{n+q-1} \\ a_{n+1} & a_{n+2} & \dots & a_{n+q} \\ \vdots & \vdots & & \vdots \\ a_{n+q-1} & a_{n+q} & \dots & a_{n+2q-2} \end{vmatrix}.$$

For $f \in \mathcal{A}$, the second Hankel determinants $H_2(1)$ and $H_2(2)$ are given by

$$H_2(1) = \begin{vmatrix} 1 & a_2 \\ a_2 & a_3 \end{vmatrix} = a_3 - a_2^2, \quad \text{and} \quad H_2(2) = \begin{vmatrix} a_2 & a_3 \\ a_3 & a_4 \end{vmatrix} = a_2 a_4 - a_3^2.$$

The second Hankel determinant $H_2(1)$ is a special case of the Fekete-Szegö functional. The next theorem gives a bound for the second Hankel determinant $H_2(2)$ for starlike functions.

Theorem 6.2.3 ***(Second Hankel Determinant)*** *If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{ST}$, then the second Hankel determinant of f satisfies*

$$|H_2(2)| = |a_2 a_4 - a_3^2| \leqslant 1.$$

Equality holds for the Koebe function κ.

Proof From (6.2.1),

$$a_2 = c_1, \quad a_3 = \frac{1}{2}c_2 + \frac{1}{2}c_1^2 \quad \text{and} \quad a_4 = \frac{1}{3}c_3 + \frac{1}{2}c_1 c_2 + \frac{1}{6}c_1^3,$$

and hence

$$H_2(2) = a_2 a_4 - a_3^2 = \frac{1}{3}c_1 c_3 - \frac{1}{4}c_2^2 - \frac{1}{12}c_1^4.$$

Since the class $\mathcal{ST}$ is rotationally invariant, it may be assumed that a_2 is real and hence c_1 is real. Suppose now that $c_1 = c$ with $0 \leqslant c \leqslant 2$. From (4.5.17) and (4.5.18) of Theorem 4.5.11, it follows that

$$\begin{aligned} H_2(2) &= \left| \frac{c_1 c_3}{3} - \frac{c_2^2}{4} - \frac{c_1^4}{12} \right| \\ &= \left| \frac{1}{24}(4 - c^2)c^2 x - \frac{1}{16}c^4 + \frac{1}{6}(4 - c^2)(1 - |x|^2)cy \right. \\ &\quad \left. - \frac{1}{48}(4 - c^2)x^2(12 + c^2) \right| \\ &\leqslant \frac{1}{16}c^4 + \frac{1}{6}c(4 - c^2) + \frac{1}{24}c^2(4 - c^2)\rho + \frac{1}{48}(4 - c^2)(c - 2)(c - 6)\rho^2 \\ &=: F(\rho), \end{aligned}$$

where $\rho = |x| \leqslant 1$ and $|y| \leqslant 1$. Since

$$F'(\rho) = \frac{c^2\left(4-c^2\right)}{24} + \frac{\left(4-c^2\right)(c-2)(c-6)\rho}{24} > 0,$$

the function F is increasing and hence $|H_2(2)| \leqslant F(\rho) \leqslant F(1) = 1$. Equality holds for the Koebe function. □

Theorem 6.2.4 *(Clunie's Theorem) If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{ST}$, then*

$$(k-1)^2 |a_k|^2 \leqslant 4 \sum_{n=1}^{k-1} n |a_n|^2, \quad a_1 = 1 \tag{6.2.2}$$

and $|a_k| \leqslant k$ for each $k \geqslant 2$ with equality if f is a rotation of the Koebe function κ.

Proof Since $f \in \mathcal{ST}$, by Theorem 6.1.4, the function

$$p(z) := \frac{zf'(z)}{f(z)}, \quad z \in \mathbb{D}, \tag{6.2.3}$$

belongs to $\mathcal{P}$. By Theorem 4.4.5, the function

$$w(z) = \frac{p(z)-1}{p(z)+1} \tag{6.2.4}$$

belongs to $\mathcal{B}_0$. Use of (6.2.3) in (6.2.4) yields

$$\left(zf'(z) + f(z)\right) w(z) = zf'(z) - f(z).$$

By writing $w(z) = \sum_{n=1}^{\infty} w_n z^n$, the last equation becomes

$$\left(2z + \sum_{n=2}^{\infty} (n+1) a_n z^n\right) \cdot \sum_{n=1}^{\infty} w_n z^n = \sum_{n=2}^{\infty} (n-1) a_n z^n. \tag{6.2.5}$$

It follows by equating the coefficients of z^k on both sides of (6.2.5) that

$$2w_{k-1} + 3a_2 w_{k-2} + \cdots + k a_{k-1} w_1 = (k-1) a_k, \quad k \geqslant 2. \tag{6.2.6}$$

This means that the coefficient a_k on the right of (6.2.6) depends only on $a_2, \ldots, a_{k-1}$ on the left of (6.2.6). Hence, for $k \geqslant 2$, write

$$\left(2z + \sum_{n=2}^{k-1} (n+1) a_n z^n\right) w(z) = \sum_{n=2}^{k} (n-1) a_n z^n + \sum_{n=k+1}^{\infty} b_n z^n. \tag{6.2.7}$$

With $z = re^{i\theta}$, integrate the square of the modulus of both sides of (6.2.7) with respect to θ. Then, using the inequality $|w(z)| < 1$ for $|z| < 1$, and applying (3.3.3),

it follows that

$$\begin{aligned}
&\sum_{n=2}^{k}(n-1)^2\left|a_n\right|^2 r^{2n}+\sum_{n=k+1}^{\infty}\left|b_n\right|^2 r^{2n}\\
&=\frac{1}{2\pi}\int_0^{2\pi}\left|\sum_{n=2}^{k}(n-1)a_n z^n+\sum_{n=k+1}^{\infty}b_n z^n\right|^2\mathrm{d}\theta\\
&=\frac{1}{2\pi}\int_0^{2\pi}\left|\left(2z+\sum_{n=2}^{k-1}(n+1)a_n z^n\right)w(z)\right|^2\mathrm{d}\theta\\
&\leqslant\frac{1}{2\pi}\int_0^{2\pi}\left|2z+\sum_{n=2}^{k-1}(n+1)a_n z^n\right|^2\mathrm{d}\theta\\
&=4r^2+\sum_{n=2}^{k-1}(n+1)^2\left|a_n\right|^2 r^{2n}.
\end{aligned}$$

Let $r\to 1^-$. Then

$$\sum_{n=2}^{k}(n-1)^2\left|a_n\right|^2\leqslant\sum_{n=2}^{k}(n-1)^2\left|a_n\right|^2+\sum_{n=k+1}^{\infty}\left|b_n\right|^2\leqslant 4+\sum_{n=2}^{k-1}(n+1)^2\left|a_n\right|^2.$$

Thus,

$$(k-1)^2\left|a_k\right|^2\leqslant 4+\sum_{n=2}^{k-1}\left((n+1)^2-(n-1)^2\right)\left|a_n\right|^2=4\left(1+\sum_{n=2}^{k-1}n\left|a_n\right|^2\right).$$

Equality clearly holds for the rotations κ_θ of the Koebe function κ given by

$$\kappa_\theta(z)=\mathrm{e}^{-\mathrm{i}\theta}\kappa(\mathrm{e}^{\mathrm{i}\theta}z)=\frac{z}{(1-\mathrm{e}^{\mathrm{i}\theta}z)^2}=z+\sum_{k=2}^{\infty}k\mathrm{e}^{\mathrm{i}(k-1)\theta}z^k,\quad\theta\in[0,2\pi].$$

The inequality $|a_k|\leqslant k$ follows from the inequality (6.2.2) by the principle of mathematical induction. For $k=2$, the inequality (6.2.2) reduces to $|a_2|^2\leqslant 4$ and this shows that $|a_2|\leqslant 2$. Assume that $|a_k|\leqslant k$ holds for all $k=2,3,\ldots,m-1$. For $k=m$, inequality (6.2.2) yields

$$(m-1)^2\left|a_m\right|^2\leqslant 4\left(1+\sum_{n=2}^{m-1}n\left|a_n\right|^2\right)\leqslant 4\left(1+\sum_{n=2}^{m-1}n^3\right)=(m-1)^2m^2$$

and so $|a_m|\leqslant m$. □

Theorem 6.2.5 *Let $f(z)=z+\sum_{k=2}^{\infty}a_k z^k\in\mathcal{ST}$, and the image $f(\mathbb{D})$ has area Δ. Then*

$$|a_k| \leqslant \frac{2}{k-1}\sqrt{\frac{\Delta}{\pi}}, \quad k \geqslant 2.$$

In particular, if $f \in \mathcal{ST}$ *and* $|f(z)| \leqslant 1$, *then* $|a_k| \leqslant 2/(k-1)$ *for* $k \geqslant 2$.

Proof Theorem 2.3.3 shows that the area Δ of the image $f(\mathbb{D})$ is

$$\Delta = \int_0^1 \int_0^{2\pi} \left|f'\left(re^{i\theta}\right)\right|^2 r \,d\theta \,dr = \pi\left(1 + \sum_{n=2}^{\infty} n\,|a_n|^2\right).$$

By Theorem 6.2.4,

$$(k-1)^2\,|a_k|^2 \leqslant 4\left(1 + \sum_{n=2}^{k-1} n\,|a_n|^2\right) \leqslant \frac{4\Delta}{\pi},$$

and so

$$|a_k| \leqslant \frac{2}{k-1}\sqrt{\frac{\Delta}{\pi}}, \quad k \geqslant 2.$$

If $|f(z)| \leqslant 1$, then $\Delta \leqslant \pi$ and $|a_k| \leqslant 2/(k-1)$. □

The image $f(\mathbb{D})$ of a function $f \in \mathcal{S}$ contains the disk $\mathbb{D}_{1/4}$. Hence, the function f admits an inverse F on a domain that contains $\mathbb{D}_{1/4}$. Let

$$F(w) = w + \gamma_2 w^2 + \gamma_3 w^3 + \cdots,$$

$\widetilde{\kappa}$ be the rotation of the Koebe function given by

$$w = \widetilde{\kappa}(z) = -\kappa(-z) = \frac{z}{(1+z)^2} = \frac{1}{4}\left(1 - \left(\frac{1-z}{1+z}\right)^2\right),$$

and K denote its inverse. Solving for z in terms of w yields

$$\begin{aligned} z &= \frac{1-\sqrt{1-4w}}{1+\sqrt{1-4w}} = \frac{1-2w-\sqrt{1-4w}}{2w} \\ &= w + 2w^2 + 5w^3 + 14w^4 + 42w^5 + 132w^6 + \cdots. \end{aligned}$$

Writing

$$K(w) = w + K_2 w^2 + K_3 w^3 + \cdots,$$

a computation gives the coefficients

$$K_k = \frac{(2k)!}{k!(k+1)!}.$$

Löwner [10] showed that $|\gamma_k| \leqslant K_k$ for each $k \geqslant 2$. The proof of this inequality relies on Löwner theory. However, in the case of starlike functions, the same result can be proved without invoking Löwner theory.

Theorem 6.2.6 *If $f \in \mathcal{ST}$ and its inverse F is given by*

$$F(w) = w + \gamma_2 w^2 + \gamma_3 w^3 + \cdots,$$

then

$$|\gamma_k| \leqslant K_k = \frac{(2k)!}{k!(k+1)!}.$$

Equality holds for the inverse of the Koebe function.

Proof The function $f \in \mathcal{ST}$ if and only if $\operatorname{Re}\big(zf'(z)/f(z)\big) > 0$, or equivalently, $\operatorname{Re}\big(f(z)/(zf'(z))\big) > 0$. This implies the existence of a function $p(z) = 1 + c_1 z + c_2 z^2 + \cdots$ satisfying $\operatorname{Re}(p(z)) > 0$ and

$$\frac{f(z)}{zf'(z)} = p(z). \tag{6.2.8}$$

Let $w = f(z)$, so that $F(w) = z$ and $f'(z)F'(w) = 1$. Substituting into (6.2.8) gives

$$\frac{wF'(w)}{F(w)} = p(F(w)),$$

or equivalently,

$$wF'(w) - F(w) = F(w)\big(p(F(w)) - 1\big) = \sum_{n=1}^{\infty} c_n (F(w))^{n+1}.$$

Since $F(w) = w + \gamma_2 w^2 + \gamma_3 w^3 + \cdots$, equating coefficients of like powers of w yields

$$(k-1)\gamma_k = \sum_{n=1}^{k-1} c_n \left(F^{n+1}(w)\right)_k,$$

where $\left(F^{n+1}(w)\right)_k$ denotes the coefficient of w^k in $F^{n+1}(w)$. In particular, for $k = 2$, this gives $\gamma_2 = c_1(F^2(w))_2 = c_1$, so that $|\gamma_2| \leqslant 2 = K_2$, establishing the result for $k = 2$.

Since

$$\left(F^{n+1}(w)\right)_k = Q_n\left(\gamma_2, \gamma_3, \ldots, \gamma_{k-1}\right),$$

where Q_n is a polynomial in $\gamma_2, \gamma_3, \ldots, \gamma_{k-1}$ with non-negative coefficients, it follows that

$$\left|\left(F^{n+1}(w)\right)_k\right| \leqslant Q_n\left(|\gamma_2|, |\gamma_3|, \ldots, |\gamma_{k-1}|\right).$$

The proof proceeds by induction. Assume the inequality $|\gamma_n| \leqslant K_n$ holds for all $n \leqslant k-1$. Then

$$\left|\left(F^{n+1}(w)\right)_k\right| \leqslant Q_n\left(K_2, K_3, \ldots, K_{k-1}\right) = \left(K^{n+1}(w)\right)_k.$$

Since $|c_n| \leqslant 2$ for each n, it follows that

$$(k-1)\,|\gamma_k| \leqslant \sum_{n=1}^{k-1} |c_n|\left|\left(F^{n+1}(w)\right)_k\right| \leqslant 2\sum_{n=1}^{k-1}\left(K^{n+1}(w)\right)_k.$$

The proof is completed by verifying the identity

$$2\sum_{n=1}^{k-1}\left(K^{n+1}(w)\right)_k = (k-1)K_k. \tag{6.2.9}$$

The rotation of the Koebe function

$$\widetilde{\kappa}(z) = \frac{z}{(1+z)^2}$$

satisfies

$$\frac{z\widetilde{\kappa}'(z)}{\widetilde{\kappa}(z)} = \frac{1-z}{1+z}.$$

Letting $w = \widetilde{\kappa}(z)$ and noting that $K(w) = z$ and $\widetilde{\kappa}'(z)K'(w) = 1$, the above identity becomes

$$\frac{K(w)}{wK'(w)} = \frac{1-K(w)}{1+K(w)}.$$

Hence,

$$wK'(w) - K(w) = 2\frac{K^2(w)}{1-K(w)} = 2\sum_{n=1}^{\infty} K^{n+1}(w).$$

This leads to the identity

$$\sum_{k=2}^{\infty}(k-1)K_k w^k = 2\sum_{n=1}^{\infty} K^{n+1}(w). \tag{6.2.10}$$

Comparing coefficients of w^k in (6.2.10) establishes (6.2.9), thereby completing the proof. □

6.3 Starlike Functions of Order α

The class of starlike functions is geometrically defined by requiring that the image of the unit disk under such functions be starlike with respect to the origin. This class admits several natural generalizations, particularly through its analytic characterization. Recall that a function $f \in \mathcal{ST}$ if the quantity $zf'(z)/f(z)$ takes values in the right half-plane. A natural generalization consists in replacing the right half-plane by other half-planes.

Definition 6.3.1 A function $f \in \mathcal{A}$ is said to be *starlike of order* α, $\alpha < 1$, if

$$\operatorname{Re} \frac{zf'(z)}{f(z)} > \alpha, \quad z \in \mathbb{D}.$$

The class of all such functions is denoted by $\mathcal{ST}(\alpha)$.

If $f \in \mathcal{A}$ is starlike of order $\alpha \geqslant 0$, then f is starlike; that is, $\mathcal{ST}(\alpha) \subset \mathcal{ST}$. For $\alpha < 0$, the class $\mathcal{ST}(\alpha)$ contains non-univalent functions. More generally, for $\beta < \alpha$, the inclusion $\mathcal{ST}(\alpha) \subset \mathcal{ST}(\beta)$ holds. The following theorem establishes a connection between two subclasses of starlike functions.

Theorem 6.3.2 *Let* $0 \leqslant \alpha < 1$, *and* $0 \leqslant \beta < 1$. *If* $f \in \mathcal{ST}(\alpha)$, *then*

$$g(z) = z \left(\frac{f(z)}{z} \right)^{\frac{1-\beta}{1-\alpha}}, \quad z \in \mathbb{D} \tag{6.3.1}$$

belongs to $\mathcal{ST}(\beta)$.

Proof Since the function f is univalent, $f(z)/z \neq 0$ and an analytic branch of logarithm of $f(z)/z$ can be defined. Therefore, the function g is analytic and $g(z)/z \neq 0$. Evidently, from (6.3.1),

$$\left(\frac{f(z)}{z} \right)^{\frac{1}{1-\alpha}} = \left(\frac{g(z)}{z} \right)^{\frac{1}{1-\beta}},$$

and thus

$$\frac{1}{1-\alpha} \left(\frac{zf'(z)}{f(z)} - 1 \right) = \frac{1}{1-\beta} \left(\frac{zg'(z)}{g(z)} - 1 \right).$$

By adding 1 to both sides yields

$$\frac{1}{1-\alpha}\left(\frac{zf'(z)}{f(z)}-\alpha\right)=\frac{1}{1-\beta}\left(\frac{zg'(z)}{g(z)}-\beta\right). \tag{6.3.2}$$

Since $f \in \mathcal{ST}(\alpha)$ if and only if

$$\frac{1}{1-\alpha}\operatorname{Re}\left(\frac{zf'(z)}{f(z)}-\alpha\right)>0,$$

it follows from (6.3.2) that $g \in \mathcal{ST}(\beta)$. □

If $f \in \mathcal{ST}(\alpha)$, then, by Theorem 6.3.2, the function $g : \mathbb{D} \to \mathbb{C}$ defined by

$$g(z)=z\left(\frac{f(z)}{z}\right)^{\frac{1}{1-\alpha}}$$

belongs to $\mathcal{ST}$. This relation gives the distortion bounds for f from the corresponding bounds for g. It also implies that the logarithmic coefficients of f (the coefficients of $\log(f(z)/z)$) are obtained by multiplying those of g by $1-\alpha$.

Theorem 6.3.3 *Let $0 \leqslant \alpha < 1$, and let $\gamma_k \in \mathbb{R}$ satisfy $\gamma_k \geqslant 0$, and $\sum_{k=1}^{n} \gamma_k \leqslant 1$. If $f_k \in \mathcal{ST}(\alpha)$ for each k, then*

$$g(z)=z\prod_{k=1}^{n}\left(\frac{f_k(z)}{z}\right)^{\gamma_k}, \quad z \in \mathbb{D}$$

belongs to $\mathcal{ST}(\alpha)$.

Proof Since each f_k is univalent in $\mathbb{D}$, the function g is analytic and belongs to $\mathcal{A}$. A direct computation yields

$$\frac{zg'(z)}{g(z)}=1+\sum_{k=1}^{n}\gamma_k\left(\frac{zf_k'(z)}{f_k(z)}-1\right).$$

Taking the real part gives

$$\operatorname{Re}\frac{zg'(z)}{g(z)}=1+\sum_{k=1}^{n}\gamma_k\operatorname{Re}\left(\frac{zf_k'(z)}{f_k(z)}-1\right)>1+\sum_{k=1}^{n}\gamma_k(\alpha-1).$$

Since $\alpha < 1$, $\gamma_k \geqslant 0$, and $\sum_{k=1}^{n} \gamma_k \leqslant 1$, it follows that

$$\operatorname{Re}\frac{zg'(z)}{g(z)}>\alpha.$$

Thus, $g \in \mathcal{ST}(\alpha)$ as claimed. □

Theorem 6.3.4 *For $k = 1, 2, \ldots, n$, let $f_k \in \mathcal{ST}(1-\alpha_k)$, $0 < \alpha_k \leqslant 1$, and let $s_n = 1 - \sum_{k=1}^{n} \alpha_k \geqslant 0$. Then the product*

$$F_n(z) = z \prod_{k=1}^{n} \frac{f_k(z)}{z}, \quad z \in \mathbb{D}$$

is starlike of order s_n. There exist functions of this type which are not starlike of any order greater than s_n.

Proof Clearly, the function F_n is analytic in $\mathbb{D}$, satisfies $F_n(0) = 0$, and $F_n'(0) = 1$, whence $F_n \in \mathcal{A}$. A straightforward computation yields

$$\operatorname{Re} \frac{zF_n'(z)}{F_n(z)} = 1 - n + \sum_{k=1}^{n} \operatorname{Re} \frac{zf_k'(z)}{f_k(z)} > 1 - n + \sum_{k=1}^{n} (1 - \alpha_k) = s_n.$$

Hence, $F_n \in \mathcal{ST}(s_n)$.

Consider the functions $f_k(z) = z \exp(-\alpha_k z)$, $k = 1, 2, \ldots, n$. Each f_k is starlike of order $1 - \alpha_k$, and the corresponding product $F_n(z)$ is not starlike of any order greater than s_n. □

For $p \in \mathcal{P}(\alpha)$, the function q defined by

$$q(z) = \frac{p(z) - \alpha}{1 - \alpha}$$

belongs to $\mathcal{P}$ and so $q(z) \prec (1+z)/(1-z)$. This gives

$$p(z) \prec \alpha + (1-\alpha)\frac{1+z}{1-z} = \frac{1 + (1-2\alpha)z}{1-z}.$$

Therefore, a function f in the class $\mathcal{A}$ belong to the class $\mathcal{ST}(\alpha)$ if and only if

$$\frac{zf'(z)}{f(z)} \prec \frac{1 + (1-2\alpha)z}{1-z}. \tag{6.3.3}$$

See Fig. 6.6.

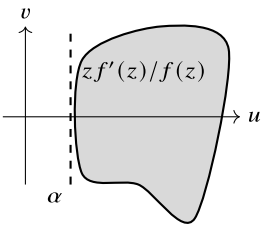


Fig. 6.6 Subordination $zf'(z)/f(z) \prec (1 + (1-2\alpha)z)/(1-z)$

Example 6.3.5 (Generalized Koebe function) For $0 \leqslant \alpha < 1$, the function

$$\kappa_\alpha(z) = \frac{z}{(1-z)^{2(1-\alpha)}}, \quad z \in \mathbb{D},$$

is called the generalized Koebe function of order α. A computation shows that

$$\operatorname{Re} \frac{z\kappa_\alpha'(z)}{\kappa_\alpha(z)} = 1 + \operatorname{Re} \frac{2(1-\alpha)z}{1-z} = \alpha + (1-\alpha) \operatorname{Re} \frac{1+z}{1-z} > \alpha,$$

and so $\kappa_\alpha \in \mathcal{ST}(\alpha)$. Moreover,

$$\frac{z\kappa_\alpha'(z)}{\kappa_\alpha(z)} = \frac{1+(1-2\alpha)z}{1-z},$$

and therefore, (6.3.3) shows that $f \in \mathcal{ST}(\alpha)$ if and only if

$$\frac{zf'(z)}{f(z)} \prec \frac{z\kappa_\alpha'(z)}{\kappa_\alpha(z)}.$$

Theorem 6.3.6 *If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{A}$ satisfies*

$$\sum_{k=2}^{\infty} (k-\alpha)\,|a_k| \leqslant 1-\alpha, \quad 0 \leqslant \alpha < 1, \tag{6.3.4}$$

then $f \in \mathcal{ST}(\alpha)$. Moreover, if $a_k \leqslant 0$ for all k, then the condition (6.3.4) is also necessary for $f \in \mathcal{ST}(\alpha)$.

Proof Since $k \leqslant (k-\alpha)/(1-\alpha)$ for each $k \geqslant 2$, (6.3.4) shows that

$$\sum_{k=2}^{\infty} k\,|a_k| \leqslant \sum_{k=2}^{\infty} \frac{k-\alpha}{1-\alpha}\,|a_k| \leqslant 1.$$

By Theorem 2.1.13, it follows that $f \in \mathcal{S}$ and so $f(z) \neq 0$ for $z \neq 0$. Using (6.3.4),

$$\begin{aligned}
&\left|zf'(z) - f(z)\right| - (1-\alpha)|f(z)| \\
&= \left|\sum_{k=2}^{\infty} (k-1)a_k z^k\right| - (1-\alpha)\left|z + \sum_{k=2}^{\infty} a_k z^k\right| \\
&\leqslant \sum_{k=2}^{\infty} (k-1)|a_k||z|^k - (1-\alpha)\left(|z| - \sum_{k=2}^{\infty} |a_k||z|^k\right) \\
&\leqslant |z|\left(\sum_{k=2}^{\infty} (k-\alpha)|a_k| - (1-\alpha)\right) \\
&\leqslant 0.
\end{aligned}$$

Consequently,

$$\left|\frac{zf'(z)}{f(z)} - 1\right| \leqslant 1 - \alpha$$

which implies

$$-\operatorname{Re}\left(\frac{zf'(z)}{f(z)} - 1\right) \leqslant \left|\frac{zf'(z)}{f(z)} - 1\right| \leqslant 1 - \alpha,$$

and hence

$$\operatorname{Re}\frac{zf'(z)}{f(z)} \geqslant \alpha.$$

An application of Theorem 1.2.12 shows that $f \in \mathcal{ST}(\alpha)$.

For the necessity, suppose that $f \in \mathcal{ST}(\alpha)$ and $a_k \leqslant 0$ for all k. For $z = r \in (0, 1)$, the inequality

$$\alpha < \operatorname{Re}\frac{zf'(z)}{f(z)} = \frac{rf'(r)}{f(r)} \tag{6.3.5}$$

holds. The condition $f \in \mathcal{S}$ ensures that $f'(z) \neq 0$ in $\mathbb{D}$, so in particular $f'(r) \neq 0$. Since $f'(0) = 1$, it follows that $f'(r) > 0$ and hence $f(r) > 0$ by (6.3.5). Therefore, the inequality (6.3.5) can be written as

$$0 < f'(r) - \alpha\frac{f(r)}{r} = 1 - \alpha + \sum_{k=2}^{\infty}(k - \alpha)a_k r^{k-1}.$$

Since each $a_k \leqslant 0$, this yields

$$0 < 1 - \alpha - \sum_{k=2}^{\infty}(k - \alpha)|a_k| r^{k-1}.$$

Taking the limit as $r \to 1^-$, the inequality $\sum_{k=2}^{\infty}(k - \alpha)|a_k| \leqslant 1 - \alpha$ is established. □

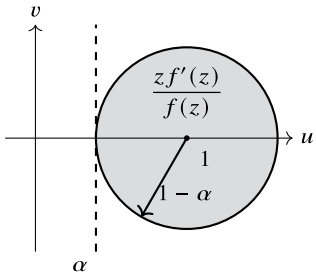


Fig. 6.7 Disk $\{w : |w - 1| < 1 - \alpha\}$ is contained in $\{w : \operatorname{Re} w > \alpha\}$

For $0 \leqslant \alpha < 1$, let $\mathcal{ST}_\alpha$ denote the class of all $f \in \mathcal{A}$ satisfying

$$\left|\frac{zf'(z)}{f(z)} - 1\right| < 1 - \alpha.$$

Clearly, $\mathcal{ST}_\alpha \subset \mathcal{ST}(\alpha)$. See Fig. 6.7.

Example 6.3.7 For $n \in \mathbb{N}$, let

$$f_n(z) = z - \frac{1-\alpha}{n-\alpha} z^n, \quad z \in \mathbb{D}.$$

By Theorem 6.3.6, $f_n \in \mathcal{ST}(\alpha)$. It can be shown that the function $f : \mathbb{D} \to \mathbb{C}$ defined by $f(z) = z + A_n z^n$ is starlike of order α if and only if

$$|A_n| \leqslant \frac{1-\alpha}{n-\alpha}.$$

Example 6.3.8 For each $k \in \mathbb{N}$, the function $f_k(z) = z - z^3/4k^2$ satisfies

$$\operatorname{Re} \frac{zf_k'(z)}{f_k(z)} = 1 - \operatorname{Re} \frac{2z^2}{4k^2 - z^2} > 1 - \frac{2}{4k^2 - 1}$$

and hence f_k is starlike of order $\alpha_k = 1 - d_k = 1 - 2/\left(4k^2 - 1\right)$. Since $\sum_{k=1}^{\infty} d_k = \sum_{k=1}^{\infty} 2/\left(4k^2 - 1\right) = 1$, Theorem 6.3.4 (with $n = \infty$) shows that

$$\frac{2}{\pi} \sin \frac{\pi}{2} z = z \prod_{k=1}^{\infty} \left(1 - \frac{z^2}{4k^2}\right)$$

is starlike. This function is not univalent in $|z| < R$ for each $R > 1$ since its derivative has a zero at $z = 1$.

6.4 Representation Theorem

Since $f \in \mathcal{ST}(\alpha)$ if and only if $\operatorname{Re}(zf'(z)/f(z)) > \alpha$, and the functions in $\mathcal{P}(\alpha)$ and $\mathcal{P}$ are similarly related, the following representation theorem holds.

Theorem 6.4.1 *Let $0 \leqslant \alpha < 1$. A function f belongs to $\mathcal{ST}(\alpha)$ if and only if*

$$f(z) = z \exp\left((1-\alpha) \int_0^z \frac{p(w) - 1}{w} \mathrm{d}w\right) \tag{6.4.1}$$

for some $p \in \mathcal{P}$.

Proof Suppose that $f \in \mathcal{ST}(\alpha)$. Then,

$$p(z) = \frac{1}{1-\alpha}\left(\frac{zf'(z)}{f(z)} - \alpha\right)$$

belongs to $\mathcal{P}$. Since $\mathcal{ST}(\alpha) \subset \mathcal{S}$, it follows that $f(z)/z$ is analytic and non-vanishing in $\mathbb{D}$ and so $\log(f(z)/z)$ is analytic in $\mathbb{D}$. Integrating

$$(1-\alpha)\frac{p(w)-1}{w} = \frac{f'(w)}{f(w)} - \frac{1}{w}$$

from 0 to z gives

$$(1-\alpha)\int_0^z \frac{p(w)-1}{w}\mathrm{d}w = \log\left(\frac{f(w)}{w}\right)\Big|_0^z = \log\left(\frac{f(z)}{z}\right).$$

This establishes (6.4.1). Notice that both $f(z)/z$ and $(p(z)-1)/z$ have removable singularities at the origin.

If (6.4.1) holds for some $p \in \mathcal{P}$, then $f(z) \neq 0$ for $z \neq 0$. Therefore, $\log(f(z)/z)$ is analytic and

$$\log\left(\frac{f(z)}{z}\right) = (1-\alpha)\int_0^z \frac{p(w)-1}{w}\mathrm{d}w.$$

Differentiation yields

$$\frac{zf'(z)}{f(z)} = \alpha + (1-\alpha)p(z).$$

Since $\operatorname{Re} p(z) > 0$, it follows that

$$\operatorname{Re}\frac{zf'(z)}{f(z)} = \alpha + (1-\alpha)\operatorname{Re} p(z) > \alpha,$$

and hence $f \in \mathcal{ST}(\alpha)$. □

For a function $p \in \mathcal{P}$, the Herglotz's representation theorem (Theorem 4.2.3) shows that there is a non-decreasing function μ on $[0, 2\pi]$ with $\int_0^{2\pi} \mathrm{d}\mu(t) = 1$ satisfying

$$p(z) = \int_0^{2\pi} \frac{1 + z\mathrm{e}^{-\mathrm{i}t}}{1 - z\mathrm{e}^{-\mathrm{i}t}}\mathrm{d}\mu(t).$$

This result leads to the following representation theorem for starlike functions of order α.

Theorem 6.4.2 *(Representation Theorem) Let $f \in \mathcal{A}$ and $0 \leqslant \alpha < 1$. A function f belongs to $\mathcal{ST}(\alpha)$ if and only if there is a non-decreasing function μ on $[0, 2\pi]$ satisfying $\int_0^{2\pi} d\mu(t) = 1$ and such that*

$$f(z) = z \exp\left(-2(1-\alpha)\int_0^{2\pi} \log(1 - e^{-it}z)d\mu(t)\right). \tag{6.4.2}$$

Proof By the Herglotz's representation theorem (Theorem 4.2.3), for a function $p \in \mathcal{P}$, there is a non-decreasing function μ on $[0, 2\pi]$ with $\int_0^{2\pi} \mathrm{d}\mu(t) = 1$ satisfying

$$p(z) = \int_0^{2\pi} \frac{1 + z\mathrm{e}^{-it}}{1 - z\mathrm{e}^{-it}}\mathrm{d}\mu(t). \tag{6.4.3}$$

Then

$$\frac{p(w) - 1}{w} = 2\int_0^{2\pi} \frac{\mathrm{e}^{-it}}{1 - \mathrm{e}^{-it}w}\mathrm{d}\mu(t).$$

An interchange of the order of integration gives

$$\begin{aligned}\int_0^z \frac{p(w) - 1}{w}\mathrm{d}w &= 2\int_0^z \int_0^{2\pi} \frac{\mathrm{e}^{-it}}{1 - \mathrm{e}^{-it}w}\mathrm{d}\mu(t)\mathrm{d}w \\ &= 2\int_0^{2\pi} \int_0^z \frac{\mathrm{e}^{-it}}{1 - \mathrm{e}^{-it}w}\mathrm{d}w\mathrm{d}\mu(t) \\ &= -2\int_0^{2\pi} \log(1 - \mathrm{e}^{-it}z)\mathrm{d}\mu(t).\end{aligned}$$

By using (6.4.1),

$$f(z) = z\exp\left(-2(1-\alpha)\int_0^{2\pi} \log(1 - \mathrm{e}^{-it}z)\mathrm{d}\mu(t)\right).$$

The converse readily follows from Theorem 6.4.1 as the function p given by (6.4.3) belongs to $\mathcal{P}$. □

6.5 Logarithmic Coefficients

Let $f \in \mathcal{A}$ with $f(z)/z \neq 0$. Then the function $\log(f(z)/z)$ is analytic in $\mathbb{D}$. Further, let

$$\log\left(\frac{f(z)}{z}\right) = 2\sum_{k=1}^{\infty} \gamma_k z^k. \tag{6.5.1}$$

The coefficients γ_k are called the *logarithmic coefficients* of the function f. The sharp bounds on these coefficients are computed for starlike functions of order α in the following result.

Theorem 6.5.1 *The logarithmic coefficients γ_k of the function f belonging to the class $\mathcal{ST}(\alpha)$ satisfies the inequality $|\gamma_k| \leqslant (1-\alpha)/k$ with equality for the generalized Koebe function κ_α.*

Proof By (6.4.2),

$$\begin{aligned}\log\left(\frac{f(z)}{z}\right) &= -2(1-\alpha)\int_0^{2\pi} \log(1-\mathrm{e}^{-\mathrm{i}t}z)\mathrm{d}\mu(t)\\ &= \int_0^{2\pi}\left(\sum_{k=1}^{\infty}\frac{2(1-\alpha)\mathrm{e}^{-\mathrm{i}kt}}{k}z^k\right)\mathrm{d}\mu(t)\\ &= 2(1-\alpha)\sum_{k=1}^{\infty}\left(\int_0^{2\pi}\frac{\mathrm{e}^{-\mathrm{i}kt}}{k}\mathrm{d}\mu(t)\right)z^k.\end{aligned}$$

Therefore, the logarithmic coefficients γ_k of the function f are given by

$$\gamma_k = (1-\alpha)\int_0^{2\pi}\frac{\mathrm{e}^{-\mathrm{i}kt}}{k}\mathrm{d}\mu(t) \quad (k \geqslant 1),$$

and hence

$$|\gamma_k| \leqslant (1-\alpha)\int_0^{2\pi}\left|\frac{\mathrm{e}^{-\mathrm{i}kt}}{k}\right|\mathrm{d}\mu(t) = \frac{1-\alpha}{k}\int_0^{2\pi}\mathrm{d}\mu(t) = \frac{1-\alpha}{k}.$$

Since

$$\log\left(\frac{\kappa_\alpha(z)}{z}\right) = -2(1-\alpha)\log(1-z) = 2(1-\alpha)\sum_{k=1}^{\infty}\frac{z^k}{k},$$

the equality $\gamma_k = (1-\alpha)/k$ holds for the generalized Koebe function κ_α. □

The bound for the logarithmic coefficients leads to the following corollary.

Corollary 6.5.2 *If $f \in \mathcal{ST}(\alpha)$ and $f(z) = z + \sum_{k=2}^{\infty} a_k z^k$, then*

$$|a_k| \leqslant A_k := \frac{1}{(k-1)!} \prod_{n=2}^{k} (n - 2\alpha).$$

Equality holds for the generalized Koebe function κ_α.

Proof The proof is by induction. The following identity is needed for the proof:

$$k A_{k+1} = 2(1-\alpha) \left(1 + \sum_{n=2}^{k} A_n \right), \quad k \geqslant 2. \tag{6.5.2}$$

Since $A_2 = 2(1-\alpha)$ and $A_3 = (1-\alpha)(3-2\alpha)$, it follows that

$$2A_3 = 2(1-\alpha)(3-2\alpha) = 2(1-\alpha)(1+A_2),$$

and therefore, (6.5.2) holds for $k = 2$.

Assume that (6.5.2) holds for $k = m$:

$$m A_{m+1} = 2(1-\alpha) \left(1 + \sum_{n=2}^{m} A_n \right). \tag{6.5.3}$$

From the definition of A_k,

$$k A_{k+1} = (k + 1 - 2\alpha) A_k,$$

and hence, by using (6.5.3),

$$\begin{aligned}
(m+1) A_{m+2} &= (m + 2 - 2\alpha) A_{m+1} \\
&= m A_{m+1} + 2(1-\alpha) A_{m+1} \\
&= 2(1-\alpha) \left(1 + \sum_{n=2}^{m} A_n \right) + 2(1-\alpha) A_{m+1} \\
&= 2(1-\alpha) \left(1 + \sum_{n=2}^{m+1} A_n \right).
\end{aligned}$$

This is equation (6.5.2) with $k = m + 1$. By induction, it follows that (6.5.2) holds for all $k \geqslant 2$.

With $f(z) = z + a_2 z^2 + a_3 z^3 + \cdots$, (6.5.1) shows that

$$1 + \sum_{k=1}^{\infty} a_{k+1} z^k = \frac{f(z)}{z} = \exp\left(2 \sum_{k=1}^{\infty} \gamma_k z^k\right),$$

which upon differentiation gives

$$\begin{aligned} \sum_{k=1}^{\infty} k a_{k+1} z^{k-1} &= \left(2 \sum_{k=1}^{\infty} k \gamma_k z^{k-1}\right) \exp\left(2 \sum_{k=1}^{\infty} \gamma_k z^k\right) \\ &= \left(2 \sum_{k=1}^{\infty} k \gamma_k z^{k-1}\right) \left(1 + \sum_{k=1}^{\infty} a_{k+1} z^k\right). \end{aligned}$$

Equating the coefficients of z^{k-1} gives

$$k a_{k+1} = 2k\gamma_k + 2 \sum_{n=2}^{k} (k + 1 - n) \gamma_{k+1-n} a_n. \tag{6.5.4}$$

Setting $k = 1$ in (6.5.4) gives $a_2 = 2\gamma_1$ and so $|a_2| \leqslant 2(1 - \alpha) = A_2$. To complete the proof by induction, assume $|a_k| \leqslant A_k$ for $k = 2, 3 \ldots, m$. By Theorem 6.5.1,

$$k|\gamma_k| \leqslant 1 - \alpha, \quad k \geqslant 1,$$

and hence, by using (6.5.4) and (6.5.2),

$$\begin{aligned} m|a_{m+1}| &\leqslant 2m|\gamma_m| + 2 \sum_{n=2}^{m} (m + 1 - n) |\gamma_{m+1-n}|\, |a_n| \\ &\leqslant 2(1 - \alpha) \left(1 + \sum_{n=2}^{m} |a_n|\right) \\ &\leqslant 2(1 - \alpha) \left(1 + \sum_{n=2}^{m} A_n\right) = m A_{m+1}. \end{aligned}$$

Thus, $|a_{m+1}| \leqslant A_{m+1}$ and completes the proof by induction. □

The following theorem extends Theorem 6.2.4 for starlike functions of order α.

Theorem 6.5.3 *(Clunie's Theorem) If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{ST}(\alpha)$, then*

$$(k - 1)^2 |a_k|^2 \leqslant 4(1 - \alpha) \sum_{n=1}^{k-1} (n - \alpha) |a_n|^2, \quad a_1 = 1$$

for each integer $k \geqslant 2$ with equality if f is a rotation of the generalized Koebe function κ_α.

Proof From (6.3.3), there is a function $w(z) = \sum_{k=1}^{\infty} w_k z^k \in \mathcal{B}_0$, satisfying

$$\frac{zf'(z)}{f(z)} = \frac{1 + (1-2\alpha)w(z)}{1 - w(z)}.$$

Then, $\big(zf'(z) + (1-2\alpha)f(z)\big)w(z) = zf'(z) - f(z)$, and

$$\left(2(1-\alpha)z + \sum_{n=2}^{\infty}(n+1-2\alpha)a_n z^n\right)\sum_{n=1}^{\infty} w_n z^n = \sum_{n=2}^{\infty}(n-1)a_n z^n. \tag{6.5.5}$$

Equating the coefficients of z^k on both sides of (6.5.5) gives

$$2(1-\alpha)w_{k-1} + (3-2\alpha)a_2 w_{k-2} + \cdots + (k-2\alpha)a_{k-1}w_1 = (k-1)a_k, \tag{6.5.6}$$

for $k \geqslant 2$. This means that the coefficient a_k on the right of (6.5.6) depends only on $a_2, \ldots, a_{k-1}$. Hence, for $k \geqslant 2$, write (6.5.5) as

$$\left(2(1-\alpha)z + \sum_{n=2}^{k-1}(n+1-2\alpha)a_n z^n\right) w(z) = \sum_{n=2}^{k}(n-1)a_n z^n + \sum_{n=k+1}^{\infty} b_n z^n. \tag{6.5.7}$$

With $z = re^{i\theta}$, integrate the square of the modulus of both sides of (6.5.7) with respect to θ, and then using $|w(z)| < 1$ for $z \in \mathbb{D}$ and (3.3.3), it follows that

$$\begin{aligned}
&\sum_{n=2}^{k}(n-1)^2|a_n|^2 r^{2n} + \sum_{n=k+1}^{\infty}|b_n|^2 r^{2n} \\
&= \frac{1}{2\pi}\int_0^{2\pi}\left|\sum_{n=2}^{k}(n-1)a_n z^n + \sum_{n=k+1}^{\infty} b_n z^n\right|^2 \mathrm{d}\theta \\
&\leqslant \frac{1}{2\pi}\int_0^{2\pi}\left|\left(2(1-\alpha)z + \sum_{n=2}^{k-1}(n+1-2\alpha)a_n z^n\right) w(z)\right|^2 \mathrm{d}\theta \\
&\leqslant \frac{1}{2\pi}\int_0^{2\pi}\left|2(1-\alpha)z + \sum_{n=2}^{k-1}(n+1-2\alpha)a_n z^n\right|^2 \mathrm{d}\theta \\
&= 4(1-\alpha)^2 r^2 + \sum_{n=2}^{k-1}(n+1-2\alpha)^2|a_n|^2 r^{2n}.
\end{aligned}$$

Letting $r \to 1^-$ yields

$$\sum_{n=2}^{k}(n-1)^2|a_n|^2 \leqslant 4(1-\alpha)^2 + \sum_{n=2}^{k-1}(n+1-2\alpha)^2|a_n|^2,$$

or equivalently,

$$(k-1)^2\,|a_k|^2 \leqslant 4(1-\alpha)\left(1-\alpha+\sum_{n=2}^{k-1}(n-\alpha)\,|a_n|^2\right), \qquad k \geqslant 2.$$

The coefficients A_k of the generalized Koebe function κ_α are given by

$$A_k = \frac{1}{(k-1)!}\prod_{n=2}^{k}(n-2\alpha).$$

It is readily established by induction that

$$(k-1)^2 A_k^2 = 4(1-\alpha)\left(1-\alpha+\sum_{n=2}^{k-1}(n-\alpha)A_n^2\right), \quad k \geqslant 2,$$

and hence equality holds for the generalized Koebe function (or its rotations). □

As in the case of starlike functions, this theorem can be employed by induction to obtain sharp estimates for the coefficients a_k for functions starlike of order α. However, in Corollary 6.5.2, the estimates are derived by using the logarithmic coefficients.

6.6 Applications of Subordination

The next theorem is a result proved by Goluzin [6, Theorem 5, p. 327] using a highly involved series of inequalities. It will be proved here by a simpler argument, namely verifying an admissibility condition. This result has numerous applications in the study of distortion properties and coefficient inequalities.

Theorem 6.6.1 *Let $f \in \mathcal{A}_n$. If $f \in \mathcal{ST}(\alpha)$, $0 \leqslant \alpha < 1$, then*

$$\operatorname{Re}\left(\frac{f(z)}{z}\right)^{\frac{n}{2(1-\alpha)}} > \frac{1}{2}, \quad z \in \mathbb{D},$$

or equivalently,

$$\frac{f(z)}{z} \prec \frac{1}{(1-z^n)^{\frac{2(1-\alpha)}{n}}}.$$

The inequality is sharp for the function $f \in \mathcal{A}_n$ defined by $f(z) = \kappa_\alpha(z^n)^{1/n}$, where κ_α is the generalized Koebe function.

Proof The function

$$p(z) := 2\left(\frac{f(z)}{z}\right)^{\frac{n}{2(1-\alpha)}} - 1, \quad z \in \mathbb{D}, \tag{6.6.1}$$

belongs to $\mathcal{H}[1, n]$. From (6.6.1), it follows that

$$f(z) = z\left(\frac{p(z)+1}{2}\right)^{\frac{2(1-\alpha)}{n}}.$$

A computation gives

$$\frac{1}{1-\alpha}\left(\frac{zf'(z)}{f(z)} - \alpha\right) = 1 + \frac{2}{n}\frac{zp'(z)}{p(z)+1} = \psi(p(z), zp'(z)) \in \Omega, \tag{6.6.2}$$

where

$$\psi(r, s) = 1 + \frac{2}{n}\frac{s}{r+1} \quad \text{and} \quad \Omega = \{w \in \mathbb{C} : \operatorname{Re} w > 0\}.$$

The function ψ satisfies

$$\operatorname{Re}\psi(\mathrm{i}\rho, \sigma) = \operatorname{Re}\left(1 + \frac{2}{n}\frac{\sigma}{1+\rho^2}(1 - \mathrm{i}\rho)\right) = 1 + \frac{2}{n}\frac{\sigma}{1+\rho^2} \leqslant 1 - 1 = 0,$$

or $\psi(\mathrm{i}\rho, \sigma) \notin \Omega$ whenever $\rho \in \mathbb{R}$ and $\sigma \leqslant -n(1+\rho^2)/2$. This shows that the function $\psi \in \mathcal{P}_n(\Omega)$. Since $\psi\big(p(z), zp'(z)\big) \in \Omega$ by (6.6.2), Theorem 5.4.2 shows that the function $p \in \mathcal{P}$. This is equivalent to

$$\operatorname{Re}\left(\frac{f(z)}{z}\right)^{\frac{n}{2(1-\alpha)}} > \frac{1}{2}.$$

The function $f \in \mathcal{A}_n$ given by

$$f(z) = \kappa_\alpha(z^n)^{1/n} = \frac{z}{(1-z^n)^{\frac{2(1-\alpha)}{n}}}$$

is the nth root transform of the generalized Koebe function κ_α and so, by Theorem 6.1.9, it is starlike of order α. A direct calculation shows that

$$\operatorname{Re}\frac{zf'(z)}{f(z)} = \operatorname{Re}\frac{1 + (1-2\alpha)z^n}{1-z^n} > \alpha.$$

Since

$$\operatorname{Re}\left(\frac{f(z)}{z}\right)^{\frac{n}{2(1-\alpha)}} = \operatorname{Re}\frac{1}{1-z^n} > \frac{1}{2},$$

and the image of $1/(1-z^n)$ is the half-plane $\{w \in \mathbb{C} : \operatorname{Re} w > 1/2\}$, the implication is sharp. □

The next lemma follows from the proof of Theorem 6.6.1.

Lemma 6.6.2 *If $p \in \mathcal{H}[1, n]$ satisfies*

$$\operatorname{Re}\left(\frac{zp'(z)}{p(z)+1}\right) > -\frac{n}{2}, \quad z \in \mathbb{D},$$

then $p \in \mathcal{P}$.

For $n = 1$, Theorem 6.6.1 reduces to the following result.

Corollary 6.6.3 *If $f \in \mathcal{ST}(\alpha)$, then*

$$\operatorname{Re}\left(\frac{f(z)}{z}\right)^{\frac{1}{2(1-\alpha)}} > \frac{1}{2}. \tag{6.6.3}$$

The inequality is sharp for the generalized Koebe function k_α.

The inequality (6.6.3) is equivalent to the subordination

$$\left(\frac{f(z)}{z}\right)^{\frac{1}{2(1-\alpha)}} \prec \frac{1}{1-z},$$

or to the subordination

$$\left(\frac{z}{f(z)}\right)^{\frac{1}{2(1-\alpha)}} \prec 1 - z.$$

Thus, (6.6.3) is equivalent to

$$\left|\left(\frac{z}{f(z)}\right)^{\frac{1}{2(1-\alpha)}} - 1\right| < 1.$$

The next corollary gives the growth, distortion, and covering estimates for starlike functions of order α.

Corollary 6.6.4 ***(Growth, Distortion, and Covering Theorem)*** *If $f \in \mathcal{ST}(\alpha)$, then*

$$-\kappa_\alpha(-r) = \frac{r}{(1+r)^{2(1-\alpha)}} \leqslant |f(z)| \leqslant \frac{r}{(1-r)^{2(1-\alpha)}} = \kappa_\alpha(r), \quad |z| = r < 1,$$

and

$$\kappa'_\alpha(-r) = \frac{1-(1-2\alpha)r}{(1+r)^{3-2\alpha}} \leqslant |f'(z)| \leqslant \frac{1+(1-2\alpha)r}{(1-r)^{3-2\alpha}} = \kappa'_\alpha(r), \quad |z| = r < 1.$$

These inequalities are sharp. Also, $f(\mathbb{D}) \supset \{w \in \mathbb{C} : |w| < 2^{-2(1-\alpha)}\}$.

Proof By Corollary 6.6.3, there is an analytic function $w : \mathbb{D} \to \mathbb{C}$ with $w(0) = 0$ and $|w(z)| < 1$ satisfying

$$\frac{f(z)}{z} = \frac{1}{(1 - w(z))^{2(1-\alpha)}}.$$

For $|z| = r < 1$, the Schwarz's Lemma (Theorem 3.1.1) shows that

$$1 - r \leqslant 1 - |w(z)| \leqslant |1 - w(z)| \leqslant 1 + |w(z)| \leqslant 1 + r,$$

and so

$$\frac{1}{(1+r)^{2(1-\alpha)}} \leqslant \frac{|f(z)|}{r} \leqslant \frac{1}{(1-r)^{2(1-\alpha)}}.$$

This proves the growth inequality. The growth inequality can also be proved directly by using Theorem 6.4.2.

To prove the distortion inequality, note that the function $p : \mathbb{D} \to \mathbb{C}$ defined by $p(z) := zf'(z)/f(z)$ belongs to $\mathcal{P}(\alpha)$ and (4.3.8) gives

$$\frac{1 - (1-2\alpha)r}{1+r} \leqslant |p(z)| \leqslant \frac{1 + (1-2\alpha)r}{1-r}.$$

Thus,

$$\left|zf'(z)\right| = |p(z)f(z)| \leqslant \frac{1 + (1-2\alpha)r}{1-r} \frac{r}{(1-r)^{2(1-\alpha)}} = \frac{r(1 + (1-2\alpha)r)}{(1-r)^{3-2\alpha}}.$$

The left distortion inequality can be similarly proved. The inequalities are sharp for the generalized Koebe function κ_α. Finally, the covering estimate is obtained by letting $r \to 1^-$ in the left hand side of the growth inequality. □

Corollary 6.6.5 *(Marx-Strohhäcker Theorem) If* $f \in \mathcal{ST}$, *then*

$$\operatorname{Re} \sqrt{\frac{f(z)}{z}} > \frac{1}{2}, \quad z \in \mathbb{D}, .$$

Equivalently,

$$\frac{zf'(z)}{f(z)} \prec \frac{z\kappa'(z)}{\kappa(z)} \Longrightarrow \frac{f(z)}{z} \prec \frac{\kappa(z)}{z},$$

where κ *is the Koebe function.*

The subordination obtained in Corollary 6.6.5 implies that

$$\log \frac{f(z)}{z} \prec \log \frac{1}{(1-z)^2} = \log \frac{\kappa(z)}{z}$$

for $f \in \mathcal{ST}$. This result can be used to prove the bound for logarithmic coefficients of $f \in \mathcal{ST}$. Observe that Theorem 6.2.1 could also be derived by using Corollary 6.6.5. Since $\operatorname{Re}\sqrt{f(z)/z} > 1/2$ for $f \in \mathcal{ST}$, it follows that $f(z) = zp(z)^2$ where $p \in \mathcal{P}(1/2)$. By comparing the coefficients of functions on both sides, and by using the bounds $|c_k| \leqslant 1$ for $p \in \mathcal{P}(1/2)$, it is readily established that $|a_k| \leqslant k$.

An application of Corollary 6.6.3 yields the following result.

Theorem 6.6.6 *If $f \in \mathcal{ST}(\alpha)$, $0 \leqslant \alpha < 1$, then*

$$\left|\arg\left(\frac{f(z)}{z}\right)\right| \leqslant 2(1-\alpha)\sin^{-1}(|z|), \quad z \in \mathbb{D}.$$

Proof Theorem 6.6.1 shows that

$$\frac{f(z)}{z} \prec \frac{1}{(1-z)^{2(1-\alpha)}},$$

and hence

$$\left(\frac{z}{f(z)}\right)^{\frac{1}{2(1-\alpha)}} \prec 1-z, \quad \text{or} \quad \left|\left(\frac{z}{f(z)}\right)^{\frac{1}{2(1-\alpha)}} - 1\right| \leqslant |z|. \tag{6.6.4}$$

If $|w-1| = r$, then $w = 1 + re^{it}$, $0 \leqslant t \leqslant 2\pi$. Since

$$\tan^{-1} x = \sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right),$$

then

$$\begin{aligned} |\arg w| &= \left|\tan^{-1}\left(\frac{r\sin t}{1+r\cos t}\right)\right| \\ &= \left|\sin^{-1}\left(\frac{r\sin t}{\sqrt{1+r^2+2r\cos t}}\right)\right| \\ &\leqslant \sin^{-1} r. \end{aligned}$$

(See Fig. 6.8.) From (6.6.4),

$$\frac{1}{2(1-\alpha)}\left|\arg\left(\frac{f(z)}{z}\right)\right| = \left|\arg\left(\frac{f(z)}{z}\right)^{\frac{1}{2(1-\alpha)}}\right| \leqslant \sin^{-1}(|z|),$$

proving the result. □

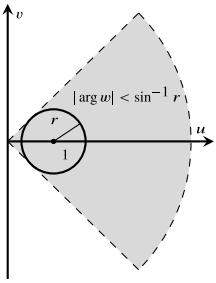


Fig. 6.8 Disk $\mathbb{D}(1; r)$ is contained in the sector $\{w : |\arg w| < \sin^{-1} r\}$

Corollary 6.6.7 *If $f \in \mathcal{ST}$, then*

$$\left|\arg\left(\frac{f(z)}{z}\right)\right| \leqslant 2\sin^{-1}(|z|), \quad z \in \mathbb{D}.$$

The above corollary should be compared with the corresponding result for a function $f \in \mathcal{S}$.

Theorem 6.6.8 *If $f \in \mathcal{S}$, then the following sharp inequalities holds:*

$$\left|\arg\left(\frac{f(z)}{z}\right)\right| \leqslant \log\frac{1+|z|}{1-|z|}, \quad z \in \mathbb{D},$$

and

$$\left|\arg\left(\frac{zf'(z)}{f(z)}\right)\right| \leqslant \log\frac{1+|z|}{1-|z|}, \quad z \in \mathbb{D}.$$

Proof The first inequality could be established by using Loewner's theory, and we shall omit the proof. The second inequality follows from the first. To see this, consider the disk automorphism of f given by

$$F(z) = \frac{f\left(\frac{\xi - z}{1-\bar{\xi}z}\right) - f(\xi)}{f'(\xi)(|\xi|^2 - 1)}, \quad z, \xi \in \mathbb{D}.$$

The function $F \in \mathcal{S}$ by Theorem 2.2.10 and

$$F(\xi) = \frac{f(\xi)}{f'(\xi)(1-|\xi|^2)}.$$

Thus,

$$\arg\left(\frac{\xi f'(\xi)}{f(\xi)}\right) = -\arg\left(\frac{F(\xi)}{\xi}\right),$$

which yields the second inequality from the first. □

Not every univalent function is starlike (of order α, $0 \leqslant \alpha < 1$). However, there exists some $\rho_f = \rho_f(\alpha) > 0$ such that

$$\operatorname{Re}\left(\frac{zf'(z)}{f(z)}\right) > \alpha$$

for all $z \in \mathbb{D}_{\rho_f}$. The largest such ρ_f is called the *radius of starlikeness of order* α of f. The number $R_{\mathcal{ST}(\alpha)}(\mathcal{S}) := \inf\{\rho_f : f \in \mathcal{S}\}$ is known as the radius of starlikeness of order α for the class $\mathcal{S}$. The radius of starlikeness of order 0 for the class $\mathcal{S}$ is simply called the radius of starlikeness for the class $\mathcal{S}$.

More generally, for a subclass $\mathcal{M}$ of $\mathcal{A}$, the *radius of starlikeness of order* α *for the class* $\mathcal{M}$ is defined by $R_{\mathcal{ST}(\alpha)}(\mathcal{M}) := \inf\{\rho_f : f \in \mathcal{M}\}$. For each $f \in \mathcal{M}$, it can easily be seen that the function $f_\rho : \mathbb{D} \to \mathbb{C}$ given by $f_\rho(z) = f(\rho z)/\rho$ is starlike of order α for each $0 < \rho \leqslant R_{\mathcal{ST}(\alpha)}(\mathcal{M})$.

Corollary 6.6.9 *The radius of starlikeness for the class* $\mathcal{S}$ *is* $\tanh(\pi/4) \approx 0.655$.

Proof Given $f \in \mathcal{S}$, let $f_\rho : \mathbb{D} \to \mathbb{C}$ be $f_\rho(z) = f(\rho z)/\rho$. Since

$$\frac{zf'_\rho(z)}{f_\rho(z)} = \frac{\rho z f'(\rho z)}{f(\rho z)},$$

Theorem 6.6.8 yields

$$\left|\arg\left(\frac{zf'_\rho(z)}{f_\rho(z)}\right)\right| = \left|\arg\left(\frac{\rho z f'(\rho z)}{f(\rho z)}\right)\right| \leqslant \log\frac{1+|\rho z|}{1-|\rho z|} \leqslant \log\frac{1+\rho}{1-\rho}.$$

Solving the inequality

$$\log\frac{1+\rho}{1-\rho} \leqslant \frac{\pi}{2}$$

for ρ gives

$$\rho \leqslant \frac{e^{\pi/2}-1}{e^{\pi/2}+1} = \tanh\left(\frac{\pi}{4}\right).$$

Therefore, for all $0 < \rho \leqslant \tanh(\pi/4) \approx 0.655$,

$$\left|\arg\left(\frac{zf'_\rho(z)}{f_\rho(z)}\right)\right| \leqslant \frac{\pi}{2}$$

or

$$\operatorname{Re}\left(\frac{zf'_\rho(z)}{f_\rho(z)}\right) > 0.$$

Hence, the radius of starlikeness for the class $\mathcal{S}$ is at least $\tanh(\pi/4)$. Verification of sharpness is left to the reader. □

If $\alpha \geqslant \beta$, trivially $\mathcal{ST}(\alpha) \subset \mathcal{ST}(\beta)$. When $\alpha < \beta$, the $\mathcal{ST}(\beta)$ radius for the class $\mathcal{ST}(\alpha)$ is computed.

Theorem 6.6.10 *Let $0 \leqslant \alpha < \beta < 1$. The $\mathcal{ST}(\beta)$ radius for the class $\mathcal{ST}(\alpha)$ is*

$$\rho_0 = \frac{1-\beta}{1+\beta-2\alpha}.$$

Proof Let $f \in \mathcal{ST}(\alpha)$. Then $zf'(z)/f(z) \in \mathcal{P}(\alpha)$ and Corollary 4.3.5 shows that

$$\operatorname{Re} \frac{zf'(z)}{f(z)} \geqslant \frac{1-(1-2\alpha)r}{1+r} > \beta, \quad |z| = r,$$

for $0 \leqslant r < \rho_0$. Sharpness is demonstrated by the generalized Koebe function. □

For $c \in \mathbb{C}$ with $\operatorname{Re} c > -1$ and $f \in \mathcal{A}$, define $F_c : \mathbb{D} \to \mathbb{C}$ by[1]

$$F_c(z) = \frac{c+1}{z^c} \int_0^z \xi^{c-1} f(\xi) \mathrm{d}\xi = (c+1) \int_0^1 t^{c-1} f(tz) \mathrm{d}t. \tag{6.6.5}$$

The operator $f \to F_c$ is known as the *Bernardi (integral) operator*. If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k$, then F_c has the series expansion given by

$$F_c(z) = z + \sum_{k=2}^{\infty} \left(\frac{c+1}{c+k} \right) a_k z^k.$$

For $c = 0$,

$$F_0(z) = \int_0^z \frac{f(\xi)}{\xi} \mathrm{d}\xi$$

is known as the *Alexander operator*. For $c = 1$,

$$F_1(z) = \frac{2}{z} \int_0^z f(\xi) \mathrm{d}\xi$$

is known as the *Libera operator*.

In 1965, Libera [7] proved that the function $F_1 \in \mathcal{ST}$ if $f \in \mathcal{ST}$. For $c \in \mathbb{N}$, Bernardi [3] proved that the function $F_c \in \mathcal{ST}$ whenever $f \in \mathcal{ST}$.

[1] Note that $z^c = e^{c \log z}$ is not analytic in $\mathbb{D}$

Theorem 6.6.11 *For $0 \leqslant \alpha < 1$ and $c \geqslant 0$, let*

$$\beta(\alpha, c) = \begin{cases} \alpha, & \alpha + c = 0, \\ \alpha - \dfrac{1-\alpha}{2(\alpha+c)}, & 0 < \alpha + c \leqslant 1 - \alpha, \\ \alpha - \dfrac{\alpha+c}{2(1-\alpha)}, & 1 - \alpha \leqslant \alpha + c. \end{cases}$$

If $f \in \mathcal{ST}(\beta(\alpha, c))$, then $F_c \in \mathcal{ST}(\alpha)$. In particular, the class $\mathcal{ST}(\alpha)$ is closed under the Bernardi integral operator $f \to F_c$.

Proof Let

$$p(z) = \frac{1}{1-\alpha}\left(\frac{zF_c'(z)}{F_c(z)} - \alpha\right), \quad z \in \mathbb{D}.$$

By (6.6.5)

$$(c+1)f(z) = zF_c'(z) + cF_c(z) = F_c(z)\big((1-\alpha)p(z) + \alpha + c\big),$$

and hence

$$\frac{zf'(z)}{f(z)} = (1-\alpha)p(z) + \alpha + \frac{(1-\alpha)zp'(z)}{(1-\alpha)p(z) + \alpha + c} = \psi\big(p(z), zp'(z)\big), \quad (6.6.6)$$

where

$$\psi(r, s) = (1-\alpha)r + \alpha + \frac{(1-\alpha)s}{(1-\alpha)r + \alpha + c}.$$

Since $f \in \mathcal{ST}(\beta(\alpha, c))$, it follows that $\operatorname{Re}\psi(p(z), zp'(z)) > \beta(\alpha, c)$ and so the proof is complete if $\operatorname{Re}\psi(\mathrm{i}\rho, \sigma) \leqslant \beta(\alpha, c)$ for all $\rho \in \mathbb{R}$ and $\sigma \leqslant -(1+\rho^2)/2$.

Clearly, for $\rho \in \mathbb{R}$ and $\sigma \leqslant -(1+\rho^2)/2$,

$$\begin{aligned} \operatorname{Re}\psi(\mathrm{i}\rho, \sigma) &= \alpha + \frac{(1-\alpha)(\alpha+c)\sigma}{(1-\alpha)^2\rho^2 + (\alpha+c)^2} \\ &\leqslant \alpha - \frac{(1-\alpha)(\alpha+c)}{2}\frac{(1+\rho^2)}{(1-\alpha)^2\rho^2 + (\alpha+c)^2}. \end{aligned} \quad (6.6.7)$$

The case $\alpha + c = 0$ is trivial and so assume $\alpha + c > 0$. The function

$$\varphi(\rho) := \frac{1+\rho^2}{(1-\alpha)^2\rho^2 + (\alpha+c)^2}, \quad \rho \geqslant 0,$$

has derivative given by

$$\varphi'(\rho) = \frac{2\big((\alpha+c)^2 - (1-\alpha)^2\big)\rho}{\big((1-\alpha)^2\rho^2 + (\alpha+c)^2\big)^2} = \frac{2(c+1)(2\alpha+c-1)\rho}{\big((1-\alpha)^2\rho^2 + (\alpha+c)^2\big)^2}.$$

If $1-\alpha \leqslant \alpha + c$, it follows that the function φ is an increasing function and so $\varphi(\rho) \leqslant 1/(1-\alpha)^2$. Similarly, if $\alpha + c \leqslant 1-\alpha$, then φ is a decreasing function and $\varphi(\rho) \leqslant 1/(\alpha+c)^2$. From (6.6.7), evidently $\operatorname{Re}\psi(\mathrm{i}\rho, \sigma) \leqslant \beta(\alpha, c)$ when $\rho \in \mathbb{R}$ and $\sigma \leqslant -(1+\rho^2)/2$.

Since $\beta(\alpha, c) \leqslant \alpha$, the second result follows from the first. □

For $c = 1$ and $\alpha = 0$, Theorem 6.6.11 reduces to the following result about the Libera operator F_1.

Corollary 6.6.12 *If $f \in \mathcal{ST}(-1/2)$, then $F_1 \in \mathcal{ST}$. In particular, the class $\mathcal{ST}$ is closed under the Libera integral operator $f \to F_1$.*

6.7 Sufficient Conditions for Starlikeness

A function f is said to be *typically real* in $\mathbb{D}$ if $(\operatorname{Im} f(z))(\operatorname{Im} z) > 0$ for all $z \in \mathbb{D} \setminus (-1, 1)$. By Theorem 2.1.6, the condition $\operatorname{Re} f'(z) > 0$ is sufficient for the univalence of f. With an additional assumption, it is also sufficient for starlikeness of f.

Theorem 6.7.1 *If $f \in \mathcal{A}$ satisfies $\operatorname{Re} f'(z) > 0$ for all $z \in \mathbb{D}$, and f' is typically real in $\mathbb{D}$, then $f \in \mathcal{ST}$.*

Proof The condition that f' is typically real implies that both $f'(tz)$ and $f'(z)$ are in the upper half-plane when $\operatorname{Im} z > 0$ and in the lower half-plane when $\operatorname{Im} z < 0$. Moreover, the inequality $\operatorname{Re} f'(z) > 0$ ensures that these values also lie in the right half-plane. This shows that $\left|\arg f'(tz) - \arg f'(z)\right| < \pi/2$, and hence $\operatorname{Re}(f'(tz)/f'(z)) > 0$. Therefore,

$$\operatorname{Re}\frac{f(z)}{zf'(z)} = \operatorname{Re}\int_0^1 \frac{f'(tz)}{f'(z)}\mathrm{d}t = \int_0^1 \operatorname{Re}\frac{f'(tz)}{f'(z)}\mathrm{d}t \geqslant 0, \quad 0 \leqslant t \leqslant 1.$$

This yields $\operatorname{Re}\left(f(z)/\left(zf'(z)\right)\right) > 0$. Since $p \in \mathcal{P}$ if and only if $1/p \in \mathcal{P}$, it follows that $\operatorname{Re}\left(zf'(z)/f(z)\right) > 0$, and hence $f \in \mathcal{ST}$. □

Theorem 6.7.2 *Let $0 < \lambda \leqslant 1$. If $f \in \mathcal{A}$ satisfies the inequality $\left|f'(z) - 1\right| < \lambda$ for all $z \in \mathbb{D}$, then*

$$\left|\arg\left(\frac{zf'(z)}{f(z)}\right)\right| < \frac{\alpha\pi}{2},$$

where

$$\alpha = \frac{2}{\pi}\sin^{-1}\left(\lambda\sqrt{1-\frac{\lambda^2}{4}} + \frac{\lambda}{2}\sqrt{1-\lambda^2}\right).$$

Proof From the subordination $f'(z) \prec 1 + \lambda z$, it follows that $|f'(z) - 1| < \lambda$, and hence

$$\left|\arg f'(z)\right| < \sin^{-1}\lambda. \tag{6.7.1}$$

Applying Theorem 5.3.6 with $F(z) = f'(z)$ gives

$$\frac{f(z)}{z} = \frac{1}{z}\int_0^z F(t)\mathrm{d}t \prec 1 + \frac{\lambda z}{2}.$$

Consequently,

$$\left|\arg\frac{f(z)}{z}\right| < \sin^{-1}\frac{\lambda}{2}. \tag{6.7.2}$$

Combining inequalities (6.7.1) and (6.7.2) yield

$$\begin{aligned}\left|\arg\frac{zf'(z)}{f(z)}\right| &= \left|\arg\frac{z}{f(z)} + \arg f'(z)\right| \leqslant \left|\arg\frac{f(z)}{z}\right| + \left|\arg f'(z)\right| \\ &< \sin^{-1}\frac{\lambda}{2} + \sin^{-1}\lambda \\ &= \sin^{-1}\left(\lambda\sqrt{1-\frac{\lambda^2}{4}} + \frac{\lambda}{2}\sqrt{1-\lambda^2}\right) = \frac{\alpha\pi}{2}.\end{aligned}$$

□

Corollary 6.7.3 *If $f \in \mathcal{A}$ satisfies the inequality $\left|f'(z) - 1\right| < 2/\sqrt{5} \approx 0.8944$ for all $z \in \mathbb{D}$, then $f \in \mathcal{ST}$.*

Theorem 6.7.4 *Let $0 < k \leqslant 1$. If $f \in \mathcal{A}$ satisfies the inequality $\left|f''(z)\right| \leqslant k$ for all $z \in \mathbb{D}$, then*

$$\left|\frac{zf'(z)}{f(z)} - 1\right| < \frac{k}{2-k}.$$

Proof Let $F(z) = zf''(z) + 1$ so that

$$\frac{1}{z}\int_0^z F(t)\mathrm{d}t = 1 + f'(z) - \frac{f(z)}{z}.$$

Since the condition $\left|f''(z)\right| \leqslant k$ is equivalent to the subordination $zf''(z) \prec kz$, Theorem 5.3.6 yields

$$\frac{1}{z}\int_0^z F(t)\mathrm{d}t \prec 1 + \frac{kz}{2},$$

or

$$f'(z) - \frac{f(z)}{z} \prec \frac{kz}{2}.$$

This subordination is equivalent to

$$z\left(\frac{f(z)}{z}\right)' \prec \frac{kz}{2},$$

and to

$$\frac{f(z)}{z}\left(\frac{zf'(z)}{f(z)} - 1\right) \prec \frac{k}{2}z. \tag{6.7.3}$$

Theorem 5.3.5 with $F(z) = f(z)/z$ shows that $f(z)/z \prec 1 + kz/2$, which in turn implies

$$1 - \frac{k}{2} < \left|\frac{f(z)}{z}\right| < 1 + \frac{k}{2}.$$

This relation and (6.7.3) shows that

$$\left(1 - \frac{k}{2}\right)\left|\frac{zf'(z)}{f(z)} - 1\right| < \left|\frac{f(z)}{z}\right|\left|\frac{zf'(z)}{f(z)} - 1\right| < \frac{k}{2},$$

and hence

$$\left|\frac{zf'(z)}{f(z)} - 1\right| < \frac{k}{2-k}.$$

□

Corollary 6.7.5 *Let $0 \leqslant \alpha < 1$. If $f \in \mathcal{A}$ satisfies the inequality*

$$\left|f''(z)\right| \leqslant \frac{2(1-\alpha)}{2-\alpha}, \quad z \in \mathbb{D},$$

then $f \in \mathcal{ST}(\alpha)$. The result is sharp.

Proof Set $k = 2(1-\alpha)/(2-\alpha)$. By Theorem 6.7.4,

$$\operatorname{Re}\frac{zf'(z)}{f(z)} \geqslant 1 - \left|\frac{zf'(z)}{f(z)} - 1\right| > 1 - \frac{k}{2-k} = \alpha,$$

and hence $f \in \mathcal{ST}(\alpha)$.

To establish sharpness, consider the function f defined by $f(z) = z + (k + \varepsilon)z^2/2$, $0 < k \leqslant 1$, $0 < \varepsilon < 1$. This function satisfies $\left|f''(z)\right| = k + \varepsilon > k$, but

$$\frac{zf'(z)}{f(z)} = \frac{2(1 + (k+\varepsilon)z)}{2 + (k+\varepsilon)z}$$

takes values less than α when z is real and close to -1. Thus, $f \notin \mathcal{ST}(\alpha)$, showing that the bound on $|f''(z)|$ is best possible. □

Geometrically, every convex function is starlike, a fact that serves as a motivation for the following generalization. For $\alpha \in \mathbb{R}$, a function $f \in \mathcal{A}$ is said to be α-*convex* (or *Mocanu convex*) if

$$\operatorname{Re}\left((1-\alpha)\frac{zf'(z)}{f(z)} + \alpha\left(1 + \frac{zf''(z)}{f'(z)}\right)\right) > 0.$$

A sufficient condition for starlikeness is next established for functions in the class $\mathcal{A}_n$ by replacing convex functions by α-convex functions.

Theorem 6.7.6 *All α-convex functions are starlike. Further, if $f \in \mathcal{A}_n$ satisfies*

$$(1-\alpha)\frac{zf'(z)}{f(z)} + \alpha\left(1 + \frac{zf''(z)}{f'(z)}\right) \in \Omega, \quad z \in \mathbb{D}, \tag{6.7.4}$$

where $\alpha \geqslant 0$ and $\Omega := \mathbb{C} \setminus \{w = it : t \in \mathbb{R}, |t| \geqslant \sqrt{\alpha n(2+\alpha n)}\}$, then $f \in \mathcal{ST}$.

Proof Let f be α-convex, and $p : \mathbb{D} \to \mathbb{C}$ be

$$p(z) := \frac{zf'(z)}{f(z)}. \tag{6.7.5}$$

Since $f \in \mathcal{A}$, and (6.7.4) holds, then $f'(z) \neq 0$ and so $f(z) \neq 0$ for $z \neq 0$. Thus, the function p is analytic in $\mathbb{D}$ and $p(0) = 1$. From (6.7.5),

$$(1-\alpha)\frac{zf'(z)}{f(z)} + \alpha\left(1 + \frac{zf''(z)}{f'(z)}\right) = p(z) + \alpha\frac{zp'(z)}{p(z)}. \tag{6.7.6}$$

If f is α-convex, then, by (6.7.6), $\operatorname{Re}(p(z) + \alpha zp'(z)/p(z)) > 0$ and hence, Theorem 5.4.9 implies that $\operatorname{Re} p(z) > 0$. This proves that α-convex functions f are starlike.

The other part of the theorem follows by an application of Theorem 5.4.10. □

Theorem 6.7.6 for $n = 1$ and $\alpha = 1$ leads to the following result.

Corollary 6.7.7 *Let $\Omega := \mathbb{C} \setminus \left\{w = it : t \in \mathbb{R}, |t| \geqslant \sqrt{3}\right\}$. If $f \in \mathcal{A}$ satisfies*

$$1 + \frac{zf''(z)}{f'(z)} \in \Omega, \quad z \in \mathbb{D},$$

then $f \in \mathcal{ST}$. In particular, if, for $z \in \mathbb{D}$,

$$\left|1 + \frac{zf''(z)}{f'(z)}\right| < \sqrt{3} \quad \text{or} \quad \left|\frac{zf''(z)}{f'(z)}\right| < 2,$$

then $f \in \mathcal{ST}$.

In fact, Miller and Mocanu [13] proved that f is starlike if $|f''(z)/f'(z)| \leqslant r_1 \approx 2.83$ where $r_1 = \sqrt{1+x_0^2}$, and x_0 is the smallest positive root of $x \sin x + \cos x = 1/e$; the sharp bound (approximately equal to 2.84116) was found by Anisiu et al. [1] as a root of certain equation.

Theorem 6.7.8 *Let $1/2 \leqslant \beta < 1$. If $f \in \mathcal{A}$ satisfies $f(z)f'(z) \neq 0$ for any non-zero z in $\mathbb{D}$, and*

$$\left|\frac{f''(z)}{f'(z)}\right| \leqslant \frac{2(1-\beta)}{\beta}\left|\frac{zf'(z)}{f(z)}\right|, \quad z \in \mathbb{D}, \tag{6.7.7}$$

then $f \in \mathcal{ST}(\beta)$ and $zf'(z)/f(z)$ is subordinate to $\beta/(\beta-(1-\beta)z)$.

Proof Since $f(z)f'(z) \neq 0$, the function

$$w(z) = \frac{\beta}{1-\beta}\left(1 - \frac{f(z)}{zf'(z)}\right)$$

is analytic in $\mathbb{D}$ and omits the value $\beta/(1-\beta)$. Moreover, $w(0) = 0$ and

$$\frac{zf'(z)}{f(z)} = \frac{\beta}{\beta-(1-\beta)w(z)}, \quad (zw(z))' = \frac{\beta}{1-\beta}\frac{f(z)f''(z)}{(f'(z))^2}. \tag{6.7.8}$$

From (6.7.7) and (6.7.8),

$$\left|(zw(z))'\right| \leqslant \left|\frac{\beta}{1-\beta}\frac{f(z)f''(z)}{(f'(z))^2}\right| \leqslant 2|z|,$$

and hence

$$|zw(z)| = \left|\int_0^z (tw(t))'\mathrm{d}t\right| \leqslant \int_0^{|z|} 2|t|d|t| = |z|^2 < 1.$$

Since $|w(z)| \leq |z| < 1$, it follows from (6.7.8) that $zf'(z)/f(z)$ is subordinate to $\beta/(\beta-(1-\beta)z)$.

Consider the function $q : \mathbb{D} \to \mathbb{C}$ defined by

$$q(z) = \frac{1}{1-\alpha z} \quad \left(\alpha = \frac{1-\beta}{\beta}, \quad \frac{1}{2} \leqslant \beta < 1\right).$$

By (4.4.7),

$$\left|q(z) - \frac{1}{1-\alpha^2 r^2}\right| \leqslant \frac{\alpha r}{1-\alpha^2 r^2}, \quad |z| = r < 1. \tag{6.7.9}$$

It follows from (6.7.9) that

$$\operatorname{Re} q(z) > \frac{1}{1-\alpha^2 r^2} - \frac{\alpha r}{1-\alpha^2 r^2} = \frac{1}{1+\alpha r} \geqslant \frac{1}{1+\alpha} = \beta.$$

Therefore, for $1/2 \leqslant \beta < 1$, the subordination $zf'(z)/f(z) \prec q(z)$ implies that $\operatorname{Re}\big(zf'(z)/f(z)\big) > \beta$ and $f \in \mathcal{ST}(\beta)$. □

Theorem 6.7.9 *If $f \in \mathcal{A}$ satisfies $f(z)f'(z) \neq 0$ for non-zero $z \in \mathbb{D}$, and*

$$\left|1 + \frac{zf''(z)}{f'(z)}\right| < \sqrt{2}\left|1 + \frac{zf'(z)}{f(z)}\right|, \quad z \in \mathbb{D}, \tag{6.7.10}$$

then $f \in \mathcal{ST}$.

Proof Let $\Omega := \{w : |w| < \sqrt{2}\} = \mathbb{D}_{\sqrt{2}}$. The function

$$p(z) := \frac{zf'(z)}{f(z)}$$

is analytic in $\mathbb{D}$, $p(0) = 1$ and

$$\frac{1 + \frac{zf''(z)}{f'(z)}}{1 + \frac{zf'(z)}{f(z)}} = \frac{p^2(z) + zp'(z)}{p(z)(1 + p(z))} = \psi(p(z), zp'(z)),$$

where

$$\psi(r, s) = \frac{r^2 + s}{r(1 + r)}.$$

The condition (6.7.10) becomes $\psi(p(z), zp'(z)) \in \Omega$. Whenever $\rho \in \mathbb{R}$ and $\sigma \leqslant -(1 + \rho^2)/2$, the function ψ satisfies

$$|\psi(\mathrm{i}\rho, \sigma)|^2 = \frac{|\sigma - \rho^2|^2}{\rho^2(1 + \rho^2)} \geqslant \frac{1}{4}\frac{(1 + 3\rho^2)^2}{\rho^2(1 + \rho^2)} := \varphi(\rho^2) \geqslant \varphi(1) = 2.$$

Thus $\psi \in \mathcal{P}_1(\Omega)$. Since $\psi(p(z), zp'(z)) \in \Omega$, Theorem 5.4.2 shows that $p \in \mathcal{P}$, whence $f \in \mathcal{ST}$. □

Theorem 6.7.10 *Let $\Omega := \mathbb{C} \setminus [3/2, \infty)$. If $f \in \mathcal{A}$ satisfies $f(z)f'(z) \neq 0$ for non-zero $z \in \mathbb{D}$, and*

$$\frac{1 + \frac{zf''(z)}{f'(z)}}{\frac{zf'(z)}{f(z)}} \in \Omega, \quad z \in \mathbb{D},$$

then $f \in \mathcal{ST}$. In particular, if a function f satisfies

$$\left|1 + \frac{zf''(z)}{f'(z)}\right| < \frac{3}{2}\left|\frac{zf'(z)}{f(z)}\right|, \quad z \in \mathbb{D},$$

or

$$\operatorname{Re}\left(\frac{1 + \frac{zf''(z)}{f'(z)}}{\frac{zf'(z)}{f(z)}}\right) < \frac{3}{2}, \quad z \in \mathbb{D},$$

then $f \in \mathcal{ST}$.

Proof The function

$$p(z) := \frac{zf'(z)}{f(z)}$$

is analytic in $\mathbb{D}$, $p(0) = 1$ and

$$\frac{1 + \frac{zf''(z)}{f'(z)}}{\frac{zf'(z)}{f(z)}} = 1 + \frac{zp'(z)}{p^2(z)} = \psi(p(z), zp'(z)) \in \Omega,$$

where

$$\psi(r, s) = 1 + \frac{s}{r^2}.$$

Whenever $\rho \in \mathbb{R}$ and $\sigma \leqslant -(1 + \rho^2)/2$, the function ψ satisfies

$$\psi(\mathrm{i}\rho, \sigma) = 1 - \frac{\sigma}{\rho^2} \geqslant 1 + \frac{1}{2} + \frac{1}{2\rho^2} \geqslant \frac{3}{2}.$$

Thus $\psi \in \mathcal{P}_1(\Omega)$. From Theorem 5.4.2, we deduce that $p \in \mathcal{P}$, or $f \in \mathcal{ST}$. □

Let G_b be the class of all functions $f \in \mathcal{A}$ satisfying the condition

$$\left| \frac{1 + \frac{zf''(z)}{f'(z)}}{\frac{zf'(z)}{f(z)}} - 1 \right| < b.$$

Then, it follows from Theorem 6.7.10 that $G_{1/2} \subset \mathcal{ST}$.

Theorem 6.7.11 *Let* $\alpha \geqslant 0$ *and* $\beta < 1$. *If* $f \in \mathcal{A}_n$ *satisfies*

$$\operatorname{Re}\left(\frac{zf'(z)}{f(z)}\left(\alpha\frac{zf''(z)}{f'(z)} + 1\right)\right) > \alpha\beta\left(\beta + \frac{n}{2} - 1\right) + \left(\beta - \frac{\alpha n}{2}\right), \quad z \in \mathbb{D},$$

then $f \in \mathcal{ST}(\beta)$.

Proof Let

$$\Omega = \left\{w : \operatorname{Re} w > \alpha\beta\left(\beta + \frac{n}{2} - 1\right) + \left(\beta - \frac{n\alpha}{2}\right)\right\},$$

and

$$p(z) = \frac{1}{1 - \beta}\left(\frac{zf'(z)}{f(z)} - \beta\right).$$

Then $p(z) = 1 + c_n z^n + \cdots$ is analytic in $\mathbb{D}$. A computation shows that

$$\frac{zf''(z)}{f'(z)} = \frac{(1 - \beta)zp'(z) + [(1 - \beta)p(z) + \beta]^2 - ((1 - \beta)p(z) + \beta)}{(1 - \beta)p(z) + \beta},$$

and hence

$$\begin{aligned}\frac{zf'(z)}{f(z)}\left(\alpha\frac{zf''(z)}{f'(z)}+1\right) &= \alpha(1-\beta)zp'(z)+\alpha(1-\beta)^2p^2(z)\\ &\quad+(1-\beta)(1+2\alpha\beta-\alpha)p(z)+\beta(\alpha\beta+1-\alpha)\\ &= \Phi(p(z),zp'(z))\in\Omega\end{aligned}$$

where

$$\Phi(r,s)=\alpha(1-\beta)s+\alpha(1-\beta)^2r^2+(1-\beta)(1+2\alpha\beta-\alpha)r+\beta(\alpha\beta+1-\alpha).$$

For all real ρ and σ satisfying $\sigma\leqslant -n(1+\rho^2)/2\leqslant -n/2$,

$$\begin{aligned}\operatorname{Re}\Phi(\mathrm{i}\rho,\sigma) &= \alpha(1-\beta)\sigma-\alpha(1-\beta)^2\rho^2+\beta(\alpha\beta+1-\alpha)\\ &\leqslant -\frac{\alpha}{2}(1-\beta)n+\beta(\alpha\beta+1-\alpha)\\ &= \alpha\beta\left(\beta+\frac{n}{2}-1\right)+\left(\beta-\frac{n\alpha}{2}\right).\end{aligned}$$

This shows that $\Phi(\mathrm{i}\rho,\sigma)\notin\Omega$ for all real ρ and $\sigma\leqslant -n(1+\rho^2)/2$, $z\in\mathbb{D}$. Since $\Phi(p(z),zp'(z))\in\Omega$, by an application of Theorem 5.4.2, it follows that $p\in\mathcal{P}$ and hence $f\in\mathcal{ST}(\beta)$. □

Further, Theorem 6.7.11 with $\beta=0$ and $n=1$ yields the following result.

Corollary 6.7.12 *If $f\in\mathcal{A}$ satisfies*

$$\operatorname{Re}\left(\frac{zf'(z)}{f(z)}\left(\alpha\frac{zf''(z)}{f'(z)}+1\right)\right)>-\frac{\alpha}{2},\quad z\in\mathbb{D},$$

for some α ($\alpha\geqslant 0$), then $f\in\mathcal{ST}$.

Also, Theorem 6.7.11 with $\beta=\alpha/2$ and $n=1$ leads to the following result.

Corollary 6.7.13 *If $f\in\mathcal{A}$ satisfies*

$$\operatorname{Re}\left(\frac{zf'(z)}{f(z)}\left(\alpha\frac{zf''(z)}{f'(z)}+1\right)\right)>-\frac{\alpha^2}{4}(1-\alpha),\quad z\in\mathbb{D},$$

for some α ($0\leqslant\alpha<2$), then $f\in\mathcal{ST}(\alpha/2)$.

Theorem 6.7.14 *If $f\in\mathcal{A}$ satisfies*

$$\left|\frac{zf'(z)}{f(z)}-1\right|^{1-\gamma}\left|\frac{zf''(z)}{f'(z)}\right|^{\gamma}<\left(\frac{3}{2}\right)^{\gamma},\quad z\in\mathbb{D},\tag{6.7.11}$$

for some $\gamma\geqslant 0$, then $f\in\mathcal{ST}$.

Proof Define a function $w : \mathbb{D} \to \mathbb{C}$ by

$$\frac{zf'(z)}{f(z)} = \frac{1+w(z)}{1-w(z)}.$$

Evidently, $w(0) = 0$. A computation shows that

$$\left(\frac{zf'(z)}{f(z)} - 1\right)^{1-\gamma} \left(\frac{zf''(z)}{f'(z)}\right)^{\gamma} = \left(\frac{2w(z)}{1-w(z)}\right)\left(1 + \frac{zw'(z)}{w(z)}\frac{1}{1+w(z)}\right)^{\gamma}.$$

Let

$$\psi(r,s) = \left(\frac{2r}{1-r}\right)\left(1 + \frac{s}{r}\frac{1}{1+r}\right)^{\gamma}$$

in $\mathbb{D}^2$, and $\Omega = \{w : |w| < (3/2)^{\gamma}\}$. Then, by (6.7.11), $\psi(w(z), zw'(z)) \in \Omega$. For $K \geqslant 1$, and $\theta \in \mathbb{R}$,

$$|\psi(e^{i\theta}, Ke^{i\theta})| \geqslant \left(1 + \frac{K}{2}\right)^{\gamma} \geqslant \left(1 + \frac{1}{2}\right)^{\gamma} = \left(\frac{3}{2}\right)^{\gamma}.$$

This shows that $\psi(e^{i\theta}, Ke^{i\theta}) \notin \Omega$ for $K \geqslant 1$. By Theorem 5.3.2, it follows that $w \in \mathcal{B}_0$ and hence $f \in \mathcal{ST}$. □

6.8 A Problem of Livingston

Recall from (6.6.5) that the Bernardi (integral) operator $F_c : \mathbb{D} \to \mathbb{C}$ for a complex number c with $\operatorname{Re} c > -1$ is

$$F_c(z) = \frac{c+1}{z^c}\int_0^z \xi^{c-1} f(\xi)\mathrm{d}\xi = (c+1)\int_0^1 t^{c-1} f(tz)\mathrm{d}t, \quad f \in \mathcal{A}. \tag{6.8.1}$$

Solving the equation (6.8.1) for f in terms of F_c gives

$$f(z) = \frac{1}{c+1}\big(zF_c'(z) + cF_c(z)\big). \tag{6.8.2}$$

The converse problem of studying properties of f from the properties of F_c is known as *Livingston problem.*

Suppose that the function $F_c \in \mathcal{ST}(\alpha)$. While the corresponding function f need not be in $\mathcal{ST}(\alpha)$, there is however a $\rho > 0$ such that

$$\operatorname{Re}\frac{zf'(z)}{f(z)} > \alpha$$

for all $z \in \mathbb{D}_\rho$. The largest such ρ is called the radius of starlikeness of order α for f. Indeed, Livingston [9] has shown that if $F_1 \in \mathcal{ST}$, then the radius of starlikeness of the function $f(z) = [zF_1(z)]'/2$ is $1/2$. Bernardi extended it for F_c when $c \in \mathbb{N}$. The following theorem determines the radius of starlikeness of order α for f when F_c is starlike of order α, and it is indeed a converse of Theorem 6.6.11.

Theorem 6.8.1 *Let $0 \leqslant \alpha < 1$, $c > -\alpha$ and f be given by (6.8.2). If $F_c \in \mathcal{ST}(\alpha)$, then the radius of starlikeness of order α of f is*

$$R_{\mathcal{ST}(\alpha)} = \frac{1+c}{2-\alpha+\sqrt{3-2\alpha+(\alpha+c)^2}}. \tag{6.8.3}$$

Proof Let

$$p(z) = \frac{zF_c'(z)}{F_c(z)}, \quad z \in \mathbb{D}.$$

Then, $p \in \mathcal{P}(\alpha)$. Since

$$(c+1)f(z) = zF_c'(z) + cF_c(z) = F_c(z)(p(z)+c),$$

it follows that

$$\frac{zf'(z)}{f(z)} = p(z) + \frac{zp'(z)}{p(z)+c}. \tag{6.8.4}$$

This equation is (6.6.6) with $\alpha = 0$.

For $p \in \mathcal{P}(\alpha)$, Corollary 4.3.5 shows that

$$\operatorname{Re} p(z) \geqslant \frac{1-(1-2\alpha)r}{1+r} \quad \text{and} \quad |p'(z)| \leqslant \frac{2(\operatorname{Re} p(z)-\alpha)}{1-r^2}, \quad |z| = r < 1.$$

The first inequality yields

$$|p(z)+c| \geqslant \operatorname{Re} p(z) + c \geqslant \frac{1-(1-2\alpha)r}{1+r} + c = \frac{1+c+(c+2\alpha-1)r}{1+r}.$$

By (6.8.4),

$$\begin{aligned}
\operatorname{Re}\frac{zf'(z)}{f(z)} - \alpha &\geqslant \operatorname{Re}(p(z)-\alpha) - \frac{|zp'(z)|}{|p(z)+c|} \\
&\geqslant \left(1 - \frac{2r}{1-r^2}\frac{1+r}{1+c+(c+2\alpha-1)r}\right)\operatorname{Re}(p(z)-\alpha) \\
&= \frac{1+c-2(2-\alpha)r+(1-c-2\alpha)r^2}{(1-r)\big(1+c+(c+2\alpha-1)r\big)}\operatorname{Re}(p(z)-\alpha).
\end{aligned}$$

Since $r = R_{\mathcal{ST}(\alpha)}$ given by (6.8.3) is the root of the equation

$$1 + c - 2(2 - \alpha)r + (1 - c - 2\alpha)r^2 = 0, \tag{6.8.5}$$

then $1 + c - 2(2 - \alpha)r + (1 - c - 2\alpha)r^2 \geqslant 0$ for $0 \leqslant r \leqslant R_{\mathcal{ST}(\alpha)}$. Clearly, $1 + c + (c + 2\alpha - 1)r > 0$ for $c > -\alpha$, whence $\operatorname{Re}(zf'(z)/f(z)) \geqslant \alpha$ for $0 \leqslant r \leqslant R_{\mathcal{ST}(\alpha)}$. Thus, the radius of starlikeness of order α of f is at least $R_{\mathcal{ST}(\alpha)}$.

To show sharpness, consider the function $F_c \in \mathcal{ST}(\alpha)$ given by

$$F_c(z) = \frac{z}{(1 - z)^{2(1-\alpha)}}, \quad 0 \leqslant \alpha < 1.$$

The corresponding f is given by

$$f(z) = \frac{1}{c + 1}(zF_c'(z) + cF_c(z)) = \frac{\big(1 + c + (1 - c - 2\alpha)z\big)z}{(1 - z)^{3-2\alpha}},$$

and hence

$$\begin{aligned}\frac{zf'(z)}{f(z)} - \alpha &= 1 - \alpha + \frac{(1 - c - 2\alpha)z}{1 + c + (1 - c - 2\alpha)z} + \frac{(3 - 2\alpha)z}{1 - z} \\ &= \frac{(1 - \alpha)\big[1 + c + 2(2 - \alpha)z + (1 - c - 2\alpha)z^2\big]}{(1 - z)\big(1 + c + (1 - c - 2\alpha)z\big)}.\end{aligned}$$

By using (6.8.5), it then follows that $(zf'(z)/f(z)) - \alpha = 0$ for $z = -R_{\mathcal{ST}(\alpha)}$. Hence, the function f is not starlike in any disk $\mathbb{D}_r$ for $r > R_{\mathcal{ST}(\alpha)}$. □

Theorem 6.8.1 for $\alpha = 0$ gives the following corollary.

Corollary 6.8.2 *Let $c > 0$. If a function $F_c \in \mathcal{ST}$, then the radius of starlikeness of f given by (6.8.2) is*

$$R_{\mathcal{ST}(\alpha)} = \frac{1 + c}{2 + \sqrt{3 + c^2}}.$$

Exercises

1. Determine a_2 such that $f(z) = z + a_2z^2$ is starlike univalent.
2. Determine α such that $f(z) = z/(1 - \alpha z)$ is starlike univalent.
3. If $f \in \mathcal{A}$ satisfies

$$\left|\frac{zf'(z)}{f(z)} - 1\right| < \left|\frac{zf'(z)}{f(z)} + (1 - 2\alpha)\right|, \quad \alpha \geqslant 0$$

show that f is starlike of order α.

4. If $f \in \mathcal{A}$ satisfies
$$\left|\frac{f(z)}{zf'(z)} - \frac{1}{2\alpha}\right| < \frac{1}{2\alpha}, \quad \alpha > 0$$
show that f is starlike of order α. In particular, if the function $f \in \mathcal{A}$ satisfies
$$\left|\frac{f(z)}{zf'(z)} - 1\right| < 1,$$
show that f is starlike of order 1/2.
5. If $f(z) = z + \sum_{n=k+1}^{\infty} a_n z^n, k \geqslant 1$ is starlike, prove that for $mk + 1 \leqslant n \leq (m + 1)k, m = 1, 2, \ldots,$
$$|a_n| \leqslant \frac{k}{(n-1)(m-1)!} \prod_{\mu=0}^{m-1} \left(\mu + \frac{2}{k}\right).$$
Show that the bound is sharp if n has the form $mk + 1$ and is also sharp if $k + 1 < n < 2k + 1$.
6. Show that the radius of starlikeness of the class of functions $f \in \mathcal{A}$ satisfying $\mathrm{Re}(f(z)/z) > 1/2$ is $1/\sqrt{2}$.
7. Show that the radius of starlikeness of the class of functions $f \in \mathcal{A}$ with $f''(0) = 0$ satisfying $|(z^2 f'(z)/f^2(z)) - 1| < 1$ is at least $1/\sqrt[4]{2}$.
8. Let $a_k \geqslant 0$ and $a_1 = 1$. If $(k + 1)a_{k+1} \leqslant k a_k$ and $(2k + 1)a_{2k+1} \leqslant (2k - 1)a_{2k-1}$ for all $k \geqslant 1$, show that $f(z) = z + \sum_{k=2}^{\infty} a_k z^k$ is starlike.
9. If $f \in \mathcal{A}$ satisfies the inequality
$$\mathrm{Re}\,\frac{zf'(z)}{f(z)} > \left|\frac{zf'(z)}{f(z)} - 1\right|,$$
show that f is starlike of order $1/2$.
10. If $f \in \mathcal{A}$ satisfies the inequality
$$\left|\arg\left(\frac{zf'(z)}{f(z)}\right)\right| < \frac{\gamma\pi}{2}, \quad 0 < \gamma \leqslant 1,$$
show that f is starlike.
11. Find conditions on the complex numbers A and B so that a function $f \in \mathcal{A}$ satisfying the subordination
$$\frac{zf'(z)}{f(z)} \prec \frac{1 + Az}{1 + Bz}$$
is starlike.
12. If $f \prec g$, show that

$$\frac{1}{2\pi}\int_0^{2\pi} |f(re^{i\theta})|^\lambda d\theta \leqslant \frac{1}{2\pi}\int_0^{2\pi} |g(re^{i\theta})|^\lambda d\theta \quad (\lambda \geqslant 0).$$

13. For a starlike function f and for each $\lambda \in \mathbb{R}$, show that

$$\frac{1}{2\pi}\int_0^{2\pi} |f(re^{i\theta})|^\lambda d\theta \leqslant \frac{r^\lambda}{2\pi}\int_0^{2\pi} |1+re^{i\theta}|^{-2\lambda} d\theta.$$

14. If a function $f \in \mathcal{ST}(\alpha)$, $0 \leqslant \alpha < 1$, show that the sharp inequality $|H_2(2)| \leqslant (1-\alpha)^2$ holds.
15. If a function $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{ST}$, show that $|a_2a_3 - a_4| \leqslant 2$. Hence show that $|H_3(1)| \leqslant 16$. Show that these inequalities are sharp.
16. If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{ST}$ satisfies $|zf'(z)/f(z) - 1| < 1$, show that $re^{-r} \leqslant |f(z)| \leqslant re^r$ and $(1-r)e^{-r} \leqslant |f'(z)| \leqslant (1+r)e^r$ where $r = |z| < 1$. Also, show that $|a_n| \leqslant 1/(n-1)$. Further show that all these inequalities are sharp.
17. Apply Theorem 5.3.2 to prove the subordination in Theorem 6.7.8.

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[2] The results in this chapter are derived from the sources in the appended list. For instance, Theorem 6.2.6 is a result in [8], while those involving subordination in Section 6.6 and Section 6.7 can be found in the text of Miller and Mocanu [12]. Section 6.7 also contains results by various scholars spanning over decades. These include Theorem 6.7.1 [20], Theorem 6.7.2 and Theorem 6.7.4 [24], Theorem 6.7.8 [19], Theorem 6.7.10 [14], Theorem 6.7.11 [17], and Theorem 6.7.14 [23].

The Libera operator [7] was generalized by Bernardi in [3]. Theorem 6.8.1 on the Livingston problem (see [4, 9, 15]) can be found in [2], with its proof in [5]. Several results for starlike functions of order $\boldsymbol{\alpha}$ can be found in [16]. Exercises 6 and 7 are respectively from [11] and [22], while Exercise 8 is from [20].

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Chapter 7
Convex Functions

Abstract A subset of $\mathbb{C}$ is said to be convex if the line segment joining any two of its points lies entirely within the set. An analytic function $f : \mathbb{D} \to \mathbb{C}$ whose image $f(\mathbb{D})$ is convex is called a convex function. The class of convex functions is studied in this chapter. An analytic characterization of convexity is established and subsequently used to derive several properties of such functions.

7.1 Analytic Characterization

A region G is said to be *convex* if the line segment $[z_1, z_2]$ joining any two points $z_1, z_2 \in G$ lies entirely within G. A function $f \in \mathcal{A}$ is said to be *convex* if the image $f(\mathbb{D})$ is convex, and the subclass of $\mathcal{S}$ of all convex functions is denoted by $\mathcal{CV}$.

Example 7.1.1 ***(Half-plane mapping)*** By Example 2.1.16, the function $\ell : \mathbb{D} \to \mathbb{C}$ defined by

$$\ell(z) = \frac{z}{1 - z}$$

belongs to $\mathcal{S}$ and maps the unit disk $\mathbb{D}$ onto the half-plane $\{w : \operatorname{Re} w > -1/2\}$. Since half-planes are convex, it follows that $\ell \in \mathcal{CV}$. This function ℓ is called a *half-plane mapping*.

It is useful to establish an analytic characterization of convex functions analogous to that for starlike functions. To obtain such a characterization, the following result is first proved.

Theorem 7.1.2 ***(Hereditary Property of Convex Functions)*** *Let $f \in \mathcal{A}$. Then $f \in \mathcal{CV}$ if and only if $f(\mathbb{D}_r)$ is convex for every r with $0 < r < 1$. Equivalently, $f \in \mathcal{CV}$ if and only if the function $f_r : \mathbb{D} \to \mathbb{C}$ defined by $f_r(z) = r^{-1} f(rz)$ belongs to $\mathcal{CV}$ for each r with $0 < r < 1$.*

Proof Suppose that f is in $\mathcal{CV}$. Let $t \in (0, 1)$ be fixed, and let $z \in \mathbb{D}$. Let $z_1, z_2 \in \mathbb{D}$ be such that $|z_1| \leqslant |z_2|$, $z_2 \neq 0$. Since $f(\mathbb{D})$ is convex, it follows that $(1 - t) f(z_1 z/z_2) + t f(z) \in f(\mathbb{D})$ and hence $f^{-1}\big((1 - t) f(z_1 z/z_2) + t f(z)\big)$ is in $\mathbb{D}$. Let

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$$w(z) = f^{-1}\left((1-t)f\left(\frac{z_1}{z_2}z\right) + tf(z)\right), \quad z \in \mathbb{D}.$$

Since f is univalent, w is a single-valued analytic function mapping $\mathbb{D}$ into itself, and satisfies $w(0) = f^{-1}((1-t)f(0) + tf(0)) = f^{-1}(0) = 0$. Thus, by the Schwarz lemma, $|w(z)| \leqslant |z|$ for each $z \in \mathbb{D}$. The identity $(1-t)f(z_1 z/z_2) + tf(z) = f(w(z))$ yields

$$\begin{aligned}(1-t)f_r(z_1) + tf_r(z_2) &= \frac{1}{r}\Big((1-t)f(rz_1) + tf(rz_2)\Big)\\ &= \frac{1}{r}f(w(rz_2)) = f_r(w_1(z_2))\end{aligned}$$

where $w_1(z) = w(rz)/r$. Since

$$|w_1(z)| = \frac{|w(rz)|}{r} \leqslant \frac{|rz|}{r} = |z|,$$

it follows that $(1-t)f_r(z_1) + tf_r(z_2) = f_r(w_1(z_2)) \in f_r(\mathbb{D})$. Hence, $f_r \in \mathcal{CV}$.

For the converse, suppose that f_r is convex for all $0 < r < 1$, so that rf_r is convex. Since $rf_r(z) = f(rz)$, the image $f(r\mathbb{D}) = rf_r(\mathbb{D})$ is convex. As

$$f(\mathbb{D}) = \bigcup_{0<r<1} f(r\mathbb{D})$$

and the family of images satisfies the inclusion $f(r\mathbb{D}) \subset f(s\mathbb{D})$ for any $0 < r < s < 1$, the union is an increasing union of convex sets. Therefore, $f(\mathbb{D})$ is convex, and hence f is a convex function. □

The hereditary property of convex functions is illustrated for $f(z) = z + z^2/4$ and $f(z) = (4/\pi)\sin(\pi z/4)$ in Fig. 7.1.

The following analytic characterization of convex functions relates a convex function to a function with positive real part.

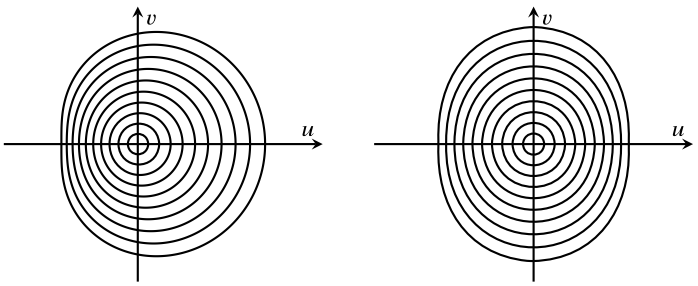


Fig. 7.1 $f(\mathbb{D}_r)$ when $f(z) = z + z^2/4$ and $f(z) = (4/\pi)\sin(\pi z/4)$

Theorem 7.1.3 *Let $f \in \mathcal{A}$. The function $f \in \mathcal{CV}$ if and only if*

$$\operatorname{Re}\left(1 + \frac{zf''(z)}{f'(z)}\right) > 0, \quad z \in \mathbb{D}. \tag{7.1.1}$$

Equivalently, $f \in \mathcal{CV}$ if and only if the function

$$p(z) = 1 + \frac{zf''(z)}{f'(z)} \tag{7.1.2}$$

belongs to $\mathcal{P}$ of Carathéodory functions.

Proof Let $f \in \mathcal{CV}$, and $f_r(z) = r^{-1} f(rz)$, $z \in \mathbb{D}$. By the hereditary property of convex functions (Theorem 7.1.2), f maps each disk $\mathbb{D}_r$ onto a convex domain for $0 < r < 1$. Consequently, the image curve C_r of the circle $|z| = r$ bounds a convex domain. Geometrically, this implies that the argument of the tangent vector to C_r is a non-decreasing function as the curve is traversed in the counterclockwise direction (see Fig. 7.2; for starlike functions, see 7.3). In other words, $\arg(\frac{\partial}{\partial\theta} f(re^{i\theta}))$ increases as θ varies from 0 to 2π. Thus,

$$\frac{\partial}{\partial\theta} \arg\left(\frac{\partial}{\partial\theta} f(re^{i\theta})\right) \geqslant 0.$$

Recall that $\log z = \log|z| + i \arg z$ and so $\arg z = \operatorname{Im} \log z$. Using this identity with $z = re^{i\theta}$, it follows that

$$\begin{aligned} 0 &\leqslant \frac{\partial}{\partial\theta} \arg\left(\frac{\partial}{\partial\theta} f(re^{i\theta})\right) \\ &= \operatorname{Im} \frac{\partial}{\partial\theta} \log\left(ire^{i\theta} f'(re^{i\theta})\right) \\ &= \operatorname{Im}\left(i + \frac{ire^{i\theta} f''(re^{i\theta})}{f'(re^{i\theta})}\right) \\ &= \operatorname{Re}\left(1 + \frac{zf''(z)}{f'(z)}\right). \end{aligned}$$

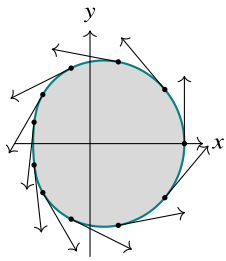


Fig. 7.2 Slope of the tangents increases on the boundary of a convex domain

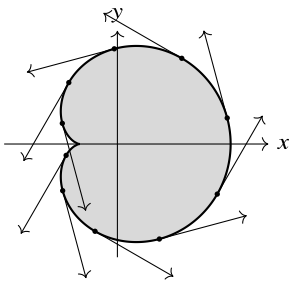


Fig. 7.3 Slope of the tangents on the boundary of a starlike domain

The function $p : \mathbb{D} \to \mathbb{C}$ defined by (7.1.2) is well-defined, since univalence of f implies $f'(z) \neq 0$ for $z \in \mathbb{D}$. Clearly, p is analytic, and by Theorem 1.2.12, $\operatorname{Re} p(z) > 0$. Thus, the inequality (7.1.1) holds.

For the converse, assume that $f \in \mathcal{A}$ satisfies the inequality (7.1.1). Let C_r be the image of the circle $|z| = r$ under f. The calculation above shows that the slope of the tangent to the curve C_r increases monotonically. As a point traverses the entire curve C_r, the argument of the tangent vector has a net change given by

$$\begin{aligned}\int_0^{2\pi} \frac{\partial}{\partial\theta}\left(\arg\left(\frac{\partial}{\partial\theta} f(re^{i\theta})\right)\right) d\theta &= \int_0^{2\pi} \operatorname{Re}\left(1 + \frac{zf''(z)}{f'(z)}\right) d\theta \\ &= \operatorname{Re}\left\{\int_{|z|=r} \left(1 + \frac{zf''(z)}{f'(z)}\right)\frac{dz}{iz}\right\} \\ &= 2\pi\end{aligned}$$

where $z = re^{i\theta}$. This shows that C_r is a simple closed curve that bounds a convex domain. By Darboux's Theorem (Theorem 2.1.7), it follows that f is univalent in $\mathbb{D}_r$, and hence univalent in $\mathbb{D}$. Since $r \in (0, 1)$ is arbitrary, this implies that f is a univalent function whose range is a convex domain. □

Example 7.1.4 The function $f : \mathbb{D} \to \mathbb{C}$ given by

$$f(z) = \frac{1}{1-x} \log\left(\frac{1-xz}{1-z}\right), \quad |x| \leqslant 1, x \neq 1$$

has derivative

$$f'(z) = \frac{1}{1-x}\left(\frac{1}{1-z} - \frac{x}{1-xz}\right) = \frac{1}{(1-z)(1-xz)}.$$

Since $\operatorname{Re}(z/(1-z)) > -1/2$ (see Example 7.1.1),

$$\operatorname{Re}\left(1+\frac{zf''(z)}{f'(z)}\right)=1+\operatorname{Re}\frac{z}{1-z}+\operatorname{Re}\frac{xz}{1-xz}>1-\frac{1}{2}-\frac{1}{2}=0.$$

Thus, f is convex.

In particular, the function

$$f(z)=\frac{1}{2}\log\left(\frac{1+z}{1-z}\right),\quad z\in\mathbb{D},$$

is convex.

Corollary 7.1.5 *If $f\in\mathcal{CV}$, then*

$$\left|1+\frac{zf''(z)}{f'(z)}-\frac{1+r^2}{1-r^2}\right|\leqslant\frac{2r}{1-r^2},\quad |z|=r<1. \tag{7.1.3}$$

For each $z\in\mathbb{D}$, the result is sharp for a rotation of the half-plane mapping ℓ.

Proof Define the function $p:\mathbb{D}\to\mathbb{C}$ by $p(z):=1+zf''(z)/f'(z)$. Then, $p\in\mathcal{P}$ and the desired result now follows from Theorem 4.3.2.

Alternatively, as in the proof of Theorem 2.5.1, the result can also be established using the bound $|a_2|\leqslant 1$ for $f\in\mathcal{CV}$ (proved in Theorem 7.2.4), by noting that the disk automorphism defined by (2.2.2) is also convex whenever f is convex. □

Corollary 7.1.6 *The class $\mathcal{CV}$ is a compact normal family.*

Proof Since $\mathcal{CV}\subset\mathcal{S}\subset\mathcal{H}(\mathbb{D})$, evidently $\mathcal{CV}$ is normal, being a subset of the normal family $\mathcal{S}$. To prove compactness, consider a sequence $\langle f_n\rangle$ of functions in $\mathcal{CV}$. By the compactness of $\mathcal{S}$, there exists a subsequence $\langle f_{n_k}\rangle$ converging to some $f\in\mathcal{S}$.

Since $f_{n_k}\to f$, Weierstrass' Theorem (Theorem 1.4.1) implies that $f'_{n_k}\to f'$ and $f''_{n_k}\to f''$. Since $f_{n_k}\in\mathcal{CV}$, and $\operatorname{Re}(1+zf''_{n_k}(z)/f'_{n_k}(z))>0$, by letting $k\to\infty$ gives $\operatorname{Re}(1+zf''(z)/f'(z))\geqslant 0$. By Theorem 1.2.12, it follows that $\operatorname{Re}(1+zf''(z)/f'(z))>0$, which shows that $f\in\mathcal{CV}$, completing the proof that the class $\mathcal{CV}$ is compact. □

7.2 Coefficients Bounds

Several properties of a convex function can be derived by relating it to a starlike function. Although every convex function is starlike, the following elementary relationship between convex functions and starlike functions is particularly useful.

Theorem 7.2.1 ***(Alexander's Theorem)*** *Suppose $f\in\mathcal{A}$, and $F(z):=zf'(z)$ in $\mathbb{D}$. Then, $f\in\mathcal{CV}$ if and only if $F\in\mathcal{ST}$.*

Proof Since $F(z) = zf'(z)$, a computation yields

$$\frac{zF'(z)}{F(z)} = 1 + \frac{zf''(z)}{f'(z)}.$$

The theorem follows immediately from this identity, together with Theorem 6.1.4 and Theorem 7.1.3. □

Example 7.2.2 If $f \in \mathcal{A}$ satisfies

$$\left|\frac{zf''(z)}{f'(z)}\right| < 1,$$

then $f \in \mathcal{CV}$. Similarly, if $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{A}$ satisfies

$$\sum_{k=2}^{\infty} k^2 |a_k| \leqslant 1,$$

then $f \in \mathcal{CV}$.

These results follow, respectively, from Example 6.1.7 and Example 6.1.8, by an application of Alexander's theorem.

Example 7.2.3 The function $f : \mathbb{D} \to \mathbb{C}$ defined by

$$f(z) = -\log(1 - z)$$

is convex.

To see this, note that

$$F(z) = zf'(z) = \frac{z}{1 - z} = \ell(z)$$

is a convex function. Since every convex function is starlike, the function F is starlike. Since $F(z) = zf'(z)$, by Alexander's theorem, f is convex.

Let $F \in \mathcal{ST}$, and $F(z) = zf'(z)$. By Alexander's theorem, f is convex. Clearly,

$$f(z) = \int_0^z \frac{F(\zeta)}{\zeta} \mathrm{d}\zeta. \tag{7.2.1}$$

Thus, an equivalent statement of Alexander's theorem is the following: if $F \in \mathcal{ST}$, then f defined by (7.2.1) belongs to $\mathcal{CV}$.

If F is the Koebe function

$$F(z) := \kappa(z) = \frac{z}{(1 - z)^2},$$

then the corresponding convex function f is

$$f(z) = \int_0^z \frac{F(\zeta)}{\zeta} \mathrm{d}\zeta = \frac{z}{1-z} = \ell(z).$$

Thus, the Koebe function κ and the half-plane mapping ℓ are related by $z\ell'(z) = \kappa(z)$.

Although the bound $|a_k| \leqslant k$ holds for convex functions, Alexander's Theorem easily yields the sharp bound $|a_k| \leqslant 1$.

Theorem 7.2.4 *If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{CV}$, then*

$$|a_k| \leqslant 1, \quad k \geqslant 2.$$

Further, this inequality is sharp for each integer $k \geqslant 2$. If equality occurs for any $k \geqslant 2$, then f is a rotation of the half-plane mapping ℓ.

Proof Let $F(z) = zf'(z)$. Then

$$F(z) = zf'(z) = z + \sum_{k=2}^{\infty} k a_k z^k \in \mathcal{ST}.$$

Theorem 6.2.1 implies that $k|a_k| \leqslant k$. Hence $|a_k| \leqslant 1$. Since

$$\ell(z) = \frac{z}{1-z} = z + \sum_{k=2}^{\infty} z^k,$$

the bound $|a_k| \leqslant 1$ is sharp. Further, if $|a_k| = 1$ for any $k \geqslant 2$, then $zf'(z)$ is a rotation of the Koebe function, and f is a rotation of ℓ. □

Since $\mathcal{CV}$ is a linear-invariant family of order $\alpha = \max\{|a_2| : f \in \mathcal{CV}\} = 1$, the inequality (7.1.3) follows from (2.5.2).

Clearly, the distortion and growth estimates for univalent/starlike functions also hold for convex functions. However, sharp bounds specific to convex functions can be obtained using Alexander's Theorem, as stated in the following result.

Theorem 7.2.5 ***(Growth and Distortion Theorem)*** *If $f \in \mathcal{CV}$, then*

$$-\ell(-r) = \frac{r}{1+r} \leqslant |f(z)| \leqslant \frac{r}{1-r} = \ell(r), \quad |z| = r < 1,$$

$$\ell'(-r) = \frac{1}{(1+r)^2} \leqslant |f'(z)| \leqslant \frac{1}{(1-r)^2} = \ell'(r), \quad |z| = r < 1,$$

and, for each $k \geqslant 2$,

$$|f^{(k)}(z)| \leqslant \frac{k!}{(1-r)^{k+1}} = \ell^{(k)}(r), \quad |z| = r < 1.$$

All of these inequalities are sharp, with equality if and only if f is a rotation of the half-plane mapping ℓ.

Proof Since $f \in \mathcal{CV}$, it follows from Alexander's Theorem (Theorem 7.2.1) that $zf' \in \mathcal{ST}$. Applying the growth inequality for functions in $\mathcal{ST}$, as given in Theorem 6.1.2, to the function $zf'(z)$, yields

$$\frac{r}{(1+r)^2} \leqslant |zf'(z)| \leqslant \frac{r}{(1-r)^2},$$

which establishes the distortion inequality.

An application of Lemma 2.5.4 to the distortion inequality yields the corresponding growth inequality. The estimate for $f^{(k)}(z)$ follows directly from the bound $|a_k| \leqslant 1$ for convex functions and the triangle inequality.

Sharpness in all cases follows from the corresponding results for starlike functions. □

Corollary 7.2.6 *If $f \in \mathcal{CV}$, then $\mathbb{D}_{1/2} \subset f(\mathbb{D})$.*

Proof Letting $r \to 1^-$ in $r/(1+r) \leqslant |f(z)|$ shows that the boundary of $f(\mathbb{D})$ is contained in the complement of $\mathbb{D}_{1/2}$ which establishes the result. However, the following is an alternative proof due to MacGregor [5].

Let c be an omitted value of f, so that $f(z) \neq c$, and $g(z) := (f(z) - c)^2$. Suppose that $z_1, z_2 \in \mathbb{D}$ are distinct and $g(z_1) = g(z_2)$. Then,

$$(f(z_1) - c)^2 - (f(z_2) - c)^2 = (f(z_1) - f(z_2))(f(z_1) + f(z_2) - 2c) = 0.$$

Therefore, either $f(z_1) = f(z_2)$, or $2^{-1}(f(z_1) + f(z_2)) = c$. The former is impossible since f is univalent in $\mathbb{D}$. The latter is also excluded because the midpoint of any two points in the convex domain $f(\mathbb{D})$ also lie in the image. Since $c \notin f(\mathbb{D})$, this is a contradiction.

Thus, the function g is univalent in $\mathbb{D}$. Consider the normalized function

$$h(z) := \frac{c^2 - g(z)}{2c}.$$

Since $f(z) \neq c$, it follows that $g(z) \neq 0$, and hence $h(z) \neq c/2$. The function $h \in \mathcal{S}$, and therefore, by the Koebe One-Quarter Theorem (Theorem 2.3.6), $|c/2| \geqslant 1/4$, which implies that $|c| \geqslant 1/2$. □

Theorem 7.2.7 ***(Clunie's Theorem)*** *If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{CV}$, then*

$$(n(n-1))^2 |a_n|^2 \leqslant 4\left(1 + \sum_{k=2}^{n-1} k^3 |a_k|^2\right),$$

and $|a_n| \leqslant 1$ for each integer $n \geqslant 2$. Equality occurs if f is a rotation of the half-plane mapping ℓ.

Proof The results follows from Theorem 6.2.4 by use of Alexander's Theorem. □

In view of Alexander's Theorem, the following result is an immediate consequence of Theorem 6.2.2. However, another proof is derived by using Theorem 4.5.4.

Theorem 7.2.8 *(**Fekete-Szegö Inequality**) If $f \in \mathcal{CV}$ and $f(z) = z + \sum_{k=2}^{\infty} a_k z^k$, then*

$$|a_3 - \mu a_2^2| \leqslant \frac{1}{3} \max\{1, 3|1 - \mu|\}, \quad \mu \in \mathbb{C},$$

and

$$|a_3 - \mu a_2^2| \leqslant \frac{1}{3} \begin{cases} 1 - (3\mu - 2)|a_2|^2, & 2/3 \leqslant \mu \leqslant 1, \\ 1 - (4 - 3\mu)|a_2|^2, & 1 \leqslant \mu \leqslant 4/3. \end{cases}$$

Equality occurs for the half-plane mapping ℓ or its square-root transformation.

Proof The function $p(z) = 1 + zf''(z)/f'(z)$ belongs to $\mathcal{P}$. Write $p(z) = 1 + c_1 z + c_2 z^2 + \cdots$. By comparing the coefficients of z and z^2 in

$$p(z) f'(z) = f'(z) + zf''(z),$$

it follows that

$$4a_2 = 2a_2 + c_1 \quad \text{and} \quad 9a_3 = 3a_3 + 2a_2 c_1 + c_2.$$

These leads to

$$a_2 = \frac{c_1}{2}, \quad a_3 = \frac{1}{6}(c_2 + c_1^2),$$

and hence

$$a_3 - \mu a_2^2 = \frac{1}{6}\left(c_2 - \left(\frac{3}{2}\mu - 1\right) c_1^2\right).$$

The results follows by an application of Theorem 4.5.4. □

For a convex function f, Theorem 7.2.8 gives

$$|a_3 - a_2^2| \leqslant \frac{1}{3}.$$

Using this inequality and the fact that the disk automorphism of a convex function produces another convex function, as in the proof of Kraus–Nehari Theorem (Theorem 2.4.4), the following theorem can be derived. However, a direct proof is given by using the Schwarz–Pick theorem (Theorem 3.1.3).

Theorem 7.2.9 *(**Nehari Theorem**) The Schwarzian derivative $\{f; z\}$ of $f \in \mathcal{CV}$ satisfies*

$$|\{f;z\}| \leqslant \frac{2}{(1-|z|^2)^2}, \quad z \in \mathbb{D}.$$

Equality holds for the function

$$f(z) = \frac{1}{2}\log\frac{1+z}{1-z}.$$

Proof The function

$$p(z) = 1 + \frac{zf''(z)}{f'(z)}$$

belongs to $\mathcal{P}$. Therefore, p is subordinate to the half-plane mapping $p_0(z) = (1+z)/(1-z)$, so that $p(z) = p_0(\omega(z))$ for some Schwarz function ω. In other words,

$$\frac{zf''(z)}{f'(z)} = \frac{1+\omega(z)}{1-\omega(z)} - 1 = \frac{2\omega(z)}{1-\omega(z)},$$

where ω is analytic and has the property $|\omega(z)| \leq |z|$ for all $z \in \mathbb{D}$. With the notation $\varphi(z) = \omega(z)/z$, this gives the representation

$$\frac{f''(z)}{f'(z)} = \frac{2\varphi(z)}{1-z\varphi(z)}$$

for the pre-Schwarzian, where φ is analytic and satisfies $|\varphi(z)| \leqslant 1$ for all $z \in \mathbb{D}$. Routine calculations now give the Schwarzian of f in the form

$$\begin{aligned}\{f;z\} &= \left(\frac{f''(z)}{f'(z)}\right)' - \frac{1}{2}\left(\frac{f''(z)}{f'(z)}\right)^2 \\ &= \frac{2(\varphi'(z)+\varphi^2(z))}{(1-z\varphi(z))^2} - \frac{2\varphi^2(z)}{(1-z\varphi(z))^2} \\ &= \frac{2\varphi'(z)}{(1-z\varphi(z))^2}.\end{aligned}$$

By the Schwarz–Pick theorem (Theorem 3.1.3),

$$\left|\varphi'(z)\right| \leqslant \frac{1-|\varphi(z)|^2}{1-|z|^2},$$

and so

$$|\{f;z\}| \leqslant 2\frac{1-|\varphi(z)|^2}{\left(1-|z|^2\right)(1-|z\varphi(z)|)^2} \leqslant \frac{2}{\left(1-|z|^2\right)^2}.$$

The last inequality follows as

$$(1 - |\varphi(z)|^2)(1 - |z|^2) \leqslant (1 - |z\varphi(z)|)^2$$

is equivalent to

$$(|z| - |\varphi(z)|)^2 \geqslant 0.$$

The bound is best possible since the horizontal strip mapping

$$f(z) = \frac{1}{2} \log \frac{1+z}{1-z}$$

has Schwarzian $\{f; z\} = 2\left(1 - z^2\right)^{-2}$. □

Alexander's Theorem and Theorem 6.4.2 immediately give the following result.

Theorem 7.2.10 ***(Herglotz's Representation Theorem)*** *Let $f \in \mathcal{A}$. Then $f \in \mathcal{CV}$ if and only if there exists a non-decreasing function μ defined on $[0, 2\pi]$ such that $\int_0^{2\pi} d\mu(t) = 1$ and satisfying*

$$f'(z) = \exp\left(\int_0^{2\pi} -2 \log\left(1 - ze^{-it}\right) d\mu(t)\right).$$

Proof If $f \in \mathcal{CV}$, then $1 + zf''(z)/f'(z) \in \mathcal{P}$. By Herglotz's representation theorem (Theorem 4.2.3), there exists a non-decreasing function μ on $[0, 2\pi]$ such that $\int_0^{2\pi} \mathrm{d}\mu(t) = 1$ and

$$1 + \frac{zf''(z)}{f'(z)} = \int_0^{2\pi} \frac{1 + z\mathrm{e}^{-\mathrm{i}t}}{1 - z\mathrm{e}^{-\mathrm{i}t}} \mathrm{d}\mu(t). \tag{7.2.2}$$

Then

$$\frac{f''(z)}{f'(z)} = \int_0^{2\pi} \frac{2\mathrm{e}^{-\mathrm{i}t}}{1 - z\mathrm{e}^{-\mathrm{i}t}} \mathrm{d}\mu(t),$$

and an integration from 0 to z with a change in the order of integration yields

$$\log f'(z) = \int_0^{2\pi} -2 \log\left(1 - z\mathrm{e}^{-\mathrm{i}t}\right) \mathrm{d}\mu(t).$$

The desired result now follows by exponentiation.

The converse is established by a logarithmic differentiation and use of the Herglotz's representation theorem. □

As an application, the following results are derived.

Corollary 7.2.11 *If $f \in \mathcal{CV}$, then*

$$\frac{1}{(1+r)^2} \leqslant |f'(z)| \leqslant \frac{1}{(1-r)^2}, \quad |z| = r < 1.$$

Proof By Theorem 7.2.10, the modulus of $f'(z)$ satisfies

$$\begin{aligned} |f'(z)| &= \exp\left(\operatorname{Re} \int_0^{2\pi} -2\log(1 - ze^{-it})\, d\mu(t)\right) \\ &= \exp\left(\int_0^{2\pi} -2\log|1 - ze^{-it}|\, d\mu(t)\right). \end{aligned}$$

Since

$$1 - r \leqslant |1 - ze^{-it}| \leqslant 1 + r$$

for $|z| = r < 1$, readily

$$\log(1 - r) \leqslant \log|1 - ze^{-it}| \leqslant \log(1 + r),$$

and integrating with respect to the probability measure $d\mu(t)$ gives

$$\log(1 - r) \leqslant \int_0^{2\pi} \log|1 - ze^{-it}|\, \mathrm{d}\mu(t) \leqslant \log(1 + r).$$

Multiplying by -2 and exponentiating, the desired result is deduced. □

Corollary 7.2.12 *If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{CV}$, then $|a_2| \leqslant 1$,*

$$\left|a_3 - \frac{2}{3}a_2^2\right| \leqslant \frac{1}{3}, \quad \textit{and} \quad \left|a_4 - \frac{3}{2}a_2 a_3 + \frac{2}{3}a_2^3\right| \leqslant \frac{1}{6}.$$

In addition, $|a_3| \leqslant 1$ and $|a_4| \leqslant 1$.

Proof Since

$$\log(1 - ze^{-it}) = -\sum_{n=1}^{\infty} \frac{z^n}{n} e^{-int},$$

Theorem 7.2.10 yields

$$\log f'(z) = \sum_{n=1}^{\infty} \left(\frac{2}{n} \int_0^{2\pi} e^{-int}\, \mathrm{d}\mu(t)\right) z^n.$$

A computation shows that

$$\log f'(z) = 2a_2 z + (3a_3 - 2a_2^2)z^2 + \left(4a_4 - 6a_2a_3 + \frac{8}{3}a_2^3\right) z^3 + \cdots .$$

Comparing coefficients of z^n between the two equations for $\log f'(z)$, and noting that μ satisfy

$$\left| \int_0^{2\pi} e^{-int} \, \mathrm{d}\mu(t) \right| \leqslant 1,$$

it follows that $|2a_2| \leqslant 2$, $|3a_3 - 2a_2^2| \leqslant 1$, and

$$\left| 4a_4 - 6a_2a_3 + \frac{8}{3}a_2^3 \right| \leqslant \frac{2}{3}.$$

Since $3|a_3| \leqslant 2|a_2|^2 + 1 \leq 3$, it follows that $|a_3| \leqslant 1$. Also,

$$4|a_4| \leqslant 2|3a_3 - 2a_2^2| + \tfrac{4}{3}|a_2|^3 + \tfrac{2}{3} \leqslant 2 + \tfrac{4}{3} + \tfrac{2}{3} = 4,$$

and hence $|a_4| \leqslant 1$. □

7.3 Curvature

Let $f : G \to \mathbb{C}$ be an analytic function, and γ_z be a curve in G. Let $\gamma_w = f(\gamma_z)$ denote the image of this curve under f. The following result provides a formula for the curvature K_w of the image curve γ_w in terms of the curvature K_z of the original curve γ_z.

Theorem 7.3.1 *Let γ_z be a twice differentiable curve $z : [a, b] \to G$ with $z'(t) \neq 0$ for all $t \in [a, b]$, and $f : G \to \mathbb{C}$ be a locally univalent analytic function. Then the curvature K_w of the image curve $\gamma_w = f(\gamma_z)$ is given by*

$$K_w = \frac{1}{|z'(t)f'(z)|} \operatorname{Im}\left(\frac{z''(t)}{z'(t)} + z'(t)\frac{f''(z)}{f'(z)}\right),$$

or equivalently

$$K_w = \frac{1}{|f'(z)|}\left(K_z + \operatorname{Im}\left(e^{i\alpha}\frac{f''(z)}{f'(z)}\right)\right),$$

where α gives the direction of γ_z and K_z is the curvature of γ_z.

Proof Note that $f'(z_0) \neq 0$ for all $z_0 \in G$. The direction to the curve γ_z given by $z(t) = x(t) + \mathrm{i}y(t)$ is determined by the increasing values of the parameter t. Let $z_0 = z(t_0)$, where t_0 is an interior point of $[a, b]$. Let ϕ denote the angle that the tangent to the image curve γ_w makes with the positive real axis. With an appropriate choice of a branch, the angle is given by

$$\phi = \arg\left(f'(z)z'(t)\right) = \operatorname{Im}\left(\log\left(f'(z)z'(t)\right)\right).$$

Let s denote arc length along γ_w. By definition, the signed curvature is given by

$$\begin{aligned} K_w &= \frac{\mathrm{d}\phi}{\mathrm{d}s} = \frac{\mathrm{d}\phi}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}s} = \frac{\mathrm{d}t}{\mathrm{d}s}\frac{\mathrm{d}}{\mathrm{d}t}\left(\operatorname{Im}\left(\log(f'(z)z'(t))\right)\right) \\ &= \frac{1}{\mathrm{d}s/\mathrm{d}t}\operatorname{Im}\left(\frac{z''(t)}{z'(t)} + z'(t)\frac{f''(z)}{f'(z)}\right). \end{aligned}$$

The arc length of the curve γ_z is given by $\int_a^t |z'(u)|\mathrm{d}u$, and the arc length of the image curve is

$$s(t) = \int_a^t |f'(z(u))||z'(u)|\mathrm{d}u.$$

Hence

$$\frac{\mathrm{d}s}{\mathrm{d}t} = |f'(z)z'(t)|.$$

Substituting this into the curvature formula yields

$$K_w = \frac{1}{|f'(z)z'(t)|}\operatorname{Im}\left(\frac{z''(t)}{z'(t)} + z'(t)\frac{f''(z)}{f'(z)}\right).$$

When $f(z) \equiv z$, this expression reduces to the usual formula for the curvature of γ_z:

$$K_z = \frac{1}{|z'(t)|}\operatorname{Im}\left(\frac{z''(t)}{z'(t)}\right) = \frac{x'y'' - y'x''}{\left[(x')^2 + (y')^2\right]^{3/2}}.$$

Combining this with the formula K_w, the curvature of the image curve can be expressed as

$$K_w = \frac{1}{|f'(z)|}\left(K_z + \operatorname{Im}\left(\mathrm{e}^{\mathrm{i}\alpha}\frac{f''(z)}{f'(z)}\right)\right),$$

where $\mathrm{e}^{\mathrm{i}\alpha} = z'(t)/\left|z'(t)\right|$, so that α represents the direction of γ_z at the point $z(t)$. □

For $f \in \mathcal{S}$, Theorem 2.5.1 gives

$$\left|\frac{f''(z)}{f'(z)}\right| \leqslant \frac{4 + 2r}{1 - r^2}$$

and Theorem 2.5.3 yields

$$|f'(z)| \geqslant \frac{1-r}{(1+r)^3}.$$

Let K_z be the curvature of a curve γ at $z_0 \in \mathbb{D}$ and K_w be the curvature of the curve $f(\gamma)$ at $w_0 = f(z_0)$. Then, for $r = |z_0|$ and $f \in \mathcal{S}$,

$$|K_w| \leqslant \frac{1}{|f'(z_0)|}\left(|K_z| + \left|\frac{f''(z_0)}{f'(z_0)}\right|\right) \leqslant \frac{(1+r)^3}{1-r}|K_z| + (4+2r)\left(\frac{1+r}{1-r}\right)^2.$$

The above bound is not sharp. The following theorem provides a sharp bound in the case of starlike functions.

Theorem 7.3.2 *Let $f \in \mathcal{ST}$. Further, let K_z be the curvature of a curve γ at $z_0 \in \mathbb{D}$ and let K_w be the curvature of the curve $f(\gamma)$ at $w_0 = f(z_0)$. Then*

$$|K_w| \leqslant \frac{(1+r)^3}{1-r}|K_z| + (4-2r)\left(\frac{1+r}{1-r}\right)^2, \quad r = |z_0|.$$

This inequality is sharp.

Proof Note that

$$|K_w| \leqslant \frac{1}{|f'(z_0)|}\left(|K_z| + \left|\frac{f''(z_0)}{f'(z_0)}\right|\right), \tag{7.3.1}$$

and the right-hand expression is invariant under rotations. Without loss of generality, set $z_0 = r \geqslant 0$. For $f \in \mathcal{ST}$, the function $zf'(z)/f(z)$ belongs to the class $\mathcal{P}$. Consequently, its reciprocal $f(z)/(zf'(z))$ also lie in $\mathcal{P}$. By the Herglotz's representation theorem for functions with positive real part (Theorem 4.2.3), there exists a non-decreasing function $\mu : [0, 2\pi] \to \mathbb{R}$ with $\int_0^{2\pi} \mathrm{d}\mu(t) = 1$ satisfying

$$\frac{f(z)}{zf'(z)} = \int_0^{2\pi} \frac{1+\mathrm{e}^{-\mathrm{i}t}z}{1-\mathrm{e}^{-\mathrm{i}t}z}\mathrm{d}\mu(t).$$

Multiplying both sides by z and differentiating yields

$$1 - \frac{f(z)f''(z)}{(f'(z))^2} = \int_0^{2\pi}\left(\frac{1+\mathrm{e}^{-\mathrm{i}t}z}{1-\mathrm{e}^{-\mathrm{i}t}z} + \frac{2\mathrm{e}^{-\mathrm{i}t}z}{(1-\mathrm{e}^{-\mathrm{i}t}z)^2}\right)\mathrm{d}\mu(t).$$

Rewriting this gives

$$\frac{f''(z)}{(f'(z))^2} = \frac{1}{f(z)}\int_0^{2\pi} -2\mathrm{e}^{-\mathrm{i}t}z\left(\frac{1}{1-\mathrm{e}^{-\mathrm{i}t}z} + \frac{1}{\left(1-\mathrm{e}^{-\mathrm{i}t}z\right)^2}\right)\mathrm{d}\mu(t).$$

By Theorem 6.1.2, for any function $f \in \mathcal{ST}$, the estimates $|f(z)| \geqslant r/(1+r)^2$ and $|f'(z)| \geqslant (1-r)/(1+r)^3$ hold. Using these bounds, one obtains

$$\left|\frac{f''(z)}{(f'(z))^2}\right| \leqslant \frac{(1+r)^2}{r} 2r\left(\frac{1}{1-r}+\frac{1}{(1-r)^2}\right) = (4-2r)\left(\frac{1+r}{1-r}\right)^2. \quad (7.3.2)$$

Substituting into inequality (7.3.1) gives the estimate

$$|K_w| \leqslant \frac{1}{|f'(z_0)|}\left(|K_z| + \left|\frac{f''(z_0)}{f'(z_0)}\right|\right) \leqslant \frac{(1+r)^3}{1-r}|K_z| + (4-2r)\left(\frac{1+r}{1-r}\right)^2.$$

Equality is attained for the rotation of the Koebe function $f(z) = z/(1+\mathrm{e}^{\mathrm{i}\theta}z)^2$, when the parameter θ is chosen such that $\mathrm{e}^{\mathrm{i}\theta}z_0 = |z_0| = r$, and the angle α is selected so that the quantity $\operatorname{Im}\left(\mathrm{e}^{\mathrm{i}\alpha} f''(z_0)/f'(z_0)\right)$ has the same sign as K_z. □

The estimate in (7.3.2) can be obtained using the estimates for functions with positive real part. Define $p(z) = f(z)/(zf'(z))$ so that $p \in \mathcal{P}$. A straightforward computation gives

$$\frac{f''(z)}{(f'(z))^2} = \frac{1}{f(z)}(1 - p(z) - zp'(z)).$$

Applying Theorem 4.3.1 yields

$$\left|\frac{f''(z)}{(f'(z))^2}\right| \leqslant \frac{(1+r)^2}{r}\left(\frac{2r}{1-r}+\frac{2r}{(1-r)^2}\right) = (4-2r)\left(\frac{1+r}{1-r}\right)^2.$$

If the curve γ_z is the circle $|z| = r$, then the curvature of the image curve $f(\gamma_z)$ at $w = f(z)$ is given by

$$K_w = \frac{1}{|f'(z)r|}\operatorname{Im}\left(\mathrm{i} + \mathrm{i}\frac{zf''(z)}{f'(z)}\right) = \frac{1}{|zf'(z)|}\operatorname{Re}\left(1 + \frac{zf''(z)}{f'(z)}\right).$$

The *radius of curvature* of $f(\gamma_z)$ at $f\left(r\mathrm{e}^{-\mathrm{i}\theta}\right)$ is the reciprocal of the curvature and is given by

$$\rho(r,\theta) = \frac{r\left|f'(z)\right|}{\operatorname{Re}\left[1 + zf''(z)/f'(z)\right]}, \quad z = r\mathrm{e}^{-\mathrm{i}\theta}.$$

Theorem 7.3.3 *If $f \in \mathcal{CV}$, then*

$$\max_{\theta} \rho(r,\theta) \leqslant \frac{r}{1-r^2}$$

with equality for the function $\ell(z) = z/(1-z)$.

Proof The Herglotz's representation for convex functions gives

$$f'(z) = \exp\left(\int_0^{2\pi} \log\left(1 - z\mathrm{e}^{-\mathrm{i}t}\right)^{-2} \mathrm{d}\mu(t)\right)$$

for some non-decreasing function $\mu(t)$ defined on $[0, 2\pi]$ with $\int_0^{2\pi} \mathrm{d}\mu(t) = 1$. Therefore,

$$\begin{aligned}\left|f'(z)\right| &= \exp\left(\mathrm{Re}\int_0^{2\pi} \log\left(1 - z\mathrm{e}^{-\mathrm{i}t}\right)^{-2} \mathrm{d}\mu(t)\right) \\ &= \exp\left(\int_0^{2\pi} \log\left|1 - z\mathrm{e}^{-\mathrm{i}t}\right|^{-2} \mathrm{d}\mu(t)\right) \\ &= \exp\left(\int_0^{2\pi} \log\frac{1}{\Delta^2}\mathrm{d}\mu(t)\right) \end{aligned} \tag{7.3.3}$$

where $z = r\mathrm{e}^{\mathrm{i}\theta}$ and $\Delta^2 = \Delta^2(r, t-\theta) = 1 - 2r\cos(t-\theta) + r^2$.

Similarly, (7.2.2) yields

$$\mathrm{Re}\left(1 + \frac{zf''(z)}{f'(z)}\right) = \int_0^{2\pi} \frac{1-r^2}{\Delta^2}\mathrm{d}\mu(t). \tag{7.3.4}$$

Using (7.3.3) and (7.3.4) in the radius of curvature expression gives

$$\rho(r,\theta) = r\left[\exp\left(\int_0^{2\pi} \log\frac{1}{\Delta^2}\mathrm{d}\mu(t)\right)\right]\left[\int_0^{2\pi} \frac{1-r^2}{\Delta^2}\mathrm{d}\mu(t)\right]^{-1}.$$

By the arithmetic geometric mean inequality, which states that

$$\exp\left(\int_0^{2\pi} f(t)\mathrm{d}\mu(t)\right) \leqslant \int_0^{2\pi} \exp(f(t))\mathrm{d}\mu(t),$$

it follows that

$$\exp\left(\int_0^{2\pi} \log\frac{1}{\Delta^2}\mathrm{d}\mu(t)\right) \leqslant \int_0^{2\pi} \frac{1}{\Delta^2}\mathrm{d}\mu(t).$$

Hence, $\rho(r,\theta) \leqslant r/\left(1-r^2\right)$.

For the half-plane mapping ℓ given by $\ell(z) = z/(1-z)$, direct computation gives

$$\rho(r,\theta) = \frac{r}{1-r^2},$$

and hence $\max_\theta \rho(r,\theta) = r/(1-r^2)$. □

7.4 Convex Functions of Order α

Analogous to the class of starlike functions, convex functions are defined by the geometric requirement that the image domain is convex. Analytically, a function f is convex in the unit disk if $\mathrm{Re}(1 + zf''(z)/f'(z)) > 0$ for all $z \in \mathbb{D}$, which means that the quantity $1 + zf''(z)/f'(z)$ lies in the right half-plane. This condition can be generalized by replacing the right half-plane with other half-planes, leading to the definition of convex functions of order α, where $0 \leqslant \alpha < 1$.

Definition 7.4.1 A function $f \in \mathcal{A}$ is said to be *convex of order* α, where $0 \leqslant \alpha < 1$, if

$$\mathrm{Re}\left(1 + \frac{zf''(z)}{f'(z)}\right) > \alpha, \quad z \in \mathbb{D}.$$

The class of all such functions is denoted by $\mathcal{CV}(\alpha)$. For $0 \leqslant \alpha < 1$, let $\mathcal{CV}_\alpha$ denote the class of all $f \in \mathcal{A}$ satisfying

$$\left|\frac{zf''(z)}{f'(z)}\right| < 1 - \alpha, \quad z \in \mathbb{D}.$$

For $\alpha = 0$, the class $\mathcal{CV}(\alpha)$ reduces to the class $\mathcal{CV} := \mathcal{CV}(0)$ of convex functions. If $f \in \mathcal{A}$ is convex of order α, then it is clearly convex; in other words, $\mathcal{CV}(\alpha) \subset \mathcal{CV}$. More generally, for $0 < \beta < \alpha$, the inclusion $\mathcal{CV}(\alpha) \subset \mathcal{CV}(\beta)$ holds.

If $f \in \mathcal{CV}_\alpha$, then

$$-\mathrm{Re}\,\frac{zf''(z)}{f'(z)} \leqslant \left|\frac{zf''(z)}{f'(z)}\right| < 1 - \alpha,$$

and hence

$$1 + \mathrm{Re}\,\frac{zf''(z)}{f'(z)} > \alpha.$$

Thus, f is a convex function of order α. This proves that $\mathcal{CV}_\alpha \subset \mathcal{CV}(\alpha)$.

Example 7.4.2 The function ℓ_α defined by

$$\ell_\alpha(z) = \begin{cases} \frac{1-(1-z)^{2\alpha-1}}{2\alpha-1}, & \alpha \neq 1/2 \\ -\log(1-z), & \alpha = 1/2 \end{cases} \tag{7.4.1}$$

is an example of a function in the class $\mathcal{CV}(\alpha)$. The function ℓ_α is related to the generalized Koebe function κ_α by Alexander's relation $z\ell'_\alpha(z) = \kappa_\alpha(z)$. The function $f(z) = (\exp((1-\alpha)z) - 1)/(1-\alpha)$ belongs to $\mathcal{CV}_\alpha$.

The following Alexander's Theorem holds between the classes $\mathcal{CV}(\alpha)$ and $\mathcal{ST}(\alpha)$.

Theorem 7.4.3 *(**Alexander's Theorem**) Suppose $f \in \mathcal{A}$, and $F(z) := zf'(z)$. Then, $f \in \mathcal{CV}(\alpha)$ if and only if $F \in \mathcal{ST}(\alpha)$.*

Proof Since $F(z) = zf'(z)$, a computation shows that

$$\frac{zF'(z)}{F(z)} = 1 + \frac{zf''(z)}{f'(z)}.$$

The result follows from this identity. □

Theorem 7.4.4 *(**Distortion, Growth, and Covering Theorem**) If $f \in \mathcal{CV}(\alpha)$, then, for $|z| = r < 1$, the following sharp inequalities hold:*

$$\ell'_\alpha(-r) = \frac{1}{(1+r)^{2(1-\alpha)}} \leqslant |f'(z)| \leqslant \frac{1}{(1-r)^{2(1-\alpha)}} = \ell'_\alpha(r),$$

$$-\ell_\alpha(-r) = \frac{(1+r)^{2\alpha-1} - 1}{2\alpha - 1} \leqslant |f(z)| \leqslant \frac{1 - (1-r)^{2\alpha-1}}{2\alpha - 1} = \ell_\alpha(r), \quad \alpha \neq 1/2$$

and

$$-\ell_{1/2}(-r) = \log(1+r) \leqslant |f(z)| \leqslant -\log(1-r) = \ell_{1/2}(r), \quad \alpha = 1/2,$$

where ℓ_α is the function given by (7.4.1). Also, the inclusion $f(\mathbb{D}) \supset \{w \in \mathbb{C} : |w| < R\}$ holds, where

$$R = \begin{cases} \frac{2^{2\alpha-1}-1}{2\alpha-1}, & \alpha \neq 1/2 \\ \log 2, & \alpha = 1/2. \end{cases}$$

Proof Since $f \in \mathcal{CV}(\alpha)$, by Alexander's Theorem, $zf' \in \mathcal{ST}(\alpha)$. Hence, Corollary 6.6.4 applied to zf' yields

$$\frac{r}{(1+r)^{2(1-\alpha)}} \leqslant |zf'(z)| \leqslant \frac{r}{(1-r)^{2(1-\alpha)}}$$

which proves the distortion inequality. An application of Lemma 2.5.4 to the distortion inequality yields the growth inequality. The covering theorem follows by letting $r \to 1^-$ in the growth inequality. Clearly the inequalities are sharp for ℓ_α. □

Corollary 6.5.2 immediately gives the following result by Alexander's Theorem.

Corollary 7.4.5 *If* $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{CV}(\alpha)$, *then* $|a_k| \leqslant A_k$, *where*

$$A_k = \frac{1}{k!} \prod_{l=2}^{k} (l - 2\alpha).$$

Equality holds for the function ℓ_α.

7.5 Radius of Convexity of Order α

Not every univalent function is convex (of order α). For example, the Koebe function is univalent and starlike but not convex. However, for each such function f, there exists some $\rho_f = \rho_f(\alpha) > 0$ such that

$$\operatorname{Re}\left(1 + \frac{zf''(z)}{f'(z)}\right) > \alpha$$

for all $z \in \mathbb{D}_{\rho_f}$. The largest such ρ_f is called the *radius of convexity of order* α or $\mathcal{CV}(\alpha)$ radius for f. The $\mathcal{CV}_\alpha$ radius is defined similarly. The number $R_{\mathcal{CV}(\alpha)}(\mathcal{S}) := \inf\{\rho_f : f \in \mathcal{S}\}$ is called the radius of convexity of order α for the class $\mathcal{S}$. In particular, the radius of convexity of order 0 for the class $\mathcal{S}$ is simply called the *radius of convexity* for the class $\mathcal{S}$.

Similarly, the number $R_{\mathcal{CV}(\alpha)}(\mathcal{ST}) := \inf\{\rho_f : f \in \mathcal{ST}\}$ is called the radius of convexity of order α of the class $\mathcal{ST}$ of starlike functions.

The next theorem determines the radius of convexity of order α for the class $\mathcal{S}$.

Theorem 7.5.1 *The radius of convexity of order* α, $0 \leqslant \alpha < 1$, *for the class* $\mathcal{S}$ *of univalent functions is*

$$R_{\mathcal{CV}(\alpha)}(\mathcal{S}) = \frac{1 - \alpha}{2 + \sqrt{3 + \alpha^2}}.$$

In particular, the radius of convexity for the class $\mathcal{S}$ *is* $2 - \sqrt{3}$.

Proof For $f \in \mathcal{S}$, by Theorem 2.5.1,

$$\left|1 + \frac{zf''(z)}{f'(z)} - \frac{1 + r^2}{1 - r^2}\right| \leqslant \frac{4r}{1 - r^2} \quad (|z| = r < 1)$$

and this shows that

$$1 + \operatorname{Re}\frac{zf''(z)}{f'(z)} \geqslant \frac{1 + r^2}{1 - r^2} - \frac{4r}{1 - r^2} = \frac{1 - 4r + r^2}{1 - r^2} =: \varphi(r).$$

Note that

$$r_0 := \frac{1-\alpha}{2+\sqrt{3+\alpha^2}}$$

is the smallest positive root of the equation $\varphi(r) = \alpha$ or the equivalent quadratic equation $(1+\alpha)r^2 - 4r + (1-\alpha) = 0$. Since

$$\varphi'(r) = -4\frac{1-r+r^2}{(1-r^2)^2} < 0$$

for $0 \leqslant r < 1$, the function is decreasing. Thus

$$1 + \operatorname{Re}\frac{zf''(z)}{f'(z)} \geqslant \varphi(r) > \varphi(r_0) = \alpha, \quad 0 \leqslant r < r_0$$

and every function $f \in \mathcal{S}$ is convex of order α in the disk $\mathbb{D}_{r_0}$. This proves that $R_{\mathcal{CV}(\alpha)}(\mathcal{S}) \geqslant r_0$.

To prove $r_0 \leqslant R_{\mathcal{CV}(\alpha)}(\mathcal{S})$, consider the following rotation f of the Koebe function κ given by

$$f(z) = \frac{z}{(1+z)^2}.$$

For this function f,

$$1 + \operatorname{Re}\frac{zf''(z)}{f'(z)} = \operatorname{Re}\frac{1-4z+z^2}{1-z^2} \leqslant 0$$

for $z = r$ and $r_0 \leqslant r < 1$, proving $r_0 \leqslant R_{\mathcal{CV}(\alpha)}(\mathcal{S})$. Thus, the radius of convexity of order α for the class $\mathcal{S}$ is r_0.

In particular, $R_{\mathcal{CV}(0)}(\mathcal{S}) = 1/(2+\sqrt{3}) = 2-\sqrt{3}$ (see Fig. 7.4). □

Since the Koebe function is starlike, the following corollary holds.

Corollary 7.5.2 *The radius of convexity of order α, $0 \leqslant \alpha < 1$, for the class $\mathcal{ST}$ of starlike functions is*

$$R_{\mathcal{CV}(\alpha)}(\mathcal{ST}) = \frac{1-\alpha}{2+\sqrt{3+\alpha^2}}.$$

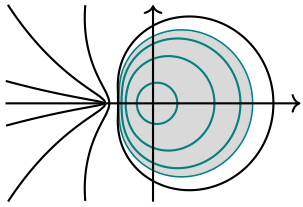


Fig. 7.4 Radius of convexity of the Koebe function is $2-\sqrt{3}$

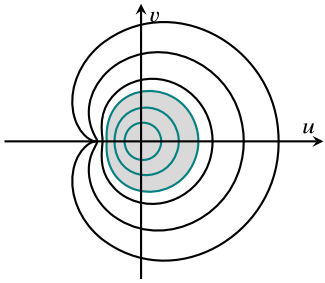


Fig. 7.5 Radius of convexity of $f(z) = z + z^2/2$ is $1/2$

In particular, the radius of convexity for the class $\mathcal{ST}$ is $2 - \sqrt{3}$.

Example 7.5.3 The radius of convexity for the function $f \in \mathcal{ST}$ given by $f(z) = z + z^2/2$ is $1/2$ (see Fig. 7.5). This can be seen as the function $p(z) = 1 + zf''(z)/f'(z) = (1 + 2z)/(1 + z)$ maps $\mathbb{D}_r$ onto

$$\left| p(z) - \frac{1 - 2r^2}{1 - r^2} \right| \leqslant \frac{r}{1 - r^2}.$$

This shows that

$$\operatorname{Re} p(z) > \frac{1 - r - 2r^2}{1 - r^2} > 0$$

for $0 \leqslant r < 1/2$.

By Example 7.2.2, the stronger condition $f \in \mathcal{CV}_0$, or equivalently $|zf''(z)/f'(z)| < 1$ implies $f \in \mathcal{CV}$. A computation shows that

$$\left| \frac{zf''(z)}{f'(z)} \right| = |p(z) - 1| = \left| \frac{z}{1 + z} \right| \leqslant \frac{r}{1 - r} < 1$$

for $r < 1/2$. Thus, the $\mathcal{CV}$ radius of f is the same $\mathcal{CV}_0$ radius of f.

The function f in Example 7.5.3 satisfies the condition $|f'(z) - 1| < 1$. For a function f satisfying this condition, the following result holds.

Theorem 7.5.4 *Let $0 \leqslant \alpha < 1$. The radius of convexity of order α and the $\mathcal{CV}_\alpha$ radius for the class of all functions $f \in \mathcal{A}$ satisfying $|f'(z) - 1| < 1$ is $\rho_0 = (1 - \alpha)/(2 - \alpha)$.*

Proof Let $f \in \mathcal{A}$ satisfy $|f'(z) - 1| < 1$. Then there is a function w in $\mathbb{D}$ with $|w(z)| \leqslant |z|$ such that $f'(z) = 1 + w(z)$. Recall that Schwarz–Pick Theorem (Theorem 3.1.3) shows that

$$|w'(z)| \leqslant \frac{1 - |w(z)|^2}{1 - |z|^2}.$$

Using this inequality and the fact that $|w(z)| \leqslant |z| = r$, it follows that

$$\left|\frac{zf''(z)}{f'(z)}\right| = \left|\frac{zw'(z)}{1+w(z)}\right| \leqslant r\frac{1-|w(z)|^2}{(1-r^2)(1-|w(z)|)} \leqslant \frac{r}{1-r}.$$

Since $r/(1-r) < 1-\alpha$ for $0 \leqslant r < (1-\alpha)/(2-\alpha)$, the function f satisfies $|zf''(z)/f'(z)| < 1-\alpha$ for $|z| < (1-\alpha)/(2-\alpha)$. This shows that the radius of convexity of order α and $\mathcal{CV}_\alpha$ radius of f is at least $(1-\alpha)/(2-\alpha)$.

Both the results are sharp for the function $f(z) = z + z^2/2$. This function satisfies $|f'(z) - 1| < 1$. At $z = -\rho_0$,

$$\operatorname{Re}\left(1 + \frac{zf''(z)}{f'(z)}\right) = \frac{1-2\rho_0}{1-\rho_0} = \alpha,$$

which proves the sharpness of the $\mathcal{CV}(\alpha)$ radius. Also, at $z = -\rho_0$,

$$\left|\frac{zf''(z)}{f'(z)}\right| = \left|\frac{z}{1+z}\right| = 1-\alpha,$$

which proves the sharpness of the $\mathcal{CV}_\alpha$ radius. □

The following result gives the $\mathcal{CV}_\alpha$ radius for the class of functions $f \in \mathcal{A}$ satisfying $f' \in \mathcal{P}$.

Theorem 7.5.5 *Let $0 \leqslant \alpha < 1$. The radius of convexity of order α and the $\mathcal{CV}_\alpha$ radius for the class of all functions $f \in \mathcal{A}$ whose derivative $f' \in \mathcal{P}$ is*

$$\rho_0 = \frac{1-\alpha}{1+\sqrt{1+(1-\alpha)^2}}.$$

Proof Since $f' \in \mathcal{P}$, Theorem 4.3.4 yields

$$\left|\frac{zf''(z)}{f'(z)}\right| \leqslant \frac{2r}{1-r^2} < 1-\alpha$$

for $0 \leqslant r < \rho_0$. Thus, the radius of convexity of order α and $\mathcal{CV}_\alpha$ radius is at least ρ_0.

The function $f_0(z) = -z - 2\log(1-z)$ satisfies $f'(z) = (1+z)/(1-z)$, and thus, $f_0' \in \mathcal{P}$. At $z = \rho_0$, it follows that

$$\frac{zf_0''(z)}{f_0'(z)} = \frac{2z}{1-z^2} = \frac{2\rho_0}{1-\rho_0^2} = 1-\alpha,$$

which proves sharpness of the $\mathcal{CV}_\alpha$ radius. At $z = -\rho_0$,

$$1 + \frac{zf_0''(z)}{f_0'(z)} = 1 + \frac{2z}{1-z^2} = \alpha,$$

which proves sharpness of the radius of convexity of order α. □

7.6 Marx-Strohhäcker Theorem

This section applies the theory of differential subordinations to establish several classical results concerning convex functions. Alexander's Theorem (Theorem 7.2.1) shows that a function f is convex if and only if zf' is starlike. The Marx-Strohhäcker theorem provides an additional connection between these two classes. These results naturally extend the geometric fact that every convex function is starlike.

The Marx-Strohhäcker theorem includes the following four differential implications for a function $f \in \mathcal{A}$, which are proved by verifying an admissibility condition through direct algebraic computation:

$$\operatorname{Re}\left(1+\frac{zf''(z)}{f'(z)}\right)>0 \quad \Rightarrow \quad \operatorname{Re}\frac{zf'(z)}{f(z)}>\frac{1}{2} \quad \Rightarrow \quad \operatorname{Re}\frac{f(z)}{z}>\frac{1}{2},$$

and

$$\operatorname{Re}\left(1+\frac{zf''(z)}{f'(z)}\right)>0 \quad \Rightarrow \quad \operatorname{Re}\sqrt{f'(z)}>\frac{1}{2} \quad \Rightarrow \quad \operatorname{Re}\frac{f(z)}{z}>\frac{1}{2}.$$

Since the functions $(1+z)/(1-z)$ and $1/(1-z)$ map $\mathbb{D}$ onto the half-planes $\{w \in \mathbb{C} : \operatorname{Re} w > 0\}$ and $\{w \in \mathbb{C} : \operatorname{Re} w > 1/2\}$, respectively, each of these four implications can be expressed in the following subordinations:

$$1+\frac{zf''(z)}{f'(z)} \prec \frac{1+z}{1-z} \quad \Rightarrow \quad \frac{zf'(z)}{f(z)} \prec \frac{1}{1-z} \quad \Rightarrow \quad \frac{f(z)}{z} \prec \frac{1}{1-z},$$

and

$$1+\frac{zf''(z)}{f'(z)} \prec \frac{1+z}{1-z} \quad \Rightarrow \quad f'(z) \prec \frac{1}{(1-z)^2} \quad \Rightarrow \quad \frac{f(z)}{z} \prec \frac{1}{1-z}.$$

The half-plane mapping $\ell \in \mathcal{A}$ given by

$$\ell(z)=\frac{z}{1-z}$$

satisfies

$$\ell'(z)=\frac{1}{(1-z)^2}, \quad 1+\frac{z\ell''(z)}{\ell'(z)}=\frac{1+z}{1-z}, \quad \text{and} \quad \frac{z\ell'(z)}{\ell(z)}=\frac{\ell(z)}{z}=\frac{1}{1-z},$$

and hence the above implications can be written in symmetric form.

Theorem 7.6.1 *(**Marx-Strohhäcker**) Let $\ell(z)=z/(1-z)$ be the half-plane mapping. For each $f \in \mathcal{A}$, the following implications hold:*

$$1+\frac{zf''(z)}{f'(z)}\prec 1+\frac{z\ell''(z)}{\ell'(z)} \quad\Rightarrow\quad \frac{zf'(z)}{f(z)}\prec\frac{z\ell'(z)}{\ell(z)} \quad\Rightarrow\quad \frac{f(z)}{z}\prec\frac{\ell(z)}{z}, \tag{7.6.1}$$

and

$$1+\frac{zf''(z)}{f'(z)}\prec 1+\frac{z\ell''(z)}{\ell'(z)} \quad\Rightarrow\quad f'(z)\prec\ell'(z) \quad\Rightarrow\quad \frac{f(z)}{z}\prec\frac{\ell(z)}{z}. \tag{7.6.2}$$

Proof To prove the first implication in (7.6.1), let

$$p(z):=2\frac{zf'(z)}{f(z)}-1, \quad z\in\mathbb{D}. \tag{7.6.3}$$

Since f is convex univalent, $f(z)\neq 0$ for $0\neq z\in\mathbb{D}$ and so the function p is analytic in $\mathbb{D}$ with $p(0)=1$. From (7.6.3),

$$\frac{zf'(z)}{f(z)}=\frac{p(z)+1}{2},$$

and a computation shows that

$$1+\frac{zf''(z)}{f'(z)}=\frac{p(z)+1}{2}+\frac{zp'(z)}{p(z)+1}=\psi\big(p(z),zp'(z)\big)\in\Omega, \tag{7.6.4}$$

where

$$\psi(r,s)=\frac{r+1}{2}+\frac{s}{r+1} \quad\text{and}\quad \Omega:=\{w\in\mathbb{C}:\operatorname{Re}w>0\}.$$

The function ψ satisfies

$$\begin{aligned}\operatorname{Re}\psi(\mathrm{i}\rho,\sigma)&=\operatorname{Re}\left(\frac{\mathrm{i}\rho+1}{2}+\frac{\sigma}{1+\rho^2}(1-\mathrm{i}\rho)\right)\\&=\frac{1}{2}+\frac{\sigma}{1+\rho^2}\leqslant\frac{1}{2}-\frac{1}{2}=0,\end{aligned}$$

whenever $\rho\in\mathbb{R}$ and $\sigma\leqslant-(1+\rho^2)/2$. This proves that $\psi\in\mathcal{P}_1(\Omega)$. Since $\psi\big(p(z),zp'(z)\big)\in\Omega$, Theorem 5.4.2 shows that $p\in\mathcal{P}$. This is equivalent to

$$\operatorname{Re}\frac{zf'(z)}{f(z)}>\frac{1}{2} \quad\text{or}\quad \frac{zf'(z)}{f(z)}\prec\frac{1}{1-z}.$$

To prove the second implication in (7.6.1), let

$$p(z):=2\frac{f(z)}{z}-1, \quad z\in\mathbb{D}. \tag{7.6.5}$$

Since $f \in \mathcal{A}$, the function p is analytic in $\mathbb{D}$ and $p(0) = 1$. From (7.6.5), it follows that $z(p(z) + 1) = 2f(z)$. A computation gives

$$\frac{zf'(z)}{f(z)} = 1 + \frac{zp'(z)}{p(z) + 1} = \psi\big(p(z), zp'(z)\big), \tag{7.6.6}$$

where

$$\psi(r, s) = 1 + \frac{s}{r + 1} \quad \text{and} \quad \Omega = \left\{ w \in \mathbb{C} : \operatorname{Re} w > \frac{1}{2} \right\}.$$

The function ψ satisfies

$$\operatorname{Re} \psi(\mathrm{i}\rho, \sigma) = \operatorname{Re} \left(1 + \frac{\sigma}{1 + \rho^2}(1 - \mathrm{i}\rho) \right) = 1 + \frac{\sigma}{1 + \rho^2} \leqslant 1 - \frac{1}{2} = \frac{1}{2}$$

whenever $\rho \in \mathbb{R}$ and $\sigma \leqslant -(1 + \rho^2)/2$. Thus, $\psi \in \mathcal{P}_1(\Omega)$. Since $\psi\big(p(z), zp'(z)\big) \in \Omega$, Theorem 5.4.2 shows that $p \in \mathcal{P}$. This is equivalent to

$$\operatorname{Re} \frac{f(z)}{z} > \frac{1}{2} \quad \text{or} \quad \frac{f(z)}{z} \prec \frac{1}{1 - z}.$$

To prove the first implication in (7.6.2), let

$$p(z) := 2\sqrt{f'(z)} - 1, \quad z \in \mathbb{D}. \tag{7.6.7}$$

Since $f \in \mathcal{A}$ and f is convex (univalent), the function p is analytic in $\mathbb{D}$ and $p(0) = 1$. From (7.6.7),

$$f'(z) = \frac{1}{4}(p(z) + 1)^2$$

and a computation shows that

$$1 + \frac{zf''(z)}{f'(z)} = 1 + \frac{2zp'(z)}{p(z) + 1} = \psi\big(p(z), zp'(z)\big),$$

where

$$\psi(r, s) = 1 + \frac{2s}{r + 1} \quad \text{and} \quad \Omega := \{ w \in \mathbb{C} : \operatorname{Re} w > 0 \}.$$

The function ψ satisfies

$$\operatorname{Re} \psi(\mathrm{i}\rho, \sigma) = 1 + \operatorname{Re} \left(\frac{2\sigma}{1 + \rho^2}(1 - \mathrm{i}\rho) \right) = 1 + \frac{2\sigma}{1 + \rho^2} \leqslant 1 - 1 = 0,$$

for $\rho \in \mathbb{R}$ and $\sigma \leqslant -(1 + \rho^2)/2$. Thus, $\psi \in \mathcal{P}_1(\Omega)$. Since $\psi\big(p(z), zp'(z)\big) \in \Omega$, Theorem 5.4.2 shows that $p \in \mathcal{P}$, or equivalently,

$$\operatorname{Re}\sqrt{f'(z)} > \frac{1}{2} \quad \text{or} \quad \sqrt{f'(z)} \prec \frac{1}{1-z}.$$

For the second implication in (7.6.2), let

$$p(z) := \frac{2f(z)}{z} - 1, \quad z \in \mathbb{D}. \tag{7.6.8}$$

Then p is analytic in $\mathbb{D}$ with $p(0) = 1$. From (7.6.8),

$$f(z) = \frac{z}{2}(1 + p(z)).$$

Now,

$$\sqrt{f'(z)} = \sqrt{\frac{zp'(z) + p(z) + 1}{2}} = \psi\big(p(z), zp'(z)\big) \in \Omega,$$

where

$$\psi(r, s) = \sqrt{\frac{r + s + 1}{2}} \quad \text{and} \quad \Omega = \left\{ w \in \mathbb{C} : \operatorname{Re} w > \frac{1}{2} \right\}.$$

It is now left to show that the function ψ satisfies

$$\operatorname{Re}\psi(\mathrm{i}\rho, \sigma) = \operatorname{Re}\sqrt{\frac{\mathrm{i}\rho + \sigma + 1}{2}} \leqslant \frac{1}{2}, \tag{7.6.9}$$

whenever $\rho \in \mathbb{R}$ and $\sigma \leqslant -(1 + \rho^2)/2$. Set $\zeta = \xi + \mathrm{i}\eta = (1 + \sigma + \mathrm{i}\rho)/2$, so that the conditions on ρ and σ becomes

$$\xi = \frac{1 + \sigma}{2} \leqslant \frac{1}{2}\left(1 - \frac{1}{2}(1 + \rho^2)\right) = \frac{1}{4}(1 - \rho^2) = \frac{1}{4} - \eta^2.$$

This implies that ζ is a point inside the parabolic region defined by the inequality $\xi \leqslant 1/4 - \eta^2$. Note that

$$\operatorname{Re}\sqrt{\zeta} = \operatorname{Re}\sqrt{\xi + \mathrm{i}\eta} = \sqrt{\frac{\xi + \sqrt{\xi^2 + \eta^2}}{2}}.$$

This last equality arises from

$$\sqrt{\alpha + \mathrm{i}\beta} = \pm\left(\sqrt{\frac{\alpha + \sqrt{\alpha^2 + \beta^2}}{2}} + \mathrm{i}\frac{\beta}{|\beta|}\sqrt{\frac{-\alpha + \sqrt{\alpha^2 + \beta^2}}{2}}\right)$$

for $\beta \neq 0$. The inequality $\xi \leqslant 1/4 - \eta^2$ shows that

$$\xi^2 + \eta^2 \leqslant \left(\frac{1}{4} - \eta^2\right)^2 + \eta^2 = \left(\frac{1}{4} + \eta^2\right)^2,$$

and hence

$$\xi + \sqrt{\xi^2 + \eta^2} \leqslant \left(\frac{1}{4} - \eta^2\right) + \left(\frac{1}{4} + \eta^2\right) = \frac{1}{2}.$$

Thus, $\operatorname{Re}\sqrt{\zeta} \leqslant 1/2$ which is exactly the admissibility condition given in (7.6.9). Hence, $\psi \in \mathcal{P}_1(\Omega)$. Since $\psi\big(p(z), zp'(z)\big) \in \Omega$, we conclude from Theorem 5.4.2 that $p \in \mathcal{P}$, or equivalently,

$$\operatorname{Re}\frac{f(z)}{z} > \frac{1}{2} \quad \text{or} \quad \frac{f(z)}{z} \prec \frac{1}{1-z}. \qquad \square$$

The bounds for the coefficients of a convex function can be derived by a simple application of the Marx-Strohhäcker theorem.

Corollary 7.6.2 *If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{CV}$, then $|a_k| \leqslant 1$ for all $k \geqslant 2$.*

Proof The Marx-Strohhäcker theorem yields

$$\operatorname{Re}\frac{f(z)}{z} > \frac{1}{2}.$$

Since $f(z)/z = 1 + \sum_{k=1}^{\infty} a_{k+1} z^k$, Theorem 4.5.1 yields $|a_{k+1}| \leqslant 1$ for all $k \geqslant 1$. $\square$

The Marx-Strohhäcker theorem also implies that $\mathcal{CV} \subset \mathcal{ST}(1/2)$. The following corollary gives the radius of starlikeness of order α, $1/2 < \alpha < 1$.

Corollary 7.6.3 *The radius of starlikeness of order α, $1/2 < \alpha < 1$ for the class $\mathcal{CV}$ is $(1-\alpha)/\alpha$.*

Proof Since every convex function is starlike of order 1/2, by Theorem 6.6.10, the radius of starlikeness of order α is at least $(1-\alpha)/\alpha$. For sharpness, consider the function f_0 defined by $f_0(z) = z/(1+z)$. $\square$

A function $f \in \mathcal{A}$ is said to be *strongly starlike of order α*, $0 < \alpha \leqslant 1$ if

$$\left|\arg\frac{zf'(z)}{f(z)}\right| < \frac{\alpha\pi}{2}, \quad z \in \mathbb{D}.$$

As another application of the Marx-Strohhäcker theorem, the radius of strongly starlikeness of convex functions is derived.

Corollary 7.6.4 *The radius of strongly starlikeness of order α, $0 < \alpha \leqslant 1$ for the class $\mathcal{CV}$ is $\sin(\alpha\pi/2)$.*

Proof The Marx-Strohhäcker theorem gives $f \in \mathcal{ST}(1/2)$, and by Corollary 4.3.5,

$$\left|\frac{zf'(z)}{f(z)} - \frac{1}{1-r^2}\right| \leqslant \frac{r}{1-r^2}, \quad |z| = r.$$

Hence,

$$\left|\arg \frac{zf'(z)}{f(z)}\right| \leqslant \sin^{-1} r < \frac{\alpha\pi}{2}$$

for $0 \leqslant r < \sin(\alpha\pi/2)$. (See the proof of Theorem 6.6.6).

For sharpness, consider the function f_0 defined by $f_0(z) = z/(1+z)$. □

The following theorem presents a further extension of the Marx-Strohhäcker result stated in Theorem 7.6.1, namely,

$$\operatorname{Re}\left(\frac{zf''(z)}{f'(z)} + 1\right) > 0 \quad \text{implies} \quad \operatorname{Re}\left(\frac{zf'(z)}{f(z)}\right) > \frac{1}{2}$$

by relaxing the condition of convexity.

Theorem 7.6.5 *If $f \in \mathcal{A}_2$ and*

$$\operatorname{Re}\left(\frac{zf''(z)}{f'(z)} + 1\right) > -\frac{1}{2}, \quad z \in \mathbb{D}, \tag{7.6.10}$$

then

$$\operatorname{Re}\frac{zf'(z)}{f(z)} > \frac{1}{2}, \quad z \in \mathbb{D}.$$

The result is sharp and the extremal function is

$$f(z) = \frac{z}{\sqrt{1-z^2}}.$$

Proof By Corollary 8.2.4 (in the next chapter), the function f satisfying the condition (7.6.10) is univalent (close-to-convex). Since $f(z) = z + a_3 z^3 + \cdots$, the function $p : \mathbb{D} \to \mathbb{C}$ defined by

$$p(z) := 2\frac{zf'(z)}{f(z)} - 1 = 1 + 4a_3 z^2 + \cdots$$

belongs to $\mathcal{H}[1, 2]$. By using (7.6.4) and (7.6.10),

$$\frac{zf''(z)}{f'(z)} + 1 = \frac{p(z)+1}{2} + \frac{zp'(z)}{p(z)+1} = \psi\big(p(z), zp'(z)\big) \in \Omega,$$

where

$$\psi(r,s) = \frac{r+1}{2} + \frac{s}{r+1} \quad \text{and} \quad \Omega = \left\{ w : \operatorname{Re} w > -\frac{1}{2} \right\}.$$

The function ψ satisfies

$$\operatorname{Re} \psi(\mathrm{i}\rho, \sigma) = \operatorname{Re} \left(\frac{\mathrm{i}\rho + 1}{2} + \frac{\sigma}{1+\rho^2}(1 - \mathrm{i}\rho) \right) = \frac{1}{2} + \frac{\sigma}{1+\rho^2} \leqslant \frac{1}{2} - 1 = -\frac{1}{2},$$

or $\psi(\mathrm{i}\rho, \sigma) \notin \Omega$ whenever $\rho \in \mathbb{R}$ and $\sigma \leqslant -(1+\rho^2)$. Thus, $\psi \in \mathcal{P}_2(\Omega)$. Since $\psi\big(p(z), zp'(z)\big) \in \Omega$, Theorem 5.4.2 shows that $p \in \mathcal{P}$. This is equivalent to

$$\operatorname{Re} \frac{zf'(z)}{f(z)} > \frac{1}{2}.$$

If $f(z) = z/\sqrt{1-z^2}$, then $f \in \mathcal{A}_2$,

$$\operatorname{Re} \left(1 + \frac{zf''(z)}{f'(z)} \right) = \operatorname{Re} \left(\frac{1+2z^2}{1-z^2} \right) > -\frac{1}{2}$$

and

$$\operatorname{Re} \left(\frac{zf'(z)}{f(z)} \right) = \operatorname{Re} \left(\frac{1}{1-z^2} \right) > \frac{1}{2}.$$

Since the image of $1/(1-z^2)$ is the half-plane $\{w \in \mathbb{C} : \operatorname{Re} w > 1/2\}$, the result is sharp. □

7.7 Sufficient Conditions for Convexity

For a function $f \in \mathcal{A}$, the Noshiro-Warschawski Theorem states that the condition $\operatorname{Re}(f'(z)) > 0$ is sufficient to ensure univalence of f. This section presents several simple sufficient conditions for convexity of functions in the class $\mathcal{A}$.

Theorem 7.7.1 *Let $0 \leqslant \alpha < 1$. If $f \in \mathcal{A}$ satisfies $|f''(z)| < (1-\alpha)/(2-\alpha)$ for all $z \in \mathbb{D}$, then $f \in \mathcal{CV}_\alpha$, and hence $f \in \mathcal{CV}(\alpha)$. The constant $(1-\alpha)/(2-\alpha)$ is best possible.*

Proof A computation shows that

$$|f'(z) - 1| \leqslant \left| \int_0^1 zf''(zt)\mathrm{d}t \right| \leqslant \frac{1-\alpha}{2-\alpha}|z| < \frac{1-\alpha}{2-\alpha},$$

and hence

$$\frac{1}{2-\alpha} = 1 - \frac{1-\alpha}{2-\alpha} < |f'(z)|.$$

Thus,

$$|f''(z)| < \frac{1-\alpha}{2-\alpha} < (1-\alpha)|f'(z)|.$$

This shows that $|zf''(z)/f'(z)| < 1-\alpha$, and hence $f \in \mathcal{CV}_\alpha$. Since $\mathcal{CV}_\alpha \subset \mathcal{CV}(\alpha)$, it follows that $f \in \mathcal{CV}(\alpha)$.

Consider the function f given by $f(z) = z + kz^2/2$, with $k > (1-\alpha)/(2-\alpha)$. Then $f''(z) = k$ and, at $z = -1$,

$$1 + \frac{zf''(z)}{f'(z)} = \frac{1+2kz}{1+kz} = \frac{1-2k}{1-k} < \alpha,$$

and

$$\left|\frac{zf''(z)}{f'(z)}\right| = \frac{k}{1-k} > 1-\alpha.$$

This shows that the constant $(1-\alpha)/(2-\alpha)$ is best possible for both radii. □

Corollary 7.7.2 *If $f \in \mathcal{A}$ satisfies $|f''(z)| < 1/2$ for all $z \in \mathbb{D}$, then $f \in \mathcal{CV}$ and*

$$\left|\frac{zf''(z)}{f'(z)}\right| < 1, \quad z \in \mathbb{D}.$$

The constant $1/2$ is best possible.

Recall that a function $f \in \mathcal{A}$ is α-convex (or Mocanu convex), $\alpha \in \mathbb{R}$, if it satisfies the inequality

$$\operatorname{Re}\left((1-\alpha)\frac{zf'(z)}{f(z)} + \alpha\left(1 + \frac{zf''(z)}{f'(z)}\right)\right) > 0, \quad z \in \mathbb{D}.$$

Theorem 7.7.3 *Every α-convex function is convex for $\alpha \geqslant 1$.*

Proof By Theorem 6.7.6, every α-convex function f is starlike. For $\alpha \geqslant 1$, it follows that

$$\alpha \operatorname{Re}\left(1 + \frac{zf''(z)}{f'(z)}\right) \geqslant (\alpha - 1) \operatorname{Re}\frac{zf'(z)}{f(z)} \geqslant 0,$$

and this proves the convexity of f. □

Theorem 7.7.4 *Let $f \in \mathcal{A}$ with $f(z)f'(z) \neq 0$, for $z \in \mathbb{D} \setminus \{0\}$, and let $3/4 \leqslant \beta < 1$. If*

$$\left|\frac{f''(z)}{f'(z)}\right| \leqslant \frac{2(1-\beta)}{\beta}\left|\frac{zf'(z)}{f(z)}\right|, \quad z \in \mathbb{D}, \tag{7.7.1}$$

then $f \in \mathcal{CV}$ and $\left|zf''(z)/f'(z)\right| < 1$, $z \in \mathbb{D}$. The constant $\frac{3}{4}$ cannot be replaced by a smaller value.

Proof Theorem 6.7.8 shows that

$$\frac{zf'(z)}{f(z)} \prec q(z),$$

where

$$q(z) = \frac{\beta}{\beta - (1-\beta)z} = \frac{1}{1-\alpha z}, \quad \alpha = \frac{1-\beta}{\beta}.$$

It follows from (6.7.9) that

$$|q(z)| \leqslant \frac{1}{1-\alpha^2 r^2} + \frac{\alpha r}{1-\alpha^2 r^2} = \frac{1}{1-\alpha r} \leqslant \frac{1}{1-\alpha} = \frac{\beta}{2\beta - 1}.$$

Hence,

$$\left|\frac{zf'(z)}{f(z)}\right| \leqslant \frac{\beta}{2\beta - 1}.$$

For $\beta \geqslant 3/4$, it follows from (7.7.1) that

$$\left|\frac{zf''(z)}{f'(z)}\right| \leqslant \frac{2(1-\beta)}{\beta}|z|\frac{\beta}{2\beta-1} \leq |z| < 1,$$

and hence $f \in \mathcal{CV}$.

For sharpness, consider the function

$$f(z) = z\left[1 - \frac{(1-\beta)}{\beta}z^3\right]^{-1/3}.$$

Then,

$$1 + \frac{zf''(z)}{f'(z)} = \frac{\beta + 3(1-\beta)z^3}{\beta - (1-\beta)z^3}$$

so that $f \in \mathcal{CV}$ if $3/4 \leqslant \beta < 1$. If $0 < \beta < 3/4$ and $z = -r$, where $\beta/(3(1-\beta)) < r^3 < 1$, then

$$\mathrm{Rc}\left(1 + \frac{zf''(z)}{f'(z)}\right) = \frac{\beta - 3(1-\beta)r^3}{\beta + (1-\beta)r^3} < 0,$$

which implies $f \notin \mathcal{CV}$. □

Exercises

1. If $f \in \mathcal{CV}$, show that $F(z) = z(f'(z))^\alpha$ belongs to $\mathcal{ST}(1-\alpha)$.
2. If $f \in \mathcal{CV}(\alpha)$, show that the ratio of the angle between two adjacent tangents to the unit circle to the angle between the two corresponding tangents to the image of the unit circle is less than $1/\alpha$.

3. Show that $f \in \mathcal{CV}(\alpha), 0 \leqslant \alpha < 1$, if and only if there is a non-decreasing function μ on $[0, 2\pi]$ with $\int_0^{2\pi} \mathrm{d}\mu(t) = 1$ such that

$$f'(z) = \exp\left(-2(1-\alpha)\int_0^{2\pi} \log(1 - \mathrm{e}^{-\mathrm{i}t}z)\mathrm{d}\mu(t)\right).$$

4. Let f be a convex mapping with Schwarzian norm

$$\|Sf\| = \sup_{z\in\mathbb{D}}(1-|z|^2)^2|Sf(z)| = 2,$$

and suppose that $(1-|z|^2)^2|Sf(z)| = 2$ for some point $z \in \mathbb{D}$. Show that f maps the disk onto a parallel strip.

5. If $f \in \mathcal{CV}(\alpha)$, show that the sharp inequality

$$\max_{\theta} \rho(r,\theta) \leqslant \frac{r}{(1-r^2)^{1-\alpha}}$$

holds.

6. For a convex function f and $\lambda \geqslant 0$, show that

$$\frac{1}{2\pi}\int_0^{2\pi} |f(r\mathrm{e}^{\mathrm{i}\theta})|^{\lambda}\mathrm{d}\theta \leqslant \frac{r^{\lambda}}{2\pi}\int_0^{2\pi} |1 + r\mathrm{e}^{\mathrm{i}\theta}|^{-\lambda}\mathrm{d}\theta.$$

7. Show that the radius of convexity of starlike functions of order $1/2$ is $\sqrt{2\sqrt{3}-3}$.
8. Show that the radius of convexity of order β of the class $\mathcal{CV}(\alpha)$ is $(1-\beta)/(1+\beta-2\alpha)$, $\beta > \alpha$.
9. If a function f is odd convex, show that $F(z) = f^2(\sqrt{z})$ is a convex function of order 1/4. If a function is odd convex of order $-1/2$, show that the function F is convex.
10. If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{CV}$, show that the sharp inequality

$$|a_{n+1} - a_n| \leqslant (2n+1)|1 - a_2|/3$$

holds for $n \geqslant 2$.

11. If $f \in \mathcal{CV}(\alpha)$, $0 \leqslant \alpha < 1$, show that the sharp inequality

$$\left|H_{2,2}(f)\right| \leqslant \frac{(1-\alpha)^2(6+5\alpha)}{48(1+\alpha)}$$

holds.

12. If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{CV}$, show that the second Hankel determinant is bounded by 1/8, that is, $|a_2a_4 - a_3^2| \leqslant 1/8$. Show that the inequality is sharp.

13. If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{CV}$, show that $|a_2 a_3 - a_4| \leqslant 1/6$. Hence show that $|H_3(1)| \leqslant 15/24$. Show that these inequalities are sharp.
14. For $f \in \mathcal{CV}$, the inverse f^{-1} exists in some disk containing $\mathbb{D}_{1/2}$. Show that the first seven coefficients of f^{-1} are bounded by 1.
15. By using the fact that $\mathrm{Re}(f(z)/z) > 1/2$ for $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{CV}$, show that $|a_n| \leqslant 1$.
16. Find the bounds for the logarithmic coefficients γ_n of $f \in \mathcal{CV}$ for $n \geqslant 1$.
17. Show that the order of starlikeness of order β of convex functions of order α is given by

$$\beta = \begin{cases} \frac{1-2\alpha}{2^{2(1-\alpha)}-2} & (\alpha \neq \frac{1}{2}), \\ \frac{1}{2\log 2} & (\alpha = \frac{1}{2}). \end{cases}$$

18. Obtain the bound for $|a_3 - \mu a_2^2|$ when $f(z) = z + \sum_{k=2}^{\infty} a_k z^k$ is a convex/starlike function of order α.

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[1] The proof of Theorem 7.2.9 is taken from [1]. Section 7.6 on the Marx-Strohhäcker Theorem is based on the monograph by Miller and Mocanu. The conclusion of Theorem 7.7.1 is stronger than that of [7, Corollary 3]. Theorem 7.7.4 is from [6]. The section on curvature is based on the works of [2] and [3]. The proof of Theorem 7.2.9 and Exercise 4 are reproduced from [1]. Exercise 5 is from [3] while Exercise 7 is from [4].

Chapter 8
Close-to-Convex Functions

Abstract A function $f \in \mathcal{A}$ is said to be close-to-convex if there exists a function $g \in \mathcal{CV}$ and $\alpha \in (-\pi/2, \pi/2)$ such that $\mathrm{Re}\left(e^{i\alpha} f'(z)/g'(z)\right) > 0$ for all $z \in \mathbb{D}$. Every close-to-convex function is shown to be univalent. A characterization involving only the function f is also established. Several subclasses of close-to-convex functions are duly investigated.

8.1 Close-to-Convex Functions

First, we establish another sufficient condition for the univalence of an analytic function by applying the Noshiro-Warschawski theorem.

Theorem 8.1.1 *If functions f and g are analytic in a domain G with g convex satisfying*

$$\mathrm{Re}\left(e^{i\alpha}\frac{f'(z)}{g'(z)}\right) > 0, \quad z \in G,$$

for some $\alpha \in (-\pi/2, \pi/2)$, then f is univalent in G.

Proof The range $G' = g(G)$ of g is a convex set. Let $h(w) = f(g^{-1}(w))$ in G'. Then,

$$h'(w) = \frac{f'(g^{-1}(w))}{g'(g^{-1}(w))} = \frac{f'(z)}{g'(z)}, \quad z = g^{-1}(w) \in G.$$

Thus, $\mathrm{Re}(e^{i\alpha}h'(w)) > 0$ for all $w \in G'$. By the Noshiro-Warschawski theorem (Theorem 2.1.6), h is univalent. Since $h(w) = f(g^{-1}(w))$, then $f(z) = h(g(z))$. As a composition of two univalent functions, f is therefore univalent. □

Theorem 8.1.1 is the Noshiro-Warschawski theorem when $g(z) = z$. In this case, the assumption that g is convex implies that $G = g(G)$ is convex. When $G = \mathbb{D}$, functions satisfying the hypothesis of Theorem 8.1.1 are referred to as close-to-convex.

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Definition 8.1.2 A function $f \in \mathcal{A}$ is said to be *close-to-convex* if there exists a convex function $g \in \mathcal{A}$ and a real number $\alpha \in (-\pi/2, \pi/2)$ such that

$$\operatorname{Re}\left(e^{i\alpha}\frac{f'(z)}{g'(z)}\right) > 0, \quad z \in \mathbb{D}. \tag{8.1.1}$$

In this case, f is said to be close-to-convex with respect to the convex function g. The class of all such functions is denoted by CC. For a fixed $\alpha \in (-\pi/2, \pi/2)$, the class CC_α consists of all functions $f \in CC$ satisfying (8.1.1).

The concept of close-to-convexity was first introduced by Kaplan [7]. Theorem 8.1.1 for the case $G = \mathbb{D}$ shows that $CC \subset \mathcal{S}$.

Corollary 8.1.3 *Every close-to-convex function is univalent.*

The factor $e^{i\alpha}$ in the definition of close-to-convexity cannot be omitted as illustrated by the following example.

Example 8.1.4 Let $0 < \beta < \pi$. Further, let $f, g : \mathbb{D} \to \mathbb{C}$ be given by

$$f(z) = \frac{z - \cos\beta e^{2i\beta}z^2}{(1 - e^{i\beta}z)^2}, \quad \text{and} \quad g(z) = \frac{z}{1 - e^{i\beta}z}.$$

Then, the function g is convex and

$$f'(z) = \frac{1 - e^{3i\beta}z}{\left(1 - e^{i\beta}z\right)^3}, \quad \text{and} \quad g'(z) = \frac{1}{(1 - e^{i\beta}z)^2}.$$

Therefore, with $\alpha = \pi/2 - \beta$ and $|z| = 1$,

$$\begin{aligned}
\operatorname{Re}\left(e^{i\alpha}\frac{f'(z)}{g'(z)}\right) &= \operatorname{Re}\left(ie^{-i\beta}\frac{1 - e^{3i\beta}z}{\left(1 - e^{i\beta}z\right)^3}\left(1 - e^{i\beta}z\right)^2\right) \\
&= -\operatorname{Im}\frac{e^{-i\beta} - e^{2i\beta}z}{1 - e^{i\beta}z} \\
&= -\operatorname{Im}\frac{\overline{\gamma} + \gamma|z|^2}{|1 - \gamma z|^2} \quad (\gamma = e^{i\beta}) \\
&= 0.
\end{aligned}$$

It then follows that

$$\operatorname{Re}\left(e^{i\alpha}\frac{f'(z)}{g'(z)}\right) > 0, \quad z \in \mathbb{D}.$$

Therefore, the function f belongs to the class CC_α. Since $e^{i\alpha}f'(z)/g'(z)$ maps $\mathbb{D}$ onto the right-half plane, f does not belong to CC_β for $\beta \neq \alpha$. In view of this, it is clear that the factor $e^{i\alpha}$ cannot be omitted in the definition of the class CC.

Example 8.1.5 Every convex function is close-to-convex; indeed, if $f \in \mathcal{CV}$, then

$$\operatorname{Re} \frac{f'(z)}{f'(z)} = 1 > 0.$$

Therefore, f satisfies the condition of close-to-convexity with $\alpha = 0$, and hence $f \in \mathcal{CC}_0$. The next example shows that every starlike function is close-to-convex.

Example 8.1.6 Every starlike function is close-to-convex; indeed, $\mathcal{ST} \subset \mathcal{CC}_0$.

Let $f \in \mathcal{ST}$. By Alexander's Theorem, there exists a function $g \in \mathcal{CV}$ such that $f(z) = zg'(z)$, and hence

$$\operatorname{Re}\left(\frac{f'(z)}{g'(z)}\right) = \operatorname{Re}\left(\frac{zf'(z)}{f(z)}\right) > 0.$$

Therefore, $f \in \mathcal{CC}_0$.

Summarizing, the following chain of inclusions holds among the standard subclasses of univalent functions:

$$\mathcal{CV} \subset \mathcal{ST} \subset \mathcal{CC}_0 \subset \mathcal{CC} \subset \mathcal{S}.$$

Example 8.1.7 The class $\mathcal{R}$ of all functions $f \in \mathcal{A}$ satisfying $\operatorname{Re} f'(z) > 0$ in $\mathbb{D}$ is a subclass of $\mathcal{CC}_0$, that is, $\mathcal{R} \subset \mathcal{CC}_0$.

Let $f \in \mathcal{R}$, and $g(z) = z$. Then, $g \in \mathcal{CV}$, and

$$\operatorname{Re}\left(\frac{f'(z)}{g'(z)}\right) = \operatorname{Re} f'(z) > 0.$$

Therefore, $f \in \mathcal{CC}_0$.

Example 8.1.8 If $f \in \mathcal{A}$ satisfies $\operatorname{Re}((1-z)^2 f'(z)) > 0$ for all $z \in \mathbb{D}$, then $f \in \mathcal{CC}_0$.

To see this, let $g(z) = z/(1-z)$. Then, $g \in \mathcal{CV}$, $g'(z) = 1/(1-z)^2$, and

$$\operatorname{Re}\left(\frac{f'(z)}{g'(z)}\right) = \operatorname{Re}((1-z)^2 f'(z)) > 0.$$

Therefore, $f \in \mathcal{CC}_0$.

In view of the Alexander's Theorem, an equivalent definition of close-to-convex functions is the following: a function $f \in \mathcal{A}$ is close-to-convex if there exists a starlike function $h \in \mathcal{A}$ and a real number $\alpha \in (-\pi/2, \pi/2)$ such that

$$\operatorname{Re}\left(e^{i\alpha} \frac{zf'(z)}{h(z)}\right) > 0, \quad z \in \mathbb{D}. \tag{8.1.2}$$

Example 8.1.9 Let $|\alpha_k| = 1$, $0 \leqslant \beta_k \leqslant 1$ for $k = 1, 2, \ldots, n$ and $\sum_{k=1}^{n} \beta_k \leqslant 2$. If $f \in \mathcal{A}$ satisfies

$$\operatorname{Re}\left(\prod_{k=1}^{n}(z-\alpha_k)^{\beta_k} f'(z)\right) > 0, \quad z \in \mathbb{D},$$

then $f \in CC_0$.

To see this, let

$$h(z) = \frac{z}{\prod_{k=1}^{n}(z-\alpha_k)^{\beta_k}}.$$

Then,

$$\frac{zh'(z)}{h(z)} = 1 - \sum_{k=1}^{n} \beta_k \frac{z}{z-\alpha_k}.$$

Since

$$\operatorname{Re}\frac{z}{z-\alpha_k} = \frac{1}{2} + \frac{1}{2}\operatorname{Re}\frac{z+\alpha_k}{z-\alpha_k} = \frac{1}{2} + \frac{1}{2}\frac{|z|^2-1}{|z-\alpha_k|^2} < \frac{1}{2},$$

then

$$\operatorname{Re}\frac{zh'(z)}{h(z)} > 1 - \sum_{k=1}^{n}\frac{\beta_k}{2} \geqslant 0.$$

Hence, h is starlike and $f \in CC_0$.

In particular, for $n = 2$, $\beta_1 = \beta_2 = 1$, the following holds: if $f \in \mathcal{A}$ satisfies $\operatorname{Re}((z-\alpha_1)(z-\alpha_2)f'(z)) > 0$ for all $z \in \mathbb{D}$, where $|\alpha_1| = |\alpha_2| = 1$, then $f \in CC_0$. The case $\alpha_1 = \alpha_2 = 1$ corresponds to Example 8.1.8.

Example 8.1.10 Suppose $g \in \mathcal{CV}$. Then the function $h(z) = \log g'(z)$ satisfies

$$\operatorname{Re}\left(1 + zh'(z)\right) > 0, \quad z \in \mathbb{D}.$$

Accordingly, a function f belongs to CC if and only if there exists an analytic function h in $\mathbb{D}$ such that $\operatorname{Re}\left(1 + zh'(z)\right) > 0$ and

$$\operatorname{Re}\left(\mathrm{e}^{\mathrm{i}\alpha - h(z)} f'(z)\right) > 0$$

for some $\alpha \in (-\pi/2, \pi/2)$.

Example 8.1.11 Functions $f, g \in \mathcal{A}$ satisfying

$$\operatorname{Re}\left(\frac{zf'(z)}{f(z)+g(z)}\right) > 0, \quad \text{and} \quad \operatorname{Re}\left(\frac{zg'(z)}{f(z)+g(z)}\right) > 0, \tag{8.1.3}$$

are called *mutually adjoint*. Such functions are close-to-convex and therefore univalent.

This follows by letting $h(z) = (f(z) + g(z))/2$. Then $h \in \mathcal{A}$ and the conditions (8.1.3) become

$$\operatorname{Re}\left(\frac{zf'(z)}{h(z)}\right) > 0, \quad \text{and} \quad \operatorname{Re}\left(\frac{zg'(z)}{h(z)}\right) > 0.$$

Summing these inequalities yields

$$\operatorname{Re}\left(\frac{zh'(z)}{h(z)}\right) = \frac{1}{2}\operatorname{Re}\left(\frac{zf'(z)}{h(z)} + \frac{zg'(z)}{h(z)}\right) > 0.$$

Therefore, the function h is starlike. Since f satisfies (8.1.2), the function f is close-to-convex and hence univalent. Similarly, the function g is close-to-convex and univalent.

Example 8.1.12 A function $f \in \mathcal{A}$ satisfying

$$\operatorname{Re}\left(\frac{zf'(z)}{f(z) - f(-z)}\right) > 0, \quad z \in \mathbb{D}, \tag{8.1.4}$$

is called *starlike with respect to symmetric points*. Every such function is close-to-convex and hence univalent.

To see this, let $h(z) = (f(z) - f(-z))/2$. Replacing z by $-z$ in inequality (8.1.4) and adding the resulting inequality to (8.1.4) leads to $\operatorname{Re}(zh'(z)/h(z)) > 0$. Since h is starlike, there exists a convex function g such that $h = zg'$. Then, (8.1.4) gives $\operatorname{Re}(f'(z)/g'(z)) > 0$. Therefore, f is close-to-convex and univalent.

Similarly, a function $f \in \mathcal{A}$ satisfying either

$$\operatorname{Re}\left(\frac{zf'(z)}{f(z) + \overline{f(\overline{z})}}\right) > 0 \quad \text{or} \quad \operatorname{Re}\left(\frac{zf'(z)}{f(z) - \overline{f(-\overline{z})}}\right) > 0$$

is close-to-convex and univalent. These functions are known, respectively, as starlike with respect to conjugate points and starlike with respect to symmetric conjugate points.

Since close-to-convex functions are univalent, the distortion, growth and covering theorems for univalent functions are applicable to close-to-convex functions as well. Sharpness of these results also carries over, as the Koebe function belongs to the class of close-to-convex functions.

Coefficient bounds for the class CC is an immediate consequence of de Branges' theorem (formerly Bieberbach conjecture). Since the proof of de Branges' theorem is omitted, a direct proof of this result is provided here.

Theorem 8.1.13 *If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in CC$, then $|a_k| \leqslant k$ for $k \geqslant 2$. Equality holds if and only if the function f is a rotation of the Koebe function.*

Proof There exists a convex function $g \in \mathcal{A}$ and $\alpha \in (-\pi/2, \pi/2)$ satisfying

$$\operatorname{Re}\left(e^{i\alpha}\frac{f'(z)}{g'(z)}\right) > 0.$$

Let

$$q(z) = e^{i\alpha}\frac{f'(z)}{g'(z)} = \sum_{k=0}^{\infty} q_k z^k. \tag{8.1.5}$$

Then, $q_0 = e^{i\alpha}$ and $\operatorname{Re} q(z) > 0$. Now Theorem 4.5.1 gives $|q_k| \leqslant 2\cos\alpha$ for $k \geqslant 1$.

Let $g(z) = z + \sum_{k=1}^{\infty} b_k z^k$ so that $|b_k| \leqslant 1$ for $k \geqslant 2$. From (8.1.5), $e^{i\alpha} f'(z) = g'(z)q(z)$, and comparing the coefficients of the corresponding power series on both sides then yields

$$e^{i\alpha} k a_k = k b_k q_0 + (k-1)b_{k-1}q_1 + (k-2)b_{k-2}q_2 + \cdots + 2b_2 q_{k-2} + q_{k-1}. \tag{8.1.6}$$

Using the bounds $|b_k| \leqslant 1$ for $k = 2, 3, \ldots$, $|q_k| \leqslant 2\cos\alpha \leqslant 2$ for $k \geqslant 1$ and $|q_0| = 1$, the following estimate is obtained:

$$k|a_k| \leqslant k + 2(k-1) + 2(k-2) + \cdots + 2.2 + 2.1 = k^2.$$

Hence, $|a_k| \leqslant k$ for $k \geqslant 2$.

Equality $|a_k| = k$ is attained for every rotation of the Koebe function. Furthermore, if $|a_k| = k$ holds for any fixed $k \geqslant 2$, then $|a_2| = 2$, which in turn implies that f must be a rotation of the Koebe function. □

Corollary 8.1.14 *If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in CC_\alpha$, then*

$$|a_k| \leqslant 1 + (k-1)\cos\alpha, \quad k \geqslant 2.$$

Proof By using (8.1.6) together with $|q_k| \leqslant 2\cos\alpha$,

$$k|a_k| \leqslant k + 2\cos\alpha((k-1) + (k-2) + \cdots + 2 + 1) = k + k(k-1)\cos\alpha,$$

and hence $|a_k| \leqslant 1 + (k-1)\cos\alpha$. □

Corollary 8.1.14 is sharp only for $k = 2$.

8.2 Kaplan's Theorem

This section examines close-to-convex functions in greater detail. By Definition 8.1.2, a normalized analytic function f belongs to the class CC of close-to-convex functions if there exists a convex function h (not necessarily normalized by $h'(0) = 1$) such that

$$\operatorname{Re}\left(\frac{f'(z)}{h'(z)}\right) > 0, \quad z \in \mathbb{D}.$$

If, in addition, the convex function h belongs to $\mathcal{A}$, then the corresponding subclass is denoted by CC_0.

Theorem 8.2.1 *If $f \in CC$, then*

$$\int_{\theta_1}^{\theta_2} \operatorname{Re}\left(1 + \frac{zf''(z)}{f'(z)}\right) d\theta > -\pi, \quad z = re^{i\theta}, \tag{8.2.1}$$

for each r $(0 < r < 1)$ and for each pair of real numbers θ_1 and θ_2 with $0 \leqslant \theta_2 - \theta_1 \leqslant 2\pi$.

Proof Let $f \in CC$ and let h be the associated convex function such that

$$\left|\arg\left(\frac{f'(z)}{h'(z)}\right)\right| < \frac{\pi}{2}.$$

Since $f'(z) \neq 0$ and $h'(z) \neq 0$, $z \in \mathbb{D}$, branches of the arguments may be chosen so that

$$\left|\arg f'(z) - \arg h'(z)\right| < \frac{\pi}{2}, \quad z \in \mathbb{D}. \tag{8.2.2}$$

Let $a(z) = \arg f'(z)$ and $b(z) = \arg h'(z)$. For $z = re^{i\theta}$, $0 < r < 1$, $\theta \in \mathbb{R}$, let

$$V(r, \theta) = a(re^{i\theta}) + \theta = \arg\left(re^{i\theta} f'(re^{i\theta})\right)$$

and

$$W(r, \theta) = b(re^{i\theta}) + \theta = \arg\left(re^{i\theta} h'(re^{i\theta})\right).$$

Then, (8.2.2) becomes

$$|V(r, \theta) - W(r, \theta)| < \frac{\pi}{2}.$$

Since h is a convex function, $W(r, \theta)$ is an increasing harmonic function of θ, and hence the partial derivative $W_\theta(r, \theta) > 0$. From this observation with the preceding inequality, it follows that for $0 \leqslant \theta_2 - \theta_1 \leqslant 2\pi$,

$$\begin{aligned} V(r, \theta_1) - V(r, \theta_2) &= (V(r, \theta_1) - W(r, \theta_1)) \\ &\quad + (W(r, \theta_1) - W(r, \theta_2)) \\ &\quad + (W(r, \theta_2) - V(r, \theta_2)) \leqslant (V(r, \theta_1) - W(r, \theta_1)) \\ &\quad + (W(r, \theta_2) - V(r, \theta_2)) < \pi. \end{aligned} \tag{8.2.3}$$

At $z = re^{i\theta}$,

$$\operatorname{Re}\left(1+\frac{zf''(z)}{f'(z)}\right) = \frac{\partial}{\partial\theta}\left(\arg\left(\frac{\partial}{\partial\theta}f(re^{i\theta})\right)\right) = \frac{\partial}{\partial\theta}\left(\arg\left(ire^{i\theta}f'(re^{i\theta})\right)\right).$$

It follows, using (8.2.3), that

$$\begin{aligned}\int_{\theta_1}^{\theta_2}\operatorname{Re}\left(1+\frac{zf''(z)}{f'(z)}\right)d\theta &= \int_{\theta_1}^{\theta_2}\frac{\partial}{\partial\theta}\left(\arg\left(\frac{\partial}{\partial\theta}f(re^{i\theta})\right)\right)d\theta\\ &= \arg\left(re^{i\theta_2}f'\left(re^{i\theta_2}\right)\right) - \arg\left(re^{i\theta_1}f'\left(re^{i\theta_1}\right)\right)\\ &= V(r,\theta_2) - V(r,\theta_1) > -\pi.\end{aligned}$$

This proves (8.2.1) . □

Close-to-convex functions can be characterized by a geometric condition somewhat similar to the defining properties of convex and starlike functions. Let f be analytic in $\mathbb{D}$ and let C_r be the image under f of the circle $|z| = r$, where $0 < r < 1$. Let T_1 and T_2 be tangent vectors to C_r at the points $f(re^{i\theta_1})$ and $f(re^{i\theta_2})$, respectively. If f is close-to-convex, then the curve C_r does not make a "reverse hairpin" turn (see Figs. 8.1 and 8.2) or, in other words, as θ increases, the tangent vector cannot turn backward through an angle larger than or equal to π or

$$\arg T_2 - \arg T_1 = V(r,\theta_2) - V(r,\theta_1) > -\pi, \quad 0 \leqslant \theta_2 - \theta_1 \leqslant 2\pi.$$

The converse is also true, and its proof requires the following lemma.

Lemma 8.2.2 *Let φ be a real-valued continuous function on $(-\infty,\infty)$, with the properties $\varphi(t+2\pi) = \varphi(t) + 2\pi$ and*

$$\varphi(t_2) - \varphi(t_1) > -\pi, \quad t_1 < t_2 .$$

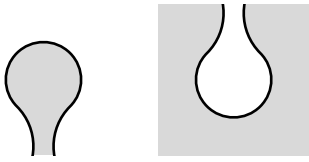


Fig. 8.1 Domains, respectively, with hairpin and reverse hairpin bend

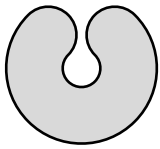


Fig. 8.2 A domain that is not close-to-convex

Then there is a continuous non-decreasing function ψ such that $\psi(t+2\pi) = \psi(t) + 2\pi$ and $|\varphi(t) - \psi(t)| \leqslant \pi/2$.

Proof Consider the function

$$\psi(t) = \sup_{s \leqslant t} \varphi(s) - \frac{\pi}{2}.$$

The function ψ is continuous and non-decreasing on $\mathbb{R}$. Since $\varphi(t+2\pi) = \varphi(t) + 2\pi$, it follows that

$$\psi(t+2\pi) = \sup_{s \leqslant t} \varphi(s+2\pi) - \frac{\pi}{2} = \psi(t) + 2\pi.$$

For any $s < t$, the inequality $\varphi(s) < \varphi(t) + \pi$ implies

$$\psi(t) \leqslant \varphi(t) + \frac{\pi}{2}.$$

Thus, the estimate

$$\varphi(t) - \frac{\pi}{2} \leqslant \sup_{s \leqslant t} \varphi(s) - \frac{\pi}{2} = \psi(t) \leqslant \varphi(t) + \frac{\pi}{2}$$

holds. □

Theorem 8.2.3 (Kaplan's Theorem) *Let $f \in \mathcal{A}$ be locally univalent in $\mathbb{D}$. Then $f \in CC$ if and only if*

$$\int_{\theta_1}^{\theta_2} \operatorname{Re}\left(1 + \frac{zf''(z)}{f'(z)}\right) d\theta > -\pi, \quad z = re^{i\theta} \tag{8.2.4}$$

for each r with $0 < r < 1$, and for each pair of real numbers θ_1 and θ_2 with $0 \leqslant \theta_2 - \theta_1 \leqslant 2\pi$.

Proof In view of Theorem 8.2.1, it remains to prove the converse. Assume that f is locally univalent in $\mathbb{D}$ and satisfies (8.2.4). To establish that f is close-to-convex, fix $\rho \in (0, 1)$ and consider

$$v_\rho(\theta) = V(\rho, \theta) = \arg\left(\rho e^{i\theta} f'\left(\rho e^{i\theta}\right)\right) = \theta + \arg\left(f'\left(\rho e^{i\theta}\right)\right).$$

The condition (8.2.4) is equivalent to

$$v_\rho(\theta_1) - v_\rho(\theta_2) < \pi, \quad \theta_1 < \theta_2.$$

Since $f'(z) \neq 0$, $z \in \mathbb{D}$, $\arg\left(f'\left(\rho e^{i\theta}\right)\right)$ must be periodic with period 2π in θ, and

$$v_\rho(\theta + 2\pi) - v_\rho(\theta) = V(\rho, \theta + 2\pi) - V(\rho, \theta) = 2\pi.$$

In view of Lemma 8.2.2, there is a non-decreasing function $w_\rho(\theta)$ such that

$$w_\rho(\theta + 2\pi) - w_\rho(\theta) = 2\pi$$

and

$$\left|w_\rho(\theta) - v_\rho(\theta)\right| \leqslant \frac{\pi}{2}, \quad \theta \in \mathbb{R}. \tag{8.2.5}$$

Now, let $P_\rho(r, \theta)$ be the Poisson kernel for the disk $\mathbb{D}_\rho$ given by

$$P_\rho(r, \theta) = \frac{\rho^2 - r^2}{\rho^2 - 2\rho r \cos\theta + r^2}, \quad r < \rho,$$

and consider the harmonic function b_ρ in $\mathbb{D}_\rho$ given by the Poisson integral

$$\begin{aligned} b_\rho(r, \theta) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} P_\rho(r, \theta - s)\left(w_\rho(s) - s\right) \mathrm{d}s \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} P_\rho(r, t)\left(w_\rho(t + \theta) - (t + \theta)\right) \mathrm{d}t. \end{aligned}$$

Also, let

$$W_\rho(r, \theta) = b_\rho(r, \theta) + \theta.$$

Using the fact that $t\, P_\rho(r, t)$ is an odd function of t, it is elementary to see that

$$W_\rho(r, \theta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P_\rho(r, t) w_\rho(t + \theta) \mathrm{d}t. \tag{8.2.6}$$

From this, it follows that $W_\rho(r, \theta)$ is a non-decreasing function of θ for each fixed $r < \rho$. Indeed, if $\theta_1 < \theta_2$, then

$$W_\rho(r, \theta_2) - W_\rho(r, \theta_1) = \frac{1}{2\pi} \int_{-\pi}^{\pi} P_\rho(r, t)\left(w_\rho(t + \theta_2) - w_\rho(t + \theta_1)\right) \mathrm{d}t \geqslant 0$$

since $w_\rho(\theta)$ is a non-decreasing function of θ. Therefore,

$$\frac{\partial W_\rho}{\partial \theta}(r,\theta) \geqslant 0,$$

where the inequality may be assumed to hold strictly because the derivative is a harmonic function.

On the other hand, from (8.2.5) it follows that

$$\left|V(\rho,\theta) - w_\rho(\theta)\right| \leqslant \frac{\pi}{2}.$$

Since $V(r,\theta) - \theta = \arg f'(re^{i\theta})$ is a (single-valued) harmonic function on $\mathbb{D}_\rho$, the considerations leading to (8.2.6) show that

$$V(r,\theta) = \frac{1}{2\pi}\int_{-\pi}^{\pi} P_\rho(r,t)V(\rho,t+\theta)\mathrm{d}t, \quad r<\rho.$$

For $r<\rho$, this relation and (8.2.6) gives

$$\left|V(r,\theta) - W_\rho(r,\theta)\right| \leqslant \frac{1}{2\pi}\int_{-\pi}^{\pi} P_\rho(r,t)\left|V(\rho,t+\theta) - w_\rho(t+\theta)\right|\mathrm{d}t \leqslant \frac{\pi}{2}.$$

Next, let $g_\rho : \mathbb{D}_\rho \to \mathbb{C}$ be the holomorphic function which satisfies $\operatorname{Re} g_\rho(0) = 0$ and $\operatorname{Im} g_\rho(z) = b_\rho(r,\theta)$ for $z = re^{i\theta}$. Also let

$$h_\rho(z) = \int_0^z e^{g_\rho(\zeta)}d\zeta, \quad z \in \mathbb{D}_\rho.$$

Then h_ρ is holomorphic on $\mathbb{D}_\rho$, $h_\rho(0) = 0$ and $\left|h'_\rho(0)\right| = 1$. Moreover, h_ρ is convex on $\mathbb{D}_\rho$. Indeed, for $z = re^{i\theta}$, $r<\rho$,

$$\operatorname{Re}\left(1 + \frac{zh''_\rho(z)}{h'_\rho(z)}\right) = 1 + \operatorname{Re}\left(zg'_\rho(z)\right) = 1 + \frac{\partial b_\rho}{\partial\theta}(r,\theta) = \frac{\partial W_\rho}{\partial\theta}(r,\theta) > 0.$$

Finally, since

$$V(r,\theta) = \theta + \arg f'(re^{i\theta}),$$

$$W_\rho(r,\theta) = \theta + b_\rho(r,\theta) = \theta + \operatorname{Im} g_\rho(re^{i\theta}) = \theta + \arg h'_\rho(re^{i\theta}),$$

and

$$\left|V(r,\theta) - W_\rho(r,\theta)\right| \leqslant \frac{\pi}{2},$$

it follows that

$$\left|\arg f'(z) - \arg h'_\rho(z)\right| \leqslant \frac{\pi}{2}, \quad z = re^{i\theta},\ r < \rho.$$

Equivalently,

$$\operatorname{Re}\left(\frac{f'(z)}{h'_\rho(z)}\right) \geqslant 0, \quad |z| < \rho.$$

The minimum principle for harmonic functions yields

$$\operatorname{Re}\left(\frac{f'(z)}{h'_\rho(z)}\right) > 0, \quad |z| < \rho.$$

Further, since $h_\rho(0) = 0$, $\left|h'_\rho(0)\right| = 1$ and h_ρ is a convex function, it follows that for each $m \in \mathbb{N}$, the family $\left\langle h_{\rho_n}\right\rangle_{n \geqslant m}$ is a normal family of convex functions on the disk $\mathbb{D}_{\rho_m}$, where $\rho_n = 1 - 1/n$. Therefore, by using Montel's theorem and a diagonal sequence argument, there is a subsequence $\langle h_{\rho_{n_k}}\rangle_{k\in\mathbb{N}}$ that converges uniformly on each disk $\mathbb{D}_\rho$ with $\rho < 1$. The limit function h is convex on $\mathbb{D}$ and satisfies

$$\operatorname{Re}\left(\frac{f'(z)}{h'(z)}\right) \geqslant 0, \quad z \in \mathbb{D}.$$

Again the minimum principle for harmonic functions yields

$$\operatorname{Re}\left(\frac{f'(z)}{h'(z)}\right) > 0, \quad z \in \mathbb{D},$$

and hence f is close-to-convex with respect to h. □

The function f belongs to CC if and only if

$$\int_C \mathrm{d}(\arg(\mathrm{d}f(z))) > -\pi,$$

for each directed arc C of the circle $|z| = r < 1$. This follows from Kaplan's Theorem (Theorem 8.2.3) because

$$\mathrm{d}(\arg(\mathrm{d}f(z))) = \operatorname{Re}\left(1 + \frac{zf''(z)}{f'(z)}\right)$$

on the circle $|z| = r$.

Corollary 8.2.4 *If $f \in \mathcal{A}$ satisfies the inequality*

$$\operatorname{Re}\left(1+\frac{zf''(z)}{f'(z)}\right) > -\frac{1}{2}, \quad z \in \mathbb{D},$$

then f is close-to-convex and hence univalent.

Proof Let $z = re^{i\theta}$. For each r with $0 < r < 1$ and for each pair of real numbers θ_1 and θ_2 with $0 \leqslant \theta_2 - \theta_1 \leqslant 2\pi$,

$$\int_{\theta_1}^{\theta_2} \operatorname{Re}\left(1+\frac{zf''(z)}{f'(z)}\right) d\theta > -\int_{\theta_1}^{\theta_2} \frac{1}{2} d\theta = \frac{\theta_1 - \theta_2}{2} \geqslant -\pi.$$

By Theorem 8.2.3, f is close-to-convex and hence univalent. □

8.3 Kaplan Classes of Type α and β

Many classes in geometric function theory can be unified using the so called Kaplan classes. These classes were first introduced by Sheil-Small in 1979 and later treated by Ruscheweyh in his book [18]. Let $\mathcal{A}_0$ be the class of all analytic functions f defined on the unit disk $\mathbb{D}$ and normalized by $f(0) = 1$. Let $\mathcal{H}$ consists of functions $f \in \mathcal{A}_0$ such that $\operatorname{Re}(e^{i\delta} f(z)) > 0$ for some $\delta \in \mathbb{R}$ and all $z \in \mathbb{D}$. For $\alpha > 0$, define

$$\mathcal{H}^\alpha = \{f^\alpha : f \in \mathcal{H}\}.$$

Note that a function $f \in \mathcal{H}$ is non-vanishing in $\mathbb{D}$, and hence f^α is analytic. For $\gamma \geqslant 0$, let

$$\mathcal{K}(0,\gamma) = \left\{ g \in \mathcal{A}_0 : \operatorname{Re}\frac{zg'(z)}{g(z)} \geqslant -\frac{\gamma}{2} \right\}.$$

The *Kaplan classes of type α and β* [18] are defined by

$$\mathcal{K}(\alpha,\beta) = \mathcal{H}^\alpha \cdot \mathcal{K}(0, \beta - \alpha) \quad \text{if} \quad 0 \leqslant \alpha \leqslant \beta,$$

and

$$\mathcal{K}(\alpha,\beta) = \left\{ g : g(z) = \frac{1}{f(z)},\ f \in \mathcal{K}(\beta,\alpha) \right\} \quad \text{if} \quad 0 \leqslant \beta \leqslant \alpha.$$

Here, $H \cdot G := \{hg : h \in H, g \in G\}$.

These sets are called Kaplan classes due to their close connection with the class of close-to-convex functions $\mathcal{CC}$. In fact

$$f \in \mathcal{CC}_0 \Leftrightarrow f' \in \mathcal{K}(1,3)$$

In addition to this, several other notable relationships exists between Kaplan classes and various subclasses of $\mathcal{S}$. The class of close-to-convex functions of order β, denoted by $\mathcal{CC}_\phi(\beta)$, consists of functions $f \in \mathcal{A}$ such that $|\arg e^{i\phi} zf'(z)/g(z)| \leqslant \beta\pi/2$ for some $g \in \mathcal{ST}$ and $\phi \in \mathbb{R}$. It is easy to verify that

$$f \in \mathcal{CC}_\phi(\beta) \Leftrightarrow f' \in \mathcal{K}(\beta, \beta + 2).$$

Another generalisation, denoted by $\overline{\mathcal{CC}}_\phi(\sigma, \lambda)$, consists of functions $f \in \mathcal{A}$ such that $\operatorname{Re} e^{i\phi} zf'(z)/g(z) > \lambda$ for some $\phi \in \mathbb{R}$ and $g \in \mathcal{ST}(\sigma)$, $0 \leqslant \sigma \leqslant 1$, $z \in \mathbb{D}$. In the case $\lambda = 0$,

$$f \in \overline{\mathcal{CC}}_\phi(\sigma, 0) \Leftrightarrow f' \in \mathcal{K}(1, 3 - 2\sigma).$$

Another class which can be related to Kaplan classes is the class V_k of functions with bounded boundary rotation. A function $f \in \mathcal{A}$ is in V_k, $k \geqslant 2$ (see [5, p. 16]), if there is a constant k such that

$$\int_0^{2\pi} \left| \operatorname{Re} \left(1 + \frac{re^{i\theta} f''(re^{i\theta})}{f'(re^{i\theta})} \right) \right| d\theta \leqslant k\pi.$$

It is known that

$$f \in V_k \Rightarrow f' \in \mathcal{K}(k/2 - 1, k/2 + 1), \quad k > 2$$

and $V_2 = \mathcal{CV}$.

Lemma 8.3.1 *Assume $0 \leqslant \alpha \leqslant \beta$. Then $f' \in \mathcal{K}(\alpha, \beta)$ if and only if there is a function $g \in \mathcal{ST}((2 - \beta + \alpha)/2)$ and a number $\varphi \in \mathbb{R}$ such that*

$$\left| \arg e^{i\varphi} \frac{zf'(z)}{g(z)} \right| \leqslant \frac{\alpha\pi}{2}, \quad z \in \mathbb{D}.$$

Proof Since $\mathcal{K}(\alpha, \beta) = \mathcal{H}^\alpha \cdot \mathcal{K}(0, \beta - \alpha)$, write $f'(z) = h(z)G(z)$ where $h \in \mathcal{H}^\alpha$ and $G \in \mathcal{K}(0, \beta - \alpha)$. Hence,

$$\operatorname{Re} \frac{zG'(z)}{G(z)} \geqslant -\frac{\beta - \alpha}{2}.$$

Define $g(z) = zG(z)$. Then

$$\operatorname{Re} \frac{zg'(z)}{g(z)} = \operatorname{Re} \frac{zG'(z)}{G(z)} + 1 \geqslant \frac{2 - \beta + \alpha}{2}.$$

Thus, $g \in \mathcal{ST}((2 - \beta + \alpha)/2)$, which established the result. □

The radius of convexity for functions f with $f' \in \mathcal{K}(\alpha, \beta)$ is next derived.

Theorem 8.3.2 *Assume $f' \in \mathcal{K}(\alpha, \beta)$, $0 \leqslant \alpha \leqslant \beta$, and $0 \leqslant \gamma < 1$. Then the radius of convexity of order γ of f is given by*

$$r_0 = \frac{2(1-\gamma)}{\alpha + \beta + \sqrt{(\alpha+\beta)^2 - 4(\beta - \alpha - 1 + \gamma)(1-\gamma)}}.$$

Proof Since $f' \in \mathcal{K}(\alpha, \beta)$, by Lemma 8.3.1, there is a function $g \in \mathcal{ST}((2-\beta+\alpha)/2)$ such that the function $h(z) = e^{i\varphi} z f'(z)/g(z)$ satisfies $|\arg(h(z))| < \alpha\pi/2$ or $h^{1/\alpha} \in \mathcal{P}$. A computation shows that

$$1 + \frac{zf''(z)}{f'(z)} = \frac{zh'(z)}{h(z)} + \frac{zg'(z)}{g(z)}. \tag{8.3.1}$$

Now $zg'/g \in \mathcal{P}((2-\beta+\alpha)/2)$, and inequality (4.3.7) in Corollary 4.3.5 shows that

$$\left| \frac{zg'(z)}{g(z)} - \frac{1 + (\beta - \alpha - 1)r^2}{1 - r^2} \right| \leqslant \frac{(\beta - \alpha)r}{1 - r^2}, \quad |z| = r < 1. \tag{8.3.2}$$

Since $h^{1/\alpha} \in \mathcal{P}$, inequality (4.3.5) of Theorem 4.3.4 yields

$$\left| \frac{zh'(z)}{h(z)} \right| \leqslant \frac{2\alpha r}{1 - r^2}, \quad |z| = r < 1. \tag{8.3.3}$$

Using inequalities (8.3.2) and (8.3.3), it follows from (8.3.1) that

$$\left| 1 + \frac{zf''(z)}{f'(z)} - \frac{1 + (\beta - \alpha - 1)r^2}{1 - r^2} \right| \leqslant \frac{(\alpha + \beta)r}{1 - r^2}, \tag{8.3.4}$$

and hence

$$\operatorname{Re}\left(1 + \frac{zf''(z)}{f'(z)}\right) \geqslant \frac{1 - (\alpha + \beta)r + (\beta - \alpha - 1)r^2}{1 - r^2}.$$

Solving the inequality

$$\frac{1 - (\alpha + \beta)r + (\beta - \alpha - 1)r^2}{1 - r^2} = \gamma$$

with respect to r gives r_0. The functions $f'(z) = (1+z)^\alpha/(1-z)^\beta$ and $g(z) = z/(1-z)^{\beta-\alpha}$ show that the result is sharp. □

8.4 Monotonic Coefficients for Close-to-Convexity

This section determines conditions on the real numbers a_n such that the function $f(z) = z + \sum_{n=2}^{\infty} a_n z^n$ is close-to-convex and hence univalent. Recall that by Theorem 1.2.12 a non-constant analytic function f with $\operatorname{Re} f(z) \geqslant 0$ in $\mathbb{D}$ must satisfy $\operatorname{Re} f(z) > 0$ in $\mathbb{D}$.

Theorem 8.4.1 *If $\langle a_n \rangle$ is a sequence of real numbers such that $a_1 = 1$, $\langle n a_n \rangle$ is non-decreasing and has a limit not exceeding 2, then f given by $f(z) = z + \sum_{n=2}^{\infty} a_n z^n$ is analytic and belongs to CC_0.*

Proof Note that $1/n \leqslant a_n \leqslant 2/n$ and $\sum z^n/n$ is convergent in $\mathbb{D}$. Thus, the power series $\sum a_n z^n$ converges in $\mathbb{D}$. Let $b_n = n a_n$ and set

$$F_k(z) = 1 + \sum_{n=2}^{k} (b_n - b_{n-1})\, z^{n-1}, \quad k = 2, 3, \ldots.$$

Since $\langle b_n \rangle$ is non-decreasing,

$$\operatorname{Re} F_k(z) \geqslant 1 - \sum_{n=2}^{k} |b_n - b_{n-1}| = 1 - \sum_{n=2}^{k} (b_n - b_{n-1}) = 2 - b_k \geqslant 0$$

for each k and $z \in \mathbb{D}$, and hence

$$\begin{aligned} \operatorname{Re}\big((1-z) f'(z)\big) &= \operatorname{Re}\left(1 + \sum_{n=2}^{\infty} (b_n - b_{n-1})\, z^{n-1}\right) \\ &= \operatorname{Re}\left(\lim_{k\to\infty} F_k(z)\right) \geqslant 0. \end{aligned}$$

Therefore, f is close-to-convex with respect to g where $g(z) = -\log(1-z)$. □

The hypothesis that $\lim n a_n \leqslant 2$ in Theorem 8.4.1, in general, cannot be replaced by $\lim n a_n \leqslant 2 + p$, where p is a positive real number. For example, suppose that the coefficients of f satisfy $n a_n = 2 + p$, where $p > 0$ and $n \geqslant 2$. Then

$$f'(z) = 1 + \sum_{n=2}^{\infty} n a_n z^{n-1} = 1 + (2+p) \sum_{n=2}^{\infty} z^{n-1} = 1 + (2+p)\frac{z}{1-z}.$$

Since $f'(z) = 0$ for $z = -1/(1+p) \in \mathbb{D}$, f is not univalent in $\mathbb{D}$.

Theorem 8.4.2 *Let $\langle a_{2n-1} \rangle$ be a sequence of real numbers such that $a_1 = 1$ and $\langle (2n-1) a_{2n-1} \rangle$ is non-decreasing with a limit not exceeding 2. Then $f(z) = z + \sum_{n=2}^{\infty} a_{2n-1} z^{2n-1}$ is analytic and belongs to CC_0.*

Proof Since $1/(2n-1) \leqslant a_{2n-1} \leqslant 2/(2n-1)$ and $\sum z^n/(2n-1)$ is convergent in $\mathbb{D}$, it follows that the power series $\sum a_{2n-1} z^{2n-1}$ converges in $\mathbb{D}$. Let $b_n = (2n-1)a_{2n-1}$ and set

$$F_k(z) = 1 + \sum_{n=2}^{k} (b_n - b_{n-1})\, z^{2n-2}, \quad k = 2, 3, \ldots.$$

Then,

$$\operatorname{Re} F_k(z) \geqslant 1 - \sum_{n=2}^{k} |b_n - b_{n-1}| = 1 - \sum_{n=2}^{k} (b_n - b_{n-1}) = 2 - b_k \geqslant 0$$

for each k and $z \in \mathbb{D}$. Hence,

$$\begin{aligned} \operatorname{Re}\left((1-z^2) f'(z)\right) &= \operatorname{Re}\left(1 + \sum_{n=2}^{\infty} (b_n - b_{n-1})\, z^{2n-2}\right) \\ &= \operatorname{Re}\left(\lim_{k\to\infty} F_k(z)\right) \geqslant 0. \end{aligned}$$

Therefore, f is close-to-convex with respect to g where

$$g(z) = \frac{1}{2} \log \frac{1+z}{1-z}. \qquad \square$$

Theorem 8.4.3 *Let $\langle a_n \rangle$ be a sequence of real numbers such that $a_1 = 1$, $a_2 \geqslant 1$ and $\langle (n+1)a_{n+1} - n a_n \rangle$ is non-decreasing with a limit not exceeding 2. Then $f(z) = z + \sum_{n=2}^{\infty} a_n z^n$ is analytic and belongs to CC_0.*

Proof Since

$$(n+1)a_{n+1} - n a_n \geqslant 2a_2 - a_1 \geqslant 1,$$

it inductively follows that $a_n \geqslant 1$. By adding the inequalities obtained by replacing k by $1, 2, 3, \ldots, n$ in

$$(k+1)a_{k+1} - k a_k \leqslant 2$$

leads to $n a_n \leqslant 2n - 1$, or

$$a_n \leqslant 2 - \frac{1}{n} < 2.$$

The inequality $1 \leqslant a_n < 2$ shows that the power series for f converges in $\mathbb{D}$ and hence f is analytic in $\mathbb{D}$.

Let $b_n = (n+1)a_{n+1} - n a_n$ and set

$$F_k(z) = 1 + (b_1 - 1)\, z + \sum_{n=3}^{k} (b_{n-1} - b_{n-2})\, z^{n-1}.$$

The sequence $\langle b_n \rangle$ is non-decreasing and hence

$$\begin{aligned}\operatorname{Re} F_k(z) &\geqslant 1 - |b_1 - 1| - \sum_{n=3}^{k} |b_{n-1} - b_{n-2}| \\ &= 1 - (b_1 - 1) - \sum_{n=3}^{k} (b_{n-1} - b_{n-2}) \\ &= 2 - b_{k-1} \geqslant 0\end{aligned}$$

for each k and $z \in \mathbb{D}$.

If $z \in \mathbb{D}$, then

$$\begin{aligned}(1-z)^2 f'(z) &= 1 + (2a_2 - 2)\, z \\ &\quad + \sum_{n=3}^{\infty} (na_n - 2(n-1)a_{n-1} + (n-2)a_{n-2})\, z^{n-1} \\ &= 1 + (b_1 - 1)\, z \quad + \sum_{n=3}^{\infty} (b_{n-1} - b_{n-2})\, z^{n-1} \\ &= \lim_{k \to \infty} F_k(z).\end{aligned}$$

Since $\operatorname{Re} F_k(z) \geqslant 0$, it follows that

$$\operatorname{Re}\left((1-z)^2 f'(z)\right) \geqslant 0$$

for $z \in \mathbb{D}$. Hence, f is close-to-convex with respect to

$$g(z) = \frac{z}{1-z}. \qquad \square$$

Theorem 8.4.4 *Let $\langle a_n \rangle$ be a sequence of real numbers such that $a_1 = 1$, $a_2 \leqslant 1$ and $\langle (n+1)a_{n+1} - na_n \rangle$ is non-decreasing with a limit not greater than $4a_2 - 2$. Then $f(z) = z + \sum_{n=2}^{\infty} a_n z^n$ is analytic and belongs to CC_0.*

Proof Let $b_n = (n+1)a_{n+1} - na_n$. The assumption implies that $b_n \leqslant 4a_2 - 2$, and so $2a_2 - 1 = b_1 \leqslant 4a_2 - 2$. Thus, $a_2 \geqslant 1/2$. Therefore,

$$b_n \geqslant 2a_2 - 1 \geqslant 0,$$

which implies that $(n+1)a_{n+1} \geqslant na_n$. Hence, $na_n \geqslant 2a_2 \geqslant 1$ or

$$a_n \geqslant \frac{1}{n} > 0.$$

Now $a_2 \leqslant 1$ yields

$$b_n \leqslant 4a_2 - 2 \leqslant 2.$$

Summing $b_k \leqslant 2$ from $k = 1, 2, \ldots, n$ leads to

$$a_n \leqslant 2 - \frac{1}{n} < 2.$$

As $a_n < 2$, the function f is analytic in $\mathbb{D}$.

Let F_k be defined as in Theorem 8.4.3. It remains to show that $\operatorname{Re} F_k(z) \geqslant 0$ for $z \in \mathbb{D}$ to complete the proof. This is now evident from

$$\begin{aligned}\operatorname{Re} F_k(z) &\geqslant 1 - |b_1 - 1| - \sum_{n=3}^{k} |b_{n-1} - b_{n-2}| \\ &= 1 - (1 - b_1) - \sum_{n=3}^{k} (b_{n-1} - b_{n-2}) \\ &= 2b_1 - b_{k-1} = 4a_2 - 2 - b_{k-1} \geqslant 0.\end{aligned}$$

□

Theorem 8.4.5 *Let $\langle a_n \rangle$ be a sequence of real numbers such that $a_1 = 1$, $a_2 \leqslant 1$, $\langle na_n \rangle$ is non-decreasing and $\langle (n+1)a_{n+1} - na_n \rangle$ is non-increasing. Then $f(z) = z + \sum_{n=2}^{\infty} a_n z^n$ is analytic and belongs to CC_0.*

Proof The hypotheses imply that $0 < a_n \leqslant 1$, and hence f is well-defined and analytic in $\mathbb{D}$. The proof is completed by proceeding as in the proof of Theorem 8.4.3. Let $b_n = (n+1)a_{n+1} - na_n$. Since $\langle b_n \rangle$ is a non-increasing sequence of non-negative real numbers, it follows that

$$\begin{aligned}\operatorname{Re} F_k(z) &\geqslant 1 - |b_1 - 1| - \sum_{n=3}^{k} |b_{n-1} - b_{n-2}| \\ &= 1 - (1 - b_1) - \sum_{n=3}^{k} (b_{n-2} - b_{n-1}) \\ &= b_{k-1} \geqslant 0.\end{aligned}$$

□

Theorems 8.4.3–8.4.5 are useful in constructing univalent functions. For example, univalent functions can be obtained through Theorem 8.4.3 in the following way: let $\langle b_n \rangle$ be any non-decreasing sequence of real numbers such that $b_1 \geqslant 1/2$ and $b_n \leqslant 2$ and then recursively define $\langle a_n \rangle$ by $a_1 = 1$ and

$$a_{n+1} = \frac{n}{n+1} a_n + \frac{b_n}{n+1}.$$

The same recursive determination of a_n holds for Theorem 8.4.4, except $0 \leqslant b_1 \leqslant 1$, and $b_n \leqslant 2b_1$. For Theorem 8.4.5 the recursive definition of a_n still holds, where $\langle b_n \rangle$ is a non-increasing sequence of real numbers satisfying $0 \leqslant b_n \leqslant 1$.

The choice $b_n = 2$ for all n in Theorem 8.4.3 yields the function

$$f(z) = \frac{2z}{1-z} + \log(1-z).$$

Other specific choices of b_n in the various theorems yield other simple functions. For example, the choices $b_n = 1$ and $b_n = 0$ in Theorem 8.4.5 again produce the functions $z/(1-z)$ and $-\log(1-z)$. For $b_1 = 1$ and $b_n = 0$ for $n \geqslant 2$ in this theorem yields the function

$$f(z) = -z - 2\log(1-z).$$

8.5 Close-to-Star Functions

Analogous to close-to-convex function is the close-to-star function.

Definition 8.5.1 A function $f \in \mathcal{A}$ is said to be *close-to-star* if there exists $g \in \mathcal{ST}$ and $\alpha \in (-\pi/2, \pi/2)$ such that

$$\operatorname{Re}\left(e^{i\alpha}\frac{f(z)}{g(z)}\right) > 0, \quad z \in \mathbb{D}. \tag{8.5.1}$$

The class of all such functions is denoted by $\mathcal{CS}$. For a fixed $\alpha \in (-\pi/2, \pi/2)$, the subclass $\mathcal{CS}_\alpha$ consists of all functions $f \in \mathcal{CS}$ satisfying (8.5.1).

The class $\mathcal{CS}$ of close-to-star functions bears the same relation with the class $\mathcal{CC}$ of close-to-convex functions in much the same way that starlike functions are related to convex functions. For a function $f \in \mathcal{A}$, one can easily see that $f \in \mathcal{CS}$ if and only if $F(z) = \int_0^z f(t)/t\,dt$ is in $\mathcal{CC}$. If F is analytic, with $F'(z) \neq 0$ for $z \in \mathbb{D}$, then $F \in \mathcal{CC}$ if and only if $f = zF' \in \mathcal{CS}$. This is the Alexander's relation between the classes $\mathcal{CC}$ and $\mathcal{CS}$.

Kaplan's Theorem for close-to-convex functions has the following analogue for close-to-star functions.

Theorem 8.5.2 *Let $f \in \mathcal{A}$ with $f(z) \neq 0$ for $z \neq 0$. The function $f \in \mathcal{CS}$ if and only if*

$$\int_{\theta_1}^{\theta_2} \operatorname{Re}\left(\frac{zf'(z)}{f(z)}\right) d\theta > -\pi, \quad z = re^{i\theta},$$

for each r with $0 < r < 1$ and for each pair of real numbers θ_1 and θ_2 with $0 \leqslant \theta_2 - \theta_1 \leqslant 2\pi$.

The next theorem on coefficient bounds follows from Theorem 8.1.13 and the Alexander's relation between the classes $\mathcal{CC}$ and $\mathcal{CS}$.

Theorem 8.5.3 *If* $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{CS}$, *then*

$$|a_k| \leqslant k^2, \quad k \geqslant 2,$$

with equality for $f(z) = (z + z^2)/(1 - z)^3$.

Proof If $f \in \mathcal{CS}$, then $F(z) = \int_0^z f(t)/t\,\mathrm{d}t = z + \sum_{k=2}^{\infty}(a_k/k)z^k$ belongs to $\mathcal{CC}$ and hence, Theorem 8.1.13 implies $|a_k/k| \leqslant k$ or $|a_k| \leqslant k^2$. The functions f, g defined by

$$f(z) = \frac{z + z^2}{(1 - z)^3}, \quad \text{and} \quad g(z) = \frac{z}{(1 - z)^2}$$

are, respectively, analytic and starlike in $\mathbb{D}$. Also,

$$\operatorname{Re} \frac{f(z)}{g(z)} = \operatorname{Re} \frac{1 + z}{1 - z} > 0.$$

Hence, the function f is close-to-star and $a_k = k^2$ for each $k \geqslant 2$. □

Theorem 8.5.4 (Growth Theorem) *If* $f \in \mathcal{CS}$, *then*

$$\frac{r(1 - r)}{(1 + r)^3} \leqslant |f(z)| \leqslant \frac{r(1 + r)}{(1 - r)^3}, \quad |z| = r < 1.$$

For each $z \in \mathbb{D} \setminus \{0\}$, *equality occurs if and only if* f *is a suitable rotation of* $f(z) = (z + z^2)/(1 - z)^3$.

Proof The function $F(z) = \int_0^z f(t)/t\,\mathrm{d}t$ belongs to $\mathcal{CC} \subset \mathcal{S}$ and $zF'(z) = f(z)$. The results follows from the distortion theorem applied to $F \in \mathcal{S}$. □

Since the coefficients of functions in the class $\mathcal{CS}$ do not satisfy $|a_k| \leqslant k$, it is evident that functions in the class $\mathcal{CS}$ need not be univalent. The radius of univalence is the same as the radius of starlikeness for the class $\mathcal{CS}_0$.

Theorem 8.5.5 *The radius of univalence for the class* $\mathcal{CS}_0$ *is* $\rho_0 = 2 - \sqrt{3}$. *The radius of starlikeness is equal to the radius of univalence.*

Proof If $f \in \mathcal{CS}_0$, then $f(z)/g(z) = p(z)$, where $g \in \mathcal{ST}$ and $p \in \mathcal{P}$. Since $g \in \mathcal{ST}$, it follows that $zg'(z)/g(z)$ is a function with positive real part and therefore, by (4.3.2),

$$\operatorname{Re} \frac{zg'(z)}{g(z)} \geqslant \frac{1 - r}{1 + r}.$$

Since $p \in \mathcal{P}$, (4.3.5) implies

$$\left| \frac{zp'(z)}{p(z)} \right| \leqslant \frac{2r}{1 - r^2}.$$

Therefore,

$$\begin{aligned}\operatorname{Re}\frac{zf'(z)}{f(z)} &= \operatorname{Re}\frac{zg'(z)}{g(z)} + \operatorname{Re}\frac{zp'(z)}{p(z)}\\ &\geqslant \frac{1-r}{1+r} - \frac{2r}{1-r^2} = \frac{1-4r+r^2}{1-r^2} \geqslant \frac{1-4\rho_0+\rho_0^2}{1-\rho_0^2} = 0\end{aligned}$$

for $r < \rho_0 = 2-\sqrt{3}$. Hence, the radius of starlikeness is ρ_0. Sharpness holds for

$$f(z) = \frac{z+z^2}{(1-z)^3}$$

because

$$\frac{zf'(z)}{f(z)} = \frac{1+4z+z^2}{1-z^2}$$

vanishes at $z = -\rho_0$. Since

$$f'(z) = \frac{1+4z+z^2}{(1-z)^4}$$

also vanishes at $z = -\rho_0$, the radius of univalence cannot exceed ρ_0. □

8.6 Symmetric Points

In this section, starlike, convex, and close-to-convex functions with respect to symmetric points are studied. Referring to Example 8.1.12, a function $f \in \mathcal{A}$ is *starlike with respect to symmetric points* if f satisfies

$$\operatorname{Re}\left(\frac{zf'(z)}{f(z)-f(-z)}\right) > 0, \quad z \in \mathbb{D}. \tag{8.6.1}$$

The class of all such functions is denoted by $\mathcal{STS}$. A function $f \in \mathcal{STS}$ also has a geometric definition.

Definition 8.6.1 A function $f \in \mathcal{A}$ is said to be *starlike with respect to symmetric points* if, for every r less than and sufficiently close to 1, and every ζ on $|z| = r < 1$, the angular velocity of f about the point $f(-\zeta)$ is positive at $z = \zeta$ as z traverses the circle $|z| = r$ in the positive direction, that is,

$$\operatorname{Re}\left(\frac{zf'(z)}{f(z)-f(-\zeta)}\right) > 0 \quad \text{for } z = \zeta,\ |\zeta| = r < 1.$$

The following theorem proves that the two definitions are equivalent.

Theorem 8.6.2 *Let $f \in \mathcal{A}$. Then $f \in \mathcal{STS}$ if and only if the inequality (8.6.1) is satisfied.*

Proof Suppose that f is univalent and starlike with respect to symmetric points. Then,

$$\operatorname{Re}\left(\frac{zf'(z)}{f(z)-f(-z)}\right) > 0, \quad |z| = r \tag{8.6.2}$$

for every r less than and sufficiently close to one. Since f is univalent, it follows that $f(z) - f(-z) \neq 0$ for $z \in \mathbb{D} \setminus \{0\}$, and hence the function $zf'(z)/(f(z) - f(-z))$ is analytic in $\mathbb{D}$. The minimum principle for harmonic functions with (8.6.2) shows that

$$\operatorname{Re}\left(\frac{zf'(z)}{f(z)-f(-z)}\right) > 0, \quad |z| \leqslant r,$$

which establishes (8.6.1).

For the converse, suppose that (8.6.1) holds. Then, f is evidently starlike with respect to symmetric points, and therefore it is sufficient to show that f is univalent in $\mathbb{D}$. However, this was already shown in Example 8.1.12. □

Corollary 8.6.3 *If $f \in \mathcal{STS}$, then $F \in \mathcal{A}$ defined by $F(z) = (f(z) - f(-z))/2$ belongs to $\mathcal{ST}$.*

Proof If $f \in \mathcal{STS}$, then the inequality (8.6.1) holds. Replacing z by $-z$ and summing the result to (8.6.1) gives $\operatorname{Re}(zF'(z)/F(z)) > 0$, which shows that F is starlike. □

Corollary 8.6.4 *Every function in the class $\mathcal{STS}$ is close-to-convex and hence univalent.*

Proof Let $f \in \mathcal{STS}$. Then, by Corollary 8.6.3, $F(z) = (f(z) - f(-z))/2$ belongs to $\mathcal{ST}$. Equation (8.6.2) yields $\operatorname{Re}(zf'(z)/F(z)) > 0$. Thus, f is close-to-convex and univalent. □

Geometrically, every convex function is starlike with respect to symmetric points. The analytic definition also implies that odd starlike functions are starlike with respect to symmetric points. If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{CV}$, its coefficients satisfy $|a_n| \leqslant 1$. The following theorem shows the same coefficient bound holds for the larger class $\mathcal{STS}$.

Theorem 8.6.5 *If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{STS}$, then*

$$|a_k| \leqslant 1, \quad k \geqslant 2,$$

with equality occurring for the function $\ell(z) = z/(1 - z)$.

Proof If $f \in \mathcal{STS}$, then

$$p(z) := \frac{2zf'(z)}{f(z) - f(-z)} = 1 + \sum_{k=1}^{\infty} c_k z^k$$

belongs to $\mathcal{P}$. Thus

$$z + \sum_{k=2}^{\infty} k a_k z^k = \left(z + \sum_{l=1}^{\infty} a_{2l+1} z^{2l+1} \right) \left(1 + \sum_{l=1}^{\infty} c_l z^l \right).$$

For even k, equating coefficients of z^k gives

$$k a_k = c_{k-1} + c_{k-3} a_3 + c_{k-5} a_5 + \cdots + c_1 a_{k-1}. \tag{8.6.3}$$

Similarly, for odd k,

$$(k-1) a_k = c_{k-1} + c_{k-3} a_3 + c_{k-5} a_5 + \cdots + c_2 a_{k-2}. \tag{8.6.4}$$

For $k = 2$, equation (8.6.3) reduces to $2a_2 = c_1$, giving $|a_2| \leqslant 1$. Since $|c_k| \leqslant 2$ for all k, the assumption $|a_n| \leqslant 1$ for $n = 2, 3, \ldots, k-1$ implies inductively from (8.6.3) and (8.6.4) that $|a_k| \leqslant 1$. Equality occurs for the function ℓ. □

Since $a_2 = c_1/2$ and $a_3 = c_2/2$,

$$a_3 - \mu a_2^2 = \frac{1}{2} \left(c_2 - \frac{\mu}{2} c_1^2 \right).$$

Applying Theorem 4.5.4 yields the following theorem.

Theorem 8.6.6 *If $f(z) = z + \sum_{n=2}^{\infty} a_n z^n \in \mathcal{STS}$, then*

$$\left| a_3 - \mu a_2^2 \right| \leqslant \max\{1, |\mu - 1|\}, \quad \mu \in \mathbb{C},$$

with equality for the functions $\ell(z) = z/(1-z)$ and $h(z) = z/(1-z^2)$.

By using the coefficients bound $|a_n| \leqslant 1$, it can be shown that

$$|f(z)| \leqslant \frac{r}{1-r} \quad \text{and} \quad |f'(z)| \leqslant \frac{1}{(1-r)^2}.$$

These inequalities are sharp for the function ℓ. However, the sharp lower bound for $|f'(z)|$ is given by

$$\frac{1}{(1+r)^2} \leqslant |f'(z)|, \quad 0 \leqslant |z| = r \leqslant 2 - \sqrt{3}$$

and

$$\frac{4(1-r^2)}{3\sqrt{3}(1+r^2)^2} \leqslant |f'(z)|, \quad 2-\sqrt{3} \leqslant r < 1.$$

The sharp lower bound for $|f(z)|$ is given by

$$\frac{r}{1+r} \leqslant |f(z)|, \quad 0 \leqslant |z| = r \leqslant 2-\sqrt{3}.$$

For a proof, see [12] and for a generalization see [13].

Theorem 8.6.7 *A function $f \in \mathcal{STS}$ if and only if there exists $p \in \mathcal{P}$ satisfying*

$$f(z) = \int_0^z p(\eta) \exp\left(\int_0^\eta \frac{p(\zeta)-p(-\zeta)-2}{2\zeta} d\zeta\right) d\eta. \tag{8.6.5}$$

Proof If $f \in \mathcal{STS}$, then

$$p(z) = \frac{2zf'(z)}{f(z)-f(-z)}$$

belongs to $\mathcal{P}$. Further,

$$p(z)-p(-z) = \frac{2z[f'(z)-f'(-z)]}{f(z)-f(-z)}.$$

Consequently,

$$\begin{aligned}
\exp\left[\frac{1}{2}\int_0^z \frac{p(\zeta)-p(-\zeta)-2}{\zeta} \mathrm{d}\zeta\right] &= \exp\left(\int_0^z \left(\frac{(f'(\zeta)-f'(-\zeta))}{f(\zeta)-f(-\zeta)} - \frac{1}{\zeta}\right) \mathrm{d}\zeta\right) \\
&= \exp\left(\log\left(\frac{f(z)-f(-z)}{z}\right) - \log 2\right) \\
&= \frac{f(z)-f(-z)}{2z}.
\end{aligned}$$

This yields

$$f'(z) = p(z)\frac{f(z) - f(-z)}{2z} = p(z)\exp\left(\int_0^z \frac{p(\zeta) - p(-\zeta) - 2}{2\zeta}\mathrm{d}\zeta\right),$$

and leads to (8.6.5).

Conversely, f given by (8.6.5) satisfies $f(0) = 0$, $f'(0) = 1$, and

$$f(z) - f(-z) = \int_0^z (p(\eta) - p(-\eta))\exp\left(\int_0^\eta \frac{p(\zeta) - p(-\zeta) - 2}{2\zeta}\mathrm{d}\zeta\right)\mathrm{d}\eta.$$

Moreover,

$$\begin{aligned}&\int_0^z (p(\eta) - p(-\eta))\exp\left(\int_0^\eta \frac{p(\zeta) - p(-\zeta) - 2}{2\zeta}\mathrm{d}\zeta\right)\mathrm{d}\eta \\ &= 2z\exp\left(\int_0^z \frac{p(\zeta) - p(-\zeta) - 2}{2\zeta}\mathrm{d}\zeta\right),\end{aligned}$$

since both sides vanish at the origin and have identical derivatives. Thus,

$$f(z) - f(-z) = 2z\exp\left(\int_0^z \frac{p(\eta) - p(-\eta) - 2}{2\eta}\mathrm{d}\eta\right).$$

From (8.6.5),

$$\begin{aligned}2zf'(z) &= 2zp(z)\exp\left(\int_0^z \frac{p(\zeta) - p(-\zeta) - 2}{2\zeta}\mathrm{d}\zeta\right) \\ &= p(z)(f(z) - f(-z)).\end{aligned}$$

Therefore,

$$\mathrm{Re}\,\frac{2zf'(z)}{f(z) - f(-z)} = \mathrm{Re}\,p(z) > 0$$

which shows that $f \in \mathcal{ST}\mathcal{S}$. □

The following is the convex analogue of starlikeness with respect to symmetric points.

Definition 8.6.8 A function $f \in \mathcal{A}$ is said to be *convex with respect to symmetric points* if, for every $r \to 1$ and every ξ on $|z| = r < 1$, the following relation holds:

$$\operatorname{Re}\left(\frac{z\left(zf'(z)\right)'}{zf'(z)+\xi f'(-\xi)}\right)>0 \quad \text{for } z=\xi,\ |\xi|=r<1.$$

The class of all such functions is denoted by $\mathcal{CVS}$.

Theorem 8.6.9 *Let $f\in\mathcal{A}$. Then $f\in\mathcal{CVS}$ if and only if*

$$\operatorname{Re}\left(\frac{\left(zf'(z)\right)'}{(f(z)-f(-z))'}\right)>0, \quad z\in\mathbb{D}. \tag{8.6.6}$$

Proof Suppose that (8.6.6) holds. Then the function f is convex with respect to symmetric points. Therefore, it is sufficient to show that $f\in\mathcal{S}$. Summing (8.6.6) with the result obtained by substituting $-z$ for z in (8.6.6), it follows that the function

$$g(z)=\frac{f(z)-f(-z)}{2}$$

is convex univalent in $\mathbb{D}$ and therefore starlike. The inequality (8.6.6) leads to

$$\operatorname{Re}\left(\frac{(zf'(z))'}{g'(z)}\right)>0.$$

By applying Theorem 5.4.8, it follows that

$$\operatorname{Re}\left(\frac{zf'(z)}{g(z)}\right)>0,$$

and hence f belongs to CC_0. This proves that f is univalent.

The necessary part is similarly proved as in Theorem 8.6.2. □

Corollary 8.6.10 *If $f\in\mathcal{CVS}$, then $F\in\mathcal{A}$ defined by $F(z)=(f(z)-f(-z))/2$ belongs to $\mathcal{CV}$.*

Proof The proof is similar to that of Corollary 8.6.3. Indeed, the first part of the proof of the previous theorem has shown this fact. If $f\in\mathcal{CVS}$, then (8.6.6) holds. Replacing z by $-z$ in (8.6.6) and summing the resulting inequality to (8.6.6) yields $\operatorname{Re}((zF')'(z)/F'(z))>0$, which shows that F is convex. □

Corollary 8.6.11 *A function $f\in\mathcal{CVS}$ if and only if $zf'\in\mathcal{ST}\mathcal{S}$.*

Proof This follows from Theorem 8.6.2 and Theorem 8.6.9. □

We next find the bound for the nth coefficient for functions in the class $\mathcal{CVS}$.

Corollary 8.6.12 *If $f(z)=z+\sum_{k=2}^{\infty}a_kz^k\in\mathcal{CVS}$, then*

$$|a_k|\leqslant 1/k, \quad k\geqslant 2,$$

with equality for

$$f(z)=-\log(1-z).$$

Proof By Corollary 8.6.11, $zf' \in \mathcal{STS}$. Applying Theorem 8.6.5 to zf' gives $|ka_k| \leqslant 1$, and hence $|a_k| \leqslant 1/k$. The result is sharp for the given function f as $a_k = 1/k$. □

Since $f \in \mathcal{CVS}$ if and only if $zf' \in \mathcal{STS}$, Theorem 8.6.7 readily leads to the following result.

Corollary 8.6.13 *A function $f \in \mathcal{CVS}$ if and only if there exists $p \in \mathcal{P}$ satisfying*

$$zf'(z) = \int_0^z p(\eta) \exp\left(\frac{1}{2}\int_0^\eta \frac{p(\zeta) - p(-\zeta) - 2}{\zeta} d\zeta\right) d\eta.$$

The following definition introduces the notation for close-to-convexity with respect to symmetric points.

Definition 8.6.14 A function $f \in \mathcal{A}$ is said to be *close-to-convex with respect to symmetric points* if there exists $g \in \mathcal{CVS}$ such that

$$\operatorname{Re}\left(\frac{f'(z)}{g'(z) + g'(-z)}\right) > 0, \quad z \in \mathbb{D}. \tag{8.6.7}$$

The class of all such functions is denoted by $\mathcal{CCS}$.

In view of Corollary 8.6.11, a function f belongs to $\mathcal{CCS}$ if there exists a function $h \in \mathcal{STS}$ such that

$$\operatorname{Re}\left(\frac{zf'(z)}{h(z) - h(-z)}\right) > 0, \quad z \in \mathbb{D}. \tag{8.6.8}$$

Proposition 8.6.15 *Every close-to-convex function with respect to symmetric points is close-to-convex and hence univalent.*

Proof Let $f \in \mathcal{CCS}$. Then (8.6.7) holds for some $g \in \mathcal{CVS}$. By Corollary 8.6.10, the function $G(z) = (1/2)(g(z) - g(-z))$ belongs to $\mathcal{CV}$. Thus (8.6.7) leads to $\operatorname{Re}(f'(z)/G'(z)) > 0$ which shows that f is close-to-convex and univalent. □

The class $\mathcal{CCS}$ contains several well-known subclasses of univalent functions, including convex functions with respect to symmetric points, starlike functions with respect to symmetric points, and functions whose derivatives have positive real parts in $\mathbb{D}$.

Coefficient bounds for functions in the class $\mathcal{CCS}$ are established in the following theorem.

Theorem 8.6.16 *If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k \in \mathcal{CCS}$, then $|a_k| \leqslant 1$ for $k \geqslant 2$ with equality for the function $\ell_\varepsilon(z) = z/(1 + \varepsilon z)$ where $|\varepsilon| = 1$.*

Proof Let $f \in CCS$. Choose a function $h \in \mathcal{ST}\mathcal{S}$ with series expansion $h(z) = z + \sum_{n=2}^{\infty} b_n z^n$ such that (8.6.8) is satisfied. Since $h \in \mathcal{ST}\mathcal{S}$, Theorem 8.6.5 implies that $|b_k| \leqslant 1$ for all $k \geqslant 2$. By (8.6.8), the function

$$F(z) = \frac{2zf'(z)}{h(z) - h(-z)} \tag{8.6.9}$$

satisfies $\operatorname{Re} F(z) > 0$ for $z \in \mathbb{D}$, $F(0) = 1$. Consequently,

$$F(z) = 1 + \sum_{k=1}^{\infty} c_k z^k,$$

and the estimate $|c_k| \leqslant 2$ holds for all $k \geqslant 1$. Substituting the series expansions of F, f and h into (8.6.9) yields

$$z + \sum_{k=2}^{\infty} k a_k z^k = \left(z + \sum_{l=1}^{\infty} b_{2l+1} z^{2l+1}\right)\left(1 + \sum_{l=1}^{\infty} c_l z^l\right).$$

For even k, equating coefficients of z^k gives

$$k a_k = c_{k-1} + c_{k-3} b_3 + c_{k-5} b_5 + \cdots + c_1 b_{k-1}. \tag{8.6.10}$$

Similarly, for odd k,

$$k a_k = c_{k-1} + c_{k-3} b_3 + c_{k-5} b_5 + \cdots + c_2 b_{k-2} + b_k. \tag{8.6.11}$$

From (8.6.10), it follows that $2a_2 = c_1$, and hence $|a_2| \leqslant 1$. Similarly, (8.6.11) implies that $3a_3 = c_2 + b_3$, and hence $3|a_3| \leqslant |c_2| + |b_3| \leqslant 3$, which shows that $|a_3| \leqslant 1$.

By assuming $|a_n| \leqslant 1$ for $n = 2, 3, \ldots, k-1$, it follows by induction from (8.6.10) and (8.6.11) that $|a_k| \leqslant 1$ for all $k \geqslant 2$. The bound is sharp for the function ℓ_ε. □

Exercises

1. Show that the image of the function f defined in Example 8.1.4 is the whole complex plane minus a vertical slit.
2. For a function $f \in \mathcal{R}$, show that the following sharp inequalities holds:

$$|a_n| \leqslant \frac{2}{n} \quad \text{for } n = 2, 3, \ldots,$$
$$\frac{1-|z|}{1+|z|} \leqslant \operatorname{Re} f'(z) \leqslant \left|f'(z)\right| \leqslant \frac{1+|z|}{1-|z|},$$
$$-|z| + 2\log(1+|z|) \leqslant |f(z)| \leqslant -|z| - 2\log(1-|z|).$$

3. For a function $f \in \mathcal{R}$, show that $f(\mathbb{D}) \supset \mathbb{D}_{2\log 2-1}$.
4. Show that the radius of convexity of the class $\mathcal{R}$ is $\sqrt{2}-1$.
5. If $f \in \mathcal{A}$ satisfies $|f'(z)-1| < \alpha|f'(z)+1|$ for some α with $0 < \alpha \leqslant 1$, show that $f'(z) \in \mathbb{D}((1+\alpha^2)/(1-\alpha^2); 2\alpha/(1-\alpha^2))$. If $f(z) = z + \sum_{k=2}^{\infty} a_k z^k$, show that $|a_k| \leqslant 2\alpha k$.
6. Show that the following nine functions with integer coefficients

$$z, \frac{z}{1 \pm z}, \frac{z}{1 \pm z^2}, \frac{z}{(1 \pm z)^2}, \frac{z}{1 \pm z + z^2},$$

are starlike univalent and hence obtain sufficient conditions for close-to-convexity for each of these functions.

7. A half-integer is half of an integer. Suppose that all the coefficients a_n of a function f in $\mathcal{S}$ are half-integers. Show that f is either one of the nine functions in the previous exercise or else one of the following twelve functions:

$$z \pm \frac{z^2}{2}, \frac{z(2 \pm z)}{2(1 \pm z)}, \frac{z\left(2 \pm z^2\right)}{2\left(1 \pm z^2\right)}, \frac{z(2 \pm z)}{2\left(1 - z^2\right)}, \frac{z(2 \pm z)}{2(1 \pm z)^2}, \frac{z\left(2 \pm z + z^2\right)}{2\left(1 \pm z + z^2\right)}.$$

Show that all these functions except the last pair are close-to-convex.

8. Show that the function $\int_0^z \mathrm{e}^{-\zeta^2} d\zeta$ is univalent and close-to-convex for $|z| < R$, where R is the largest positive root of the equation: $\arctan\sqrt{4R^4-1} - \sqrt{4R^4-1} = -\pi$.
9. If $f \in \mathcal{A}$ satisfies $\operatorname{Re}(z^2 f'(z)/(g(z)g(-z))) < 0$ for some starlike function g of order 1/2, show that f is close-to-convex. Prove that the coefficients of f are bounded by 1. Show further that f satisfies the sharp inequality

$$\frac{1-r}{(1+r)(1+r^2)} \leqslant |f'(z)| \leqslant \frac{1}{(1-r)^2}.$$

Determine the corresponding bound for $|f(z)|$.

10. Show that the starlike functions with respect to conjugate points and symmetric conjugate points are close-to-convex and univalent.
11. Show that every odd starlike function is starlike with respect to symmetric points.
12. Show that every starlike function with real coefficients is starlike with respect to conjugate points. Prove a similar result for starlike functions with respect to symmetric conjugate points.
13. Show that functions f and g in the class $\mathcal{A}$ are mutually adjoint if there are functions p, q in the class $\mathcal{P}$ satisfying

$$f(z) = \int_0^z p(\eta) \exp\left(\frac{1}{2}\int_0^\eta \frac{p(\zeta)+q(\zeta)-2}{\zeta}\mathrm{d}\zeta\right)\mathrm{d}\eta$$

and

$$g(z) = \int_0^z q(\eta) \exp\left(\frac{1}{2}\int_0^\eta \frac{p(\zeta)+q(\zeta)-2}{\zeta}\mathrm{d}\zeta\right)\mathrm{d}\eta.$$

14. If f is starlike, it is mutually adjoint to itself. If f is starlike with respect to symmetric points, then it is mutually adjoint with respect to $-f(-z)$. Hence obtain the integral representation from the previous exercise for starlike functions, and starlike functions with respect to symmetric points.

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[1] The first two sections in this chapter are based on Kaplan [7] and the book by Duren. The papers of [14, 20] elucidate the Noshiro-Warschawski theorem. Section 8.3 is grounded on [15], Sect. 8.4 on MacGregor [11], and Sect. 8.5 on [16]. Theorem 8.5.5 is a result in [10]. The section on symmetric points is based on [19] and [2]. Exercises 2–4 are taken from MacGregor [9], Exercise 5 from [1], Exercises 6 and 7 are, respectively, from [3] and [6], Exercise 8 from [17], Exercise 9 from [4] and finally Exercise 13 from [8].

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